

Power System Transient Stability Analysis

Using the Transient Energy Function Method

A. A. Fouad/Vijay Vittal

**POWER SYSTEM
TRANSIENT STABILITY ANALYSIS
USING THE TRANSIENT ENERGY
FUNCTION METHOD**

A. A. Fouad

Vijay Vittal

*Department of Electrical Engineering
and Computer Engineering
Iowa State University*

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*If I have seen further it is by standing
upon the shoulders of Giants.*

Sir Isaac Newton

The authors dedicate this book to the eminent scholars whose pioneering
work has been a source of inspiration and guidance.

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PREFACE

This book details the development and application of the transient energy function (TEF) method as a tool for power system transient stability assessment. The need for such a manuscript stems from the extensive and diverse literature on the topic and mainly from the increased interest in the TEF method by the electric utility industry and research groups.

The contents of the book have been selected to meet the needs of a diverse range of groups interested in this topic. They include university faculty and graduate students working in power system direct stability analysis, engineers in research organizations involved in research and development projects on power system dynamics, and electric utility engineers who are interested in the use of the TEF method as a tool for dynamic security assessment.

The focus of the book is exclusively on the TEF method and its application to power systems. This choice is guided by the following reasons. First, good references that survey various techniques of direct power system transient stability analysis are currently available. Second, transient stability analysis has become very complex with regards to the problems studied, the type of answers sought, and the kind of issues dealt with. Finally, in the last decade, the TEF method has successfully dealt with these problems and the associated issues.

The subject matter presented in the book provides both the analytical foundations of the TEF method and the various practical issues which arise in the application of the method to analyze power systems. The treatment provides a blend of system-theoretic approach and reasoning based on time simulation. The interested reader is then guided toward further material on a given topic. The material presented acquaints the reader with the state of the art in the TEF method development as of 1990.

A significant portion of the material presented in the book represents contributions from sponsored projects in which the authors have had personal involvement. These included research and development efforts which were generally supported by funding agencies such as the Electric Power Research Institute, the U.S. National Science Foundation, and by Iowa State University. Since these projects dealt with problems encountered in segments of the North American interconnected network, engineers of the concerned utilities have had substantial involvement. The authors wish to acknowledge the significant contributions made by the engineers of Ontario Hydro and British Columbia Hydro to the development and testing of the TEF method as a tool for direct stability analysis.

This book would not have been possible without the help of the students in the Power System Dynamics Group at Iowa State University. In addition to participating in the various research projects on the TEF method, they were very helpful in the painstaking task of data preparation and verification of results. They were also generous with their comments and critique of the first draft of the book. To them, the authors wish to express their thanks and gratitude.

Finally, the authors wish to thank Ms. Gloria Oberender for the excellent job of typing this rather difficult manuscript.

Ames, Iowa

TABLE OF SYMBOLS AND PARAMETERS

Symbol	Variable
Network Parameters	
i_k, I_k	Current injection at node k
v_k, V_k	Voltage at a node or a bus k
d, q (subscript)	Direct and quadrature axes components of a variable, respectively
E_i / δ_i	Internal generator EMF in magnitude and angle of generator i (with respect to a synchronous reference frame)
P, Q	Real and reactive powers, respectively
Y	Short-circuit admittance matrix (usually expressed to the internal generator nodes)
Y_{ij}	The ij th element of Y
G_{ij}, B_{ij}	Real and imaginary components of Y_{ij}
C_{ij}, D_{ij}	$E_i E_j B_{ij}, E_i E_j G_{ij}$
δ_{ij}	$\left[\delta_i - \delta_j \right]$
ω	Angular velocity $\left[= \frac{d\delta}{dt} \right]$
ω_R	Reference or base value of ω
$*$ (superscript)	Conjugate of a variable
Center of Inertia Variables	
COI	Center of inertia
δ_o	Center of inertia angle

ω_o	Center of inertia angular speed $\left[= \dot{\delta}_o \right]$
θ_i	Angle of generator i with respect to the COI $\left[= \delta_i - \delta_o \right]$
$\tilde{\omega}_i$	$= \frac{d\theta_i}{dt}$
P_{COI}	COI accelerating power

Generator Parameters

x'_d, x'_q	d -and q -axes transient reactances
H	Stored kinetic energy, per MVA, at rated speed
M	Moment of inertia $\left[= \frac{2H}{\omega_R} \right]$
x_d, x_q	Synchronous d -and q -axes reactances
τ'_{do}, τ'_{qo}	Open circuit d -and q -axes time constants
E'	Internal generator EMF (two-axis model)
E'_d, E'_q	d -and q -axes components of E'
E_{FD}	Field voltage converted to an equivalent stator EMF
P_m	Mechanical power
P_e	Electrical power
I_{GL}	Current injection at generator node due to nonlinear loads
D	Damping factor in per unit (pu)

Transient Energy Function Variables

$\theta^{cl}, \tilde{\omega}^{cl}$	Angle and speed at fault clearing (COI frame)
SEP, θ^s	Stable equilibrium point (usually for the postdisturbance network)

UEP, θ^u	Unstable equilibrium point
V	Transient energy function (TEF)
V_{PE} , V_{KE}	Transient potential energy, transient kinetic energy
cr, sys (subscripts)	Refer to the critical group of generators and the rest of the generators in the system respectively
$corr$ (subscripts)	Corrected (usually used with V_{KE})
ΔV , ΔV_{PE}	Transient energy margin, potential energy margin
ΔV_n	Normalized energy margin ($\Delta V / V_{KEcorr}$)
MOD	Mode of disturbance; identifies the severely disturbed generators, usually their angles in the UEP are advanced ($> 90^\circ$)
I_{ij}	A term in V obtained from a linear trajectory approximation of a path-dependent integral

Chapter 1

POWER SYSTEM TRANSIENT STABILITY

1.1 INTRODUCTION

Power system stability deals with the power system's response to disturbances. The period of interest is the transient period before the new steady-state conditions are reached. Before we define these terms, it is important to give the context in which power system stability studies are made. It is one of the important aspects of power system reliability.

The electric utility industry in North America, in the words of its North American Electric Reliability Council (NERC), uses reliability in a bulk electric system to indicate "the degree to which the performance of the elements of that system results in electricity being delivered to customers within accepted standards and in the amount desired. The degree of reliability may be measured by the frequency, duration, and magnitude of adverse effects on the electric supply."¹ The council also suggests that reliability can be addressed by considering two basic and functional aspects of the bulk electric system—adequacy and security. Power system stability is among the concerns in power system security.

Again quoting NERC, "Security is the ability of the bulk electric system to withstand sudden disturbances such as electric short circuits or unanticipated loss of system components." This rather important concern includes, among others, that following a disturbance stability is maintained. It is time now to define these terms.

¹ This statement is to be found inside the cover page of many of the North American Electric Reliability Council publications.

1.1.1 Definitions

A task force of the Power System Engineering Committee of the Power Engineering Society of IEEE has suggested the definitions given below [1]. They are endorsed, and recommended, for use by the society.

A disturbance is a sudden change in an operating condition or an operating parameter of the power system. A disturbance is considered small when linearization of the system equations, for the purpose of analysis, can be justified. When the linearization of the system equations cannot be justified, the disturbance is considered large.

The power system is considered at steady state when the operating parameters are considered constant for the purpose of analysis.

When a disturbance occurs in a power system initially considered at steady state, stability deals with whether an *acceptable* steady-state condition can be reached, following the transient. If the disturbance is large, the stability of concern is called "transient stability." If the disturbance is small, the stability of concern is "steady-state stability."

Since transient stability is the subject of concern in this book, we note that we are dealing with a phenomenon described by a set of nonlinear differential and algebraic equations.

1.2 CONVENTIONAL TRANSIENT STABILITY STUDIES

In this section we briefly describe how a "conventional" transient stability study is conducted. At this point we must stress that the equations describing the dynamic behavior of a power system are highly nonlinear. These equations primarily describe the dynamic behavior of the synchronous generators in the system. Of particular interest is the behavior of such generator parameters as rotor angles, internal electromotive forces (EMFs), terminal voltages, currents, etc. These, in turn, determine the behavior of other network parameters such as the voltage at key buses, power and reactive power flow in

transmission lines, and so on. Other questions investigated in such studies are the operation of various controls (used as stability aids).

From the above, we can see that a transient stability study is often more than an investigation of whether the synchronous generators, following the occurrence of the disturbance, will remain in synchronism. It can be a general-purpose transient analysis, in which the "quality" of the dynamic system behavior is investigated. The transient period of primary interest is the electromechanical transient, usually lasting up to a few seconds in duration. If growing oscillations are of concern, or if the behavior of special controls is of interest, a longer transient period may be covered in the study.

A transient stability study is conducted for a given initial network configuration and system operating condition and for a particular disturbance sequence. For example, if the disturbance is a fault, its type, location, and duration and how that fault is cleared (i.e., the post disturbance network) must be specified. The system data for this scenario are collected, this includes generators' parameters, network impedances, loads, control parameters and settings, various constraints and limits, relay settings, etc. All this information must be provided to the transient stability analysis software.

The software, with the proper instructions, then analyzes the system by solving the system equations to obtain the resulting dynamic behavior. This usually includes the time response of variables and parameters, the operation of various controls, and similar pertinent information. The output is available to the user (i.e., the analyst); it can be printed on paper or microfiche or displayed graphically—part of it, or all of it—as a function of time. Other important data, e.g., relay operation, can be extracted or derived from it. However, general conclusions about the quality of the dynamic system response must be arrived at or inferred by the user.

At this point it must be clearly stated that the results obtained, from a transient stability run, are valid only for the particular scenario analyzed. Any change in the data,

in the initial operating condition, or in the transient stability scenario, will require repeating the stability run. This could mean a change in any or combinations of network configuration (pre, during, or postdisturbance); initial loading and/or flows; the location, duration and type of disturbance; any parameter which significantly influences the results (e.g., excitation control parameter); and many others. Therefore, a transient stability run by itself is seldom very meaningful. A series of runs are usually needed to "do the job," i.e., to satisfy the engineer that the bulk power supply network being analyzed is being properly planned or will be safely operated.

With the enormous number of possible transient stability scenarios to be analyzed in a large interconnected power network, a judicious choice of the credible scenarios to be analyzed is among the important decisions to be made by the analyst. This choice has become increasingly difficult in today's environment.

1.3 REASONS FOR CONDUCTING TRANSIENT STABILITY STUDIES

It has already been pointed out that transient stability studies represent an important aspect of power system reliability. Further elaboration is in order. There are reliability criteria, established by the appropriate reliability councils, which must be satisfied. Among other things the criteria ascertain that the bulk power supply in North America is planned and operated with security in mind. This means that if specified disturbances occur, no serious widespread interruptions (called cascading outages) would result. The reliability councils determine the type of disturbances which the power networks must withstand, and the conditions for which the system is to be tested [2]. Until recently, transient stability studies were conducted in connection with planning new bulk power supply facilities, i.e., new generating units or new high-voltage (HV) transmission lines. This also applies to major modification of the bulk power supply and design of special stability controls (e.g., generation tripping, fast valving, and braking resistors).

A variety of scenarios would then be selected (by the planning engineers) to test the strength of the network and to ascertain that the reliability criteria, which apply to dynamic system performance, are met. The robust transmission network which then existed essentially guaranteed that these criteria were met, at all times, when the system was operated.

Since the mid-1970s, we have witnessed a significant increase in the use of transient stability studies, to ascertain that the existing bulk power supply facilities are properly utilized. Such studies, therefore, support system operating functions, and are often conducted by engineers in system operation (operations planning groups). The purpose of these studies can be to check that a given system configuration and/or operating condition meets the reliability criteria, to check (or change) relay settings, and to set the parameters for special stability control devices and check their settings.

While the above represent the vast majority of transient stability studies conducted by the industry, they are by no means the only ones. Other special transient stability studies made are to test the system when subjected to very severe (possible but improbable) disturbances, to study and analyze a known system upset, and to devise or analyze system restoration plans.

These considerations mean that, for transient stability-limited networks, numerous transient stability studies (thousands) have to be made to find the answers sought regarding prevention of cascading outages. The answers, whether they concern system design or system operation, are translated into such things as proper operation of controls and of protective equipment, appropriate remedial actions to be taken when needed, establish safe operating limits in all (credible) situations, adequacy of sources of power and reactive power when needed, and so on. Therefore, they are part of the decision-making process: by the management of the operating entities and by regulators. In a

stability-limited network, the dollar savings can be considerable (if economic loading can be increased safely), and the consequences of judgment errors can be serious.

1.4 CHANGING CONDITIONS IN THE NORTH AMERICAN INTERCONNECTION

There are many factors which now make transient stability of daily concern in many parts of the North American interconnected network. Among these are trends which have been with us for some time, while other factors are more recent. Some of the important factors are

1. Power generation trends which adversely affect system stability, namely, increased size units, lower inertia constant, and higher short-circuit ratio.
2. Fewer HV transmission lines are being built because of right-of-way restrictions, the regulatory process, and the climate of the capital costs as well as cost of money being unfavorable.
3. Heavier use of existing transmission facilities because of policy to conserve oil and natural gas, causing limited operation of some power plants, and availability of remote large blocks of economy power and their transmission to load center over high-capacity transmission circuits.

Accompanying the heavier use of the HV transmission network we note such adverse trends as more dependence on controls. Examples of such controls are faster, high-ceiling exciters; power system stabilizers; fast-valving, generation-tripping, braking resistors; new switching arrangements, etc. We also note that the competing requirements of computer control of many functions (e.g., in power plants) require delicate balancing, which can be easily upset when system disturbances occur. Other adverse trends to system stability are that some transmission lines which had been built primarily for reliability (i.e., for emergency use) are now often used for normal operation, and that we now have much larger contingencies to deal with, e.g., loss of high-capacity transmission or generation.

These trends have resulted in transmission-limited power networks in several areas of North America; some are stability-limited.

1.5 STABILITY IMPLICATIONS OF CHANGING CONDITIONS

Today's transient stability studies are often quite complex. They involve large networks, typically including the networks of several utilities extending over large geographical areas. Taking into account such complexities as designing and checking the operation of controls and protective relays, analysis of the response of key plants and/or loads, and of the adequacy of voltage support, etc., it becomes evident that the concern with transient stability can be a major undertaking. The number of variables that are characteristic to a particular issue, the number of key parameters which influence the outcome, the number of weak points in the system, all these can be quite large. Among the key issues are the following:

1. Transient analysis becomes complex in range and scope. To capture the important features of the transient behavior, the size of the system to be analyzed may have to be large; a longer transient period may need to be examined; and a variety of dynamic problems may need to be analyzed.
2. A power network stressed by heavier loading tends to exhibit different dynamic characteristics and problems. Among them are the possibility of occurrence of system splits involving large areas with many plants. This system breakup, following a disturbance, may take place away from the location of the disturbance. Its location may not be readily predictable. Power system damping, which tends to be inherently weak, becomes even weaker.

As indicated, among the consequences of heavier transmission loading is that some parts of the North American interconnection have become stability-limited. This trend has made transient stability of primary concern to the system operator. For stability-limited power systems, transient stability studies are made to establish stability-limited conditions. Examples of such conditions are the loading of a critical

power plant, the power flow on a key transmission interface, the voltage at an important bus (e.g., at a nuclear power plant), etc. Conditions at these key elements are monitored in real time to make certain that the established limits are obtained with off-line studies. This requires numerous stability studies to obtain all the needed data. The problem occurs when conditions are encountered, which is often the case nowadays, for which stability limits had not been computed. Further discussion of these conditions is in order.

Until a few years ago, power system analysis had been primarily the domain of the system planner (i.e., system designer) who planned and built robust power networks; normally and in the vast majority of the situations the systems were operated well below their established limits. Relatively few studies (few by today's standards) were needed to establish the limits. What we are witnessing is that heavily loaded and stressed networks are now commonplace. The importance of capturing the conditions that prevail in the system at various times has become evident [3]. That is the reason why assumptions on system conditions, on models that capture the phenomena, etc., are important. Wrong assumptions, wrong models, and wrong system conditions by analysts are no longer acceptable by the system operator. Therefore, when conditions are encountered for which stability-limited conditions had not been derived by off-line studies, conservative assumptions may have to be made to obtain a "safe" operating limit which does not endanger the power system security by causing cascading outages when disturbances occur. The alternative is to have a scheme by which on-line derivation of transient stability limits are made. The tools for such a scheme are just emerging. Thus, the changing conditions have also had an impact on the requirements of the tools of stability analysis.

Obtaining stability limits using time simulation requires solving for the generator angles as well as other variables, at various time instants to examine the system behavior and to determine if it is acceptable. By adjusting critical system parameters, the process is repeated to determine the limits. This procedure is time consuming. State-of-the-art time

simulation techniques would not permit on-line computation of stability limits in the time frame required for the stability monitoring scheme. The transient energy function (TEF) method, which is the subject of this book, has the potential of meeting the desired specification for on-line derivation of transient stability limits. By using a combination of stability assessment and sensitivity techniques, the limits can be directly computed.

While the changing power system conditions seem to focus our attention on the shortcomings of existing tools of analysis, they also set high standards of performance which a new transient stability tool must meet. It must deal not only with the complex issues outlined in the preceding sections, but it must also match the performance given by the well-tested tool now in use: in accuracy and reliability of results and the speed with which a solution can be obtained.

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Chapter 2

THE POWER SYSTEM MODEL

2.1 THE POWER NETWORK

In power system stability analysis the key mathematical equations are those which describe the dynamic behavior of the synchronous generators. The power system model used, therefore, is that which emphasizes modeling of the important torques influencing the motion of the generators' rotors and their associated controls. The rest of the power network is modeled in sufficient detail only to the extent that it influences the electrical torques of the generators. Occasionally the power network may contain important system components which may be substantially affected by the dynamic behavior of the rotors of the synchronous generators during the transient. They, in turn, would be included in the power network model, and their behavior is kept track of during the transient. This view is reflected in the power system model presented in this chapter. A schematic diagram of the power network is shown in Fig. 2.1.

The nodes shown in Fig. 2.1 are the nodes of interest. They include n generator terminal nodes (nodes 1, 2, ..., n) and m additional nodes of interest which are usually load buses (nodes $n + 1$, $n + 2$, ..., $n + m$). The latter network nodes may include some nodes the identities of which are retained for special purposes, e.g., their voltages may be of interest. All other network nodes are not of interest. They can be, or are assumed to have been, eliminated and included in the network admittance matrix containing the retained $(n + m)$ nodes.

The generators shown in Fig. 2.1 are each represented by a terminal voltage V , an internal EMF E/δ , and a reactance x' in between. The internal EMF will depend on the

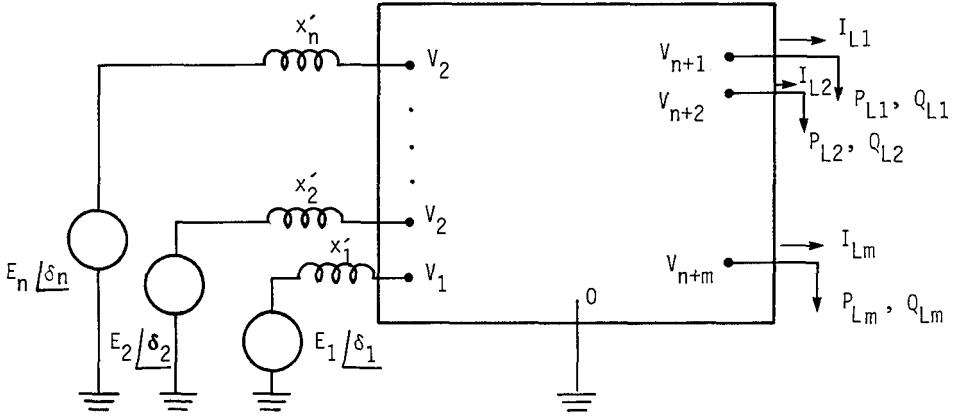


Figure 2.1 Schematic diagram of a power network.

model used. The loads are shown to be drawing certain powers and reactive powers.

Again, these loads may be represented by different models. The angles of the network node voltages shown in Fig. 2.1 are to a synchronous reference frame.

As is usually the case with all transient stability studies, the power network shown in Fig. 2.1 is the positive sequence network. If the disturbance is a fault, it is always modeled as a three-phase fault. If the disturbance of interest is an unbalanced fault, it is represented by a shunt impedance between the faulted bus and the reference node. Thus the positive sequence currents and voltages are correctly obtained, and the generators' positive sequence torques are correctly computed. Other torques, i.e., negative and zero sequence torques, are ignored.

2.2 THE CLASSICAL POWER SYSTEM MODEL

This is the "simplest" power system model used in stability studies. It is usually

limited to analysis of first swing transients. The assumptions commonly made are (see Chapter 2 of [1]) given here:

For the synchronous generators,

1. Mechanical power input is constant.
2. Damping or asynchronous power is neglected.
3. The generator is represented by a constant EMF behind the direct axis transient (unsaturated) reactance.
4. The mechanical rotor angle of a synchronous generator can be represented by the angle of the voltage behind transient reactance.

Assumptions 1 and 2 are not the source of substantial error in the results of the stability analysis, unless there is an appreciable change in the generator speeds during the first swing transient. They can be improved upon somewhat by assuming a linear variation of each with the change in the generator speed (see Section 2.3 of [1]).

In the classical model the load is usually represented by passive impedances (or admittances), determined from the initial (predisturbance) conditions. These impedances are then held constant throughout the stability study. This assumption can be improved upon by using the so-called nonlinear load models discussed later on in this chapter.

With the loads represented by constant impedances the load nodes and the terminal voltage nodes of the generators are eliminated. The resulting network, shown in Fig. 2.2, contains only the internal generator nodes. The generator reactances and the constant impedance loads are included in the network bus impedance (or admittance) matrix.

The n -port power network shown in Fig. 2.2 is made up of passive elements. The network's admittance matrix $\mathbf{Y}^1$ has the diagonal elements $\bar{Y}_{ii}$, and the off-diagonal elements $\bar{Y}_{ij}$.

¹ The following notation will be used consistently: a symbol in boldface indicates a matrix, a bar on top indicates a phasor.

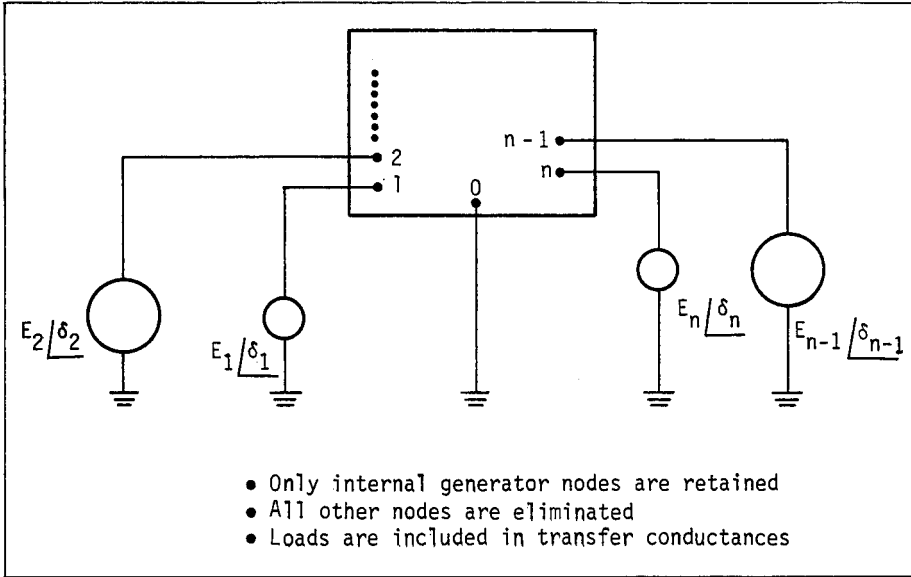


Figure 2.2 Reduced power network (classical model).

where

$$\bar{Y}_{ii} = Y_{ii} / \theta_{ii} = \text{driving point admittance for node } i$$

$$= G_{ii} + j B_{ii}$$

and

$$\bar{Y}_{ij} = Y_{ij} / \theta_{ij} = \text{negative of the transfer admittance between nodes } i \text{ and } j$$

$$= G_{ij} + j B_{ij}$$

The vector of node currents, which is also the generator currents, is given by

$$\bar{I} = \bar{Y} \bar{E} \quad (2.1)$$

The power into the network at node i , which is the electrical power output of machine i , is given by $P_{ei} = \text{Re } \bar{E}_i \bar{I}_i^*$ (2.2)

$$\begin{aligned} P_{ei} &= E_i^2 G_{ii} + \sum_{\substack{j=1 \\ j \neq i}}^n E_i E_j Y_{ij} \cos(\theta_{ij} - \delta_i + \delta_j) \quad i = 1, 2, \dots, n \\ &= E_i^2 G_{ii} + \sum_{\substack{j=1 \\ j \neq i}}^n E_i E_j [B_{ij} \sin(\delta_i - \delta_j) + G_{ij} \cos(\delta_i - \delta_j)] \quad i = 1, 2, \dots, n \quad (2.3) \\ &= E_i^2 G_{ii} + \sum_{\substack{j=1 \\ j \neq i}}^n [C_{ij} \sin(\delta_i - \delta_j) + D_{ij} \cos(\delta_i - \delta_j)] \quad i = 1, 2, \dots, n \end{aligned}$$

where

$$C_{ij} = E_i E_j B_{ij} \quad \text{and} \quad D_{ij} = E_i E_j G_{ij}$$

In (2.3) the admittance matrix terms used must correspond to the network conditions for which the electrical power P_{ei} is computed. For example, if the disturbance is a fault, the prefault, faulted, and postfault networks have different $\bar{Y}$ matrices.

The generators' equations of motion are then given by

$$\begin{aligned} \frac{2H_i}{\omega_R} \frac{d\omega_i}{dt} + D_i \omega_i &= P_{mi} - \left[E_i^2 G_{ii} + \sum_{\substack{j=1 \\ j \neq i}}^n E_i E_j Y_{ij} \cos(\theta_{ij} - \delta_i + \delta_j) \right] \\ \frac{d\delta_i}{dt} &= \omega_i - \omega_R \quad i = 1, 2, \dots, n \end{aligned} \quad (2.4)$$

where H_i is the inertia constant for generator i in MWs/MVA and ω_i and ω_R are the speed of generator i and the reference speed, respectively.

It should be noted that prior to the disturbance ($t = 0^-$) $P_{mi0} = P_{ei0}$

$$P_{mi0} = E_i^2 G_{ii0} + \sum_{\substack{j=1 \\ j \neq i}}^n E_i E_j Y_{ij0} \cos(\theta_{ij0} - \delta_{i0} + \delta_{j0}) \quad (2.5)$$

The subscript 0 is used to indicate the pretransient conditions.

2.3 LOAD MODEL

2.3.1 Classical Model

Consider load bus k . Let the predisturbance conditions at bus k be voltage $\bar{V}_k$ and the load power and reactive power be P_{Lk} and Q_{Lk} , respectively. Then the equivalent load admittance is given by

$$P_{Lk} + j Q_{Lk} = V_k^2 (G_{Lk} - j B_{Lk}) \quad (2.6)$$

$$S_{Lk} = P_{Lk} + j Q_{Lk} = V_k I_{Lk}^* = V_k (V_k Y_{Lk})^* = V_k^2 Y_{Lk}^* = V_k^2 (G_{Lk} - j B_{Lk})$$

or

$$G_{Lk} = P_{Lk} / V_k^2, \quad (2.7)$$

$$B_{Lk} = - Q_{Lk} / V_k^2$$

As stated in the previous section, the load admittances are included in the network admittance matrix. A reduced network admittance matrix is then obtained by eliminating all the network nodes, including the load buses.

2.3.2 Nonlinear Load Model

If the load at bus k is represented as a constant current or a constant MVA load,

$$P_{Lk} + j Q_{Lk} = V_k \angle \psi_k \bar{I}_{Lk}^* \quad (2.8)$$

or

$$\bar{I}_{Lk} = \frac{P_{Lk} - j Q_{Lk}}{V_k \angle -\psi_k} \quad (2.9)$$

When constant current and constant power loads are included, a special network procedure is used to incorporate these loads in the classical model solution. This procedure must meet the requirement that the current $\bar{I}_{Lk}$ needs to be appropriately preserved during the transient. When the network is reduced to the internal generator nodes, $\bar{I}_{Lk}$ is preserved by determining equivalent current injections at the internal generator nodes. A procedure is also developed to update the value of the current injection during the transient.

This procedure is discussed in some detail in Chapter 9.

2.4 CENTER OF INERTIA FORMULATION

In the system equations (2.4) the generators' angles and speeds are given with respect to a synchronous frame of reference. To obtain the center of inertia (COI) coordinates the center of inertia is defined by

$$\delta_o = \frac{1}{M_T} \sum_{i=1}^n M_i \delta_i \quad (2.10)$$

$$\dot{\delta}_o = \frac{1}{M_T} \sum_{i=1}^n M_i \dot{\delta}_i$$

where M_i = inertia constant for generator $i = 2H_i / \omega_R$,

and

$$M_T = \sum_{i=1}^n M_i$$

Motion of the COI is defined by the equations

$$M_T \dot{\omega}_o = \sum_{i=1}^n P_{mi} - P_{ei} = \sum_{i=1}^n P_i - 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^N D_{ij} \cos \delta_{ij} \triangleq P_{COI} \quad (2.11)$$

$$\dot{\delta}_o = \omega_o \quad i = 1, 2, \dots, n$$

where

$$P_i = P_{mi} - E_i^2 G_{ii} \quad i = 1, 2, \dots, n$$

We now define the generators' angles and speeds relative to the COI by

$$\left. \begin{aligned} \theta_i &= \delta_i - \delta_o \\ \tilde{\omega}_i &= \dot{\delta}_i - \dot{\delta}_o \end{aligned} \right\} \quad i = 1, 2, \dots, n \quad (2.12)$$

The system equations of motion become

$$M_i \ddot{\omega}_i = P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} \quad (2.13)$$

$$\dot{\theta}_i = \tilde{\omega}_i \quad i = 1, 2, \dots, n$$

We note that the center of inertia variables satisfy the constraints

$$\sum_{i=1}^n M_i \theta_i = \sum_{i=1}^n M_i \tilde{\omega}_i = 0 \quad (2.14)$$

2.5 HIGHER-ORDER GENERATOR MODEL WITH EXCITATION CONTROL

2.5.1 Synchronous Generator Model

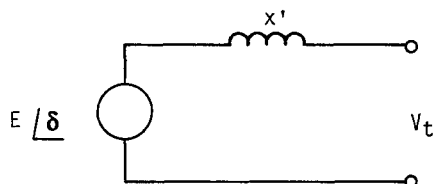


Figure 2.3 Synchronous generator model.

When excitation control is of interest, the synchronous generator is assumed to be represented by the so-called two-axis model (see Section 4.15.3 of [1]). Fig. 2.3 shows a schematic of the model; x' is the transient reactance (usually the average of the d -axis transient reactance x'_d and q -axis transient reactance x'_q). The voltage E' is the transient EMF or the voltage behind x' ; its components E'_d and E'_q vary and have different time constants (see below). The angle δ is now the generator rotor angle (or torque angle).

For generator k , the equations describing the dynamic behavior of the generator EMFs are given by

$$\tau'_{dok} \dot{E}'_{qk} = E_{FDk} - E'_{qk} + (x_{dk} - x'_{dk}) I_{dk} \quad (2.15)$$

$$\tau'_{qok} \dot{E}'_{dk} = -E'_{dk} - (x_{qk} - x'_{qk}) I_{qk} \quad (2.16)$$

and the equations for the rotor motion, with respect to a synchronous frame, are given by

$$M_k \dot{\omega}_k = P_{mk} - E'_{dk} I_{dk} - E'_{qk} I_{qk} + (x'_{qk} - x'_{dk}) I_{dk} I_{qk} \quad (2.17)$$

$$\delta = \omega - 1 \quad k = 1, 2, \dots, m$$

where

P_m = mechanical power

m = number of generators represented in detail

τ'_{do}, τ'_{qo} = open-circuit direct and quadrature axes time constants, respectively

E_{FD} = stator EMF corresponding to field voltage

x_d, x'_d = direct axis synchronous and transient reactances

x_q, x'_q = quadrature axis synchronous and transient reactances

E'_d, E'_q = direct and quadrature axes stator EMFs corresponding to rotor flux components

I_d, I_q = direct and quadrature axes stator currents

When the COI formulation is used, (2.17) becomes

$$M_k \dot{\omega}_k = P_{mk} - E'_{dk} I_{dk} - E'_{qk} I_{qk} + (x'_{qk} - x'_{dk}) I_{dk} I_{qk} - \frac{M_k}{M_T} P_{COI} \quad (2.18)$$

where

$$P_{COI} = \sum_{j=1}^n P_{mj} - \sum_{k=1}^m [E'_{dk} I_{dk} + E'_{qk} I_{qk} - (x'_{qk} - x'_{dk}) I_{dk} I_{qk}] - \sum_{\ell=m+1}^n E_{\ell} I_{q\ell} \quad (2.19)$$

where $\ell = m + 1, \dots, n$ represents the number of generators represented classically.

2.5.2 Generator Currents—Constant Impedance Loads

When the loads are represented by the classical model, the generator currents and voltages are represented by the following equations (see Chapter 9 of [1]):

$$I_{qk} = \sum_{j=1}^m [F_{G+B}(\theta_{kj}) E'_{qj} - F_{B-G}(\theta_{kj}) E'_{dj}] + \sum_{\ell=m+1}^n F_{G+B}(\theta_{k\ell}) E_{\ell} \quad (2.20)$$

$$I_{dk} = \sum_{j=1}^m [F_{B-G}(\theta_{kj}) E'_{qj} + F_{G+B}(\theta_{kj}) E'_{dj}] + \sum_{\ell=m+1}^n F_{B-G}(\theta_{k\ell}) E_{\ell}$$

$$k = 1, 2, \dots, m$$

$$I_{qi} = \sum_{j=1}^m F_{G+B}(\theta_{ij}) E'_{qj} + \sum_{\ell=m+1}^n F_{G+B}(\theta_{i\ell}) E_{\ell} - \sum_{j=1}^m F_{B-G}(\theta_{ij}) E'_{dj}$$

$$i = m + 1, \dots, n \quad (2.21)$$

where

$$F_{G+B}(\theta_{ij}) = G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij} \quad (2.22)$$

$$F_{B-G}(\theta_{ij}) = B_{ij} \cos \theta_{ij} - G_{ij} \sin \theta_{ij}$$

2.5.3 Generator Currents–Nonlinear Loads

To accommodate both constant impedance and nonlinear loads, the synchronous generator model shown in Fig. 2.4 is used

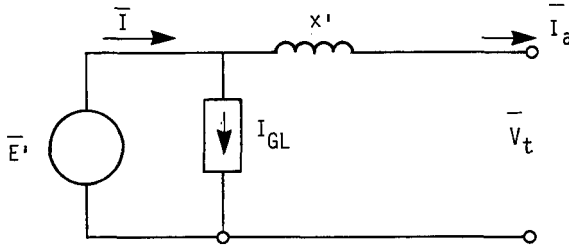


Figure 2.4 Synchronous generator model with current injection representing nonlinear load.

where

- $\bar{I}$ = current in the generator
- $\bar{I}_a$ = current into the network
- I_{GL} = current for classical model portion of the network
- $\bar{I}_{GL}$ = constant current or constant MVA component reflected at the generator node

In the following, the capital subscripts D and Q will refer to the network (i.e., synchronous) reference frame, while lowercase symbols, e.g., q and d , will refer to the generator rotor reference frame. In addition, the terms $[(x'_q - x'_d) I_d I_k]$ in the power equations are neglected.²

² This is done to simplify the expressions without loss of generality. However, these terms can be easily added if deemed desirable.

Steady state:

We can easily show that the generator electrical power P_e is given by

$$\begin{aligned} P_e &= (E'_Q I_{aQ} + E'_D I_{aD}) + (E'_Q I_{GLQ} + E'_Q I_{GLD}) \\ &= P'_e + E' I_{GL} \cos(\theta - \phi) \end{aligned} \quad (2.23)$$

where

P'_e = electrical power corresponding to classical portion of the network, i.e.,
constant impedance loads

$E' \angle \theta$ = generator EMF

$I_{GL} \angle \phi$ = current injection at generator node due to nonlinear loads

Transient state:

We can show that the generator i current, to the d - and q -frames, are given by

$$I_{qi} + j I_{di} = (I_{aqi} + j I_{adi}) + I_{GLi} \epsilon^{j \phi_i} \quad (2.24)$$

Where phasor notation is used for convenience (as in Chapter 9 of [1]), I_{aqi} and I_{adi} can be obtained from equations similar to (2.20) and (2.21).

Then the electrical power for generator i is given by

$$\begin{aligned} P_{ei} &= P'_{ei} + (E'_{qi} I_{GLqi} + E'_{di} I_{GLdi}) \\ &= P'_{ei} + P_{eGLi} \end{aligned} \quad (2.25)$$

and P_{COI} becomes

$$P_{COI} = P'_{COI} - \sum_{i=1}^n P_{eGLi} \quad (2.26)$$

Note:

The generator equations (2.15–2.17) will become

$$\tau'_{d0k} \dot{E}'_{qi} = E_{FDi} - E'_{qi} + (x_{di} - x'_{di}) I_{adi} \quad (2.27)$$

$$\tau'_{qoi} \dot{E}'_{di} = -E'_{di} - (x_{qi} - x'_{qi}) I_{aqi}$$

$$M_i \ddot{\omega}_i = P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} \quad (2.28)$$

$$= \left[P_{mi} - P'_{ei} - \frac{M_i}{M_T} P'_{COI} \right] - \left[P_{eGLi} + \frac{M_i}{M_T} \sum_{j=1}^n P_{eGLj} \right]$$

2.5.4 Excitation System Model

The exciter is assumed to be represented by an equivalent one gain, one time constant, and one limiter model, as shown in Fig. 2.5. This model is used to standardize the treatment of incorporation of the effect of excitation control on the flux variation in the synchronous generator, and a higher-order exciter model can be used if desired. Standard techniques are available in the literature to reduce a higher-order system to an equivalent one gain, one time constant in the frequency range of interest (see for example Section 3.5 of [3]). By proper choice of the parameters K_A , τ_E , E_{FDmax} , and E_{FDmin} in Fig. 2.5, most modern excitation systems can be modeled.

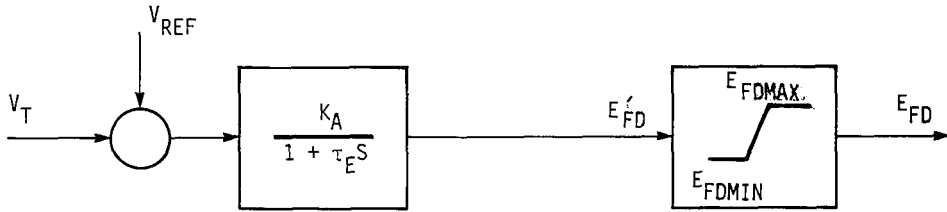


Figure 2.5 Exciter model.

The exciter equations for generator k are given by

$$\begin{aligned} \tau_{Ek} \dot{E}'_{FDk} &= g(E'_{FDk}, V_{tk}, V_{REFk}) \\ &= -E'_{FDk} + K_{Ak}(V_{REFk} - V_{tk}) \end{aligned} \quad (2.29)$$

E_{FD}, E'_{FD} = stator EMF corresponding to field voltage

E_{FD} = value after the limiter

E'_{FD} = value before the limiter

V_t = generator terminal voltage

V_{REF} = exciter reference voltage

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Chapter 3

TRANSIENT STABILITY ANALYSIS USING ENERGY FUNCTIONS – AN INTRODUCTION

3.1 THE BASIC IDEA

As was discussed in detail in Chapter 2, a power system is described mathematically by a set of differential equations together with a set of algebraic equations. These sets of equations are large in number (for a medium-size power system, their number may be in the hundreds), they are nonlinear (they include several types of nonlinearities), and they are coupled.

In contrast to linear systems, where analytical solutions are usually possible, there are only a few classes of nonlinear systems where general statements can be made about their dynamic behavior and their stability characteristics. Among these are certain power system models where stability behavior can be analyzed by the so-called direct methods, including the transient energy function (TEF) method.

To determine transient stability of a power system, the conventional, and still the standard, method is to solve the system equations to obtain a time solution of the system variables and parameters, for a given scenario of events (e.g., occurrence of a fault, clearing of the fault, breaker reclosing). From this time solution all pertinent variables of interest are determined as a function of time. From such information, the analyst decides whether the system dynamic response will result in an acceptable steady-state performance. Depending on the type of problem under investigation, the information of interest may be the rotor angles of certain generators (most common), power flow over a certain group of transmission lines, voltage at a certain bus, and so on.

The alternate method, which is explored in this book, is to determine stability "directly." A certain function, in this case describing the system transient energy, is computed at the end of the disturbance period. This value is compared with a critical (threshold) value of the energy for transient stability assessment. The difference between the critical value of energy and the energy at the end of the disturbance is the energy margin. It is related to the parameter of interest in transient stability assessment and is used as an indication of stability.

Power system transient stability assessment by the so-called direct methods has been around for many years. Many functions have been proposed and used by various investigators. The transient energy function method, using as critical energy the potential energy at the controlling unstable equilibrium point (UEP), is the method used exclusively in this book.

3.2 AN EXAMPLE

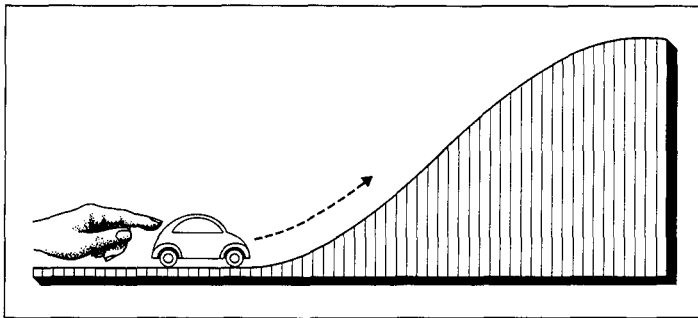


Figure 3.1 A vehicle is given a sudden sharp push up a hill.

Figure 3.1 shows a schematic illustration of the basic idea behind the use of the TEF method for transient stability assessment. A vehicle, initially at the bottom of a hill, is

given a sudden sharp push up the hill. If the push is large enough, the vehicle will go over the hill; presumably it will tumble and become unstable. If the push is not large, the vehicle will climb only part way up the hill and return; it is assumed that the vehicle's motion will be damped out to come to rest; i.e., it will be stable.

Let us assume that we wish to determine, for a given set of conditions (size of the vehicle, height of the hill, magnitude of the push, etc.), what the outcome of disturbing the vehicle's equilibrium will be. We can easily see that two different methods can be used to calculate whether or not the vehicle will go over the hill.

1. Obtain a time solution of the equations describing the position of the vehicle. From this solution, examine how far the vehicle will travel. (This method resembles taking a series of pictures and tracking the changing positions as if watching a movie of the vehicle's path.)
2. Compute the energy put into the vehicle by the push and compare it with the energy needed to climb the hill. (In this method, there is no need to track the position of the vehicle as it moves.)

These methods are simple to use if the calculations only involve one vehicle and one hill. Complications occur if there are several vehicles involved. Then it is necessary to determine (1) which vehicles will be pushed the hardest, (2) how much of the available energy will go into pushing each of them, (3) which direction will they move, and (4) how high a hill must they climb before they will go over the top.

This simple example summarizes the difference between direct transient stability analysis of a one-machine – infinite-bus system and that of a multimachine power system using the TEF method. The former is equivalent to the well-known equal area criterion.

3.3 THE EQUAL AREA CRITERION [1, 2]

Consider the one-generator – infinite-bus power network shown schematically in Fig. 3.2. If the generator is represented by the classical model (see Chapter 2), we can

show the dynamic behavior of the generator rotor angle δ is described by

$$M \frac{d^2\delta}{dt^2} = P_i - P_m \sin(\delta - \gamma) \quad (3.1)$$

where

M = inertia constant

P_i , P_m , and γ are constants

Multiplying both sides of (3.1) by $\frac{d\delta}{dt}$, integrating and noting that $\omega = \frac{d\delta}{dt}$, we get

$$\int M \omega \frac{d\omega}{dt} dt = \int [P_i - P_m \sin(\delta - \gamma)] \frac{d\delta}{dt} dt \quad (3.2)$$

integrating again we get

$$\frac{1}{2} M \omega^2 \bigg|_{\omega_1}^{\omega_2} = P_i \delta \bigg|_{\delta_1}^{\delta_2} + P_m \cos(\delta - \gamma) \bigg|_{\delta_1}^{\delta_2} \quad (3.3)$$

If the limits are between initial conditions ($\delta_1 = \delta_0$, $\omega_1 = 0$) and at clearing ($\delta_2 = \delta_c$, $\omega_2 = \omega_c$), then the L.H.S. of (3.3) gives the kinetic energy of the generator at fault clearing. This is represented by the area A_1 in Fig. 3.3.

Taking the limits between fault clearing ($\delta_1 = \delta_c$, $\omega_1 = \omega_c$) and at maximum angle swing ($\delta_2 = \delta_m$, $\omega_2 = 0$), then the L.H.S. is the negative of the kinetic energy at clearing. It is represented by the area A_2 in Fig. 3.3.

We also note that at δ_m , since $\omega_m = 0$, all the kinetic energy which had been present at δ_c has now been converted to potential energy, and the areas A_1 and A_2 in Fig. 3.3 are equal (this is the origin of the name "equal area criterion").

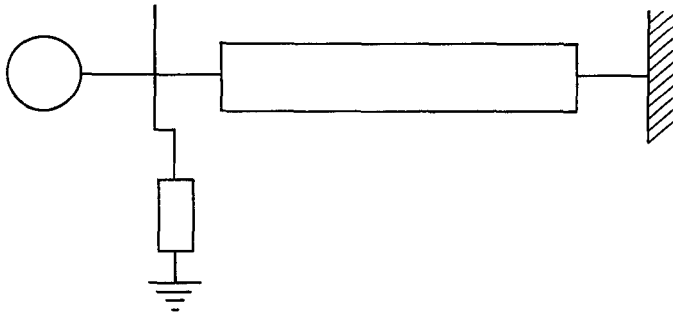


Figure 3.2 One machine–infinite – bus–network.

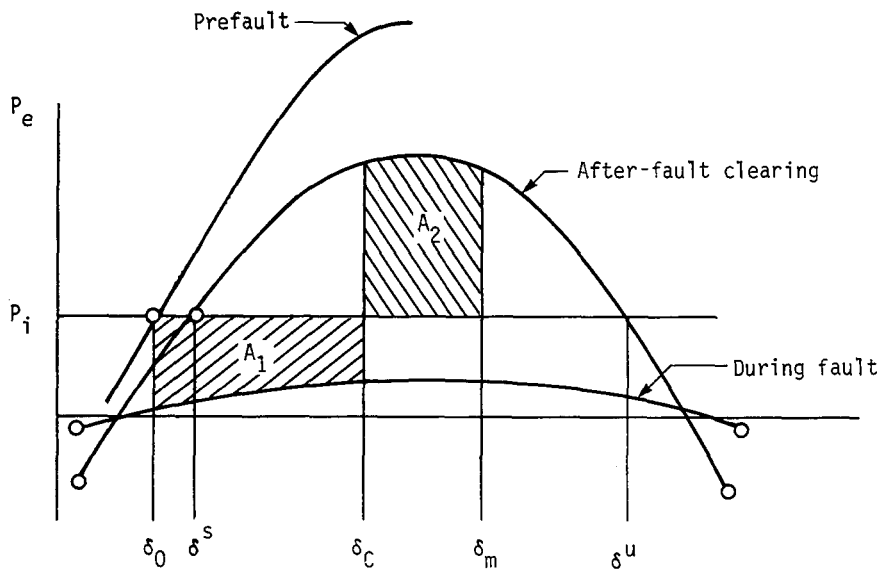


Figure 3.3 Power–angle curves.

Upon further examination of Fig. 3.2, we note that

- Depending on size of A_1 , δ_m can be somewhere between δ_c and δ^u if δ reaches a maximum and decreases.
- Therefore the maximum value of A_2 is when δ_m coincides with δ^u .
- This value is characteristic of the network and the operating condition.

Some additional statements can be made about the energy balance when the disturbance is removed:

- Since areas A_1 and A_2 represent kinetic energy (KE) and potential energy (PE), respectively, after disturbance is removed (δ_c) there is conversion of energy from KE to PE.
- This conversion process depends on δ^u , and on the area between P_i and the postfault $P-\delta$ curve.

Relating the above discussion to the material presented in Section 3.2, we note that we have just introduced the concepts of

- Transient energy V
- Critical value of $V(V_{cr}, V^u)$
- Equilibrium conditions (δ^s, δ^u)
- We can easily show that any two-generator power systems can be reduced to an equivalent one-machine – infinite-bus system (see Chapter 2 of [1])

3.4 MULTIMACHINE ENERGY FUNCTIONS

Energy criteria for transient stability are the forerunners of all direct methods of multimachine power system transient stability assessment. They attempted to extend the results of the equal area criterion, discussed in Section 3.3, to the case of a power system

containing more than two generators represented by the classical model. Early investigations of the subject were conducted in the Soviet Union (1930s and 1940s). Very few results appeared in the technical literature in the West in the 1940s and 1950s. It was in the early 1960s when the application of Lyapunov's second method to power systems generated considerable interest in the academic community. The classical power system model was used in most of these investigations.

Early work on energy criteria covered the investigation of the two main issues involved in direct stability analysis, namely, (1) the system energy and (2) the critical value of that energy. Investigation of the second issue invariably involves the investigation of equilibrium conditions, i.e., the stable and unstable solutions. The following summary is intended only as a very brief review of some of the main contributions which are considered representative of the early work in this area.

3.4.1 Gorev's Energy Criteria [3, 4]

Gorev considered the classical power system model with zero transfer conductances. A function ϕ_i equal to P_{ei} , the electrical power of machine i , was defined. Thus, for generator i

$$\phi_i = P_{ei} \quad (3.4)$$

and the system equations become

$$M_i \frac{d^2 \delta_i}{dt^2} = \phi_i \quad (3.5)$$

$$\frac{d\delta_i}{dt} = \omega_i \quad i = 1, \dots, n$$

where δ_i and ω_i are the angle and speed of generator i . The *first energy criterion* is obtained by defining the energy function U such that

$$\phi_i \triangleq \frac{\partial U}{\partial \delta_i} \quad (3.6)$$

then

$$\frac{1}{2} \sum_{i=1}^n M_i d\omega_i^2 = \sum_{i=1}^n \frac{\partial U}{\partial \delta_i} d\delta_i \quad (3.7)$$

and if we define

$$T \triangleq \frac{1}{2} \sum_{i=1}^n M_i \omega_i^2 \quad (3.8)$$

then $dT = dU$, and we obtain for the energy function U ,

$$U = \sum_{i=1}^{n-1} P_{ii} \delta_i + \sum_{i>j} P_{ij} \cos \delta_{ij} \quad (3.9)$$

Gorev shows that for the restorative phase of the transient, i.e., after the last switching, the energy criterion for stability is given by

$$T = U - U_0 + T_0 \geq 0 \quad (3.10)$$

Equation (3.10) gives Gorev's first energy criterion. In (3.10), subscript 0 denotes the

initial conditions. This criterion is difficult to apply because of the computational burden required to obtain the surfaces satisfied by (3.9).

Gorev's *second energy criterion* of stability states that "a sufficient condition of stability is for the value of T at the highest saddle point of the surface T to be less than zero". The highest saddle point is the saddle nearest to the point of T_{max} , which is the stable equilibrium point.

Determination of the highest saddle, and hence, its energy function, determines a critical energy value U_c such that

$$U_c = U_o + T_o \leq 0 \quad (3.11)$$

This is the mathematical form of the energy criterion. It is equivalent to the determination of a *region* of stability for the equilibrium point.

Gorev's second energy criterion influenced the thinking of power system direct stability researchers for a long time. For many years the position of the saddle point nearest the equilibrium point remained the principal means to determine the region of stability.

Determination of U_c is by solving the set

$$\frac{\partial U}{\partial \delta_{in}} = 0 \quad (3.12)$$

which again is far from being a simple task.

3.4.2 Magnusson's Method of Transitional Energy [3, 5]

Magnusson's approach, which appeared in 1947, is very similar to that of Gorev's, including a detailed investigation of the three-machine system. The significant difference

between Magnusson's formulation and that of Gorev's is that Magnusson derived a potential energy function *with respect to the (posttransient) equilibrium point*. This energy function V is given by

$$\begin{aligned}
 V = & P_{12} [\cos (\delta_{13}^* - \delta_{23}^*) - \cos (\delta_{13} - \delta_{23})] \\
 & - (\delta_{13} - \delta_{13}^* - \delta_{23} + \delta_{23}^*) \sin (\delta_{13}^* - \delta_{23}^*) \\
 & + P_{13} [\cos (\delta_{13}^* - \cos \delta_{13} - (\delta_{13} - \delta_{13}^*) \sin \delta_{23}^*] \\
 & + P_{23} [\cos \delta_{23}^* - \cos \delta_{13} - (\delta_{23} - \delta_{23}^*) \sin \delta_{23}^*] \quad (3.13)
 \end{aligned}$$

where δ^* is the stable equilibrium angle.

Magnusson's potential function V and Gorev's energy function U are similar, and the procedure for determining the criteria of stability is the same in both cases. A region of stability is determined by the potential of the "nearest" saddle to the equilibrium point. Again the critical points are determined from solution of a set of equations similar to (3.12). The roots of these equations give the critical points: the maximum, the minimum, and minimaximum. *The lower of the minimaxima is the root of interest.* The value of V at that point is V_1 . If the total transient energy, kinetic energy plus potential energy, is equal to a value *less than* V_1 , then the system will be stable.

The process is to calculate the initial total transient energy and compare it with V_1 . For multiswitching transients, the method is applied to the *final* transient.

3.4.3 Aylett's Energy Integral Criterion [3, 6]

In his work, Aylett studied the phase-plane trajectories of a multimachine system using the classical mathematical model of the machines and the network. Starting with a

two-machine system, he pointed out that there is a "critical" phase-plane trajectory, passing through the saddle point, which he called the separatrix. Trajectories of transients representing closed curves inside the separatrix will form oscillatory trajectories. Phase trajectories outside the separatrix represent nonoscillatory (unbounded) trajectories. Critical switching of faulted system is decided by determining the phase trajectory of the faulted system. If it is cleared at or inside the separatrix the system is stable. Aylett also developed an energy integral for the two-machine system.

For the multimachine system, Aylett obtained a set of $\frac{n(n-1)}{2}$ differential equations in the intermachine angle coordinates. He then obtained the energy integral expression, which in the case of zero transfer conductances is given by

$$\sum_{r=1}^{r=n-1} \sum_{p=r+1}^{p=n} \frac{M_p M_r}{\sum_{r=1}^{r=n} M_r} \delta_{rp}^2$$

$$= \sum_{r=1}^{r=n-1} \sum_{p=r+1}^{p=n} \left\{ \frac{V_r V_p}{Z_{rp}} [\sin \delta_{rpc} (\delta_{rp} - \delta_{rpc}) + \cos \delta_{rp} - \cos \delta_{rpc}] \right\} \quad (3.14)$$

In (3.14), the L.H.S. is the kinetic energy and the R.H.S. is the potential energy. The total energy integral is obtained from both terms. Again Aylett states that in the critical case $PE = KE$; in stable situations the $PE > KE$, and instability occurs when $KE > PE$. This is evaluated at the instant of switching to clear the fault.

An important aspect of Aylett's work is the formulation of the system equations based on the intermachine movements. This is in accord with the physical dynamic behavior of the machines which determines whether synchronism is maintained. This mathematical formulation was used by other authors such as Ribbens-Pavella and Tavora whose work will be discussed in later sections.

3.4.4 Tavora and Smith's Investigation of the Transient Energy [3, 7]

In 1972 *Tavora and Smith* dealt with the transient energy of a multimachine system, and examined equilibrium conditions. The mathematical model used for the machines and the network was the classical model with zero transfer conductances. Two mathematical formulations of the swing equation were used:

1. Referring the rotor angles to a center of angle a set of $(n - 1)$ equations was obtained. The equations for the center of angle and the resulting swing equations are similar to those given in Section 2.5. An interesting aspect of their work is that they defined synchronous equilibrium by conditions where the generators' speeds and acceleration, in the COI frame, are zero.
2. Using the internode coordinates which is similar to equation (3.14) used by Aylett (with zero mutual conductances), except that the reference node is node n . A coordinate transformation was used to obtain a set of $(n - 1)$ equations in the new coordinates.

Tavora and Smith obtained expressions for the *total* kinetic energy of the system E_k and the *transient* (or intermachine) kinetic energy E_{ks} which the authors say determines stability. They suggested that for stability the trajectory after the last phase of the disturbance be confined to a bounded region. This region is bounded by a "potential well" determined by the Hamiltonian of the motion. Thus, for stability, the total energy of the disturbed motion must be less than that needed to escape from this potential well. The potential well has at its rim a point which determines the least energy level for escape.

In their work, Tavora and Smith suggest a two-part procedure for locating the lowest saddle point of the potential energy function.

3.5 THE MODERN TRANSIENT ENERGY FUNCTION

3.5.1 Introduction

In the early 1960s, research work on power system transient stability analysis by direct methods became almost exclusively dependent on the use of Lyapunov functions.

There was a great deal of interest in the subject as evidenced by the numerous publications (see, for example, the survey in [3] and [8]) until the mid-1970s and beyond. The formalism and rigor associated with the application of Lyapunov's second method to the power system transient stability problem appealed to the academic community. The investigations covered two general areas (1) development of new functions, mostly energylike functions, which satisfy Lyapunov conditions, and (2) finding the region of stability of the stable equilibrium point associated with these functions.

Based on the examination of the literature of the 1960s–1970s on the application of Lyapunov's second method to power system transient stability, one can make two interesting observations (1) that greater emphasis was given to the development of new Lyapunov functions and (2) that the application of these functions to power system problems yielded "conservative" results, e.g., predicted smaller critical clearing time than that predicted by conventional time simulation method. In retrospect, one is inclined to think that researchers assumed that the conservatism in the results was largely due to the use of the "wrong" functions and that better results can be obtained with improved Lyapunov functions. Relatively less attention was given to the determination of the boundary of the region of stability and to the critical value of the function against which transient stability assessment is made.

The authors of this book consider that a detailed treatment of the Lyapunov's second method and its application to power system transient stability is outside the main thrust of this book, since our main focus is on the transient energy function method. We wish to emphasize, however, that out of the numerous literature on the second method of Lyapunov in the power system journals came a deeper understanding of the dynamic behavior of synchronous generators during transients, of the transient energy associated with their motion, and of the region of stability and its boundary. While we are bypassing the treatment of these subjects in detail, we are duly acknowledging the significance of their contributions to the field of power system direct transient stability analysis.

We conclude these introductory remarks by singling out some specific advances which had greater impact on the transient energy function as we now know it. These developments are

- Development of the center of angle formulation (called COI here), and the recognition of the energy associated with its motion (see [3] and the references cited in it, e.g., [7] of this chapter)
- The first recognition by El-Abiad and his coworkers [9] that the UEP of interest is not the one with the lowest energy, but rather the UEP closest to the system trajectory
- The suggestion by Uyemura [10] to approximate the path-dependent terms in the Lyapunov functions by path-independent terms, e.g., by linear trajectory approximation
- In terms of theoretical justification of some of the concepts used in the TEF method, extending Lyapunov's results using the invariance principle and other recent analytical results to be discussed in some detail in Chapter 4

3.5.2 The SCI Work

The work by Athay, Podmore, and colleagues, in the 1976–1979 period (first at SCI, Inc., and later at ESCA Corp.) is the basis for the transient energy function method used today. They took advantage of the mentioned improvements and/or developments. In addition, they addressed many other analytical and practical issues. Their 1979 report [11] is considered a benchmark reference material in the TEF area of investigation.¹ Their work, which will be covered in some detail in later chapters, included

- COI formulation and linear approximation of path-dependent terms.
- Search for UEP, and hence critical transient energy, in the direction of system trajectory.
- Investigation of the potential energy boundary surface (PEBS); and invoking the invariance principle.

¹ The authors make it required reading for their students.

- Investigation of more practical power system sizes than previously used in direct stability analysis research.
- Use of extensive computer simulations to investigate energy behavior as well as other system functions at different instants in the transient.
- Preliminary investigation of higher-order models for synchronous generators.

3.6 MOTIVATION FOR CURRENT DIRECT STABILITY WORK

In this chapter we have briefly reviewed the development of the transient energy function method for use in direct transient stability analysis of power systems. We can also anticipate the considerable improvements recently achieved which will be discussed in detail in the remaining chapters. Now that the early conservatism has been overcome, it would be interesting to examine direct methods' original premise against the future promise of the TEF method.

Having been associated with direct transient stability analysis for some time (10 and 25 years), the authors note that there is considerable change in our perception of what direct methods can do. In the 1960s it was often mentioned that direct methods promised much faster solution time to determine critical clearing times than what conventional time solution methods then offered. That claim in itself is rather outdated. First, computers, and solution methods, of the 1960s were much less advanced than what is now available. Second, if critical clearing times were of interest then other, and more complex, stability-related problems are now of interest, as discussed in Chapter 1 of this book. Yet the motivation for pursuing direct stability analysis methods are now greater than ever; why?

Our experience indicates that the present motivation for pursuing work on direct methods is less to achieve faster solution times than that which can be obtained by conventional techniques, even though that is still among the advantages of direct methods of interest: to obtain qualitative information on system stability behavior, to identify the critical generators which are severely affected by the disturbance, to determine sensitivity

to variation of system parameters, and to achieve faster computational of stability limits. These features do not necessarily make the new direct method a candidate to replace the conventional time simulation method, but rather a tool to augment it. This new tool is more suited for some of the applications than the conventional tools.

It must also be recognized that the potential advantages of the new tool in themselves place added constraints on the development of that tool. It must deal with actual industry problems, in a setting that is representative of the industry environment. Thus, the research and development efforts must go hand in hand.

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Chapter 4

REVIEW OF STABILITY THEORY APPLIED TO THE TRANSIENT ENERGY FUNCTION METHOD

4.1 INTRODUCTION

4.1.1 Notational Conventions

We will first provide a brief description of the notational convention used in this chapter.

$x \in A$	means x is an element of the set A
$x \notin A$	means x is not an element of the set A
R	is the set of all real numbers
$R \times R$	denotes the Cartesian product, i.e., if $x, y \in R$, the ordered pair $(x, y) \in R \times R$
R^n	is the n -dimensional real space
$\ \cdot\ $	is an appropriately defined norm
B_r	ball of radius r , $r > 0$, and centered at the origin
$B = \{x \in A: P(x) \text{ is true}\}$	means the set B consists of all elements $x \in A$ such that $P(x)$ is true
$B \subset A$	means B is a subset of the set A
$A \cap B$	denotes the intersection of the sets A and B , the set of elements in both A and B
$A \cup B$	denotes the union of the sets A and B , the set of element in A or B
ϕ	denotes the set with no elements, the empty set

4.1.2 Nonlinear Physical Systems

Nonlinear physical systems differ from linear systems in two basic characteristics:

1. In general "closed-form" expressions for solutions of nonlinear systems cannot be obtained, whereas one can usually obtain such solutions for linear systems. In many instances, analysis of nonlinear systems is performed by obtaining sequences of approximating functions that converge to the exact solution or by determining estimates for the true solution. As a consequence, it is difficult to obtain a physical feel for the behavior of a nonlinear system, compared with a linear system.
2. The analytical tools used in the analysis of nonlinear systems are generally more advanced in concept and involve more detail than are those used to analyze linear systems.

Most physical systems are described by their mathematical model for purposes of analysis. A mathematical model that describes a wide variety of physical nonlinear systems, with p inputs and k outputs is described by a set of k ordinary differential equations of the form

$$\frac{d^{m_i} y_i(t)}{dt^{m_i}} = g_i[t, y_1(t), \dot{y}_1(t), \dots, y_1^{(m_1-1)}(t), y_2(t), \dots, y_2^{(m_2-1)}(t), \dots, y_k(t), \dots, y_k^{(m_k-1)}(t), u_1(t), \dots, u_p(t)] \quad i = 1, 2, \dots, k \quad (4.1)$$

where $u_1(\cdot), \dots, u_p(\cdot)$ are the input functions and $y_1(\cdot), \dots, y_k(\cdot)$ are the output functions.

Define

$$x_{m_{i+j}}(t) = \frac{d^{j-1} y_{i+1}(t)}{dt^{j-1}} \quad j = 1, \dots, m_{i+1}, \quad i = 0, \dots, k-1 \quad (4.2)$$

$$\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_n(t)]^T \quad (4.3)$$

$$\mathbf{u}(t) = [u_1(t), u_2(t), \dots, u_p(t)]^T \quad (4.4)$$

where we take $m_0 = 0$, and define

$$n = m_1 + m_2 + \dots + m_k \quad (4.5)$$

Then $\mathbf{x}(t)$ is a state vector for the system defined by (4.1), and the system of equations (4.1) can be equivalently represented by a single first-order vector differential equation of the form

$$\dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x}(t), \mathbf{u}(t)) \quad t \geq 0 \quad (4.6)$$

4.1.3 Well Posedness

For equation (4.6) to truly represent a physical system, it should be well posed. To satisfy the property of well posedness, equation (4.6) should

1. Have at least one solution for every $\mathbf{x}_0 = \mathbf{x}(t_0)$ and every $\mathbf{u}$ (existence).
2. Have exactly one solution for every $\mathbf{x}_0 = \mathbf{x}(t_0)$ and every $\mathbf{u}$ (uniqueness).
3. Have exactly one solution for every $\mathbf{x}_0 = \mathbf{x}(t_0)$ and every $\mathbf{u}$ which exists for all $t \geq t_0$.
4. Have exactly one solution for every $\mathbf{x}_0 = \mathbf{x}(t_0)$ and every $\mathbf{u}$ which exists for all $t \geq t_0$ and which depends continuously on the initial data $\mathbf{x}(t_0)$.

For a more detailed discussion on the aspect of well posedness the readers are referred to [1, 2, 3, 4].

4.2 AUTONOMOUS SYSTEMS, EQUILIBRIUM POINTS

In this section, we shall introduce a few definitions which are commonly used in analytical work related to stability theory. Before proceeding to these definitions, a small point needs to be cleared up. Many of the definitions, theorems, etc., that follow are stated for differential equations of the type

$$\dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x}(t)) \quad (4.7)$$

Comparing (4.7) with (4.6), we see that in (4.6) the dependence of the right-hand side on an input $\mathbf{u}(\cdot)$ is explicitly identified, whereas this dependence if any, is suppressed in (4.7). This is due to the fact that in system analysis as opposed to synthesis, one is generally concerned with the behavior of a system of the form (4.6) under a known input $\mathbf{u}(\cdot)$.

Then (4.6) can be written as

$$\dot{\mathbf{x}}(t) = \mathbf{f}_{\mathbf{u}}(t, \mathbf{x}) = \mathbf{f}(t, \mathbf{x}, \mathbf{u}(t)) \quad (4.8)$$

which is of the form (4.7). Therefore, (4.7) can represent either an "unforced" system or a system with a fixed input.

4.2.1 Autonomy and Nonautonomy

Definition 4.1

The system described by (4.7) is said to be autonomous if $\mathbf{f}(t, \mathbf{x}) \equiv \mathbf{f}(\mathbf{x})$, i.e., independent of t and is said to be nonautonomous otherwise.

4.2.2 Equilibrium Points

Definition 4.2

A point $\mathbf{x}_0 \in R^n$ is called an equilibrium point for system (4.7) at time t_0 if $\mathbf{f}(t, \mathbf{x}_0) \equiv 0$ for all $t \geq t_0$.

If $\mathbf{x}_o$ is an equilibrium point of (4.7) at time t_o , then it is clear that $\mathbf{x}_o$ is also an equilibrium point of (4.7) at all times $t \geq t_o$. Also if (4.7) is autonomous, then $\mathbf{x}_o$ is an equilibrium point of (4.7) at all times. Conversely, if a solution of a differential equation turns out to be a constant vector, then it is an equilibrium point.

Example 4.1

Consider the motion of a simple pendulum described by the equations

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -k \sin x_1 \quad k > 0\end{aligned}\tag{4.9}$$

This system is autonomous. Also, $\mathbf{x}_o = [x_{10} \ x_{20}]^T$ is an equilibrium point of (4.9) if and only if

$$\left. \begin{aligned}x_{20} &= 0 \\ \sin x_{10} &= 0\end{aligned} \right\} \text{ simultaneously}\tag{4.10}$$

i.e., the set of equilibrium points of (4.9) is the set of points in R^2 of the form

$$(\pi n, 0), \quad n = 0, \pm 1, \pm 2, \dots$$

Physically, the pendulum has two equilibrium points. One of those is located as shown in Fig. 4.1a and the second point is located as shown in Fig. 4.1b. However, the model of this pendulum, described by (4.9), has countably infinitely many equilibrium points which are located in R^2 as shown.

Definition 4.3

An equilibrium point $\mathbf{x}_e$ of (4.7) is said to be an isolated equilibrium point if there exists some neighborhood N of $\mathbf{x}_e$ which does not contain any other equilibrium point of (4.7).



Figure 4.1 Equilibrium points of a pendulum.

Example 4.2

A simple model representing the spreading of a disease in a given population is represented by the equations

$$\dot{x}_1 = -a x_1 + b x_1 x_2 \quad (4.11)$$

$$\dot{x}_2 = -b x_1 x_2$$

where x_1 denotes the density of infected individuals, x_2 denotes the density of noninfected individuals, and $a > 0$ and $b > 0$ are parameters. These equations are valid only for the case $x_1 \geq 0$ and $x_2 \geq 0$.

In this system, every point on the positive x_2 -axis is an equilibrium point, as a result none of the equilibrium points are isolated.

4.3 STABILITY IN THE SENSE OF LYAPUNOV

We now give precise definitions of some stability concepts. Throughout this section,

we consider systems of equations (4.7), and we assume that (4.7) possesses an isolated equilibrium at the origin. Thus, $f(t, o) = o$ for all $t \geq o$.

4.3.1 Stability Definitions

Definition 4.4

The equilibrium $x = o$ of (4.7) is said to be stable in the sense of Lyapunov, or simply stable, if for every real number $\epsilon > o$ and initial time $t_o > o$ there exists a real number $\delta(\epsilon, t_o) > o$ such that for all initial conditions satisfying the inequality

$$\|x(t_o)\| = \|x_o\| < \delta$$

the motion satisfies

$$\|x(t)\| < \epsilon \text{ for all } t \geq t_o$$

The symbol $\|\cdot\|$ stands for a norm. Several norms can be defined on an n -dimensional vector space. For further reading on this topic, refer to [3, 4].

Note that stability in the sense of Lyapunov is a local concept. It does not tell us a priori how small δ had to be chosen in the definition.

The equilibrium $x = o$ at time t_o is unstable at time t_o if it is not stable at time t_o . This means that the equilibrium point $x = o$ is unstable if for some $\epsilon > o$, no $\delta > o$ can be found, such that when $\|x_o\| < \delta$, $\|x(t)\| < \epsilon$ for all $t \geq t_o$. Physically, this means that the equilibrium point $x = o$ is unstable if there is some ball B_ϵ of radius ϵ centered at $x = o$ such that for every $\delta > o$, no matter how small, there is a nonzero initial state $x(t_o)$ in B_δ such that the trajectory starting at $x(t_o)$ eventually leaves B_ϵ .

In Fig. 4.2 we depict the behavior of the trajectories in the vicinity of stable equilibrium point for the case $x \in R^2$. By choosing the initial points in a sufficiently small

spherical neighborhood of radius δ , we can force the graph of the solution for $t \geq t_0$ to lie entirely inside a cylinder of radius ϵ .

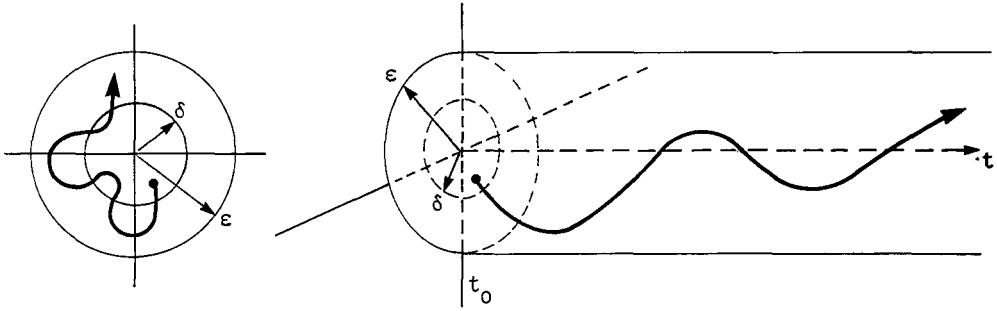


Figure 4.2 Stability.

The definition of stability given is unsatisfactory from an engineering viewpoint, where one is more interested in a stricter requirement of the system behavior to eventually return to some equilibrium point.

4.3.2 Asymptotic Stability

Definition 4.5

The equilibrium $\mathbf{x} = \mathbf{o}$ of (4.7) is asymptotically stable at time t_0 , if

1. $\mathbf{x} = \mathbf{o}$ is stable at $t = t_0$
2. For every $t_0 \geq 0$ there exists an $\eta(t_0) > 0$ such that

$$\lim_{t \rightarrow \infty} \|\mathbf{x}(t)\| \rightarrow 0 \quad \text{whenever} \quad \|\mathbf{x}(t_0)\| < \eta$$

(ATTRACTIVITY)

It is important to note that this definition combines the aspect of stability as well as attractivity of the equilibrium. The concept is again a local one, because the region containing all the initial conditions which converge to the equilibrium is some portion of

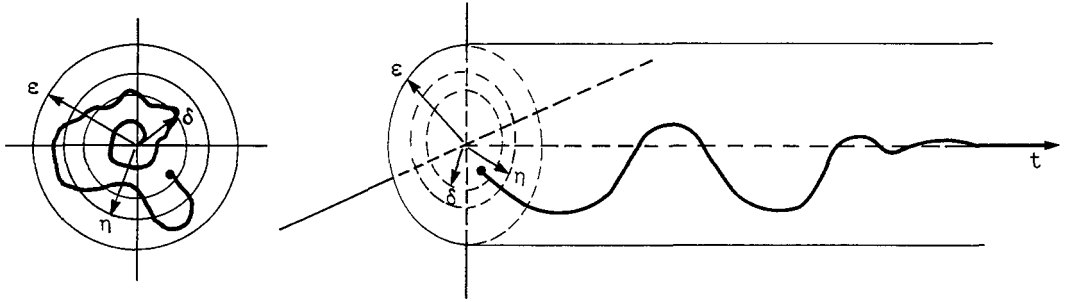


Figure 4.3 Asymptotic stability.

the state space. Figure 4.3 provides a pictorial description of the concept of asymptotic stability.

4.3.3 Lyapunov's Theorems

In stating Lyapunov's theorems, certain sign definiteness properties of functions need to be established. In doing this we use the following definition.

Definition 4.6

A continuous function $x : [0, r_1] \rightarrow R^+$ (where $R^+ = [0, \infty)$) (or a continuous function $\psi : R^+ \rightarrow R^+$) is said to belong to class K , i.e., $\psi \in K$ if

1. $\psi(0) = 0$
2. $\psi(a_1) > \psi(a_2)$ whenever $a_1 > a_2$

A continuous function $\psi : R^+ \rightarrow R^+$ is said to belong to class KR , i.e., $\psi \in KR$ if

1. $\psi \in K$
2. $\lim_{r \rightarrow \infty} \psi(r) = \infty$

Definition 4.7

A continuous function $v : R^+ \times R^n \rightarrow R$ is said to be positive definite if there exists a function $\psi \in K$ such that

1. $v(t, \mathbf{x}) \geq \psi(\|\mathbf{x}\|)$ for all $t \geq 0$
for all $\mathbf{x} \in B_r$; $r > 0$
2. $v(t, \mathbf{0}) = 0$ for all $t \geq 0$

where B_r is a ball of radius r defined by the set $\{\mathbf{x} \in R^n : \|\mathbf{x}\| < r\}$.

Definition 4.8

A continuous function $v : R^+ \times R^n \rightarrow R$ is said to be radially unbounded if there exists a function $\psi \in KR$ such that

1. $v(t, \mathbf{x}) \geq \psi(\|\mathbf{x}\|)$ for all $t \geq 0$
for all $\mathbf{x} \in R^n$
2. $v(t, \mathbf{0}) = 0$ for all $t \geq 0$

Definition 4.9

v is negative definite if $-v$ is positive definite.

Definition 4.10

A continuous function $v : R^+ \times R^n \rightarrow R$ is said to be decrescent if there exists a function $\psi \in K$ such that

$$|v(t, \mathbf{x})| \leq \psi(\|\mathbf{x}\|) \text{ for all } t \geq 0 \text{ for all } \mathbf{x} \in B_r, r > 0$$

Definition 4.11

Let $v : R^+ \times R^n \rightarrow R$ (resp., $v : R^+ \times B_r \rightarrow R$) be continuously differentiable with respect to all of its arguments and let ∇v denote the gradient of v with respect to $\mathbf{x}$. Then $\dot{v} \text{ (4.7)} : R^+ \times R^n \rightarrow R$ (resp. $v : R^+ \times B_r \rightarrow R$) is defined by

$$\begin{aligned}
 \dot{v}_{(4.7)}(t, \mathbf{x}) &= \frac{\partial v}{\partial t}(t, \mathbf{x}) + \sum_{i=1}^n \frac{\partial v}{\partial x_i}(t, \mathbf{x}) f_i(t, \mathbf{x}) \\
 &= \frac{\partial v}{\partial t}(t, \mathbf{x}) + \nabla v(t, \mathbf{x})^T f(t, \mathbf{x})
 \end{aligned} \tag{4.12}$$

We call $\dot{v}_{(4.7)}$ the derivative of v (with respect to t) along the solutions of (4.7) (or along the trajectories of (4.7)).

Before we state the principal Lyapunov-type stability results, we present a geometric interpretation of some of these results in R^2 derived from [3]. To this end, we consider the system of equations

$$\begin{cases} \dot{x}_1 = f_1(x_1, x_2) \\ \dot{x}_2 = f_2(x_1, x_2) \end{cases} \tag{4.13}$$

and we assume that f_1 and f_2 are such that for every $(t_o, \mathbf{x}_o)$, $t_o \geq 0$, (4.13) has a unique solution $\mathbf{x}(t, t_o, \mathbf{x}_o)$ with $\mathbf{x}(t_o, t_o, \mathbf{x}_o) = \mathbf{x}_o$. We also assume that $(x_1, x_2)^T = (o, o)^T$ is the only equilibrium in B_h for some $h > 0$. Then $v(\mathbf{x}) = c$, ($c \geq 0$) defines for sufficiently small constants $c > 0$ a family of closed curves c_i which cover the neighborhood B_h as shown in Fig. 4.4. Note that the origin $\mathbf{x} = o$ is located in the interior of each such curve and in fact $c_o = \{o\}$.

Now suppose that all trajectories of (4.13) originating from points on the circular disc $\|\mathbf{x}\| \leq r_1 \leq h$ cross the curves $v(\mathbf{x}) = c$ from the exterior toward the interior when we proceed along these trajectories in the direction of increasing values of t . Then we can conclude that these trajectories approach the origin as t increases, i.e., the equilibrium $\mathbf{x} = o$ is in this case asymptotically stable.

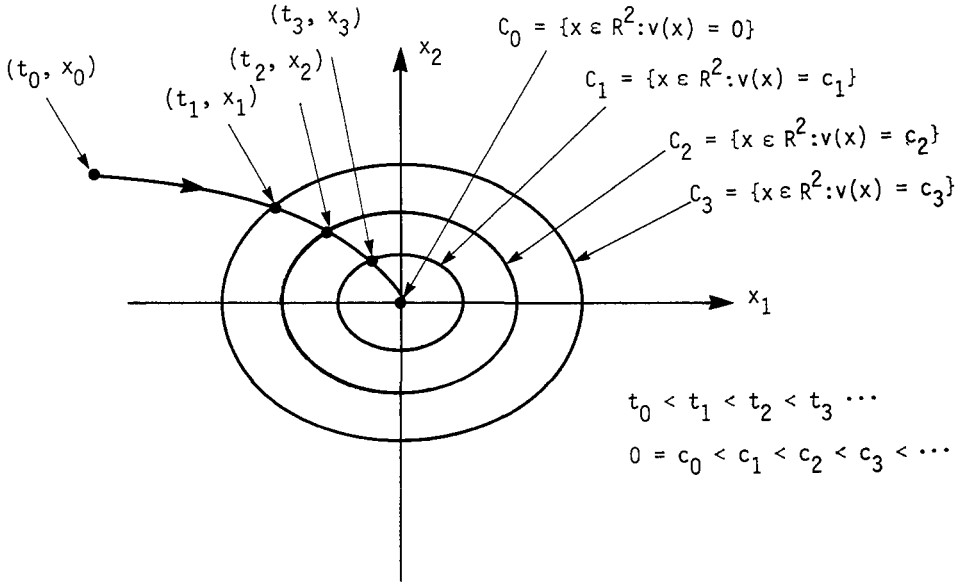


Figure 4.4 Asymptotic stability.

In terms of the given v function, this means that the function decreases monotonically as t increases. This implies that the derivative $\dot{v}_{(4.13)}$ must be negative definite in some B_r for $r > 0$ sufficiently small.

THEOREM 4.1

If there exists a continuously differentiable positive definite function v with a negative semidefinite (or identically zero) derivative $\dot{v}_{(4.7)}$, then the equilibrium $x = 0$ of (4.7) is stable.

Example 4.3

Consider the system shown in Fig. 4.5, with $f(\cdot)$, $g(\cdot)$ continuously differentiable and satisfying the following conditions.

$\sigma g(\sigma) > 0$ for $\sigma \neq 0$ and

$$\sigma \in [-\sigma_0, +\sigma_0], \sigma_0 > 0$$

and $\sigma f(\sigma) \geq 0$ for $\sigma \in [-\sigma_0, +\sigma_0]$.

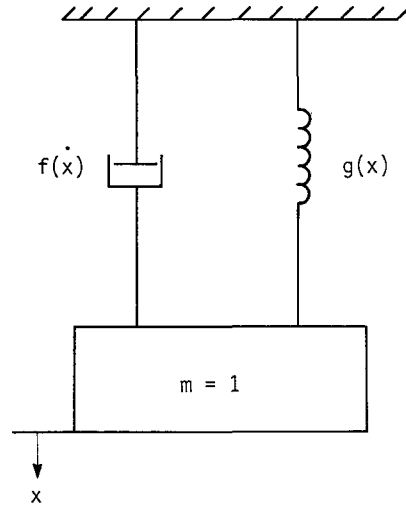


Figure 4.5 Schematic diagram of a spring mass system.

The equations describing the motion of this system are given by

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -f(x_2) - g(x_1) \end{aligned} \quad (4.14)$$

The system (4.14) has an isolated equilibrium at $x = 0$. The total energy for the system is the sum of the kinetic energy and potential energy, given by

$$v(x) = \frac{1}{2} x_2^2 + \int_0^{x_1} g(\eta) d\eta$$

Note that this function is continuously differentiable, that $v(0) = 0$, and that $v(x)$ is positive definite in the region determined by $-\infty < x_2 < \infty$, $-\sigma_0 < x_1 < \sigma_0$, shown in Fig. 4.6.

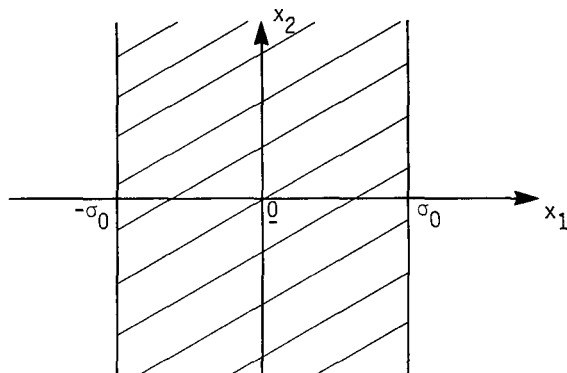


Figure 4.6 Stability region.

$$\begin{aligned}\dot{v}_{(4.14)}(x) &= \nabla v(x)^T f(x) \\ &= [g(x_1) \ x_2] \begin{bmatrix} x_2 \\ -f(x_2) - g(x_1) \end{bmatrix}\end{aligned}$$

$$\dot{v}_{(4.14)}(x) = -x_2 f(x_2) \leq 0$$

In accordance with Theorem 4.1, the equilibrium $x = o$ of (4.14) is stable.

THEOREM 4.2

If there exists a continuously differentiable, positive definite function v with a negative definite derivative $\dot{v}_{(4.7)}$, then the equilibrium $x = o$ of (4.7) is asymptotically stable.

Example 4.4

Consider the system described in Example 4.3. For the v function, $\dot{v}_{(4.14)}$ is only negative semidefinite for all values of x in the region shown in Fig. 4.6; as a result, no statement can be made regarding the asymptotic stability of the system using Theorem 4.2

as the condition of $\dot{v}_{(4.14)}$ being negative definite is not satisfied. This is one of the limitations of Lyapunov's original results. We shall relax this constraint later using the invariance principle.

Example 4.5

Consider the system

$$\begin{aligned}\dot{x}_1 &= (x_2 - c_2 x_2) (x_1^2 + x_2^2 - 1) \\ \dot{x}_2 &= (c_1 x_1 + x_2) (x_1^2 + x_2^2 - 1)\end{aligned}\tag{4.15}$$

which has an isolated equilibrium at the origin $\mathbf{x} = \mathbf{o}$. Choosing

$$v(\mathbf{x}) = c_1 x_1^2 + c_2 x_2^2,$$

we obtain

$$\dot{v}_{(4.15)}(\mathbf{x}) = 2(c_1 x_1^2 + c_2 x_2^2)(x_1^2 + x_2^2 - 1)$$

If $c_1 > 0$, $c_2 > 0$, then v is positive definite and radially unbounded and $\dot{v}_{(4.15)}$ is negative definite in the region $x_1^2 + x_2^2 < 1$. Theorem 4.2 is applicable and we conclude that the equilibrium $\mathbf{x} = \mathbf{o}$ is asymptotically stable.

4.3.4 Invariance Theory

In this subsection, we present some results based on the invariance theory of ordinary differential equations and apply it to autonomous systems of the form

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})\tag{4.16}$$

Here we assume that $\mathbf{f}$ and $\partial \mathbf{f} / \partial x_i$, $i = 1, 2, \dots, n$, are continuous in a region D , which is a subset of R^n , denoted by $D \subset R^n$ (where D may be all of R^n), and we assume that $\mathbf{x} = \mathbf{o}$ is in the interior of D . We also assume that $\mathbf{x} = \mathbf{o}$ is an isolated equilibrium.

Definition 4.12

A set Γ of points in R^n is invariant with respect to (4.16) if every solution of (4.16) starting in Γ remains in Γ for all time.

Example 4.6

A set consisting of any equilibrium point of (4.16) is an invariant set with respect to (4.16).

THEOREM 4.3

Let v be a continuously differentiable, real-valued function defined on some domain $D \subset R^n$ containing the origin and assume that $\dot{v}_{(4.16)} \leq 0$ on D . Assume that $v(0) = 0$. For some real constant $k \geq 0$, let H_k be the component of the set $S_k = \{x : v(x) \leq k\}$ which contains the origin. Suppose that H_k is a closed and bounded subset of D . Let $E = \{x \in D : \dot{v}_{(4.16)} = 0\}$. Let M be the largest invariant subset of E with respect to (4.16). Then every solution of (4.16) starting on H_k at $t = 0$ approaches the set M as $t \rightarrow \infty$.

This theorem by itself does not provide a stability result, but describes the qualitative behavior of the system. Using this theorem we can now establish the following stability result.

THEOREM 4.4

Assume that for system (4.16) there exists a continuously differentiable, real-valued, positive definite function v defined on some set $D \subset R^n$ containing the origin. Assume that $\dot{v}_{(4.16)} \leq 0$ on D . Suppose that the origin is the only invariant subset with respect to (4.16) of the set $E = \{x \in D : \dot{v}_{(4.16)} = 0\}$. Then the equilibrium $x = 0$ of (4.16) is asymptotically stable.

We shall now demonstrate how the invariance principle helps provide a stronger stability result in autonomous systems via an example.

Example 4.6

Consider the system analyzed in Examples 4.3 and 4.4. From 4.3, we could ascertain that the equilibrium point is stable. However, no conclusion regarding asymptotic stability could be derived using Theorem 4.2 in Example 4.4, as $\dot{v}_{(4.14)}$ was only negative semidefinite. We will now apply the result of Theorem 4.4 to this system.

From Example 4.3,

$$\dot{v}_{(4.14)}(x) = -x_2 f(x_2) \leq 0$$

$$E = \{x \in R^2: \dot{v}_{(4.14)}(x) = 0\} = x_1 - \text{axis}$$

The set E is thus described by the entire x_1 - axis. In order to apply Theorem 4.4, we then have to show that the only invariant set in E is the origin $x = o$. To do this consider the slope function for this system given by

$$\frac{dx_2}{dx_1} = \frac{-f(x_2) - g(x_1)}{x_2}$$

In E where $x_2 = 0$, $\frac{dx_2}{dx_1} = \pm \infty$. The direction of these slopes is shown in Fig. 4.7. They clearly show that no trajectory of (4.14) can lie in E for all time except the origin $x = o$. As a result all the conditions of Theorem 4.4 are satisfied; hence, the equilibrium $x = o$ is asymptotically stable.

If by some method we can show that all solutions of (4.16) remain bounded as $t \rightarrow \infty$, then the following result concerning bounded solutions of (4.16) will be useful.

THEOREM 4.5

Let $v: R^n \rightarrow R^+$ be a continuously differentiable and let $v(o) = 0$. Suppose that

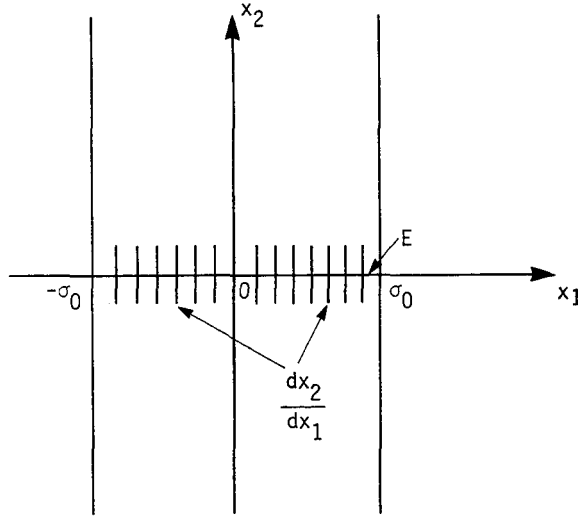


Figure 4.7 Illustration of invariance principle.

$\dot{v}_{(4.16)} \leq 0$ for all $x \in R^n$. Let $E = \{x \in R^n : \dot{v}_{(4.16)}(x) = 0\}$. Let M be the largest invariant subset of E . Then all bounded solutions of (4.16) approach M as $t \rightarrow \infty$.

4.3.5 Domain of Attraction

Definition 4.13

The set of all $x(t_0) \in R^n$ such that $x(t, t_0, x(t_0)) \rightarrow 0$ as $t \rightarrow \infty$ for some $t_0 \geq 0$ is called the domain of attraction of the equilibrium $x = 0$ of (4.7).

Many practical systems possess more than one equilibrium point. In such cases, one is usually very interested in knowing the extent of the domain of attraction of an asymptotically stable equilibrium. In this subsection, we briefly address the problem of obtaining the estimates of the domain of attraction of the equilibrium $x = 0$ of the autonomous system (4.16). For a detailed discussion of this topic refer to [3, Section 5.12].

We assume that f and $\partial f / \partial x_i$, $i = 1, 2, \dots, n$, are continuous in a region $D \subset R^n$, and we assume that $\mathbf{x} = \mathbf{o}$ is in the interior of D . We also assume that $\mathbf{x} = \mathbf{o}$ is an isolated equilibrium point, and we let $\mathbf{x}(t, \mathbf{x}_0)$ be a solution of (4.16) satisfying $\mathbf{x}(\mathbf{o}) = \mathbf{x}_0$.

Let us assume that there exists a continuously differentiable and positive definite function v such that

$$\dot{v}_{(4.16)}(\mathbf{x}) \leq 0 \text{ for all } \mathbf{x} \in D$$

Let $E = \{\mathbf{x} \in D : \dot{v}_{(4.16)}(\mathbf{x}) = 0\}$ and suppose that $\{\mathbf{o}\}$ is the only invariant subset of E with respect to (4.16). Let $n = 2$ and suppose that $S_\ell = \{\mathbf{x} : v(\mathbf{x}) \leq \ell\}$ is a closed and bounded subset of D . Let H_ℓ be the component of S_ℓ which contains the origin for $\ell \geq 0$. (Note that when $\ell = 0$, $H_\ell = \{\mathbf{o}\}$.) From Theorem 4.3 we can say that every closed and bounded region H_ℓ which is contained in D will also be contained in the domain of attraction of the origin $\mathbf{x} = \mathbf{o}$. Thus, we can compute $\ell = \ell_m$, so that H_{ℓ_m} has this property, as the largest value of ℓ for which the component of $S_{\ell_m} = \{\mathbf{x} : v(\mathbf{x}) = \ell_m\}$ containing the origin, actually meets the boundary of D . Note that even when D is unbounded, all sets of H_ℓ which are completely contained in D are invariant sets with respect to (4.16). Thus, every bounded solution of (4.16) which starts in H_ℓ tends to the origin by Theorem 4.5.

4.3.6 The Potential Energy Boundary Surface

The concept of the potential energy boundary surface (PEBS) was first proposed by Kakimoto and colleagues [5–7], and later extended by Athay and colleagues, as well as other researchers [8–13]. For a given disturbance trajectory, the PEBS describes a "local" approximation of the stability boundary. The process of finding this local approximation is associated with the determination of the stability boundary of a lower dimensional system. Fig. 4.8, which was taken from [8], shows the PEBS for a three-machine system. It is

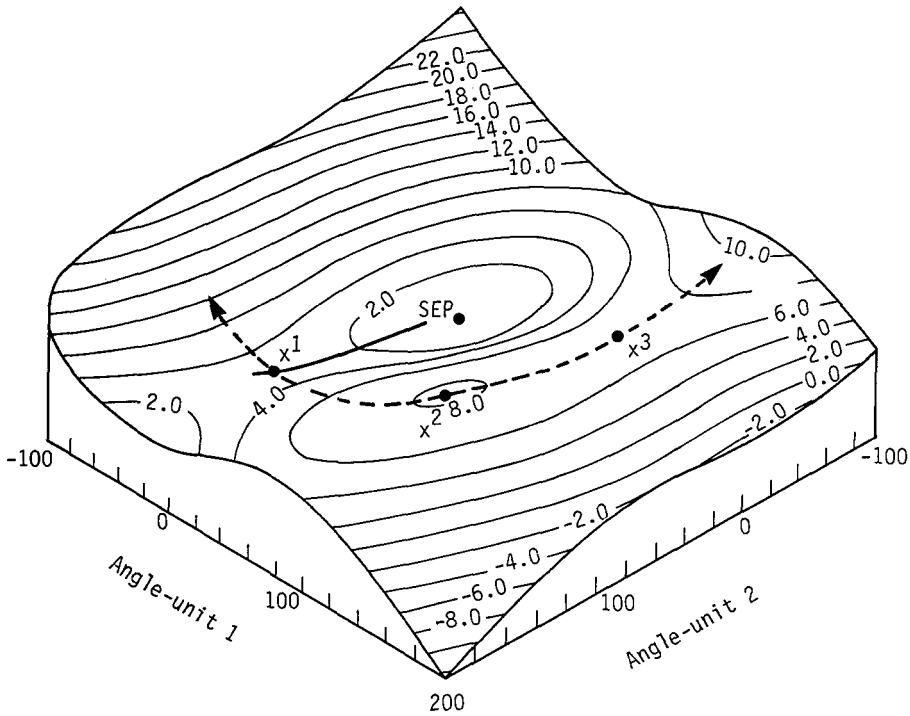


Figure 4.8 The PEBS. Reprinted from © *IEEE Transactions on Power Apparatus and Systems* 98 (1979).

formed by joining the points of maximum potential along any direction originating from the stable equilibrium point x^s shown in Fig. 4.8. The PEBS constructed in this manner is orthogonal to the equipotential curves. Furthermore, along the direction orthogonal to the PEBS the potential energy achieves a local maximum at the PEBS.

In [9], the concept of the PEBS was characterized by means of a constructive procedure for a power system with transfer conductances. From equation (2.13) we have for a n -machine power system

$$f_i(\theta) = P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI}$$

The potential energy is then given by

$$V_{PE}(\theta) = - \sum_{i=1}^n \int_{\theta_i^{s2}}^{\theta_i} f_i(\theta) d\theta_i \quad (4.17)$$

where θ^{s2} is the postdisturbance stable equilibrium point (SEP).

Invoking the linear path assumption,

$$\theta = (\theta - \theta^{s2}) \alpha + \theta^{s2} = \Delta \theta \alpha + \theta^{s2} \quad (4.18)$$

leads to V_{PE} being approximated as

$$V_{PEapprox}(\theta) = - \sum_{i=1}^n \int_0^{\alpha} f_i(\Delta \theta \alpha + \theta^{s2}) \Delta \theta_i d\alpha \quad (4.19)$$

$$\frac{dV_{PEapprox}}{d\theta} = 0 \text{ implies } - \sum_{i=1}^n f_i(\Delta \theta \alpha + \theta^{s2}) \Delta \theta_i = 0 \quad (4.20)$$

If (4.20) is satisfied for some α , i.e., some θ^* then it means that θ^* satisfies

$$\sum_{i=1}^n f_i(\theta) (\theta - \theta_i^{s2}) = 0 \quad (4.21)$$

This constructive procedure, characterizes a point on the PEBS with angle-space coordinates θ^* via the following property. Along the straight path (in θ -space) emanating from the postdisturbance SEP θ^{s2} , the potential energy function is maximized for the first time at θ^* . In other words, the PEBS can be characterized in θ -space by setting the directional derivative of the potential energy function equal to zero (4.21) along rays emanating from the postdisturbance SEP θ^{s2} .

4.4 ANALYTICAL JUSTIFICATION FOR THE CONTROLLING UEP AND THE PEBS

Most of the results dealing with the transient energy function method applied to power systems have been derived largely based on heuristics and simulations. In [14] and [15] a complete topological characterization of the stability boundary of a stable equilibrium point is derived. We will now present some results from [14], where the domain of attraction for a multimachine power system is characterized and the concept of controlling UEP analytically justified.

4.4.1 Characterization of the Stability Boundary

Consider the stability boundary of the system described by (4.16). Some standard terminology is established first to describe the concept. The derivative of f at a point x^a is called the Jacobian matrix at x^a and is denoted by $J(x^a)$. An equilibrium point (Definition 4.2) is hyperbolic if the Jacobian has no eigenvalues with zero real part. The stability region or domain of attraction (Definition 4.13) of a stable equilibrium point x^s is denoted by $A(x^s)$. The stability boundary (boundary of the stability region) and the closure of $A(x^s)$ are denoted by $\partial A(x^s)$ and $\bar{A}(x^s)$. A closure of a set is formed by adjoining to the set all its boundary points. Thus, $\bar{A}(x^s) = A(x^s) \cup \partial A(x^s)$. E denotes the set of all equilibrium points of (4.16). The type of the equilibrium point x^a is defined according to the number of

eigenvalues of $J(x^a)$ with positive real part. If the Jacobian of the equilibrium point x^a has exactly one eigenvalue with positive real part, we call it a type-one equilibrium point.

Definition 4.14

Let x^a be a hyperbolic equilibrium point of (4.16); its stable manifold $W^s(x^a)$ and unstable manifold $W^u(x^a)$ are defined as

$$W^s(x^a) = \left\{ x_o : x(t, x_o) \rightarrow x^a \text{ as } t \rightarrow \infty \right\}$$

$$W^u(x^a) = \left\{ x_o : x(t, x_o) \rightarrow x^a \text{ as } t \rightarrow -\infty \right\}$$

The stable and unstable manifolds are invariant sets.

Definition 4.15

Consider two manifolds A, B in R^n . We say that the intersection of A and B denoted by $A \cap B$ satisfies the transversality condition if (1) at every point of the intersection, $x \in (A \cap B)$, the tangent spaces of A and B at x span the tangent space of R^n at x

$$T_x(A) + T_x(B) = R^n \text{ for } x \in A \cap B$$

or (2) they do not intersect at all.

Furthermore, the following assumptions concerning (4.16) are made.

- (A1) All the equilibrium points on the stability boundary of (4.16) are hyperbolic.
- (A2) The intersection of $W^s(x^a)$ and $W^u(x^b)$ satisfies the transversality condition, for all the equilibrium point x^a, x^b on the stability boundary.
- (A3) There exists a continuously differentiable function $V : R^n \rightarrow R$ for (4.16) such that
 1. $\dot{V}_{(4.16)}(x) \leq 0$ at $x \in E$.
 2. If x is not an equilibrium point, then the set $\{t \in R : \dot{V}_{(4.16)}(x) = 0\}$ has measure 0 in R ; i.e., its length is zero.

3. $V(\mathbf{x})$ is bounded.

We shall refer to this function as the "energy function" for later reference. Based on these assumptions the following three theorems, characterizing the stability boundary of system (4.16), are proved in [14].

THEOREM 4.6

For the dynamical system (4.16) satisfying assumptions (A1)–(A3), $\mathbf{x}^i$ is an unstable equilibrium point on the stability boundary $\partial A(\mathbf{x}^s)$ of a stable equilibrium point $\mathbf{x}^s$ if and only if $W^u(\mathbf{x}^i) \cap A(\mathbf{x}^s) \neq \emptyset$.

THEOREM 4.7

For the dynamical system (4.16) satisfying assumptions (A1)–(A3), let $\mathbf{x}^i$, $i = 1, 2, \dots$, be the unstable equilibrium points on the stability boundary $\partial A(\mathbf{x}^s)$ of a stable equilibrium point $\mathbf{x}^s$; then

$$\partial A(\mathbf{x}^s) = \bigcup_{\mathbf{x}^i \in E \cap \partial A} W^s(\mathbf{x}^i)$$

i.e., the stability boundary ∂A is the union of all the stable manifolds of those unstable equilibrium points contained in the intersection of E and ∂A .

THEOREM 4.8

If the dynamical system (4.16) satisfies the assumption (A3), then on the stable manifold $W^s(\mathbf{x}^i)$ of an equilibrium point $\mathbf{x}^i$, the point at which the energy function achieves the minimum is the equilibrium point $\mathbf{x}^i$ itself; i.e.,

$$V(\mathbf{x}^i) = \min_{\mathbf{x} \in W^s(\mathbf{x}^i)} V(\mathbf{x})$$

For the classical power system model developed in Chapter 2, for the case with transfer conductances neglected, these theorems can be applied [14] and the concept of the region of stability can be rigorously justified. Furthermore, the concept of the controlling UEP can also be justified.

Consider the fault-on trajectory $\mathbf{x}(t)$, starting from $\mathbf{x}_0$ at the time of the fault, as shown in Fig. 4.9. If UEP $\mathbf{x}^3$ was selected to be the controlling UEP, then the set of points on the dashed curve denoted by $R(v_{\mathbf{x}^3})$ would represent the boundary of stability ∂A . It is obvious that, for this case, the set $R(v_{\mathbf{x}^3}) = \{\mathbf{x} : V(\mathbf{x}) < v_{\mathbf{x}^3}\}$ is an underestimate of the "true" critical energy. Let us call the point $\mathbf{x}^e$ where the fault-on trajectory $\mathbf{x}(t)$ crosses the stability boundary $\partial A(\mathbf{x}^s)$, the exit point of the trajectory $\mathbf{x}(t)$. The value of the energy function at the exit point is actually the true critical energy. Suppose that $\mathbf{x}^e$ belongs to the stable manifold $W^s(\mathbf{x}^{co})$ of the UEP $\mathbf{x}^{co}$ on the stability boundary and $V(\mathbf{x}_0) < V(\mathbf{x}^{co})$. Since $V(\mathbf{x})$ assumes its minimum on the stable manifold $W^s(\mathbf{x}^{co})$ at the equilibrium point $\mathbf{x}^{co}$ (Theorem 4.8), the fault-on trajectory must pass through the constant energy surface $\{\mathbf{x} : V(\mathbf{x}) = V(\mathbf{x}^{co})\}$ before it reaches the exit point $\mathbf{x}^e$. Let us define $v_{co} = V(\mathbf{x}^{co})$. Clearly, $v_{co} \geq v_{\mathbf{x}^3}$; hence, v_{co} is a better estimate of the critical energy. For a given fault-on trajectory $\mathbf{x}(t)$, if the exit point $\mathbf{x}^e$ lies on the stable manifold of $\mathbf{x}^{co}$, then we call $\mathbf{x}^{co}$ the controlling UEP. To find the controlling UEP is a nontrivial problem. An algorithm to determine it will be developed in Chapter 5 and detailed in Chapter 6.

This completes the characterization of the region of stability for a multi-machine power system.

4.4.2 Characterization of the PEBS

In [5–13], the concept of the PEBS had been developed based on heuristic arguments and simulation. A theoretical foundation for the PEBS was provided in [16] based on the

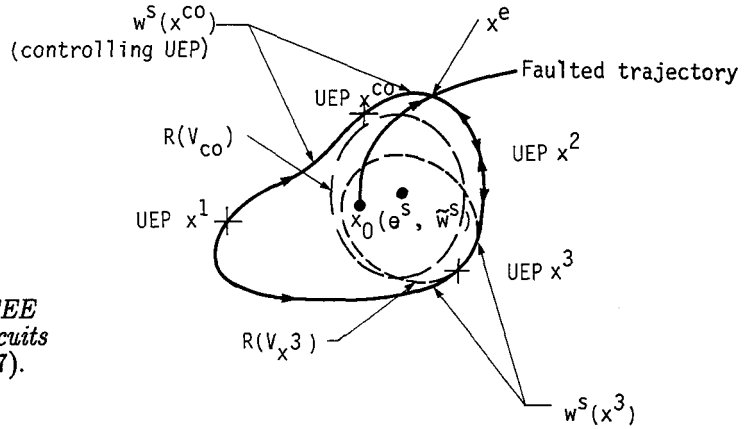


Figure 4.9 The concept of controlling UEP. Reprinted from © IEEE Transactions on Circuits and Systems 34 (1987).

approach suggested in [17–18], to view the PEBS as the stability boundary of the associated gradient system.

From the equations developed in Chapter 2, the equations of motion for a classical power system, with transfer conductances neglected, can be represented by

$$\begin{aligned} \dot{\theta}_i &= \tilde{\omega}_i \\ M_i \dot{\tilde{\omega}}_i &= -D_i \tilde{\omega}_i - \frac{\partial V_{PE}(\theta)}{\partial \theta_i} \quad i = 1, 2, \dots, n \end{aligned} \quad (4.22)$$

where D_i is the damping coefficient and $V_{PE}(\theta)$ is given by (4.17).

The associated gradient system is described by

$$\dot{\theta}_i = -\frac{\partial V_{PE}(\theta)}{\partial \theta_i} \quad i = 1, 2, \dots, n \quad (4.23)$$

The following assumptions are also needed to state some results.

- (C1) The intersection of the stable and unstable manifolds of the equilibrium points on the stability boundary satisfy the transversality condition (Definition 4.15).
- (C2) The number of equilibrium points on the stability boundary is finite.

The following results [16] can then be proved.

THEOREM 4.9

Let $\mathbf{x}^s$ be a stable equilibrium point of (4.23) and $\mathbf{x}^i$, $i = 1, 2, \dots, n$ be the equilibrium points lying on $\partial A_{(4.23)}(\mathbf{x}^s)$. If assumption (C1) is satisfied, then

$$\partial A_{(4.23)} = \bigcup_{\mathbf{x}^i \in E \cap \partial A_{(4.23)}(\mathbf{x}^s)} W_{(4.23)}^s(\mathbf{x}^i)$$

THEOREM 4.10

$\mathbf{x}^s$ is an equilibrium point of (4.23) if and only if $(\mathbf{x}^s, \mathbf{o})$ is an equilibrium point of (4.22).

Definition 4.16

The stability boundary $\partial A(\theta^s)$ (Theorem 4.9) of (4.23) is the PEBS.

THEOREM 4.11

The PEBS intersects the level surface $\{\theta: V_{PE}(\theta) = c\}$ orthogonally.

We are now in a position to discuss under what conditions the PEBS method gives a good approximation of the relevant local stability boundary of the original system (4.22). Recall, from the previous subsection that the point that the fault-on trajectory leaves the

stability boundary (of the postdisturbance system) is called the exit point. The stability boundary of system (4.22) (or (4.23)) is the union of stable manifolds of the equilibrium points on the stability boundary. The relevant local stability boundary is the stable manifold on which the exit point of the fault-on trajectory lies. Given a fault-on trajectory $(\theta(t), \tilde{\omega}(t))$, let $\theta^e, \tilde{\omega}^e$ denote the exit point on the stability boundary of the original system (4.22), and let $\bar{\theta}^e$ denote the exit point of the projected trajectory, i.e., the projection of the fault-on trajectory in the θ -space of the reduced system (4.23). Let $\partial V(\mathbf{x})$ denote the connected constant energy surface of $V(\cdot)$ containing the stable equilibrium point (θ^s, o) passing through the point $\mathbf{x}$, i.e., $\partial V(\mathbf{x}) =$ the connected component of the set $\{(\theta, \tilde{\omega}) : V(\theta, \tilde{\omega}) = V(\mathbf{x})\}$ containing the SEP (θ^s, o) . The idea of the PEBS method is to use the set $\partial V(\bar{\theta}^e, o)$ to approximate the relevant local stability boundary of the original system (4.22).

To determine the effectiveness of $\partial V(\bar{\theta}^e, o)$ in approximating the relevant local stability boundary of the original system (4.22), the following result is used. The notation $V_c(\mathbf{x})$ is used to denote the connected component of the set $\{(\theta, \tilde{\omega}) : V(\theta, \tilde{\omega}) < V(\mathbf{x})\}$ containing the stable equilibrium point (θ^s, o) .

THEOREM 4.12

Let $(\hat{\theta}, o)$ be an equilibrium point on the stability boundary $\partial A(\theta^s, o)$ of (4.22).

Then,

1. The connected constant energy surface $\partial V(\hat{\theta}, o)$ intersects with the stability manifold $W^s(\hat{\theta}, o)$ only at the point $(\hat{\theta}, o)$; moreover, the set $V_c(\hat{\theta}, o)$ does not contain any point which belongs to the stable manifold $W^s(\hat{\theta}, o)$.
2. Suppose $(\theta^1, \tilde{\omega}^1) \in W^s(\hat{\theta}, o)$, $(\theta^1, \tilde{\omega}^1) \neq (\hat{\theta}, o)$ and the equilibrium point $(\hat{\theta}, o)$ is the type $-k$. Then the dimension of the set which is the intersection between the connected constant energy surface $\partial V(\theta^1, \tilde{\omega}^1)$ and the stable manifold $W^s(\hat{\theta}, o)$

is $2n - k - 1$; moreover, the set $V_c(\theta^1, \tilde{\omega}^1)$ does contain points which belong to the stable manifold $W^s(\hat{\theta}, o)$.

These results essentially state that if the following two conditions are satisfied:

1. $(\bar{\theta}^e)$ is on the stable manifold of $(\hat{\theta})$ implies $(\theta^e, \tilde{\omega}^e)$ is on the stable manifold of $(\hat{\theta}, o)$.
2. The fault-on trajectory $(\theta(t), \tilde{\omega}(t))$ passes through the constant energy surface $\partial V(\theta^e, o)$ before it passes through the stable manifold $W^s(\hat{\theta}, o)$.

Then the PEBS gives a good approximation of the relevant local stability boundary of the original system for the given fault-on trajectory $(\theta(t), \tilde{\omega}(t))$.

The foregoing discussion is depicted pictorially in Figs. 4.10, 4.11 derived from [16].

In Fig. 4.10a we observe that the connected constant energy surface $\partial V(\hat{\theta}, o)$ can be used to approximate part of the stability boundary $\partial A(\theta^s, o)$, i.e., the stable manifold $W^s(\hat{\theta}, o)$ part, where as in Fig. 4.10b it is obvious that it is inappropriate to approximate the relevant part of the stability boundary (i.e., the stable manifold $W^s(\hat{\theta}, o)$ part, by the constant energy surface $\partial V(\theta^1, \tilde{\omega}^1)$ because part of the connected energy surface $\partial V(\theta^1, \tilde{\omega}^1)$ clearly lies outside the stability boundary $\partial A(\theta^s, o)$ as a result of which the fault-on trajectory passes through the surface $\partial V(\theta^1, \tilde{\omega}^1)$ after it passes through the stable manifold $W^s(\hat{\theta}, o)$ as shown in Fig. 4.11.

The above discussion illustrates the following salient features of the controlling UEP concept based on the analysis of the stable manifold:

1. The exit point $x^e(\theta^e, \tilde{\omega}^e)$ on the stable manifold is unique. If the clearing instant is extended beyond critical clearing, the faulted trajectory always intersects the manifold at the same point x^e .

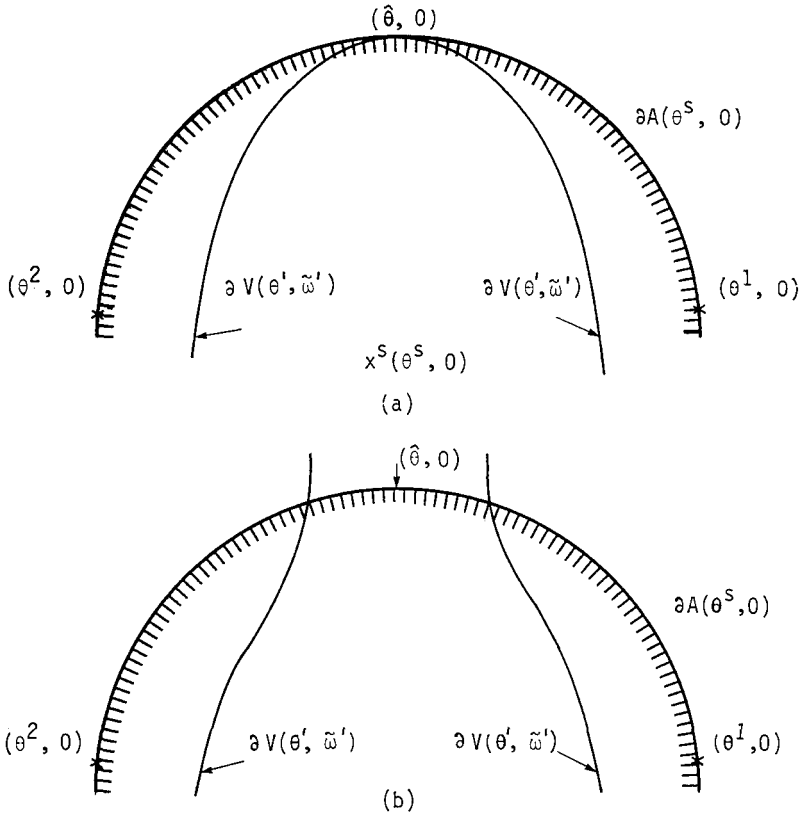


Figure 4.10 Approximation of the stability boundary. Reprinted from © IEEE Transactions on Circuits and Systems 35 (1988).

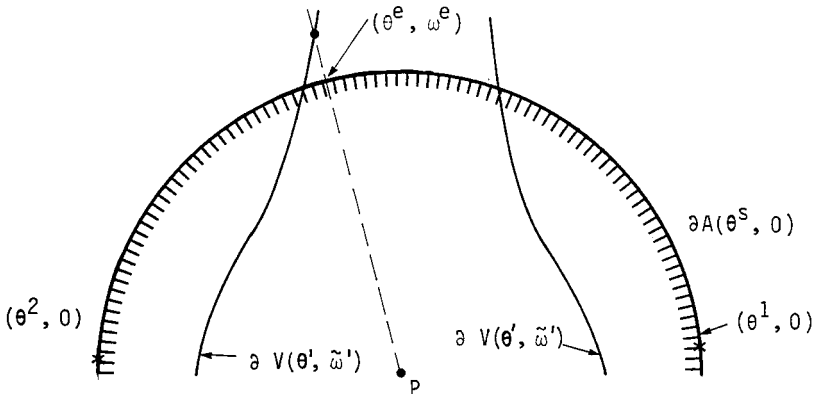


Figure 4.11 Approximation of the exit point. Reprinted from © IEEE Transactions on Circuits and Systems 35 (1988).

2. θ^e denotes the projection of the exit point on the reduced system, i.e., the projection of the fault-on trajectory in the θ -space of the reduced system. If the fault-on trajectory $(\theta(t), \tilde{\omega}(t))$ passes through the constant energy surface $\partial V(\theta^e, o)$ before it passes through the stable manifold of the controlling UEP, then the PEBS gives a good approximation of the relevant local stability boundary of the original system.
3. If an estimate of the critical energy is made by the so-called "Repetitive PEBS method" [5, 7], i.e., to clear the fault before the fault-on trajectory reaches the exit point x^e , the projection on the θ -space of the reduced system may not give an accurate estimate of the critical energy. In this case the projection on the reduced system will not be an accurate representation of the point on the cleared trajectory which approaches the stable manifold when the potential energy is a maximum, thus, an erroneous estimate of the critical energy may be obtained.

A recently proposed technique [19] uses items (1) and (2) to determine the controlling UEP. This technique is described in Chapter 6.

4.4.3 Concluding Remarks

These analytical results clearly indicate that the PEBS in certain cases can overestimate the domain of attraction. This has been observed by other researchers when stability analysis has been conducted on realistic systems [6, 9]. The analytical results also indicate that the controlling UEP if correctly determined will provide a more accurate estimate of the domain of attraction, and will give stability assessment results which will always be on the conservative side.

Based on the analysis of the authors' experience with the PEBS results on several systems and the analytical justification of the controlling UEP provided in [14], the results presented in the rest of the chapters in this book will estimate the domain attraction based solely using the controlling UEP.

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Chapter 5

THE TRANSIENT ENERGY FUNCTION METHOD APPLIED TO THE CLASSICAL POWER SYSTEM MODEL

5.1 INTRODUCTION

In this chapter we present a general overview of the transient energy function (TEF) method applied to the classical power system model developed in Chapter 2. The TEF using the center of inertia (COI) formulation will be developed for a multimachine power system. The various steps involved in conducting the transient stability assessment procedure using the TEF method will be described.

In detailing the aspects of the TEF method, the various components of the energy function will be described, and the energy contributing to system separation will be identified. A measure of the degree of stability (or instability) in terms of the transient energy margin will be introduced. In addition, the concept of the controlling unstable equilibrium point will be developed. The issues which determine trajectory behavior will be discussed and the notion of the mode of disturbance (MOD) will be introduced. The criterion to determine the correct MOD will be developed and illustrated via examples. This criterion will be compared and contrasted with other criteria in the literature.

5.2 THE TRANSIENT ENERGY FUNCTION

5.2.1 Derivation

The transient energy function can be derived directly from the dynamic equations written with respect to the COI. These equations developed in Chapter 2 are repeated here for convenience

$$M_i \ddot{\omega}_i = P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} \quad (5.1)$$

$$\dot{\theta}_i = \tilde{\omega}_i \quad i = 1, 2, \dots, n$$

These equations represent the dynamics of the postdisturbance system. The energy function for this system is derived as follows:

- Multiply the i th postdisturbance dynamic equation by $\dot{\theta}_i$ and form the sum

$$\sum_{i=1}^n \left[M_i \ddot{\omega}_i - P_i + P_{ei} + \frac{M_i}{M_T} P_{COI} \right] \dot{\theta}_i \quad (5.2)$$

- Substituting the expression for P_{ei} from (2.3), this sum takes the form

$$\sum_{i=1}^n \left[M_i \ddot{\omega}_i - P_i + \sum_{\substack{j=1 \\ j \neq i}}^n C_{ij} \sin \theta_{ij} + \sum_{\substack{j=1 \\ j \neq i}}^n D_{ij} \cos \theta_{ij} + \frac{M_i}{M_T} P_{COI} \right] \dot{\theta}_i \quad (5.3)$$

- Invoking the relationship $C_{ij} = C_{ji}$ and $D_{ij} = D_{ji}$ $i, j = 1, 2, \dots, n$

$$\sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n C_{ij} \sin \theta_{ij} \dot{\theta}_i = \sum_{i=1}^{n-1} \sum_{j=i+1}^n C_{ij} \sin \theta_{ij} (\dot{\theta}_i + \dot{\theta}_j) \quad (5.4)$$

and

$$\sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n D_{ij} \cos \theta_{ij} \dot{\theta}_i = \sum_{i=1}^{n-1} \sum_{j=i+1}^n D_{ij} \cos \theta_{ij} (\dot{\theta}_i + \dot{\theta}_j) \quad (5.5)$$

- Substituting (5.4) and (5.5) into (5.3) and integrating the resulting expression with respect to time using as a lower limit $t = t^s$ where $\tilde{\omega}(t^s) = 0$ and $\theta(t^s) = \theta^{s2}$ the postdisturbance stable equilibrium point, we obtain

$$\begin{aligned}
 V = & \frac{1}{2} \sum_{i=1}^n M_i \tilde{\omega}_i^2 - \sum_{i=1}^n P_i (\theta_i - \theta_i^{s2}) \\
 & - \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[C_{ij} (\cos \theta_{ij} - \cos \theta_{ij}^{s2}) - \int_{\theta_i^{s2} + \theta_j^{s2}}^{\theta_i + \theta_j} D_{ij} \cos \theta_{ij} d(\theta_i + \theta_j) \right]
 \end{aligned} \tag{5.6}$$

Note that in (5.3) the contribution of the last term is zero, since

$$\sum_{i=1}^n \frac{M_i}{M_T} P_{COI} \dot{\theta}_i = P_{COI} \sum_{i=1}^n \frac{M_i}{M_T} \dot{\theta}_i = 0$$

5.2.2 Description of Components

The terms of the transient energy function can be physically interpreted in the following manner (all changes being calculated with respect to the postdisturbance stable equilibrium point).

- $$\begin{aligned}
 \frac{1}{2} \sum_{i=1}^n M_i \tilde{\omega}_i^2 &= \frac{1}{2} \sum_{i=1}^n M_i (\omega_i - \omega_o)^2 \\
 &= \frac{1}{2} \sum_{i=1}^n M_i (\omega_i^2 - 2 \omega_i \omega_o + \omega_o^2) \\
 &= \frac{1}{2} \sum_{i=1}^n M_i \omega_i^2 - \omega_o \sum_{i=1}^n M_i \omega_i + \omega_o^2 \frac{1}{2} \sum_{i=1}^n M_i
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \sum_{i=1}^n M_i \omega_i^2 - \omega_o (M_T \omega_o) + \frac{1}{2} M_T \omega_o^2 \\
&= \frac{1}{2} \sum_{i=1}^n M_i \omega_i^2 - \frac{1}{2} M_T \omega_o^2
\end{aligned}$$

This represents the total change in kinetic energy of all generator rotors relative to the COI. It is equal to the change in kinetic energy of all generator rotors minus the change in the kinetic energy associated with the COI.

$$\bullet \quad \sum_{i=1}^n P_i (\theta_i - \theta_i^{s2}) = \sum_{i=1}^n P_i (\delta_i - \delta_i^{s2}) - \left[\sum_{i=1}^n P_i \right] (\delta_o - \delta_o^{s2})$$

This represents the change in position energy of all rotors relative to the COI. It is equal to the change in position energy of all rotors minus the change in the position of energy associated with the COI.

$$\bullet \quad C_{ij} (\cos \theta_{ij} - \cos \theta_{ij}^{s2})$$

This represents the change in the stored magnetic energy of branch ij .

$$\bullet \quad \int_{\theta_i^{s2} + \theta_j^{s2}}^{\theta_i + \theta_j} D_{ij} \cos \theta_{ij} d(\theta_i + \theta_j)$$

This represents the change in the dissipation energy of branch ij .

The description given clearly demonstrates the advantage of the TEF formulated relative to COI. In the rotor kinetic energy and position energy terms the component of energy contributing to the motion of COI and not affecting the stability of the system is subtracted. This aspect of the formulation improves the description of the transient energy used in stability assessment and accounts for the components of energy responsible for stability (or instability) more accurately.

5.2.3 Approximation of Dissipation Energy Component

It is to be noted that the term in the energy expression corresponding to the dissipation component consists of a path dependent integral. This cannot be evaluated unless the system trajectory is known. Uyemura and colleagues [1] developed an approximation to this term based on a linear angle trajectory assumption. This approximation was then used by Athay and colleagues [2] and compared with the actual dissipation energy component evaluated along the trajectory. From the experience gained from numerous simulations, it has been found that for a first swing transient, this approximation is fairly accurate and has become standard practice in the TEF method to incorporate the effect of transfer conductances.

The approximation is derived as follows: let $\theta_i(t)$ and $\theta_j(t)$ represent the angular trajectories of machines i and j with respect to the time t , respectively. Let θ^a and θ^b represent the initial and final vectors, of angular positions for the n machines. The linear angle trajectory assumption is invoked between θ^a and θ^b and is expressed by

$$\theta_i = \theta_i^a + (\theta_i^b - \theta_i^a) t \quad 0 \leq t \leq 1 \quad (5.7)$$

$$i = 1, 2, \dots, n$$

At $t = 0$, $\theta = \theta^a$, while at $t = 1$, $\theta = \theta^b$.

Taking the derivative of (5.7) we get

$$\begin{aligned} d\theta_i &= (\theta_i^b - \theta_i^a) dt \\ d\theta_j &= (\theta_j^b - \theta_j^a) dt \end{aligned} \quad (5.8)$$

Adding the two equations in (5.8)

$$d(\theta_i + \theta_j) = (\theta_i^b - \theta_i^a + \theta_j^b - \theta_j^a) dt \quad (5.9)$$

and by subtracting $d\theta_j$ from $d\theta_i$ in (5.8)

$$d(\theta_i - \theta_j) = d\theta_{ij} = (\theta_i^b - \theta_i^a - \theta_j^b + \theta_j^a) dt \quad (5.10)$$

Eliminating dt from (5.9) and (5.10) we get

$$d(\theta_i + \theta_j) = \frac{\theta_i^b - \theta_i^a + \theta_j^b - \theta_j^a}{\theta_{ij}^b - \theta_{ij}^a} d\theta_{ij} \quad (5.11)$$

Replacing $d(\theta_i + \theta_j)$ in the dissipation energy term in (5.6), the expression can now be integrated with respect to θ_{ij} between any two points and is given by

$$I_{ij} = D_{ij} \frac{\theta_i^b - \theta_i^a + \theta_j^b - \theta_j^a}{\theta_{ij}^b - \theta_{ij}^a} (\sin \theta_{ij}^b - \sin \theta_{ij}^a) \quad (5.12)$$

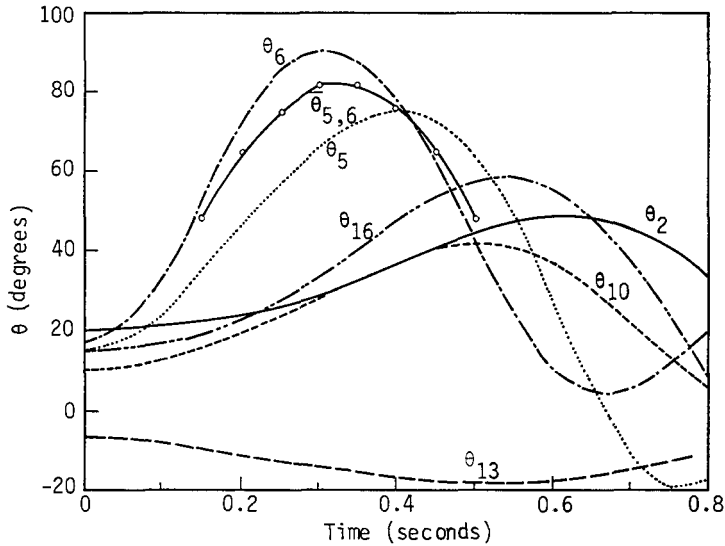
5.2.4 Energy Not Contributing to System Separation

In [3] and [4] a detailed analysis of the energy behavior along the time domain trajectory was conducted using a 17-generator equivalent of the network of the state of Iowa. A figure showing the main high-voltage lines in the system is given in the appendix. The data for this system are also provided. Numerous cases of disturbances were analyzed for this system.

The various components of transient energy, developed in the previous subsections, were carefully analyzed along the time domain trajectories. A detailed analysis of these components provided a significant characteristic of the energy behavior. It was observed, that in all cases where the system was *stable* following the removal of the disturbance, a certain amount of the total kinetic energy in the system is *not* absorbed. This clearly indicates that *not all* the transient kinetic energy, created by the disturbance, contributes to the instability of the system. Some of this kinetic energy is responsible for the intermachine motion between the generators, and does not contribute to the separation of the severely disturbed generators from the rest of the system.

To illustrate this point more clearly, consider the case of a three-phase fault at bus 372 of the 17-generator Iowa System cleared in 0.150s. Figure 5.1 shows a plot of the rotor angles (in COI reference frame) of the severely disturbed generators. This plot clearly shows that these generators (which are the most severely disturbed) are stable.

The various components of the transient energy obtained for this fault case are shown in Fig. 5.2. This figure shows the pattern of energy exchange during the transient period. After the fault is removed at 0.15 sec, we note that no new energy is injected into the system and that the total energy remains constant. The kinetic energy V_{KE} decreases since it is converted to potential energy V_{PE} (position, magnetic, and dissipation) which increases. The variation of these two components of energy is of interest. The instant at which V_{PE} becomes a maximum approximately corresponds to the instant at which V_{KE} becomes a *minimum*, and these instants coincide with the instant at which the inertial



*Figure 5.1 Seventeen-generator system
fault at bus 372 cleared in 0.150 sec. Rotor angles.*

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center of the most severely disturbed (critical) machines, i.e., $\bar{\theta}_{5,6}$ (in Fig. 5.1), peaks and begins to recover.

The important point to note is that the total kinetic energy V_{KE} only becomes a minimum and not zero, even though the system is stable. This clearly substantiates the earlier observation, that not all the kinetic energy at the end of the disturbance contributes to system instability. As a result, using the kinetic energy of the system at the end of the disturbance will result in a more conservative stability assessment. It is now apparent that the kinetic energy at the end of the disturbance needs to be corrected to account for the

portion of energy involved in intermachine oscillation and not contributing to the notion of the severely disturbed generators away from the rest of the system.

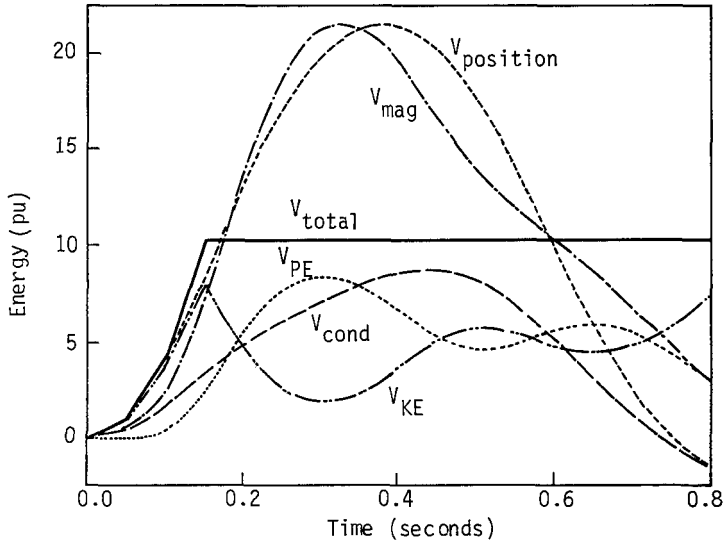


Figure 5.2 Seventeen-generator system
fault at bus 372 cleared in 0.150 sec. Energy. Reprinted from
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5.2.5 Corrected Kinetic Energy

From the discussion presented, it is obvious that for accurate transient stability assessment using the TEF, the component of kinetic energy not contributing to instability should be subtracted from the energy that needs to be absorbed by the system for stability to be maintained. This component of energy is given by the minimum of kinetic energy achieved on a given trajectory. From the viewpoint of the TEF method, this component will not be available as the trajectory is not simulated. Hence, it becomes essential to estimate this component from the conditions available at the end of the disturbance when

the stability assessment is made. This procedure makes an implicit assumption that the kinetic energy correction at the end of the disturbance is equal to the kinetic energy correction when the transient kinetic energy becomes a minimum. The authors have found from numerous simulations conducted during the course of their work that accuracy was not much sacrificed by this assumption.

From the observations conducted in [3, 4], and the example depicted in Figs. 5.1, 5.2, an important feature of the dynamic behavior and energy exchange process for multimachine power systems can be extracted. This characteristic shows that when the critical group of generators (most severely disturbed) tends to lose synchronism from the rest of the system, instability is determined by the gross motion of these machines, i.e., by the motion of their COI. This is depicted in Figs. 5.1, 5.2 where the instant of maximum of V_{PE} coincides with the instant of the peak of $\bar{\theta}_{5,6}$ (the COI of the critical group of machines 5 and 6). Based on this salient dynamic characteristic, it is postulated that the transient kinetic energy which is responsible for the separation of the critical generators from the rest of the system is that associated with the gross motion of the critical generators, i.e., that of their inertial center with regard to the inertial center of the other generators. The remaining portion of the transient kinetic energy need not be absorbed by the network for stability to be maintained. It is identified with the inter-machine motion in each of the groups separating from each other.

The kinetic energy associated with the gross motion of a group of k machines having angular speeds $\tilde{\omega}_1, \tilde{\omega}_2, \dots, \tilde{\omega}_k$ is the same as the kinetic energy of their inertial center. The speed of the inertial center of that group and its kinetic energy are given by

$$\tilde{\omega}_{cr} = \frac{\sum_{i=1}^k M_i \tilde{\omega}_i}{\sum_{i=1}^k M_i} \quad (5.13)$$

$$V_{KE_{cr}} = \frac{1}{2} \left[\sum_{i=1}^k M_i \right] (\tilde{\omega}_{cr})^2 \quad (5.14)$$

If the system contains very large inertias, i.e., the center of inertia of the whole power system is nearly stationary, the kinetic energy given by equation (5.14) tends to separate that group of generators from the rest of the system. If these are the "critical machines", i.e., the generators tending to separate from the system, then the kinetic energy given by equation (5.14) is the correct V_{KE} contributing to system separation. The kinetic energy term in the energy function should be corrected accordingly.

If the system inertias are finite, i.e., if the system's inertial center is not stationary, the kinetic energy that contributes to system separation is a little more complex. Essentially, the disturbance splits the generators of the system into two groups: the critical machines and the rest of the generators. Their inertial centers have inertia constants and angular speeds M_{cr} , ω_{cr} and M_{sys} , ω_{sys} , respectively. The gross motion of the two groups approximates that of a two-machine system. The kinetic energy causing the separation of the two groups is the same as that of an equivalent one-machine – infinite-bus system having inertia constant M_{eq} and angular speed ω_{eq} given by

$$M_{eq} = \frac{M_{cr} \cdot M_{sys}}{M_{cr} + M_{sys}} \quad (5.15)$$

$$\tilde{\omega}_{eq} = (\tilde{\omega}_{cr} - \tilde{\omega}_{sys})$$

and the corresponding kinetic energy is given by

$$V_{KE_{corr}} = \frac{1}{2} M_{eq} (\tilde{\omega}_{eq})^2 \quad (5.16)$$

Again the kinetic energy term in (5.6) should be replaced by (5.16)

5.3 TRANSIENT STABILITY ASSESSMENT

Transient stability assessment using the TEF method is made only for the last transient. Thus, for a case where there are multiple transients or switchings, a stability assessment using the TEF method can only be done for the last transient or switching.

The stability assessment is done by comparing two values of the transient energy V . The value of V is computed at the end of the last disturbance, e.g., if the disturbance is a simple fault the value of V at fault clearing, V_{cl} .

The other value of V which to a great extent determines the accuracy of the stability assessment is the critical value of V , V_{cr} which is the potential energy at the controlling unstable equilibrium point (UEP), for the particular disturbance under investigation.

If $V_{cl} < V_{cr}$, the system is stable, and if $V_{cl} > V_{cr}$, the system is unstable. Alternatively the assessment made by computing the energy margin ΔV given by

$$\Delta V = V_{cr} - V_{cl} \quad (5.17)$$

Substituting for V_{cr} and V_{cl} from (5.6) and using the linear angle path assumption for the dissipation terms between the conditions at the end of the disturbance and the controlling UEP, we have

$$\begin{aligned} \Delta V = & -\frac{1}{2} M_{eq} \tilde{\omega}_{eq}^{cl^2} - \sum_{i=1}^n P_i (\theta_i^u - \theta_i^{cl}) \\ & - \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[C_{ij} (\cos \theta_{ij}^u - \cos \theta_{ij}^{cl}) - D_{ij} \frac{\theta_i^u - \theta_i^{cl} + \theta_j^u - \theta_j^{cl}}{(\theta_{ij}^u - \theta_{ij}^{cl})} (\sin \theta_{ij}^u - \sin \theta_{ij}^{cl}) \right] \end{aligned} \quad (5.18)$$

where $(\theta^{cl}, \tilde{\omega}^{cl})$ are the conditions at the end of the disturbance and (θ^u, o) represents the controlling UEP.

If ΔV is greater than zero the system is stable, and if ΔV is less than zero, the system is unstable.

Transient stability assessment using the energy margin has two advantages:

1. As shown in (5.18), the energy margin can be evaluated by evaluating V between the conditions at the end of the disturbance $(\theta^{cl}, \tilde{\omega}^{cl})$ and those at the controlling UEP (θ^u, o) . This invokes the linear angle assumption for the dissipation term only between $(\theta^{cl}, \tilde{\omega}^{cl})$ and (θ^u, o) and improves the accuracy of the assessment.
2. A qualitative measure of the degree of stability (or instability) can be obtained if ΔV is normalized with respect to the corrected kinetic energy at the end of the disturbance [3]

$$\Delta V_n = \Delta V / V_{KEcorr} \quad (5.19)$$

The significance of (5.19) can be described as follows. Consider the case where the system is stable. In this case, the transient energy margin (5.18) represents a "margin of safety" before instability occurs. Thus, a more severe disturbance, as indicated by the transient energy at the end of the disturbance, could be tolerated until the margin is used up. It would be logical, therefore, to relate the margin ΔV to the amount of transient energy directly responsible for instability.

The component of transient energy at the end of the disturbance that must be converted to other forms of energy for stability to be maintained is the corrected kinetic energy (V_{KEcorr}) . In other words, a true measure of the severity of the disturbance is the ratio $(\Delta V / V_{KEcorr})$.

5.4 CONCEPT OF THE CONTROLLING UNSTABLE EQUILIBRIUM POINT

The analytical concepts behind the controlling UEP and its importance in determining an accurate estimate of the region of stability were discussed earlier in Chapter 4. In this section we present a technique to determine the correct controlling UEP based on the physical behavior of the power system. As described in Section 5.3, the determination of the correct controlling UEP is *essential* for accurate transient stability assessment using the TEF method. The assessment is made by comparing the system transient energy at the end of the disturbance to the critical value of energy, which is the potential energy at the correct controlling UEP. Hence, the accuracy of the assessment hinges on obtaining the correct controlling UEP.

The UEP represents a set of generators' angles which satisfy equation (5.20)

$$f_i = P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} = 0 \quad i = 1, 2, \dots, n \quad (5.20)$$

For a multi-machine power system with n -generators, there are $(2^{n-1} - 1)$ solutions possible. Each solution will give a certain value of potential energy. Thus, identifying the "correct" UEP for a given disturbance means identifying the correct energy level against which the stability assessment is made. Failure to do so was among the main reasons for the conservative results obtained earlier with the direct method.

During the early stages of development of the direct method [5], the UEP having the lowest energy value was used to determine the critical energy. This resulted in stability assessment results which were too conservative for practical power systems. In [6] Gupta and El-Abiad took the first steps toward removing some of this conservatism when they made a detailed study of the trajectories of the severely disturbed generators following the disturbance. They observed that the trajectory of these generators passes close to a UEP

different from that having the lowest energy level. They called this UEP the "closest" UEP and suggested that its energy level is important for stability assessment.¹

An approach based on accelerations of machines was suggested in [7]. It consisted of determining the machine or machines with largest acceleration at the instant of disturbance removal, and using this information to determine the UEP. This criterion as the one in [6] was based only on one aspect of the physical behavior of the system, i.e., the effect of the disturbance on the generators; but does not account for the influence of the postdisturbance network on their motion.

In [2], the concepts proposed in [6, 7] were further enhanced. Using the disturbed trajectory, the point of intersection θ^{ss} of the disturbed trajectory and the postdisturbance principle singular surface was determined. This was then used to form a direction vector h with the postdisturbance stable equilibrium point θ^{s2} . Along this direction vector h a one dimensional minimization problem was solved to minimize

$$F(\theta) = \sum_{i=1}^n f_i^2(\theta) = \sum_{i=1}^n \left[P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} \right]^2 \quad (5.21)$$

and obtain $\hat{\theta}^u$. With $\hat{\theta}^u$ as a starting point, a suitable numerical technique was used to obtain the controlling UEP. Figure 5.3 obtained from [2] summarizes the construction procedure. This procedure was a significant development in the sense that based on an understanding of the system behavior, it was recognized that there are two aspects to determining the controlling UEP (1) the effect of the disturbance itself and (2) the effect of the postdisturbance network.

In the criteria developed in [6, 7, 2] it was assumed that the generators whose rotor angles were advanced in the controlling UEP actually lost synchronism. However, from the

¹ The choice of the name "closest UEP" has caused some confusion to this day because it is sometimes interpreted to mean the UEP with the lowest energy level.

detailed trajectory analysis done in [3, 4], it was shown that the controlling UEP could be among a group of UEPs located in the direction in which the severely disturbed generators following a disturbance move. Another important observation made in [3] showed that the potential energy surfaces at and near the controlling UEP are largely shaped by the angular positions of the advanced generators. Some of these generators may not actually lose synchronism initially, at the critically unstable conditions.

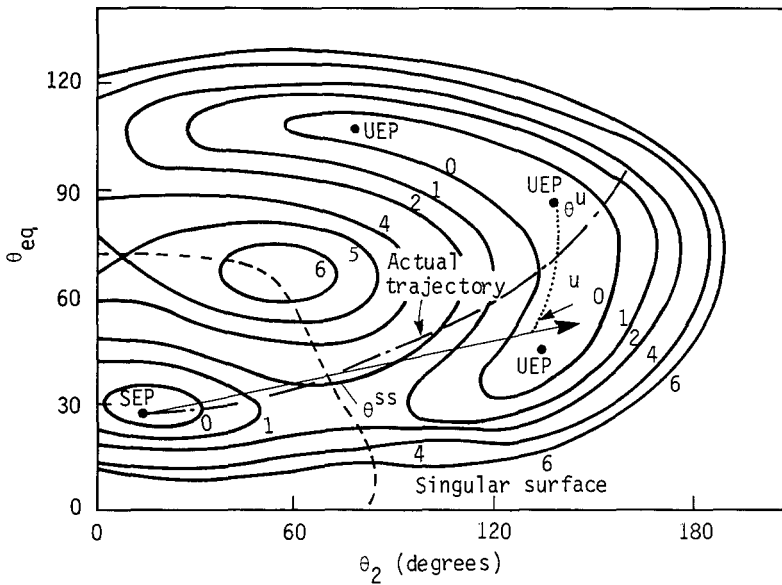


Figure 5.3 Determination of UEP [2].

Determining the advanced generators for the controlling UEP is referred to as determining the mode of disturbance.² For a given UEP the components of the rotor positions of the advanced generators generally have angles greater than 90° . Hence, if for a

² In the early work done by the authors [3, 4, 8] it was called the "mode of instability".

given UEP machines j and k have their angles greater than 90° , the MOD for this UEP is referred to as " jk ".

In the simulations conducted in [3, 4] the following modes were encountered:

1. Simple modes; in this case the MOD specified by the controlling UEP actually corresponds to that with the angle of machines initially losing synchronism being greater than 90° .
2. A mode where the critical energy level is determined by two or more severely disturbed machines, even though only one machine initially loses synchronism. The UEP for this case would have the angles of several machines advanced greater than 90° .
3. With faults at locations electrically close to a cluster of generators, a number of candidate UEPs with similar energy levels exist. Again only one generator may initially lose synchronism, but the UEPs would have more than one generator with advanced angles.

These results indicated that identifying the controlling UEP was a more complex undertaking than it had been realized in earlier investigations.

Based on the evidence provided by [6, 7, 2, 3, 4] the authors of [8] carefully reexamined the physical behavior of a multimachine power system subjected to a disturbance. They determined that during the ensuing transient, the more severely disturbed generators may or may not lose synchronism with the rest of the system, depending on whether the potential energy absorbing capacity of the network is adequate to convert the transient kinetic energy at the end of the disturbance into potential energy. Therefore, the controlling UEP must properly account for two aspects of the transient phenomena (1) *the effect of the disturbance on the various generators* and (2) *information on the energy absorbing capacity of the postdisturbance network*. These two aspects must be incorporated in the criterion to determine the controlling UEP.

In order to develop a criterion, the concept of normalized energy margin ΔV_n (5.19) was invoked. It was found that the relative magnitudes of ΔV_n give a reliable indication

of the *degree of stress* exerted on different generators by the disturbance. Using a similar reasoning, the following criterion was proposed in [8] for determining the controlling UEP.

The postdisturbance trajectory approaches (if the disturbance is large enough) the controlling UEP. This is the UEP with the lowest normalized potential energy margin at the instant the disturbance is removed.

Stated mathematically,

$$\Delta V_{PEn} = \frac{\Delta V_{PE}}{V_{KEcorr}} \quad (5.22)$$

where

$$\begin{aligned} \Delta V_{PE} &= V_{PE}^u - V_{PE}^{c\ell} \\ &= -\sum_{i=1}^n P_i (\theta_i^u - \theta_i^{c\ell}) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[C_{ij} (\cos \theta_{ij}^u - \cos \theta_{ij}^{c\ell}) \right. \\ &\quad \left. - D_{ij} \left[\frac{\theta_i^u - \theta_i^{c\ell} + \theta_j^u - \theta_j^{c\ell}}{\theta_{ij}^u - \theta_{ij}^{c\ell}} \right] (\sin \theta_{ij}^u - \sin \theta_{ij}^{c\ell}) \right] \end{aligned} \quad (5.23)$$

V_{KEcorr} = corrected kinetic energy (5.16) based on the MOD of the UEP chosen.

It is to be noted that the quantity ΔV_{PEn} given by (5.22) captures all the characteristics required to determine the controlling UEP.

The numerator ΔV_{PE} measures the potential energy absorbing capacity of a given UEP, in the postdisturbance period, and the denominator provides a measure of the transient energy gained by the system during the disturbance. Hence, the ratio combines

aspects of both the disturbance and the postdisturbance network. The candidate UEP having the lowest ΔV_{PEN} , represents the configuration of the system in which the ratio of the potential energy absorbing capacity (of the postdisturbance network) to the corrected kinetic energy (gained during the disturbance) is the least; i.e., this configuration of the system is most vulnerable and represents the weakest link in the system.

To clarify the concept of the MOD further, we present the following example. Consider the reduced Iowa System for which the data are given in the appendix. Consider the case of a three-phase fault at bus 773 (Ft. Calhoun), cleared by opening the line between bus 773 and bus 779.

Note from Fig. A1 there are several generators close to the fault location, e.g., 2, 5, 6, 10, 12, 17. All these generators are affected by the fault.

Time domain simulations conducted for this case showed that the system is critically unstable at a clearing time of $t_{cl} = 0.357$ secs, and in this situation only machine 16 loses synchronism with the rest of the system as shown in Fig. 5.4. If the fault lasts a little longer and is cleared at $t_{cl} = 0.423$ secs, machines 2, 5, 6, 10, 12, 16, 17 lose synchronism with the rest of the system as shown in Fig. 5.5.

To analyze this case we will consider the following candidate MODs:

1. 16
2. 10, 12, 16, 17
3. 2, 10, 12, 16, 17
4. 2, 5, 6, 10, 12, 16, 17

These were chosen based on the electrical proximity of the machines to the disturbance. The exact UEPs for these modes were determined and the critical energy V_{cr} for each UEP was evaluated. These results are displayed in Table 5.1.

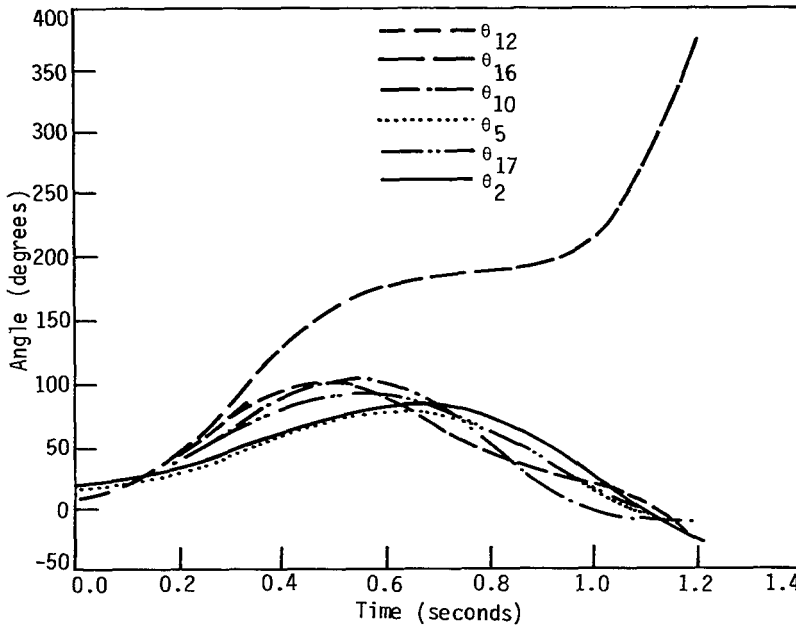


Figure 5.4 Seventeen-generator test system. Swing curves of generators 2, 5, 10, 12, 16, and 17 for fault at Ft. Calhoun: $t_c = 0.357$ sec. Reprinted from © IEEE Transactions on Power Apparatus and Systems 100 (1981).

An energy analysis was also conducted on the time domain trajectory for this case. For this fault cleared at 0.354 secs, the system is critically stable, and the maximum value of potential energy attained is 31.002 pu, and at this point the angular positions of the machines are given in Table 5.2.

The point of maximum V_{PE} on the critically stable trajectory represents the point closest to the PEBS (Chapter 4), and the potential energy provides an estimate of the critical energy V_{cr} . It is to be noted that for this fault case MC 16 loses synchronism in the critically unstable case as depicted by Fig. 5.4, yet the critical energy of the UEP corresponding to MOD 16 (12.1425 pu) is much smaller than the potential energy level obtained on the critical trajectory. Hence, it is clear that the UEP corresponding to MOD

16 is not the controlling UEP. This example also illustrates that the MOD of the controlling UEP does not necessarily correspond to the machines initially losing synchronism.

Further examination of the data in Tables 5.1 and 5.2 shows that the remaining three UEPs have critical energy close to each other and correspond well with the energy profile on the critical trajectory. The controlling UEP must be one among these UEPs. We will now apply the MOD test to these three UEPs at a clearing time $t_{cl} = 0.340$ secs. The results of this test are shown in Table 5.3. From Table 5.3 the MOD having the least ΔV_{PEN} is the MOD 2, 5, 6, 10, 12, 16, 17. Hence, this is the MOD corresponding to the controlling UEP.

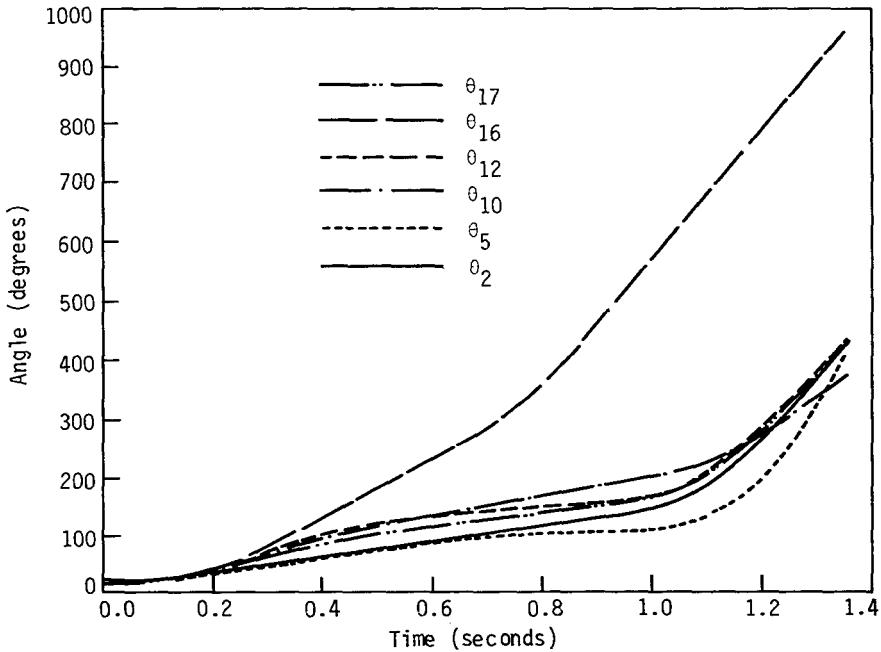


Figure 5.5 17—generator test system. Swing curves of generators 2, 5, 10, 12, 16, and 17 for fault at Ft. Calhoun: $t_c = 0.423$ sec. Reprinted from ©IEEE Transactions on Power Apparatus and Systems 100 (1981).

TABLE 5.1 Angular Positions at Candidate UEPs and Critical Energies

Generator	MOD 16	MOD 10, 12, 16, 17	MOD 2, 10, 12, 16, 17	MOD 2, 5, 6, 10, 12, 16, 17
1	-9.1456	-11.6885	-16.0597	-13.5622
2	23.2536	68.6021	126.3161	131.8890
3	2.6699	5.5650	10.0445	19.5067
4	-8.4681	-14.4465	-15.1491	-17.7257
5	21.7616	45.9962	64.0932	121.5156
6	25.2343	53.6607	73.9197	130.4571
7	-7.3026	-19.8439	-24.0178	-26.7323
8	-7.3522	-11.2320	-11.7188	-9.4003
9	2.91047	-10.2419	-14.3583	-16.9889
10	19.7421	172.9903	113.4385	119.6882
11	-2.3664	6.0545	12.5712	30.7321
12	21.0896	92.4636	116.5378	121.8247
13	-14.8644	-32.2606	-38.4943	-42.8104
14	-13.8741	-31.4391	-37.9748	-41.6679
15	-4.3304	-12.0388	-13.3027	-15.4053
16	159.8411	183.8073	203.3842	123.2795
17	23.7862	163.6709	124.9800	127.5906
V_{cr}	12.1425	27.8737	30.14091	32.2092

To demonstrate further the importance of the correct controlling UEP, we will now conduct a stability assessment using the three MODs considered in Table 5.3 and show that only the controlling UEP gives the correct stability assessment.

The results of the assessment are shown in Table 5.4. Two clearing times $t_{cl} = 0.340$ secs and $t_{cl} = 0.370$ secs are considered to bracket the stable and unstable cases. Of the three candidate UEPs considered, only the controlling UEP MOD 2, 5, 6, 10, 12, 16, 17³ as determined by the MOD test provides the correct stability assessment.

To emphasize the importance of incorporating the information on the effect of the disturbance and the effect of the energy absorbing capacity of the postdisturbance network, we continue the same example by presenting data on the generators' acceleration.

³ In some of the earlier work done by the authors the MOD chosen was either 10, 12, 16, 17 or 2, 10, 12, 16, 17, this may have been caused by the type of solution method used to determine the UEPs. The results obtained in Table 5.4 use a vary small error tolerance and the UEPs are evaluated with this error tolerance.

For the fault cleared at $t_{cl} = 0.340$ secs the accelerations of the machines in the system at fault clearing are given in Table 5.5 sorted in descending order.

TABLE 5.2 Angular Positions at V_{PEmax} on Critically Stable Trajectory

Generator	θ at V_{PEmax}
1	-9.9837
2	62.1242
3	2.932
4	-14.71
5	59.79
6	72.32
7	-13.61
8	-5.78
9	-4.02
10	89.232
11	9.47
12	97.169
13	-24.94
14	-24.72
15	-12.44
16	128.124
17	82.38

TABLE 5.3 MOD Test at $t = 0.340$ s

MOD	ΔV_{PE}	V_{KEcorr}	ΔV_{PEN}
10, 12, 16, 17	17.51693	8.0427	2.17799
2, 10, 12, 16, 17	16.86897	8.20337	2.05635
2, 5, 6, 10, 12, 16, 17	15.05770	9.9079	1.51977

TABLE 5.4 Stability Assessment Using Candidate UEPs

MOD	Stable from Time Simulation $t_{cl} = 0.340$ sec		Unstable from Time Simulation $t_{cl} = 0.370$ sec	
	ΔV	Comments	ΔV	Comments
10, 12, 16, 17	4.48714	Stable	0.59047	Stable
2, 10, 12, 16, 17	5.46921	Stable	1.37853	Stable
2, 5, 6, 10, 12, 16, 17	3.68858	Stable	-0.59497	Unstable

The data in Table 5.5 show that the acceleration of generator 16 clearly dominates the accelerations of all other machines. Hence, a criterion solely based on the accelerations of generators at fault clearing will choose a UEP corresponding to MOD 16. This will result

TABLE 5.5 Acceleration at $t_{cl} = 0.340$ sec.

Generator	Acceleration (in pu)
16	29.4685
10	12.6319
2	10.6910
1	10.2391
5	9.0723
12	7.9951
11	8.6172
3	8.311
13	8.0693
4	7.8487
15	6.9546
9	4.1770
17	4.0452
14	3.8909
7	3.8734
6	2.1202

in an erroneous stability assessment as discussed earlier in the example. Furthermore, the accelerations alone may give an erroneous indication of which generators are severely disturbed, as evidenced by the positions of generators 1 and 6 in Table 5.5. The example also demonstrates the importance of including the characteristics of the postdisturbance network in any criterion to determine the MOD of the controlling UEP.

This completes the conceptual development of the MOD test and the associated calculations in the TEF method. The practical and numerical aspects of these evaluations will be detailed in Chapter 6.

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Chapter 6

COMPUTATIONAL ASPECTS OF THE TRANSIENT ENERGY FUNCTION METHOD FOR THE CLASSICAL GENERATOR MODEL

6.1 TEF PROCEDURE OUTLINE

In this chapter we provide a detailed description of the numerical and computational aspects of the various steps involved in the TEF method. A discussion of the philosophy behind the selection of appropriate computational techniques will also be presented.

We begin by providing an outline of the various computational steps used in the TEF method procedure. Figure 6.1 provides a schematic diagram of these computational steps.

In the step dealing with the input data, the appropriate information from the predisturbance power flow solution, the synchronous machine parameters, and disturbance specification is read. From these data, the initial values of the machine internal node voltages and rotor angle positions are calculated.

The sparse network admittance matrix is constructed from the connectivity information derived from the predisturbance power flow data. Sparsity is maintained to save on storage and improve the speed of the procedure. Based on the disturbance and switching information, the admittance matrix is then optimally ordered and reduced to eliminate all the fixed buses. The optimal ordering is done to retain sparsity to the extent possible by reducing the fill-in terms during reduction.

Based on the disturbance data, the disturbance is appropriately represented. In the case where the disturbance is a fault, the fault is represented in the network, and the

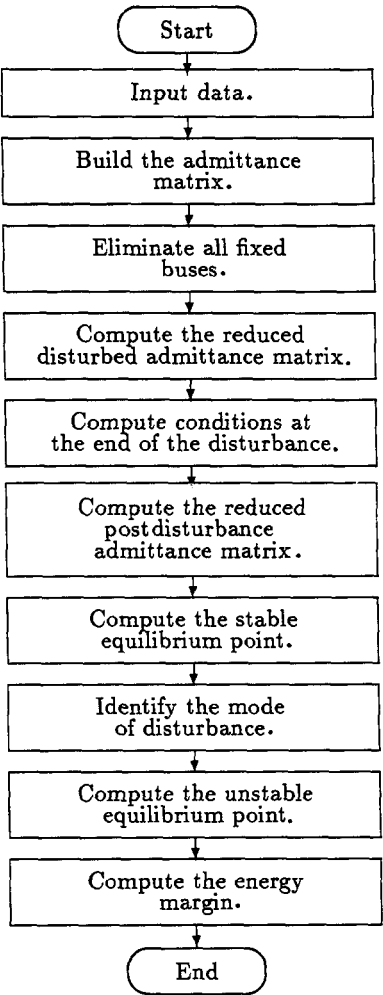


Figure 6.1 Schematic diagram of TEF procedure outline.

admittance matrix obtained in the previous step is augmented with the internal generators nodes and reduced to the internal generator nodes.

The fault is then simulated, and the conditions at the end of the fault are evaluated.

Using the information on the switching action, the reduced postdisturbance admittance matrix is obtained. The stable equilibrium point is then computed.

The conditions at the end of the fault, the postdisturbance stable equilibrium point, and the postdisturbance network are then used to determine the relevant mode of disturbance described in Section 5.4.

Knowing the mode of disturbance, the exact controlling unstable equilibrium point is then computed.

The transient energy margin is then calculated and transient stability assessment is made.

A detailed description of the computational aspects of the various steps is presented in the following sections.

6.2 CALCULATION OF INITIAL CONDITIONS

6.2.1 Data Requirement

The input data required for the TEF analysis are identical to that required by any conventional time domain simulation technique. The data requirement can be classified into three categories:

1. *Predisturbance network information:* This consists of the solved predisturbance power flow solution providing the level of generation, the network connectivity and parameters, and the magnitude and angles of the voltages at all buses. It would also include the amount and composition of loads at each bus, i.e., the fractions of the loads to be represented by constant current, constant impedance and constant MVA; and the total predisturbance real and reactive power load.
2. *Synchronous generator data:* For the classical generator model, this consists of the generator inertia (H) and the direct axis transient reactance (x'_d).

3. *Fault specification and switching data:* This specifies the type, location, and duration of the fault or fault sequence. The parameters associated with it, e.g., equivalent symmetrical component impedances at the fault point for asymmetrical faults. The data will also include the switching operation required to isolate or remove the disturbance.

6.2.2 Internal Voltage Calculations

From the input power flow data and the synchronous generator data, which are obtained on a common system MVA base, the internal voltages of the generators are computed from the predisturbance generator terminal voltages and currents as follows. Let the terminal voltage be used temporarily as a reference as shown in Fig. 6.2.

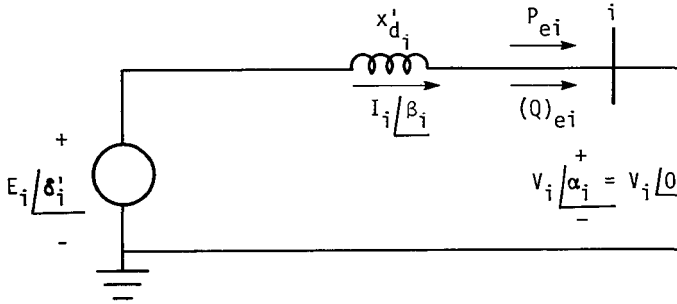


Figure 6.2 Generator representation for computing δ_{i0} .

If we define $\bar{I}_i = I_{i1} + j I_{i2}$, then from the relation $P_{ei} + jQ_{ei} = \bar{V}_i \bar{I}_i^*$ we have $I_{i1} + j I_{i2} = (P_{ei} - jQ_{ei}) / V_i$. The internal EMF of generator i is computed from $E_i / \delta_i' = \bar{V}_i + j x'_{di} \bar{I}_i$,

or

$$E_i / \delta_i' = (V_i + Q_{ei} x'_{di} / V_i) + j(P_{ei} x'_{di} / V_i) \quad i = 1, 2, \dots, n \quad (6.1)$$

The initial generator angle δ_{i0} is then obtained by adding the pre-disturbance terminal voltage angle α_i to δ_i' , or

$$\delta_{io} = \delta'_i + \alpha_i \quad i = 1, 2, \dots, n \quad (6.2)$$

These angles are then converted into the center of inertia reference frame as described in Section 2.4 to obtain

$$\delta_o = \frac{1}{M_T} \sum_{i=1}^n M_i \delta_{io} \quad (6.3)$$

where

$$M_T = \sum_{i=1}^n M_i$$

and

$$\theta_i = \delta_{io} - \delta_o \quad i = 1, 2, \dots, n \quad (6.4)$$

The internal voltage of each machine is then given by $E_i / \underline{\theta_i}$, $i = 1, 2, \dots, n$.

6.2.3 Calculation of Initial Nonlinear Load Current Component

When nonlinear load components are included in the model, the initial component of each type of nonlinear load has to be calculated (see Section 2.3.2). This calculation needs to be done only for the constant current and constant MVA components of the load. The component of constant impedance load is folded into the admittance matrix using the following procedure. From the predisturbance power flow data, knowing the composition of the constant impedance loads at each bus, the constant impedance representation can be evaluated as

$$R_i + j x_i = \frac{|V_i|^2}{(P_{Li} p_{li})^2 + (Q_{Li} q_{li})^2} (P_{Li} p_{li} + j Q_{Li} q_{li}) \quad (6.5)$$

$$i = 1, 2, \dots, \ell$$

where p_{li}, q_{li} = fraction of constant impedance MW and MVAR load components at bus i .

For the constant current and constant MVA portions of the load, we evaluate the initial current components as

$$I_{CI_i}^o = \left[\frac{(P_{Li}^o p_{2i} + j Q_{Li}^o q_{2i})}{\bar{V}_i^o} \right]^* \quad i = 1, 2, \dots, \ell \quad (6.6a)$$

$$I_{CM_i}^o = \left[\frac{(P_{Li}^o p_{3i} + j Q_{Li}^o q_{3i})}{\bar{V}_i^o} \right]^* \quad i = 1, 2, \dots, \ell \quad (6.6b)$$

where

ℓ = number of buses with nonlinear loads, and for bus i ,

P_{Li}^o = predisturbance MW load

Q_{Li}^o = predisturbance MVAR load

$\bar{V}_i^o$ = predisturbance load bus voltage

p_{2i}, q_{2i} = fraction of constant current MW and MVAR load components, respectively

p_{3i}, q_{3i} = fraction of constant MVA, MW, and MVAR load components, respectively

6.3 CONSTRUCTION OF ADMITTANCE MATRICES

The admittance matrices used in the TEF analysis are evaluated systematically using the input power flow data and generator data. The first step in this procedure consists of building the predisturbance network admittance matrix. This matrix is modified to obtain the other network matrices.

6.3.1 Predisturbance Network Admittance Matrix $\bar{Y}_{PR}^A$

This matrix is formed from the power flow data, using the line, transformer, and bus data including the constant impedance loads. The elements of the matrix are identified as follows: $\bar{Y}_{PRii}^A$ is the sum of all the admittances corrected to node i , and $\bar{Y}_{PRij}^A$ is the negative of the sum of the admittances between nodes i and j . Transformers with off-nominal turns ratio will have to be appropriately represented in the admittance matrix. See [1, p. 216] and [2, p. 21] for the procedure to incorporate the transformers with off-nominal turns ratio in the admittance matrix. The admittance matrix is computed using sparsity techniques and the storage is sparse. The matrix is then ordered as shown in Fig. 6.3. The ordered groups in Fig. 6.3 are formed as follows:

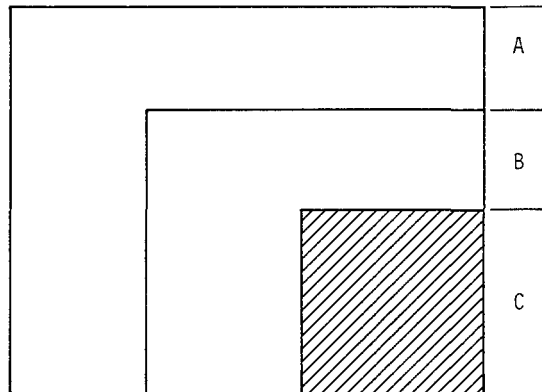


Figure 6.3 Admittance matrix ordering.

- A – Buses which are free to move during optimal ordering.
- B – Buses which are involved in the disturbance and switching. These are fixed during the ordering process.
- C – Generator terminal buses. These are fixed during the ordering process.

6.3.2 Disturbed Network Admittance Matrix $\bar{Y}_F$

The predisturbance network admittance matrix $\bar{Y}_{PR}^A$ described in Fig. 6.3 is suitably altered depending on the disturbance; this then provides the disturbed network admittance matrix $\bar{Y}_F^A$. The use of this matrix in the incorporation of nonlinear loads will be described in Chapter 8. The procedure to reduce the matrix $\bar{Y}_F^A$ to the internal generator nodes is then initiated. The buses in group C of Fig. 6.3 are augmented with the internal generator nodes, and the terminal buses of the generators are eliminated providing an admittance matrix denoted by $\bar{Y}_F^B$. The use of these matrices in the representation of nonlinear loads will be described in Chapter 8. The remaining buses are then eliminated. In eliminating the buses corresponding to group A, optimal ordering is done to reduce the number of fill-in terms. The matrix is then fully reduced to the internal generator nodes to obtain the reduced disturbed network admittance matrix $\bar{Y}_F^C$.

6.3.3 Postdisturbance Network Admittance Matrix $\bar{Y}_{PO}$

The procedure to obtain the postdisturbance network admittance matrix is very similar to the procedure to determine the disturbed admittance matrix. The predisturbance network admittance matrix $\bar{Y}_{PR}^A$ described in Fig. 6.3 is suitably altered depending on the postdisturbance switching operation to obtain the postdisturbance network admittance matrix $\bar{Y}_{PO}^A$. This matrix is then reduced to the internal generator nodes. The buses in group C are again augmented with the internal generator nodes, and the terminal nodes are eliminated to obtain $\bar{Y}_{PO}^B$. The remaining buses are then eliminated to obtain the reduced postdisturbance network admittance matrix $\bar{Y}_{PO}^C$.

The use of the matrices $\bar{Y}_{PO}^A$ and $\bar{Y}_{PO}^B$ in representing the nonlinear loads will be described in Chapter 8.

6.4 CONDITIONS AT THE END OF THE DISTURBANCE

Disturbances in general could be classified into two categories: fault and nonfault. Based on the type of the disturbance, the procedure to evaluate the conditions at the end of the disturbance and its effect on the system energy could be vastly different. We begin by discussing the case where the disturbance is a fault.

6.4.1 Conditions at Fault Clearing

Two approximate techniques have been developed to determine the conditions at fault clearing. A detailed description of these two methods will now be provided.

Constant acceleration approach

In this technique, the acceleration of each machine is held constant over each time step of the faulted period. The time period from the instant of application of the fault t_o to the instant of fault clearing t_{cl} is divided into several intervals, each of length Δt . The following procedure is then used to determine the conditions at t_{cl} .

1. Using the reduced faulted network admittance matrix $\bar{Y}_F^C$ parameters, the electrical power output of each generator $P_{ei}(t_o^+)$ at the instant the fault is applied is determined where P_{ei} is given by (2.3). The accelerating power for each machine is then computed using

$$P_{ai}(t_o^+) = P_{mi} - P_{ei}(t_o^+) \quad i = 1, 2, \dots, n \quad (6.7)$$

2. Assuming a constant acceleration over the first time interval, the change in angular speed is given by

$$\Delta \omega_i(t_1) = \frac{P_{a_i}(t_0) \Delta t}{2H} \text{ pu} \quad i = 1, 2, \dots, n \quad (6.8)$$

3. The angular change is given by

$$\Delta \delta_i(t_1) = (2\pi f \times \frac{180}{\pi}) \left[\frac{P_{a_i}(t_1) (\Delta t)^2}{4H} \right] \text{ degrees} \quad i = 1, 2, \dots, n \quad (6.9)$$

and

$$\delta_i(t_1) = \delta_i(t_0) + \Delta \delta_i(t_1) \quad i = 1, 2, \dots, n \quad (6.10)$$

$$\omega_i(t_1) = 0.0 + \Delta \omega_i(t_1) \quad i = 1, 2, \dots, n \quad (6.11)$$

where the initial speeds are assumed to be zero and f is the system frequency.

4. Using the updated angles from step 3, the accelerating power of each machine at time t_1 , $P_{ai}(t_1)$, $i = 1, 2, \dots, n$, is evaluated as described in step 1.

5. The parameters at the end of the next interval are given by

$$\Delta \omega_i(t_2) = \frac{P_{a_i}(t_1) \Delta t}{2H} \text{ pu} \quad i = 1, 2, \dots, n \quad (6.12)$$

and

$$\Delta \delta_i(t_2) = (2\pi f \times \frac{180}{\pi}) \left[\frac{P_{a_i}(t_1) (\Delta t)^2}{4H} \right] \quad i = 1, 2, \dots, n \quad (6.13)$$

6. The procedure is then repeated until the instant of fault clearing is reached.

Modified constant acceleration approach

In this approach, the procedure follows the steps described. At the end of each time interval, a one-step correction procedure is applied to improve the estimate of the variable without substantially increasing the computational burden. As a result, the variable at the $(n + 1)$ th interval is given by

$$x_i(n+1) = x_i(n) + \frac{1}{2} \left[\dot{x}_i(n) + \dot{x}_i^0(n+1) \right] \cdot \Delta t \quad i = 1, 2, \dots, n \quad (6.14)$$

where $\dot{x}_i^0$ is used to indicate the initial estimate of the derivative obtained from the procedure described earlier.

6.4.2 Disturbances Other than Faults

In relation to the TEF method, the conditions at the end of the disturbance are evaluated to determine the transient energy due to the disturbance. In the case of a fault, as described in Section 6.4.1, the rotor angle positions and rotor angle velocities at the end of the fault are evaluated to obtain the energy at fault clearing. In this type of disturbance the change in the transient energy occurs during the fault period. After the disturbance is removed, the transient energy is constant.

For disturbances other than faults, it is essential to analyze the effect of the disturbance on the system transient energy. In some cases the disturbance itself is a sudden change in the potential energy (e.g., a sudden opening of a loaded transmission line). Again after the disturbance, the total system energy is constant.

From a stability assessment viewpoint, the situation boils down to determining the effect of the disturbance on the energy margin ΔV given by (5.18), repeated here for convenience.

$$\Delta V = -\frac{1}{2} M_{eq} \tilde{\omega}_{eq}^{cl^2} - \sum_{i=1}^n P_i (\theta_i^u - \theta_i^{cl})$$

$$- \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[C_{ij} (\cos \theta_{ij}^u - \cos \theta_{ij}^{cl}) - D_{ij} \frac{\theta_i^u - \theta_i^{cl} + \theta_j^u - \theta_j^{cl}}{(\theta_{ij}^u - \theta_{ij}^{cl})} (\sin \theta_{ij}^u - \sin \theta_{ij}^{cl}) \right] \quad (6.15)$$

In terms of the components of the energy margin, the equation can be written as

$$\Delta V = \Delta V_{KE} + \Delta V_{PE} \quad (6.16)$$

where $\Delta V_{KE} = -\frac{1}{2} M_{eq} \tilde{\omega}_{eq}^{cl^2}$ represents the kinetic energy margin and ΔV_{PE} consisting of the remaining terms in (6.15) represents the potential energy margin. In order to model a disturbance, the effect of the disturbance on these two components of the energy margin will have to be determined.

The physical nature of the disturbance should be carefully analyzed to determine how it affects the components ΔV_{KE} and ΔV_{PE} . In turn, this will result in determining the vectors θ^{cl} and $\tilde{\omega}^{cl}$ at the end of the disturbance. For certain disturbances, e.g., loss of generation, or change in network configuration, the disturbance immediately affects only the potential energy component. In these cases, the system is at equilibrium before the disturbance, the disturbance occurs, and the system begins to move. There is no finite period of time for which the disturbance lasts. The disturbance does not inject any kinetic energy but alters the potential energy absorbing capacity of the system. Hence, this aspect of the disturbance needs to be modeled.

Based on this description any disturbance can be modeled by appropriately representing its effect on the ΔV_{KE} and ΔV_{PE} components. In Chapter 8 a detailed description of the loss of generation disturbance will be presented.

To clarify this issue, consider the situation where the system is initially at equilibrium and a sudden change in the network occurs, e.g., a transmission line is tripped. In this situation, for the purpose of analysis, the disturbance has occurred instantaneously. As a result, the kinetic energy V_{KE} does not initially change. However, the potential energy-absorbing capacity of the system has been altered and the potential energy V_{PE} will change. However, after the disturbance, the total energy in the system remains constant. The type and manner of conversion among V_{KE} and V_{PE} depends on the type

of disturbance. A fault disturbs the energy balance over its duration; on the other hand, other types of disturbances affect the system instantaneously.

In simulating the example considered of suddenly losing a transmission line, the appropriate entries of the predisturbance network admittance matrix will be modified and reduced to obtain the reduced postdisturbance network admittance matrix. Careful analysis of this transient reveals that the predisturbance SEP θ^{s1} itself will constitute the angular positions of the generators at the end of the disturbance. The postdisturbance SEP θ^{s2} will be different from θ^{s1} because of the change in the network admittance. This value of θ^{s2} will affect the value of θ^u , but will not in itself appear in the expression for the change in energy. The net effect is that when the potential energy change is computed, its value is the same as the value of the potential energy evaluated between θ^u and θ^{s1} using the postdisturbance Y -matrix parameters.

6.5 DETERMINATION OF THE CONTROLLING UEP

The most important component of the TEF procedure consists of determining the controlling UEP. This in turn involves the following key steps:

1. Identifying the correct UEP.
2. Obtaining a starting point for the UEP solution close to the exact UEP.
3. Calculation of the exact UEP.

It should be pointed out that for many practical situations, especially in stressed system conditions, step 2 is very important for the UEP solution to converge to the correct UEP.

In the following sections we detail two techniques which encompass steps 1 and 2. The calculation of the exact UEP is described in Section 6.8.

6.6 UEP DETERMINATION USING THE MODE OF DISTURBANCE PROCEDURE

As described in Section 5.4, the determination of the mode of disturbance (MOD) constitutes a very important step in the TEF procedure. The criterion developed in equation (5.22) repeated here for convenience requires the determination of the normalized potential energy margin ΔV_{PEN} :

$$\Delta V_{PEN} = \frac{\Delta V_{PE}}{V_{KEcorr}} \quad (6.17)$$

The controlling UEP would be that among the possible ones in the direction of the postdisturbance trajectory having the lowest ΔV_{PEN} .

Based on this criterion it is evident that an automated practical technique to determine the MOD would first have to identify the candidate modes which need to be tested. Furthermore, calculating ΔV_{PEN} requires knowledge of the UEP θ^u for all the candidate modes. Computationally, this can be an expensive and time-consuming task that may make the large-scale application of the TEF method impractical. Keeping these issues in mind, the following automated procedure has evolved based on several of the authors' experiences [3, 4, 5].

6.6.1 Selection of the Candidate Modes

The candidate modes to be tested by the MOD test to a large extent depend on how the disturbance affects the system. Hence, the selection of the candidate modes is based on several measures of the severity of the disturbance. Using the information available at the end of the disturbance (t_{cl}), the following procedure is adopted to select the candidate modes:

1. A list is obtained of machines ranked in the descending order of their kinetic energy at t_{cl} , i.e., based on the quantities $\frac{1}{2} M_i \tilde{\omega}_i^{cl^2}$, where $\tilde{\omega}_i^{cl}$, $i = 1, 2, \dots, n$, is the speed with respect to the COI at t_{cl} . In forming this list, the machines that belong to the same power station are grouped together, once a machine in the station has appeared in the list, irrespective of the actual positions of the other machines in the station. In order to make the procedure more efficient, only those machines whose kinetic energy is within 0.5% of the maximum kinetic energy are included in the list. This list provides a measure of the impact of the disturbance in terms of the kinetic energy gained by machine (reflected by the speed $\tilde{\omega}_i^{cl}$ of each machine) at the end of the disturbance.
2. A second list of machines is obtained, ranked in the descending order of their acceleration at t_{cl} , i.e., based on the quantities $f_i(\theta^{cl})/M_i$, where $f_i(\theta^{cl})$, $i = 1, 2, \dots, n$, is the accelerating power with respect to the COI of machine i at t_{cl} . As in the kinetic energy list, machines belonging to the same station are grouped together. Only those machines whose acceleration is within 0.5% of the maximum acceleration are retained in the list. This list provides a measure of the severity of the disturbance in terms of the acceleration of each machine at the end of the disturbance.
3. Based on the information obtained in the two sorted lists, a final list of key machines of interest is generated by grouping the machines as follows. This combined list characterizes the consolidated measure of the effect of the disturbance on the various machines. The machine on the top of the list will be most affected and the one at the bottom of the list, least affected.
4. The final list obtained in step 3 provides only a list of the machine numbers severely impacted by the disturbance. Based on this information, further analysis needs to be done to determine what groups of machines are more affected than others. This analysis will provide the candidate modes to be tested by the MOD test.

Machines appearing in both the kinetic energy and acceleration lists
=====
Machines appearing in the kinetic energy list, but not the acceleration list
=====
Machines appearing in the acceleration list, but not the kinetic energy list

In order to determine the groups of machines most affected by the disturbance, the corrected kinetic energy V_{KEcorr} given by (5.16) is used as a measure. Starting from the list obtained in step 3, the machine or group of machines at the bottom of the list (if the machines at the bottom of the list belong to the same plant) are included in the group forming the rest of the generators (subscript *sys* in (5.15)) and all the other generators are included in the critical group (subscript *cr* in (5.15)). The corrected kinetic energy for this mode of system separation is then calculated and the information regarding the machines included in the critical group is also stored. In a sequential manner, the machine or group of machines at the bottom of the critical group chosen is then moved into the group forming the rest of the generators and the corrected kinetic energy and critical group information is again stored. In this manner, we successively add machines to the group forming the rest of the generators and move to the top of the list obtained in step 3. The V_{KEcorr} obtained for the various groups is then sorted in descending order. In this list we retain only those groups whose V_{KEcorr} is within 10% of the maximum V_{KEcorr} in the list. The retained groups and the information regarding the machines participating in these groups provide the list of candidate modes to be tested using the MOD test.

The procedure just described exhaustively tests the candidate groups obtained from the list of machines in step 3. In a large system this could be computationally burdensome. In order to improve the computational efficiency of this procedure, the technique is modified as follows. From the list obtained in step 3, start from the top. Include the first machine or group of machines (if all the machines at a plant are included) in the critical group (*cr*). The remaining machines in the list are included in the group forming the remaining generators (*sys*). The V_{KEcorr} is then evaluated, and the information regarding the machine groupings is stored. The next machine or group of machines from the list generated in step 3 is included in the critical group and V_{KEcorr} is evaluated. The V_{KEcorr} is then sorted in descending order. The procedure of including more machines in the critical group is continued until V_{KEcorr} is within 10% of the

maximum V_{KEcorr} obtained until the consideration of the present grouping. The information regarding the machines participating in each group then provides the candidate modes to be tested by the MOD test.

6.6.2 A Practical Mode of Disturbance Test

The sorted V_{KEcorr} list and the associated critical group information available from step 4 of the candidate mode selection procedure provides the candidates to be tested using the MOD test. The MOD test requires the evaluation of ΔV_{PEN} given by (6.17). Calculating ΔV_{PEN} requires knowledge of θ^u for all the candidate modes. Computationally this can be an expensive and time-consuming task that may make large scale application of the TEF method impractical. Based on the work done in [3–5], the following simplified procedure is proposed to reduce the computational burden. The method is based on computing ΔV_{PEN} for the candidate modes using an approximate value of θ^u . However, once the candidate mode with the lowest value of approximate ΔV_{PEN} is determined, the corresponding θ^u is computed accurately.

1. Construction of Corner Point

For each candidate mode, obtained in the final V_{KEcorr} list generated in step 4 of Section 6.5.1, the information regarding the machines contained in the critical group is used to construct the approximate θ^u for that particular mode. The approximate positions of the θ^u for each candidate mode are first estimated from a knowledge of the postdisturbance stable equilibrium point θ^{s2} . This is based on the fact [6, 7] that, in the synchronous reference frame, the UEP lies in the proximity of the corner point of a polytope.

For a given candidate mode, where machines i and j are contained in the critical group, an estimate of the corner point for an n -machine system is given by

$$\left[\hat{\theta}_{ij}^u \right]^T = \left[\theta_1^{s2}, \theta_2^{s2}, \dots, \left[\pi - \theta_i^{s2} \right], \dots, \left[\pi - \theta_j^{s2} \right], \dots, \theta_n^{s2} \right] \quad (6.18)$$

where the postdisturbance SEP θ^{s2} is expressed in radians. It is to be noted that the components of the vector θ^{s2} satisfy the inertial center constraint $\sum_{i=1}^n M_i \theta_i^{s2} = 0$. On the other hand, the components of the vector $\hat{\theta}_{ij}^u$ given by equation (6.18) do not; hence a correction needs to be made to account for the motion of the inertial center from θ^s to $\hat{\theta}^u$. A procedure to determine the corner point and correct for the motion of the COI in one step is given next.

2. Corner Point COI Correction

For each candidate mode the corner point obtained using (6.18) is corrected for the inertial center motion as follows.

a. A machine is first classified as belonging to either of two groups

Group I: Critical machines, as described by the information characterizing each candidate mode. These mainly consist of the advanced machines (whose angles are generally greater than $\pi/2$ radians).

Group II: The rest of the machines.

b. Compute

$$\theta_I^{s2} = \frac{1}{M_I} \sum_{i \in I} M_i \theta_i^{s2}, \quad (6.19a)$$

$$\theta_{II}^{s2} = \frac{1}{M_{II}} \sum_{i \in II} M_i \theta_i^{s2}, \text{ and} \quad (6.19b)$$

$$\theta_{I-II}^{s2} = \theta_I^{s2} - \theta_{II}^{s2}, \quad (6.19c)$$

where

$$M_I = \sum_{i \in I} M_i, \quad M_{II} = \sum_{i \in II} M_i$$

The quantity θ_{I-II}^{s2} represents the angular separation between the inertial centers of the advanced group of generators and the rest of the generators at the postdisturbance SEP.

- c. At the corner point, this angular separation is given by $(\pi - \theta_{I-II}^{s2})$. As a result, the motion of the relative positions of the inertial centers of the two groups between the corner point and the postdisturbance SEP is given by $(\pi - 2\theta_{I-II}^{s2})$. This motion is allocated to each group in the inverse ratio of their total inertia. Hence,

$$\Delta\theta_I = (\pi - 2\theta_{I-II}^{s2}) \frac{M_{II}}{M_I + M_{II}} \quad (6.20a)$$

$$\Delta\theta_{II} = (\pi - 2\theta_{I-II}^{s2}) \frac{M_I}{M_I + M_{II}} \quad (6.20b)$$

- d. Calculate

$$\hat{\theta}_i^u \text{ for } i \in I = \theta_i^{s2} + \Delta\theta_I \quad (6.21)$$

$$\hat{\theta}_i^u \text{ for } i \in II = \theta_i^{s2} - \Delta\theta_{II} \quad (6.22)$$

Equations (6.21) and (6.22) give the corner point corrected for the motion of the inertial center. For the candidate mode selected in (6.18) with machines i, j advanced, it will be shown that angles obtained using the procedure given satisfy the inertial center constraints.

Example

- a. Group I $\rightarrow i, j$

Group II $\rightarrow k = 1, \dots, n, \quad k \neq i, j$

$$\text{b. } \theta_I^{s2} = \frac{M_i \theta_i^{s2} + M_j \theta_j^{s2}}{M_i + M_j} = \frac{M_i \theta_i^{s2} + M_j \theta_j^{s2}}{M_I}$$

$$\theta_{II}^{s2} = \frac{\sum_{k=1, k \neq i, j}^n M_k \theta_k^{s2}}{M_T - (M_i + M_j)} = \frac{\sum_{k=1, k \neq i, j}^n M_k \theta_k^{s2}}{M_{II}}$$

where

$$M_T = \sum_{i=1}^n M_i$$

we can show that

$$\begin{aligned} \theta_I^{s2} - \theta_{II}^{s2} &= \frac{M_T (M_i \theta_i^{s2} + M_j \theta_j^{s2}) - M_I \sum_{k=1}^n M_k \theta_k^{s2}}{M_I M_{II}} \\ &= \frac{M_T \theta_I^{s2}}{M_{II}} \end{aligned}$$

$$\text{as } \sum_{k=1}^n M_k \theta_k^{s2} = 0$$

$$\text{c. } \Delta \theta_I = \left[\pi - \frac{2M_T \theta_I^{s2}}{M_{II}} \right] \frac{M_{II}}{M_T}$$

$$\Delta \theta_{II} = \left[\pi - \frac{2M_T \theta_I^{s2}}{M_{II}} \right] \frac{M_I}{M_T}$$

$$\text{d. } \hat{\theta}_i^u = \theta_i^{s2} + \left[\pi - \frac{2M_T \theta_I^{s2}}{M_{II}} \right] \frac{M_{II}}{M_T}$$

$$\hat{\theta}_j^u = \theta_j^{s2} + \left[\pi - \frac{2M_T \theta_I^{s2}}{M_{II}} \right] \frac{M_{II}}{M_T}$$

$$\hat{\theta}_k^u = \theta_k^{s2} - \left[\pi - \frac{2M_T \theta_I^{s2}}{M_{II}} \right] \frac{M_I}{M_T}, \quad k = 1, 2, \dots, n \\ \neq i, j$$

$$\sum_{i=1}^n M_i \hat{\theta}_i^u = \sum_{i=1}^n M_i \theta_i^{s2} + M_i \left[\pi - \frac{2M_T \theta_I^{s2}}{M_{II}} \right] \frac{M_{II}}{M_T}$$

$$+ M_j \left[\pi - \frac{2M_T \theta_I^{s2}}{M_{II}} \right] \frac{M_{II}}{M_T}$$

$$- \sum_{\substack{k=1 \\ k \neq i, j}}^n M_k \left[\pi - \frac{2M_T \theta_I^{s2}}{M_{II}} \right] \frac{M_{II}}{M_T}$$

$$= 0 + \left[\pi - \frac{2M_T \theta_I^{s2}}{M_{II}} \right] \frac{M_I M_{II}}{M_T}$$

$$- \left[\pi - \frac{2M_T \theta_I^{s2}}{M_{II}} \right] \frac{M_I M_{II}}{M_T} = 0$$

The corrected corner point thus obtained is further improved to get a better estimate of the approximate θ^u .

3. *Further Improvement of Corrected Corner Point – Ray Point*

For each candidate mode, after having determined the corrected corner point, the following procedure is adopted to obtain a better estimate of the approximate θ^u . In angular space, a ray from θ^{s2} to the corrected corner point $\hat{\theta}^u$ obtained in step 2 is formed as shown in Fig. 6.4.

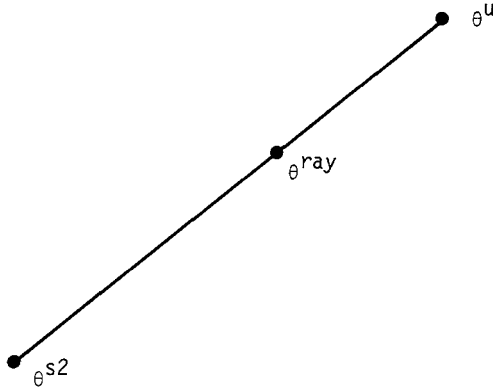


Figure 6.4 Maximum potential energy point on ray from θ^{s2} to $\hat{\theta}^u$.

Along this ray, the potential energy is maximized. Any point on the ray emanating from θ^{s2} and passing through $\hat{\theta}^u$ can be denoted as :

$$\theta = \theta^{s2} + \alpha \Delta \theta \quad (6.23)$$

where $\alpha \geq 0$ is a scalar and $\Delta \theta = \hat{\theta}^u - \theta^{s2}$. Along this ray, the potential energy varies only as a function of the scalar α , and as shown in (4.19) is given by

$$V_{PE_{RAY}}(\alpha) = - \sum_{i=1}^n \int_0^{\alpha} f_i (\Delta \theta \alpha + \theta^{s2}) \Delta \theta_i \cdot d\alpha \quad (6.24)$$

and its derivative with respect to the parameter α is given by

$$\frac{dV_{PE_{RAY}}(\alpha)}{d\alpha} = - \sum_{i=1}^n f_i (\Delta \theta \alpha + \theta^{s2}) \cdot \Delta \theta_i \quad (6.25)$$

Hence, the problem of determining the point of maximum potential energy along the ray is simplified to a one-dimensional minimization problem with respect to the variable α .

Let the value of α , where the potential energy reaches the first relative maximum, be α^* and the corresponding angular position vector will be

$$\theta^{ray} = \theta^{s2} + \alpha^* \Delta \theta \quad (6.26)$$

The point θ^{ray} as described in Section 4.3.6 will be situated on the potential energy boundary surface (PEBS). In order to compute α^* we iterate using a standard one-dimensional minimization approach [8,9] to find the value of α where $dV_{PE_{RAY}}/d\alpha$ given by (6.25) is equal to zero.

The rationale behind this procedure is that θ^{ray} , the point on the PEBS provides a more accurate approximation to the UEP θ^u . Having obtained θ^{ray} , it is used as an approximation to the UEP, and ΔV_{PE} and hence ΔV_{PEN} are evaluated for each candidate mode. The candidate mode which has the lowest value of ΔV_{PEN} is the MOD for the

controlling UEP for the given disturbance and postdisturbance condition (see Section 6.8.4). Knowing the advanced machines in the MOD, the exact UEP will be determined.

6.7 UEP DETERMINATION USING THE EXIT POINT METHOD

6.7.1 Introduction

The exit point method to determine the controlling UEP has been derived from [10] and modified to incorporate the COI formulation and to provide robustness and reliability in obtaining a starting point close to the controlling UEP. The method is based on the concept of the stable manifold of the controlling UEP and the associated gradient system developed in Section 4.4.2. The procedure is algorithmic in nature and combines both steps 1 and 2 described in Section 6.5. The main steps in the procedure consist of obtaining the exit point which lies on the stable manifold of the controlling UEP, and then moving along the projection of the manifold in the θ -space using the associated gradient system to obtain a point very close to the controlling UEP.

6.7.2 Modified Exit Point Procedure

The procedure consists of the following steps:

1. Using the faulted trajectory, described by the COI formulation (equation (2.13)), determine the exit point $\mathbf{x}^e(\theta^e, \tilde{\omega}^e)$ on the stable manifold of the controlling UEP. This is done by obtaining the first maximum of the potential energy along the faulted trajectory with respect to the postdisturbance network. The faulted trajectory is obtained by integrating equations (6.27).

$$M_i \dot{\tilde{\omega}}_i = P_{mi}^f - P_{ei}^f - \frac{M_i}{M_T} P_{COI}^f \quad (6.27)$$

$$\dot{\theta}_i = \tilde{\omega}_i \quad i = 1, 2, \dots, n$$

Values of θ obtained from the solution of (6.27) are substituted into the postfault mismatch equations given by

$$f_i = P_{mi}^{pf} - P_{ei}^{pf} - \frac{M_i}{M_T} P_{COI}^{pf} \quad i = 1, 2, \dots, n \quad (6.28)$$

$\mathbf{x}^e$ is then obtained by satisfying the condition $\sum_{i=1}^n -f_i \tilde{\omega}_i = 0$. The robustness and reliability of the method is extremely sensitive to accuracy of the check provided. The zero crossing along the faulted trajectory is usually detected between two time steps by a change in sign of the quantity $\sum_{i=1}^n -f_i \tilde{\omega}_i$. Once the change in sign is detected, the solution at the previous time step is stored and the time step is reduced by an order of magnitude. The equations are integrated until the quantity $\sum_{i=1}^n -f_i \tilde{\omega}_i$ is less than a tolerance of 0.00001. This ensures that the exit point determined accurately describes a point on the stable manifold of the controlling UEP.

2. Using θ^e as a starting point, integrate the associated gradient system equations given by

$$\dot{\theta}_i = P_{mi}^{pf} - P_{ei}^{pf} - \frac{M_i}{M_T} P_{COI}^{pf} \quad i = 1, 2, \dots, n-1 \quad (6.29)$$

$$\dot{\theta}_n = - \frac{\sum_{i=1}^{n-1} M_i \dot{\theta}_i}{M_n}$$

This set of equations is often numerically ill conditioned and stiff. In order to make the integration robust and reliable, an automatic step-length and solution order selection procedure is essential. To satisfy this requirement the use of specialized software packages for solving ordinary differential equations are essential. The authors have used a package called ODEPACK [11]. A subroutine called LSODE is used to integrate the system of equations described in (6.29). The integration algorithm is based on the gear and gear B method. Reference [11] provides a detailed description of the algorithm. In performing this integration starting from the exit point, we will be determining how the rotor angle positions change along the gradient surface. At each step of the integration evaluate $\sum_{i=1}^n |f_i| = F$ and determine the first minimum of F along the gradient surface. Let θ^* be the vector of rotor angles at this point.

3. Use θ^* as a starting point and obtain the UEP solution.

Detailed testing of the above technique on several systems has provided consistently robust and reliable results.

6.8 CALCULATION OF EQUILIBRIUM POINTS

The technique to calculate the equilibrium points is a general solution technique which can be used to evaluate both the postdisturbance SEP and the controlling UEP. The only difference between these two solutions is the starting point provided for the solution procedure.

6.8.1 Problem Formulation

The procedure to obtain the equilibrium points consists of determining a solution to a system of nonlinear algebraic equations given by (5.20) and repeated here for convenience

$$f_i = P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} \quad i = 1, 2, \dots, n \quad (6.30)$$

In general, there are two approaches to solving such a system of equations. They are :

1. *The direct solution approach* : In this case, equation (6.30) is directly solved using an iterative technique. The Newton–Raphson method had been previously used. However, it has in some cases, exhibited divergent characteristics and has not been completely reliable [4, 13]. In [12] a modified Newton–Raphson technique has been used with greater success. The modification consists of incorporating a cubic interpolation with the Newton–Raphson iterations to choose an optimum step length. This adds greater numerical stability.
2. *Indirect solution approach* : In this case, the solution technique is formulated as a nonlinear least squares minimization problem [13]. The objective function to be minimized is given by

$$F(\theta) = \sum_{i=1}^n f_i^2(\theta) \quad (6.31)$$

where f_i is given in equation (6.30). In [13], the Davidon–Fletcher–Powell (DFP) method of unconstrained minimization was used. This technique was

carefully tested in [4]. It was found that for systems of small dimension, the technique was reliable and efficient. However, for large systems, the method is computationally inefficient and unreliable.

In [4] a careful search of the various methods available to solve the minimization problem was made, keeping in consideration the system size dealt with and the stressed nature of the system. Based on this search, the corrected Gauss–Newton (CGN) method [14] was selected as a replacement of the DFP technique. The CGN method is a very robust and reliable technique; it, however, requires a singular value decomposition of the Jacobian matrix at every iteration. This results in severe computational burden; as a result the method has been replaced by the full Newton method [14]. The Newton method, like the CGN method, uses second–derivative information. It does not require any singular value decomposition and is very fast and reliable.

6.8.2 Solution Algorithms

Modified Newton–Raphson technique

The modified Newton–Raphson method is used to determine the solution to the system of equations given by (6.30). Even though there are n equations, the number of unknowns is only $(n-1)$ because knowing $(n-1)$ angle positions in the inertial center reference frame the inertial center constraint $\sum_{i=1}^n M_i \theta_i = 0$ can be used to obtain the n th angle. As a result, a $(n-1) \times (n-1)$ Jacobian, obtained as $\left[\frac{\partial f_i}{\partial x_j} \right]$, $i = 1, 2, \dots, n-1$, and $j = 1, 2, \dots, n-1$ is utilized. The detailed expressions for the Jacobian terms are given in Section 6.7.3. The algorithm can then be summarized as follows.

1. Select $\mathbf{x}^0$, the starting point for the solution procedure, and calculate $f_i^{(0)}$
 $i = 1, 2, \dots, n$ and $J^{(0)} = \left[\frac{\partial f_i}{\partial x_j} \right] \bigg|_{\mathbf{x}=\mathbf{x}^0}$, $i = 1, 2, \dots, n-1$, $j = 1, 2, \dots, n-1$.

Set the iteration count $k = 0$.

2. Evaluate $F(\mathbf{x}) = \sum_{i=1}^n f_i^2(\mathbf{x})$. If $F(\mathbf{x}) \leq \epsilon$ (ϵ is the precision index), the algorithm is terminated; otherwise, continue at step 3.

3. Compute the direction of search

$$\Delta \mathbf{x}^{(k)} = - \left[J^{(k)} \right]^{-1} \mathbf{f}^{(k)}$$

4. Compute the step length α using cubic interpolation. Then

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha \Delta \mathbf{x}^{(k)}$$

The cubic interpolation procedure minimizes $F(\mathbf{x})$ with respect to α in the direction $\Delta \mathbf{x}$. Figure 6.5 summarizes the procedure.

$$\alpha = \alpha_y - \frac{(\alpha_y - \alpha_w) [F'_y + (r-s)]}{F'_y - F'_w + 2r}$$

where $F' = \partial F / \partial \alpha$

$$r = (s^2 - F'_w F'_y)^{1/2}$$

$$s = 3 \frac{(F_w - F_y)}{(\alpha_y - \alpha_w)} + F'_w + F'_y$$

5. Evaluate $x_n^{(k+1)} = \frac{\sum_{i=1}^{n-1} M_i x_i^{(k+1)}}{M_n}$ and return to step 2.

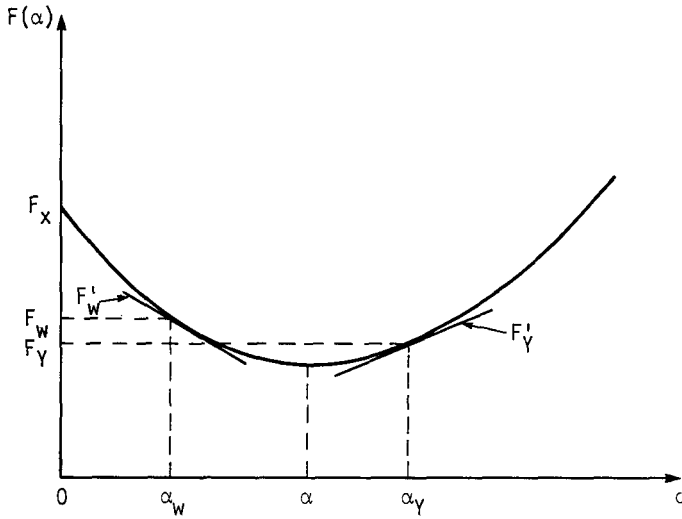


Figure 6.5 Cubic interpolation to minimize $F(x)$ with respect to α in the direction Δx .

The Newton's method

The Newton's method obtains a point $\mathbf{x}^*$ which minimizes the sum of squares of nonlinear functions given by

$$F(\mathbf{x}) = \sum_{i=1}^n f_i^2(\mathbf{x}) \quad \mathbf{x} \in R^{n-1}$$

As described in the modified Newton–Raphson method, we need to solve for only $(n - 1)$ angular variables in the inertial center formulation. The gradient vector $\mathbf{g}(\mathbf{x})$ and Hessian matrix $\mathbf{G}(\mathbf{x})$ of $F(\mathbf{x})$ are given by $2\mathbf{J}(\mathbf{x})^T \mathbf{f}(\mathbf{x})$ and $2[\mathbf{J}(\mathbf{x})^T \mathbf{J}(\mathbf{x}) + \mathbf{B}(\mathbf{x})]$, respectively, where $\mathbf{J}(\mathbf{x})$ is the $n \times n - 1$ Jacobian matrix of $\mathbf{f}(\mathbf{x})$ whose i th row is $\nabla f_i(\mathbf{x}) = (\partial f_i / \partial x_1, \partial f_i / \partial x_2,$

..., $\partial f_i / \partial x_{n-1}$), $B(\mathbf{x}) = \sum_{i=1}^n f_i(\mathbf{x}) G_i(\mathbf{x})$, and $G_i(\mathbf{x})$ is the Hessian matrix of $f_i(\mathbf{x})$. The

algorithm can then be summarized as follows:

1. Select the starting point for the solution procedure $\mathbf{x}^{(0)} \in R^{n-1}$; calculate

$$x_n^{(0)} = \frac{-\sum_{i=1}^{n-1} M_i x_i^{(0)}}{M_n}, \quad f_i^{(0)}, \quad i = 1, 2, \dots, n,$$

$$J^{(0)} = \left[\frac{\partial f_i}{\partial x_j} \right] \Big|_{\mathbf{x}=\mathbf{x}^{(0)}}, \quad i = 1, 2, \dots, n \quad j = 1, 2, \dots, n-1$$

$$g^{(0)} = 2J^{(0)T} f^{(0)} \quad \text{and} \quad B^{(0)} = \sum_{i=1}^n f_i^{(0)} G_i^{(0)}$$

Set iteration count $k = 0$.

2. Evaluate $F(\mathbf{x}) = \sum_{i=1}^n f_i^2(\mathbf{x})$. If $F(\mathbf{x}) \leq \epsilon$, the algorithm is terminated; otherwise, continue at step 3.

3. Compute the direction of search using the equation

$$(J^{(k)T} J^{(k)} + B^{(k)}) \Delta \mathbf{x}^{(k)} = -J^{(k)T} f^{(k)}$$

4. Compute the step-length α using cubic interpolation (see above). Then

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha \Delta \mathbf{x}^{(k)}$$

5. Evaluate $x_n^{(k+1)} = \frac{-\sum_{i=1}^{n-1} M_i x_i^{(k+1)}}{M_n}$ and return to step 2.

6.8.3 Analytical Expressions for the Jacobian and Hessian Matrices

In this subsection, we provide the analytical details for the Jacobian matrix and the Hessian matrix used in the equilibrium point solutions. The mismatch equation given in (6.30) can be expressed as

$$f_i(\theta) = P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} \quad i = 1, 2, \dots, n \quad (6.32)$$

where θ_i , $i = 1, 2, \dots, n$ are the rotor angles in the inertial center reference frame and

$$P_{ei} = \sum_{\substack{j=1 \\ j \neq i}}^n [D_{ij} \cos \theta_{ij} + C_{ij} \sin \theta_{ij}] + E_i^2 G_{ii} \quad (6.33)$$

where the constants C_{ij} , D_{ij} , M_i , M_T , and P_i (below) have all been defined in Section 2.4. $\theta_{ij} \triangleq \theta_i - \theta_j$ and

$$P_{COI} = \sum_{i=1}^n P_{mi} - P_{ei} = \sum_{i=1}^n P_i - 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n D_{ij} \cos \theta_{ij} \quad (6.34)$$

Due to the inertial center constraints, only $(n - 1)$ rotor angles are independent variables, and the n th angles can be obtained as

$$\theta_n = - \sum_{i=1}^{n-1} \frac{M_i}{M_n} \theta_i.$$

The Jacobian matrix of f , denoted by $[J_{ij}]$, is given by

$$J = \left[\frac{\partial f_i}{\partial \theta_j} \right] \quad i = 1, 2, \dots, n, \quad j = 1, 2, \dots, n-1 \quad (6.35)$$

The gradient of $F(\theta)$ equation (6.21), denoted by the vector (g_i) , is given by

$$g = \left[\frac{\partial F}{\partial \theta_i} \right] \quad i = 1, 2, \dots, n-1$$

$$= 2J^T f \quad (6.36)$$

The Hessian matrix of f_i is denoted by the matrix G_i for each machine and is given by

$$G_i = \left[\frac{\partial^2 f_i}{\partial \theta_j \partial \theta_k} \right] \quad j = 1, 2, \dots, n-1, \quad k = 1, 2, \dots, n-1 \quad (6.37)$$

for a given i

The matrix B is defined as

$$B = \sum_{i=1}^n f_i G_i \quad (6.38)$$

The ij th element of the Jacobian matrix J_{ij} is calculated explicitly as

$$J_{ij} = \frac{\partial f_i}{\partial \theta_j} = - \frac{\partial P_{ei}}{\partial \theta_j} - \frac{M_i}{M_T} \frac{\partial P_{COI}}{\partial \theta_j} \quad (6.39)$$

$$i = 1, 2, \dots, n \quad \text{and} \quad j = 1, 2, \dots, n-1$$

The ijk th element of the Hessian matrix, G_k is defined as

$$G_{kij} = - \frac{\partial^2 P_{ek}}{\partial \theta_i \partial \theta_j} - \frac{M_k}{M_T} \frac{\partial^2 P_{COI}}{\partial \theta_i \partial \theta_j} \quad (6.40)$$

$$i = 1, 2, \dots, n-1, \quad j = 1, 2, \dots, n-1$$

$$\text{for a given } k = 1, 2, \dots, n$$

Thus, the first derivatives to be computed in the closed form are $\partial P_{ei} / \partial \theta_j$, $\partial P_{COI} / \partial \theta_j$ terms in equation (6.39). The second derivatives to be computed in closed form are the $\partial^2 P_{ek} / \partial \theta_i \partial \theta_j$, $\partial^2 P_{COI} / \partial \theta_i \partial \theta_j$ terms in equation (6.40).

Computation of first derivatives

Define

$$d(\theta_i, \theta_j) \triangleq -D_{ij} \sin \theta_{ij} + C_{ij} \cos \theta_{ij} \quad (6.41)$$

$$\frac{\partial P_{ei}}{\partial \theta_i} = \sum_{\substack{j=1 \\ j \neq i}}^{n-1} d(\theta_i, \theta_j) + d(\theta_i, \theta_n) \left[1 + \frac{M_i}{M_n} \right] \quad (6.42a)$$

$$i = 1, n-1$$

$$\frac{\partial P_{ei}}{\partial \theta_j} = -d(\theta_i, \theta_j) + d(\theta_i, \theta_n) \frac{M_j}{M_n} \quad (6.42b)$$

$$i \neq j, j = 1, n-1$$

and

$$\frac{\partial P_{en}}{\partial \theta_i} = \sum_{\substack{j=1 \\ j \neq i}}^{n-1} \left[-d(\theta_n, \theta_j) \frac{M_i}{M_n} - d(\theta_n, \theta_i) \cdot \left[1 + \frac{M_i}{M_n} \right] \right] \quad (6.42c)$$

$$i = 1, 2, \dots, n-1$$

$$\begin{aligned}
\frac{\partial P_{COI}}{\partial \theta_i} = & 2 \left[\sum_{\substack{j=1 \\ j \neq i}}^{n-1} D_{ij} \sin \theta_{ij} + \sum_{\substack{j=1 \\ j \neq i}}^{n-1} D_{jn} \sin \theta_{jn} \frac{M_i}{M_n} \right. \\
& \left. + D_{in} \sin \theta_{in} \left[1 + \frac{M_i}{M_n} \right] \right]
\end{aligned} \quad (6.43)$$

Computation of second derivatives

$$\text{Define } e(\theta_i, \theta_j) = -D_{ij} \cos \theta_{ij} - C_{ij} \sin \theta_{ij} \quad (6.44)$$

$$\begin{aligned}
\frac{\partial^2 P_{ei}}{\partial \theta_i \partial \theta_i} = & \sum_{\substack{j=1 \\ j \neq i}}^{n-1} e(\theta_i, \theta_j) + e(\theta_i, \theta_n) \cdot \left[1 + \frac{M_i}{M_n} \right]^2 \\
& i = 1, 2, \dots, n-1
\end{aligned} \quad (6.45a)$$

$$\begin{aligned}
\frac{\partial^2 P_{ei}}{\partial \theta_i \partial \theta_j} = & -e(\theta_i, \theta_j) + e(\theta_i, \theta_n) \cdot \left[1 + \frac{M_i}{M_n} \right] \frac{M_j}{M_n} \\
& i, j = 1, 2, \dots, n-1 \\
& i \neq j
\end{aligned} \quad (6.45b)$$

$$\begin{aligned}
\frac{\partial^2 P_{ek}}{\partial \theta_i \partial \theta_i} = & e(\theta_k, \theta_i) + e(\theta_k, \theta_n) \left[\frac{M_i}{M_n} \right]^2 \\
& k \neq i, k, i = 1, 2, \dots, n-1
\end{aligned} \quad (6.45c)$$

$$\begin{aligned}
\frac{\partial^2 P_{ek}}{\partial \theta_i \partial \theta_j} = & e(\theta_k, \theta_n) \frac{M_i}{M_n} \cdot \frac{M_j}{M_n} \\
& k \neq i, j, k, i, j = 1, 2, \dots, n-1 \\
& i \neq j
\end{aligned} \quad (6.45d)$$

$$\frac{\partial^2 P_{en}}{\partial \theta_i \partial \theta_i} = \sum_{\substack{j=1 \\ j \neq i}}^{n-1} \left[d(\theta_n, \theta_j) \left[\frac{M_i}{M_n} \right]^2 \right] + e(\theta_n, \theta_i) \left[1 + \frac{M_i}{M_n} \right]^2 \quad (6.45e)$$

$i = 1, 2, \dots, n-1$

$$\frac{\partial^2 P_{en}}{\partial \theta_i \partial \theta_j} = \sum_{\substack{k=1 \\ k \neq i, j}}^{n-1} e(\theta_n, \theta_k) \frac{M_i}{M_n} \cdot \frac{M_j}{M_n} + e(\theta_n, \theta_i) \left[1 + \frac{M_i}{M_n} \right] \cdot \frac{M_j}{M_n} \\ + e(\theta_n, \theta_j) \left[1 + \frac{M_j}{M_n} \right] \cdot \frac{M_i}{M_n} \quad (6.45f)$$

$i \neq j, i, j = 1, 2, \dots, n-1$

$$\frac{\partial^2 P_{COI}}{\partial \theta_i \partial \theta_i} = 2 \left[\sum_{\substack{j=1 \\ j \neq i}}^{n-1} D_{ij} \cos \theta_{ij} + \sum_{\substack{j=1 \\ j \neq i}}^{n-1} D_{jn} \cos \theta_{jn} \left[\frac{M_i}{M_n} \right]^2 \right. \\ \left. + D_{in} \cos \theta_{in} \left[1 + \frac{M_i}{M_n} \right]^2 \right] \quad (6.46a)$$

$i = 1, 2, \dots, n-1$

$$\frac{\partial^2 P_{COI}}{\partial \theta_i \partial \theta_j} = 2 \left[-D_{ij} \cos \theta_{ij} + \sum_{\substack{k=1 \\ k \neq i, j}}^{n-1} D_{kn} \cos \theta_{kn} \cdot \frac{M_i}{M_n} \cdot \frac{M_j}{M_n} \right. \\ \left. + D_{in} \cos \theta_{in} \left[1 + \frac{M_i}{M_n} \right] \cdot \left[\frac{M_j}{M_n} \right] \right. \\ \left. + D_{jn} \cos \theta_{jn} \left[1 + \frac{M_j}{M_n} \right] \cdot \left[\frac{M_i}{M_n} \right] \right] \quad (6.46b)$$

$i \neq j, i, j = 1, 2, \dots, n-1$

6.8.4 Computational Issues

In obtaining equilibrium point solutions, four computational issues play a crucial role in obtaining accurate and reliable results quickly. Three of these issues relate to the reliability of the solution; the other issue deals with its speed. These issues are discussed below.

Selection of a starting point for UEP solution

The selection of an appropriate starting point for the UEP solution is important in two situations (1) when several UEPs of similar energy levels are clustered close to each other and (2) when the system is stressed and demonstrates a complex dynamic behavior.

In the first situation, selecting a good starting point is essential. If the procedure starts very far away from the controlling UEP, it could converge to one of the neighboring UEPs, resulting in an error in computation of the critical energy.

In the case of large, highly stressed power systems, the problem is accentuated by the fact that the potential energy surface around a UEP may be "very steep" in certain directions and "shallow" in other directions. (Reference [15] provides a graphical analysis of the potential energy surface around the UEP of a stressed power system.) This characteristic makes the convergence to the desired UEP a very difficult numerical task.

Accurate transient stability assessment to a large extent depends on reliably converging to the "correct" or "desired" UEP determined by the MOD test detailed in Section 6.6. For the solution methods described in 6.8.2, reliable convergence to the "correct" UEP is heavily dependent on the starting point for the iterative solution procedure. Selection of a poor starting point may result in convergence to the "wrong" solution.

On analyzing the needs for a criterion to determine a good starting point, two characteristics have to be considered (1) in angular space, the starting point should be close to the desired UEP and (2) the starting point should have a similar energy level to the

desired UEP. If one examines these requirements more closely, it is apparent that these were exactly the same features used in determining the approximation to the UEP in the MOD test in Section 6.6.

As a result, the point on the ray θ^{ray} between θ^{s2} and $\hat{\theta}^u$ at which the potential energy becomes a maximum is used as a starting point for the solution methods. In the authors' experience, this choice of the starting point has worked successfully on several realistic test systems.

Convergence to the desired UEP

This constitutes the most important analytical issue in the large scale application of the TEF method. The reliability of the transient stability assessment largely hinges on the accurate determination of the desired UEP and evaluation of the critical energy at the UEP.

To a great extent, the solution to this issue lies in selecting a robust and safeguarded solution technique. The techniques described in Section 6.8.2 have proved to be very successful, reliable, and efficient. Another factor aiding the convergence to the desired UEP is the selection of the starting point. The ray point θ^{ray} has a large impact in convergence to the correct UEP.

Postdisturbance conditions

In large power networks under stressed conditions, the convergence to the postdisturbance SEP may encounter some problems. The steps taken to improve the convergence to the desired UEP to a large extent help in obtaining the SEP. The same algorithms used to obtain the UEP are also used to obtain the SEP with the predisturbance SEP θ^{s1} as the starting point for the solution procedure.

Another issue which arises is the question of the existence of the stable equilibrium point. Some systems are operated so close to stability limits that in fact the

postdisturbance equilibrium may be steady-state unstable. In such cases, the solution technique will have converged on to a UEP rather than to a SEP, and the TEF method cannot be applied. Hence, it is essential to check whether the postdisturbance equilibrium is stable or unstable. This can be done very easily by evaluating the system Jacobian at the postdisturbance equilibrium point and calculating its eigenvalues. If all the eigenvalues have negative real parts, the equilibrium is indeed stable. If even one eigenvalue has a positive real part, the equilibrium point is unstable.

In certain cases when disturbances last for a longer period of time or when systems are heavily stressed, it may occur that the state of the system at the instant the disturbance is removed lies outside the domain of attraction of the SEP. In such cases, the normalized potential energy margin used to determine the mode of disturbance will be negative. In such cases we again choose the candidate mode with the lowest normalized potential energy margin ΔV_{PEN} .

Speed of equilibrium point solution

The determination of the UEP and SEP constitute important steps in the TEF procedure. In order to make the procedure computationally efficient and fast, it is essential that UEP and SEP solutions should be obtained quickly and efficiently.

The modified Newton–Raphson procedure and the Newton method of nonlinear least squares minimization described in Section 6.8.2 have been found to be the most efficient. Although the methods require the evaluation and manipulation of full (nonsparse) Jacobian and Hessian matrices, the computational efficiency and speed have been vastly improved via efficient calculation of the matrix elements (using, e.g., rectangular coordinate arithmetic operations to evaluate trigonometric terms) and exploitation of symmetric operations.

6.9 ENERGY MARGIN CALCULATION

Having obtained the conditions at the end of the disturbance $(\theta^{cl}, \tilde{\omega}^{cl})$, the controlling UEP θ^u , and knowing the postdisturbance system parameters C_{ij} and D_{ij} , $i = 1, 2, \dots, n$, $j = 1, 2, \dots, n$, the energy margin can be obtained by directly substituting these variables in the energy margin ΔV expression given by (5.18) repeated here for convenience

$$\Delta V = -\frac{1}{2} M_{eq} \tilde{\omega}_{eq}^{cl^2} - \sum_{i=1}^n P_i (\theta_i^u - \theta_i^{cl})$$

$$- \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[C_{ij} (\cos \theta_{ij}^u - \cos \theta_{ij}^{cl}) - D_{ij} \frac{\theta_i^u - \theta_i^{cl} + \theta_j^u - \theta_j^{cl}}{(\theta_{ij}^u - \theta_{ij}^{cl})} (\sin \theta_{ij}^u - \sin \theta_{ij}^{cl}) \right] \quad (6.47)$$

In certain situations the last term in the energy margin expression obtained by the linear trajectory approximation of the dissipation components denoted by

$$I_{ij} = D_{ij} \frac{\theta_i^u - \theta_i^{cl} + \theta_j^u - \theta_j^{cl}}{\theta_{ij}^u - \theta_{ij}^{cl}} (\sin \theta_{ij}^u - \sin \theta_{ij}^{cl}) \quad (6.48)$$

$$\begin{matrix} i = 1, 2, \dots, n-1 \\ j = 2, 3, \dots, n \end{matrix}$$

can attain extremely large values as a result of the difference $(\theta_{ij}^u - \theta_{ij}^{cl})$ in the denominator being small. To correct this numerical problem, the expression for the dissipation term (6.48) is modified as follows:

$$I_{ij} = D_{ij} \frac{\theta_i^u - \theta_i^{cl} + \theta_j^u - \theta_j^{cl}}{\theta_{ij}^u - \theta_{ij}^{cl}} \left[2 \cos \left[\frac{\theta_{ij}^u + \theta_{ij}^{cl}}{2} \right] \sin \left[\frac{\theta_{ij}^u - \theta_{ij}^{cl}}{2} \right] \right] \quad (6.49)$$

$$\begin{aligned} i &= 1, 2, \dots, n-1 \\ j &= 2, 3, \dots, n \end{aligned}$$

$$\begin{aligned} &= D_{ij} (\theta_i^u - \theta_i^{cl} + \theta_j^u - \theta_j^{cl}) \cos \left[\frac{\theta_{ij}^u + \theta_{ij}^{cl}}{2} \right] \left[1 - \frac{\left[\frac{\theta_{ij}^u - \theta_{ij}^{cl}}{2} \right]^2}{3!} \right. \\ &\quad \left. + \frac{\left[\frac{\theta_{ij}^u - \theta_{ij}^{cl}}{2} \right]^4}{5!} \dots \right] \quad \begin{aligned} i &= 1, 2, \dots, n-1 \\ j &= 2, 3, \dots, n \end{aligned} \end{aligned} \quad (6.50)$$

The expression developed in (6.50) does not contain the $(\theta_{ij}^u - \theta_{ij}^{cl})$ term in the denominator, and also depending on its magnitude, the series can be appropriately truncated.

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Chapter 7

APPLICATIONS OF THE TEF METHOD

– CLASSICAL POWER SYSTEM MODEL

7.1 INTRODUCTION

The basic elements of transient stability assessment of a classically modeled power system, using the transient energy function method, have been presented in Chapter 5. The key step is the determination of the transient energy margin ΔV (equation (5.18)) and its normalized value ΔV_n (equation (5.19)). Stability information is obtained from the value of ΔV or ΔV_n . The procedure for obtaining ΔV and ΔV_n is outlined in the flow-chart in Fig. 7.1. The computational issues involved are presented in Chapters 5 and 6.

In Chapter 1 the complexity of modern interconnected power networks and the types of studies conducted to ensure that these networks are built and operated to meet the reliability criteria were reviewed. These must be kept in mind as we discuss the applications of the TEF method for transient stability assessment of today's power networks. These applications must address the types of problems to be analyzed (e.g., types of disturbances), the types of power system conditions to be dealt with (e.g., system size, loading level), the types of results needed (e.g., stability classification, power limits), and the quality of the results obtained (e.g., accuracy, reliability).

In this chapter we will deal with what might be considered "simple" applications, i.e., when such complexities as multiswing instability, complex disturbance scenarios, etc., are not of concern. Therefore, in this chapter we are concerned with first swing transient stability, i.e., whether the rotor angles of the severely disturbed generators reach a maximum and recover. The classical power system is assumed.

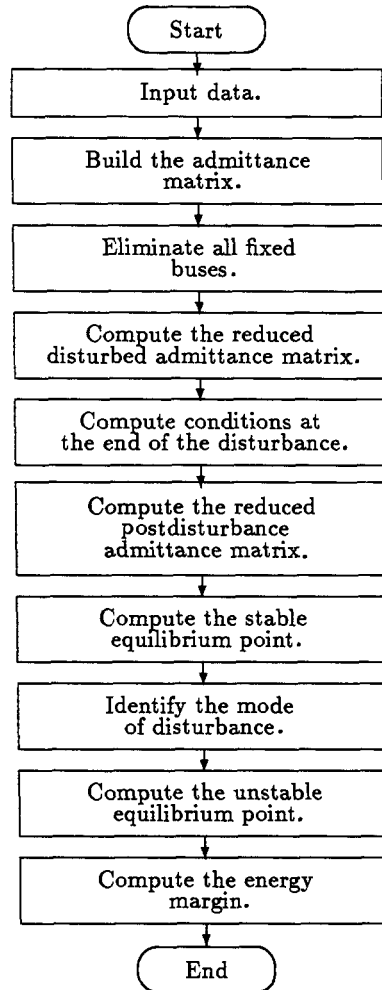


Figure 7.1 Flowchart for stability assessment calculations.

Even when we focus our attention on first swing stability, we find that a wide range of practical questions need to be addressed. Some are related to the type of stability information sought, i.e., the kind of answer obtained from the stability analysis. Others are related to the quality of the answer and its usefulness (its reliability, accuracy, etc.). In this chapter these practical questions will be addressed in the simplest terms, i.e., using the classical power system model.

7.2 PLANT MODE VERSUS INTERAREA MODE STABILITY

Some remarks about the dominant modes of oscillation in a power network may be in order. For an n -generator system there are $(n - 1)$ inertial modes of oscillation, i.e., modes which are primarily governed by the machine inertias and the synchronizing torques in the system associated with these machines. There are numerous other modes originated by the control equipment: excitation systems, turbine-governors, automatic generation control equipment, etc. All these modes of oscillation are excited whenever a disturbance, large or small, is introduced in the system. The situation is further complicated by the fact that these modes may interact in a variety of ways, substantially influencing the transient system behavior. Why is this subject of concern to us in the study of the applications of TEF? Actually it is a matter that *should always be* of interest to analysts dealing with power system response to disturbances.

A rather modest size power network to be analyzed (say, 100 generators) would experience hundreds of modes of oscillations — which ones are of interest? The answer to this question is that this depends on the conditions of the system, e.g., whether there are weak or strong ties between groups of generators, the loading level of the transmission network, and the flow distribution on key transmission interfaces. It is not our intent here to determine *when* certain modes become dominant. Rather, we merely point out that nowadays in transient stability studies two conditions are encountered:

1. *A very small group of generators (tightly coupled) swinging against the rest of the generators in the power system:* The most common situation is when this group of generators is in the same plant and the disturbance is electrically close to that plant. However, the following situations may be (and have often been) encountered: the group may include generators in neighboring plants and when the generator most affected by a disturbance is not the one electrically closest to it.

We will call this type of transient "plant mode" transient stability. We note that such a study precludes the oscillations of the various generators in the group against each other (these oscillations are of interest primarily in the proper "tuning" of the automatic control equipment of these generators).

To illustrate the plant mode, time plots of the rotor angles of generators 5 and 6 (in the COI frame of reference) in the reduced Iowa System (see the appendix for description of the system) are shown in Fig. 7.2. The disturbance is a three-phase fault at bus 372, near these generators. The critically stable and critically unstable cases are displayed. The MOD test (see Section 5.4) for this disturbance indicates that these generators are severely disturbed.

An interesting plant mode is encountered in the same system when the fault is near generator 16 (Ft. Calhoun). For this case, the severely disturbed generators

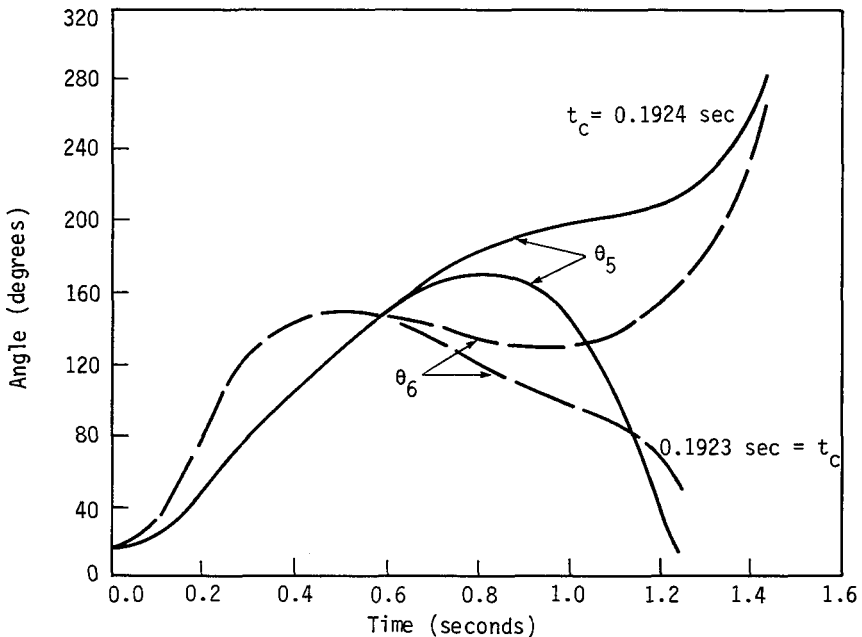


Figure 7.2 Seventeen-generator test system. Swing curves of generators 5 and 6 for RAUN fault; $t_c = 0.1923$ and 0.1924 sec.

are 2, 5, 6, 10, 12, 16, and 17. However, only generator 16 initially loses synchronism when the fault is cleared at 0.357 sec. Time plots of the rotor angles, in the COI frame, were shown in Figs. 5.4, 5.5.

2. *A large group of generators, not necessarily tightly coupled electrically, tending to split from the rest of the system:* The boundary of separation may be large and may be electrically far from the disturbance. This is a relatively recent phenomenon which has been encountered in different parts of the North American interconnection. It has been associated with the heavier loading of the transmission network, especially under heavy power transfers between areas. We will call this type of transient "Interarea Mode." When this phenomenon is encountered, the generators which appear to be most severely disturbed, initially experience "stable" oscillatory behavior. However, they may later on (typically 1–2 sec later) join a larger group of generators in separating from the interconnected network.

When the transient stability study is conducted using the conventional time simulation technique, knowledge of the presence of the interarea mode is important only for the analyst to run the time simulation long enough to capture the phenomenon of area separation if it occurs. When the TEF method is used, however, stressed system conditions are expected to be encountered (see Chapter 6).

The interarea mode of stability is illustrated with the 50-generator equivalent of the Ontario Hydro network near the Nanticoke power plant (see Section VII, [2] for a description of the system). For a heavily loaded condition, the system is so stressed such that faults at certain buses cause the system to "split," which causes large groups of generators to lose synchronism with the rest of the interconnected system. The result of such a stressed case, taken from [2], is shown in Figs. 7.3, 7.4. Time plots of the angles of the Nanticoke generators, in the COI frame, are shown, together with the inertial center of the 29-generator group. Figure 7.4 shows that while the Nanticoke generators are stable in the first swing (their own), the whole 29-generator group separates from the rest of the system at a later instant.

7.3 PLANT MODE STABILITY

As explained in the previous sections, in plant mode stability (or instability) a relatively small group of generators, electrically close to the disturbance location, are severely disturbed. When the disturbance is sufficiently large, this group separates from the rest of the generators in the system. This group of generators, called the critical group, is usually located at a plant or plants very close to the disturbance, and their number is small. This critical group is determined by the mode of disturbance test. They form the generator grouping representing the minimum ratio of the network's potential

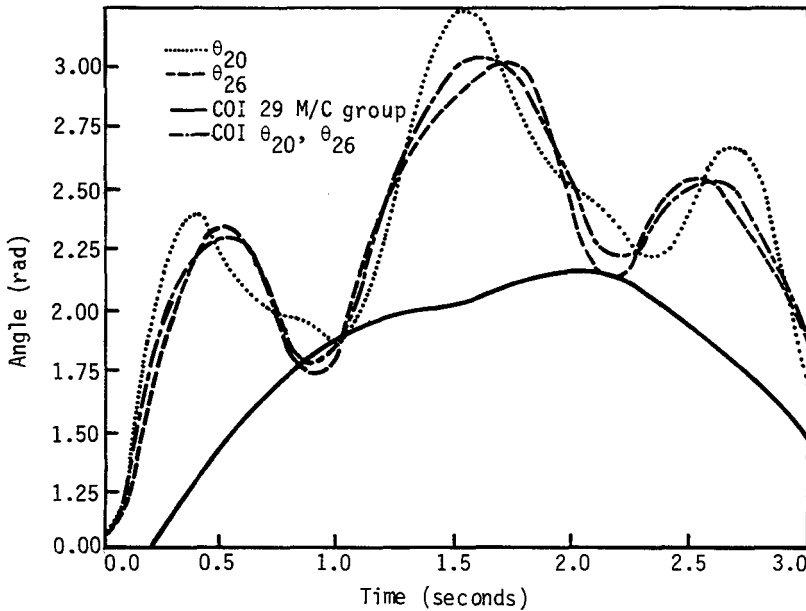


Figure 7.3 Rotor angle plots of the 50-generator stressed system — stable case. Reprinted from © IEEE Transactions on Power Systems 3 (1988).

energy-absorbing capacity to the corrected kinetic energy associated with that group at the end of the disturbance.

In this section some of the issues commonly dealt with in plant mode stability are discussed.

7.3.1 Degree of Stability (or Instability)

As previously explained, stability or instability at the end of the disturbance are determined based on whether $\Delta V \geq 0$. However, the question may arise as to how the severity of two disturbances, for the same power network and the same initial condition, may compare.

Early investigations revealed that the ratio of the transient energy margin to the

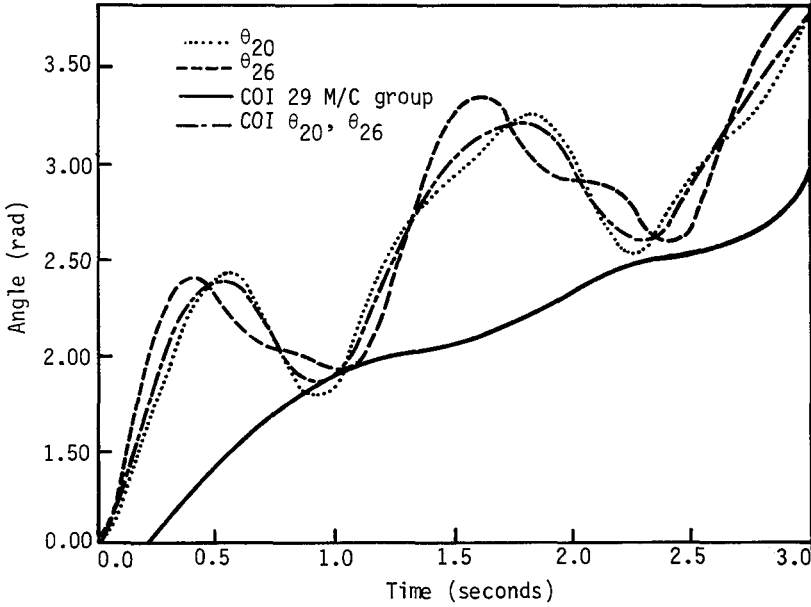


Figure 7.4 Rotor angle plots of the 50-generator stressed system – unstable case. Reprinted from ©IEEE Transactions on Power Systems 3 (1988).

corrected kinetic energy is indicative of the degree of stability or instability. This is called the *normalized energy margin* ΔV_n :

$$\Delta V_n = \frac{\Delta V}{V_{KE|corr}} \quad (7.1)$$

Thus while for $\Delta V > 0$ the system is stable and for $\Delta V < 0$ the system is unstable, more information can be obtained from the value of ΔV_n . For example,

$$\Delta V_n \geq 0, \quad \text{system is } \begin{cases} \text{stable} \\ \text{unstable} \end{cases} \quad (7.2)$$

$$\Delta V_n^{(1)} > \Delta V_n^{(2)} > 0, \quad \text{system is more stable in scenario 1 than in scenario 2} \quad (7.3a)$$

and

$$\Delta V_n^{(1)} < \Delta V_n^{(2)} < 0, \quad \text{system is more unstable in scenario 1 than in scenario 2} \quad (7.3b)$$

To illustrate the significance of the normalized energy margin concept, the following data obtained from the 17-generator reduced Iowa network¹ [1] are offered. Three-phase and single-line-to-ground faults at different locations on the 345-kilovolt network, all cleared at 0.15 sec, were analyzed using the TEF method. Values of the transient energy margin ΔV and their normalized values ΔV_n were computed. The data, reproduced from [1], are shown in Tables 7.1, 7.2. The severity of the disturbance, as given by the values of ΔV_n , were found to be different for some cases from those given by the values of ΔV . For example, in Table 7.1, the fault at Cooper cleared by opening line 6–774 gives an energy margin $\Delta V = 5.2$, while the Raun fault cleared by opening line 3.72–482 gives $\Delta V = 9.159$. These values of ΔV may seem to indicate that the Raun fault is less severe than the Cooper fault. However, the values of ΔV_n of 1.332 and 1.647, respectively, indicate the opposite is true. Several cases in Tables 7.1, 7.2 show that the fault severity as indicated by the relative values of ΔV_n may be different from that indicated by the values of ΔV .

To verify the relative severity of the faults as indicated by ΔV_n , time simulation results were obtained for the three-phase faults. These results, also given in Table 7.1, confirm the validity of the relative severity of the disturbance as given by the values of ΔV_n ; i.e., lower values of ΔV_n correspond to lower values of critical fault clearing times.

¹ See the appendix for description of the network.

The experience of the authors, derived from applying the TEF method to several power networks, seems to substantiate the investigation conducted in [1]. This experience can be summarized as follows:

- For the same network, same initial condition, and the same disturbance location, the relative effect of the *fault duration* on the degree of stability can be indicated by the value of ΔV_n .
- The same conclusion can be extended to the effect of the fault impedance, as reflected on the type of fault (e.g., three-phase versus SLG, etc.)
- For the same network, same initial condition, same type and duration of fault, but different fault locations, ΔV_n is indicative of the relative severity of the disturbance, as illustrated by the results in Tables 7.1, 7.2.

While the authors are confident of the validity of the concept of ΔV_n , it should be stated that no "hard" data are available on such questions as the following situations:

- ΔV_n for the same system, same disturbance, but *different initial conditions*
- ΔV_n for different systems
- ΔV_n for complex sequences of disturbance

7.3.2 Critical Clearing Time

For a given power system, a given initial operating condition, a given type of fault, and a given fault location, the critical fault-clearing time is "conceptually" very simple to determine. It corresponds to the fault clearing time at which the value of V given by

$$V = V_{cr} = V^u \quad (7.4)$$

This can be obtained (again, conceptually) by increasing the fault duration until the critical value of the transient energy V is reached. Alternatively, the critical value of V can be obtained by interpolation between two values slightly below and slightly above

TABLE 7.1
Ranking of Three-Phase Faults
17-Generator System

Rank	ΔV Corrected KE	ΔV (pu)	Fault Location	Critical Clearing Time (sec)	
				Stable	Unstable
1	0.691	4.756	Raun, line 372-773	0.177	0.180
2	1.080	7.432	Raun, line 372-193	0.1923	0.1924
3	1.332	9.159	Raun, line 372-482	0.192	0.196
4	1.530	9.870	Raun, transformer 372-800	0.196	0.200
5	1.620	6.208	C.B. no. 3, line 436-771	0.200	0.204
6	1.647	5.200	Cooper, line 6-774	0.204	0.216
7	1.988	6.278	Cooper, line 6-16	0.212	0.212
8	2.059	6.502	Cooper, line 6-439	0.216	0.220
9	2.089	6.596	Cooper, line 6-393	—	—
10	3.283	12.579	C.B. no. 3, line 436-439	—	—

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PAS-101 (1982).

TABLE 7.2
Ranking of SLG Faults
17-Generator System

Rank	ΔV Corrected KE	ΔV (pu)	Disturbance	
			Fault Bus	Lines Removed
1	0.962	0.963	Cooper	6-774, 6-393
2	2.057	3.408	C.B. no. 3	436-771, 436-439
3	3.321	3.217	Cooper	6-439, 6-393
4	4.200	7.523	Raun	372-482, 372-193
5	4.319	7.879	Raun	372-193, 372-482
6	4.354	4.350	C.B. no. 3	436-439, 436-771
7	5.550	5.561	Cooper	6-774, 6-16
8	34.205	21.980	Ft. Calhoun	773-372, 773-775
9	54.379	27.097	Ft. Calhoun	773-779, 773-775

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PAS-101 (1982).

V_{cr} .² The current TEF program, however, computes ΔV and ΔV_n . The critical fault clearing time is then given by

$$\Delta V = \Delta V_n = 0 \quad (7.5)$$

The recommended procedure is to obtain ΔV_n for two fault durations; by interpolation or extrapolation the fault-clearing time corresponding to $\Delta V_n = 0$ is obtained. Because of the nonlinear nature of the relationship between fault-clearing time and ΔV_n , it is important that at least one of the fault durations is selected to give $|\Delta V_n|$ approximately equal to zero.

7.3.3 Power Limits

This is typical of the type of practical answers sought by the analyst in a stability study performed in an operations planning environment. The power network condition is given, i.e., the load level, distribution of generation, transmission connectivity (which lines are not in service), etc. The breaker operating times are known (e.g., successful clearing at first relay zone, stuck breaker fault). The loading at a key power plant, or the power flow at a critical transmission interface (i.e., group of transmission lines connecting two areas of the network) may be transient stability limited.³ The analyst then determines those limits and recommends that the system be operated at a safe margin below them.

To arrive at the desired answer using the TEF method, different stability runs are made until two cases are obtained: one for $\Delta V > 0$ and the other for $\Delta V < 0$. The power limit is obtained by interpolation, i.e., that corresponding to $\Delta V = 0$ (or $\Delta V_n = 0$).

To illustrate this procedure, some data, obtained from [5], are given below. Tables 7.3, 7.4 give the TEF results for the so-called Bruce 100-generator and the Nanticoke

² In the early days of direct stability analysis, this procedure was used to obtain critical fault-clearing times.

³ As explained in Chapter 1 of this book, in several areas of the North American interconnection the heavy transmission loadings have made these areas stability limited.

50-generator networks. The power limits were obtained by interpolation, i.e., corresponding to $\Delta V = 0$.

Again, it should be pointed out that since an interpolation procedure is used, the data points should correspond to values of $|\Delta V|$ or $|\Delta V_n|$ close to zero for the results to be considered accurate. In addition, analytical sensitivity techniques have recently been developed to obtain power stability limits directly (see Chapter 9).

7.3.4 Other Stability Information

It is recognized that an experienced analyst can learn a lot about a power network from the time simulation results, e.g., the width and the height of the generator rotor angle swings and the initial angle spread across the network. Similarly with TEF, the analyst can find much useful information about the power system's dynamic behavior. Furthermore, this information is quantitative.

First, let us take a second look at the MOD concept. We have been keenly interested in the concept because it is used to determine the critical energy against which transient stability assessment is to be made. Having determined that, what else do we learn about the system? The generators with advanced rotor angles in the MOD are the *severely disturbed* generators. This means that to obtain power limits, in a stability-limited power network, these are the generators to be concerned about. Therefore, the effect of increased loading on these generators would be of primary interest. By the same reasoning, if "relief" is to be sought, it would be most effective if it affects these generators. Thus, corrective action should be used primarily with these generators.

The significance of the above should not be overlooked. It is true that in some systems the severely disturbed generator(s) can be identified by inspection. However, many situations are encountered in today's complex power networks in which it is not obvious which generators will be most affected by the disturbance (situations have been encountered in which the generator initially losing synchronism is not that electrically

nearest to the disturbance). How the generators in a power network respond to a disturbance will depend *both* on the severity of the disturbance and the robustness of the postdisturbance network.⁴ Therefore, there is much to learn about the power network from the condition of the postdisturbance network.

TABLE 7.3
TEF Results for Bruce 100—Generator System

Case #	Bruce Output (MW)	Normalized Energy Margin	Comments
1	1500	0.150	Stable
2	1513	0.025	Stable
3	1514	-0.008	Unstable
4	1530	-0.150	Unstable

Stability limit = 1514 MW.

TABLE 7.4
TEF Results for Nanticoke—Normal System

Case #	Nanticoke Output (MW)	Normalized Energy Margin	Comments
1	3800	0.113	Stable
2	3840	0.001	Stable
3	3841	-0.002	Unstable
4	3900	-0.158	Unstable

Stability limit = 3840.5 MW.

⁴ Unfortunately this point is often overlooked by some researchers.

Let us take a second look at Fig. 3.1, depicting a simple direct assessment of whether a vehicle will or will not go over a hill when given a push. When this example is made to depict a simple power network, i.e., a one-generator–infinite–bus system, the equilibrium condition is the postdisturbance condition. We note that the height of the hill can be effectively reduced if the level of the "valley" is raised. In the power system analogy, the postdisturbance network would be stressed either by heavy loading or by a weakened transmission.

By closely examining the information given by the TEF method, we note that it includes the following (1) the postdisturbance equilibrium angles θ^{s2} , (2) the UEP angles θ^u , and (3) the energy at the UEP V^u . This information merits further examination.

An important indicator of stressed conditions in the postdisturbance network is the values of the angles θ^{s2} . Of particular concern is when the angles of the severely disturbed generators approach 90° . Furthermore, the conditions favoring the occurrence of interarea mode instability can be observed (1) a large number of generator rotor angles become advanced (i.e., approach 90°), and (2) the values of many angles in θ^{s2} approach those of θ^u . Of equal significance to the analyst is that the pattern of θ^{s2} behavior may be indicative of where the system "split" will occur, if interarea mode instability takes place.

Another key indicator of stressed postdisturbance system conditions is that the value of the potential energy at the UEP V^u , which is the same as V_{cr} , becomes very small in magnitude.

Example

The above indications of stress are illustrated using the 50-generator test system. Two cases are presented: an unstressed system case and a stressed system case. For the two cases, the total system load and generation are the same, and the loading of the plant near the fault is the same; the only difference is that the loading of a neighboring plant is substantially increased (in the stressed case) causing a much heavier loading of the transmission network in the area, especially in the postdisturbance network.

Data for the postdisturbance network stable equilibrium angles θ^{s2} , as well as for the controlling unstable equilibrium angles θ^u , are presented for both cases in Tables 7.5, 7.6.

We note that the angle spread for θ^{s2} in the stressed case is much greater than that for the unstressed case. In addition, while the largest generator angle in the unstressed case is 58° , in the stressed case several generator angles are at or near 90° .

For θ^u , while for the unstressed case only generators 20 and 26 have angles greater than 90° , in the stressed case, there are 29 generator angles which are advanced. It is also interesting to note values of the potential energies at the UEP. They are :

$$V^u \text{ unstressed} = 8.61889 \text{ pu}$$

$$V^u \text{ stressed} = 6.62251 \text{ pu}$$

Again, the stressed conditions are clearly demonstrated by the reduced value of $V_{critical}$.

7.4 THE INTERAREA MODE

Again as explained in Section 7.2, this phenomenon is associated with the tendency of a large group of generators to separate from the rest of the system. The separation boundary may occur away from the disturbance location. This phenomenon is associated with a stressed postdisturbance transmission network due to heavy loading and/or large transmission impedances.

As with plant mode stability, analysis of stability-limited conditions may take place in a variety of forms: stability classification with qualitative assessment of the degree of stability or instability, power limits at a plant or across a transmission interface, as well as other information on the generators which influence the dynamic system behavior. The analysis, however, is much more complex than with the case of plant mode instability.

TABLE 7.5 50—Generator Nanticoke (Unstressed)

INT. BUS#	LF. BUS#	STATION NAME	STABLE EQUIL. PT. (SEP) (RADIAN)	PT. (SEP) (DEGREES)	UNSTABLE EQUIL. PT (UEP) (RADIAN)	PT (UEP) (DEGREES)	
1	187	LAK***2A	0.55280	31.67290	1.01060	57.90326	
2	287	BEC***2A	0.19508	11.17737	0.82045	47.00843	
3	1032	MCK***	0.29960	17.16592	0.98360	56.35620	
4	1036	HIG***LS	0.78152	44.77777	1.45385	83.29959	
5	1262	CHF***T	0.06114	3.50294	0.75109	43.03445	
6	1751	CAN***LT	0.29884	17.12238	1.11598	63.94092	
7	1752	AGU***LT	0.83964	48.10800	1.35062	77.38462	
8	1754	ARN***LT	0.17286	9.90412	0.91081	52.18549	
9	1771	BRU***T3	0.31294	17.92997	1.15362	66.09781	
10	1776	CAM***T1	0.69410	39.76926	1.15286	66.05428	
11	1777	CAR***T1	1.01601	58.21305	1.29217	74.03569	
12	1780	CHE***T1	0.14827	8.49549	0.96561	55.32526	
13	1782	DEC***LT	0.36049	20.65445	1.02206	58.55982	
14	1783	DES***T1	0.39213	22.46723	1.24249	71.18952	
15	1793	DOU***LT	0.45005	25.78613	1.30550	74.79961	
16	1796	HOL***T1	0.31889	18.27110	1.15215	66.01309	
17	1806	LAK***T8	0.17492	10.02209	1.02028	58.45769	
18	1807	LAM***T1	0.40256	23.06486	0.68270	39.11580	
19	1815	LON***LT	0.30395	17.41511	1.13081	64.79061	
20	1820	NAN***T5	0.95897	54.94510	2.94468	168.71793	→ Advanced
21	1825	PIC***T2	0.45907	26.30290	1.34309	76.95319	
22	1826	PIC***T3	0.46497	26.64079	1.34576	77.10612	
23	1831	SAU***T1	-0.01492	-0.85457	0.56809	32.54890	
24	1843	STW***5	0.01077	0.61721	0.70122	40.17692	
25	1853	BRU***T1	0.32655	18.70980	1.16654	66.83794	
26	1856	NAN***T3	0.89206	51.11147	2.75485	157.84113	→ Advanced
27	1863	LAK***T7	0.33020	18.91920	1.19911	68.70406	
28	2007	18C***BE	-0.02355	-1.34932	-0.02989	-1.71273	
29	2016	18H***T0	-0.03105	-1.77881	0.04427	2.53630	
30	2051	19B***VR	-0.00319	-0.18267	0.14900	8.53729	
31	2079	19M***12	-0.03861	-2.21244	0.06762	3.87432	
32	2152	AK***	-0.82052	-47.01254	-0.69036	-39.55490	
33	2184	CHA***UG	-0.08535	-4.89029	0.37042	21.22346	
34	2203	DUN***N3	0.28575	16.37208	0.75634	43.33484	
35	2264	HNT***7G	0.28636	16.40717	0.87631	50.20901	
36	2459	BR***3	-0.48050	-27.53038	-0.29240	-16.75319	
37	2601	BER***1	-0.65740	-37.66611	-0.57670	-33.04226	
38	2609	CLV***LF	-0.19648	-11.25728	-0.15400	-8.82368	
39	2616	ERI***O.	0.12431	7.12263	0.26335	15.08911	
40	2651	ALL***	0.04267	2.44479	-0.06024	-3.45172	
41	2652	BD***4	0.74141	42.47993	0.63602	36.44129	
42	2654	CAL***1	0.29090	16.66729	0.17694	10.13797	
43	2655	CAP***	-1.07758	-61.74094	-1.36292	-78.08962	
44	2679	O1A***TR	0.04937	2.82854	0.04827	2.76543	
45	2669	SPP***CH	-0.22677	-12.99286	-0.42575	-24.39375	
46	2674	VER***ON	0.04545	2.60436	-0.00925	-0.52979	
47	2699	O3D***BE	0.07115	4.07682	0.12884	7.38223	
48	2719	3BR***	0.08637	4.94871	0.08097	4.63936	
49	2739	880***8	0.47354	27.13183	0.34209	19.60015	
50	2666	SPP***KN	0.00592	0.33912	-0.14703	-8.42426	

TABLE 7.6 50-Generator Nanticoke (Stressed)

INT. BUS#	LF BUS#	STATION NAME	STABLE EQUIL. PT. (SEP)		UNSTABLE EQUIL. PT (UEP)	
			(RADIANS)	(DEGREES)	(RADIANS)	(DEGREES)
1	187	LAK***2A	0.91938	52.67669	2.17839	124.81264
2	287	BEC***2A	0.66044	37.84056	1.80140	103.21271
3	1032	MCK***	0.83883	48.06145	2.15037	123.20695
4	1036	HIG***LS	1.32479	75.90466	2.62385	150.33582
5	1262	CHF***T	0.59623	34.16130	1.79932	103.09319
6	1751	CAN***LT	0.93632	53.64705	2.27248	130.20381
7	1752	AGU***LT	1.27833	73.24321	2.49642	143.03442
8	1754	ARN***LT	0.73538	42.13417	1.98345	113.64338
9	1771	BRU***T3	1.33313	76.38251	2.74334	157.18176
10	1776	CAM***T1	1.06271	60.88883	2.32445	133.18146
11	1777	CAR***LT	1.23921	71.00146	2.52410	144.62007
12	1780	CHE***T1	0.76026	43.55979	2.07533	118.90747
13	1782	DEC***LT	0.83862	48.04923	2.00279	114.75117
14	1783	DES***T1	1.03441	59.26727	2.39105	136.99733
15	1793	DOU***LT	1.24534	71.35299	2.65392	152.05856
16	1796	HOL***T1	0.95631	54.79280	2.30475	132.05257
17	1806	LAK***T8	0.80349	46.03667	2.14234	122.74683
18	1807	LAM***T1	0.63819	36.56535	0.97056	55.60904
19	1815	LON***LT	0.94033	53.87722	2.28470	130.90349
20	1820	NAN***T5	1.57398	90.18227	3.13299	179.50728
21	1825	PIC***T2	1.06966	61.28718	2.43165	139.32333
22	1826	PIC***T3	1.06477	61.00706	2.42368	138.86678
23	1831	SAU***T1	0.45480	26.05797	1.55720	89.22083
24	1843	STW***.5	0.54910	31.46086	1.75292	100.43504
25	1853	BRU***T1	1.36921	78.45021	2.79073	159.89711
26	1856	NAN***T3	1.44798	82.96341	2.97086	170.21794
27	1863	LAK***T7	0.95582	54.76437	2.30928	132.31195
28	2007	18C***BE	-0.00470	-0.26930	-0.04186	-2.39828
29	2016	18H***T0	0.06341	3.63308	0.11684	6.69433
30	2051	19B***VR	0.15662	8.97345	0.29653	16.99016
31	2079	19M***12	0.08299	4.75510	0.17499	10.02598
32	2152	AK ***	-0.65566	-37.56666	-0.46832	-26.83295
33	2184	CHA***UG	0.28873	16.54313	1.26947	72.73515
34	2203	DUN***N3	0.64540	36.97873	1.63624	93.74941
35	2264	HNT***7G	0.72552	41.56942	1.83986	105.41604
36	2459	BR.*** 3	-0.25906	-14.84310	0.06321	3.62174
37	2601	BER*** 1	-0.53359	-30.57249	-0.41570	-23.81779
38	2609	CLV***LF	-0.11244	-6.44247	-0.04907	-2.81166
39	2616	ERI***O.	0.26751	15.32702	0.49831	28.55089
40	2651	ALL***	-0.07319	-4.19358	-0.22338	-12.79847
41	2652	BD *** 4	0.67680	38.77765	0.52312	29.97256
42	2654	CAL*** 1	0.21152	12.11912	0.04101	2.34946
43	2655	CAP***	-1.26102	-72.25122	-1.69138	-96.90888
44	2679	O1A***TR	0.08093	4.63686	0.07141	4.09144
45	2669	SPP***CH	-0.40237	-23.05422	-0.69520	-39.83209
46	2674	VER***ON	0.01572	0.90050	-0.07507	-4.30096
47	2699	O3D***BE	0.15135	8.67163	0.21033	12.05108
48	2719	3BR***	0.11779	6.74881	0.11111	6.36602
49	2739	880***8	0.22093	12.65815	0.02777	1.59101
50	2666	SPP***KN	-0.16564	-9.49022	-0.39132	-22.42077

7.4.1 Analytical and Numerical Issues

These issues are associated with the determination of the critical energy against which stability assessment is to be made. This means that the controlling UEP must be identified and solved for. Since a large number of generators are involved, a scheme for automatic determination of the advanced generators (i.e., the MOD) is needed. In addition, since the stressed postdisturbance network is usually very ill conditioned, a robust solution algorithm is needed to solve for the controlling UEP.

Controlling UEP determination

If the MOD determination procedure is followed (see Section 6.5), a list characterizing the effect of the disturbance on the various machines is formed as follows [2]:

- After ranking the machines according to the kinetic energies ($M \tilde{\omega}_{cl}^2$) at fault clearing, and then ranking them according to their acceleration ($f(\theta)_{cl} / M$), the combined list is formed.

M/Cs appearing in both the $M \tilde{\omega}_{cl}^2$ and $f(\theta)_{cl} / M$ lists.
M/Cs appearing in $M \tilde{\omega}_{cl}^2$, but not in $f(\theta)_{cl} / M$ list.
M/Cs appearing in $f(\theta)_{cl} / M$ list, but not in $M \tilde{\omega}_{cl}^2$ list.

- Generator groupings are checked for their values of $V_{KE}|_{corr}$ at fault clearing. A number of groupings which have a selected band of kinetic energy is selected.
- Each of the groupings is checked for ΔV_{PE} at the ray point and ΔV_{PEN} .
- The correct MOD is that with the lowest value of ΔV_{PEN} .
- For that mode, the ray point is used as the starting point for the UEP solution.

Note: If needed, a procedure is offered in [2] to justify the UEP obtained by the above procedure.

If the analytical procedure to determine the controlling UEP, discussed in Section 6.6, is used, special attention is given to the solution of the set (6.29) for a large stressed power network. Numerical problems may be encountered which may lead to slow convergence (or in some cases to no convergence) special solution techniques may be needed for such systems.

7.4.2 Stability Classification

As previously stated, transient stability classification is based on the computed value of the normalized energy margin ΔV_n for the disturbance sequence under investigation. To obtain ΔV_n , the value of the corrected kinetic energy is required. An interesting aspect of the interarea mode phenomenon is that the value of the corrected kinetic energy to be used is that corresponding to the grouping having the maximum $V_{KE}|_{corr}$ for the advanced generators in the interarea mode.

As an illustration of this for the stressed 50-generator system, the maximum $V_{KE}|_{corr}$ corresponds to the two Nanticoke generators advanced (generators 20 and 26). This value of $V_{KE}|_{corr}$ is used to obtain the normalized energy margin ΔV_n . For a given system condition, i.e., a Nanticoke plant loading of 3800 megawatts (MW), and a given disturbance, a value of $\Delta V_n = 0.001$ is obtained (indicating a critically stable condition).

Time simulation results verify that the stability assessment using the TEF method is correct, although somewhat conservative (see Table 7.7).

7.4.3 Power Limits

Two examples, taken from [2], illustrate the use of the TEF method to obtain power limits for stability-limited conditions in a stressed power network.

In the same 50-generator test system discussed, the stability-limited loading of the Nanticoke power plant (generators 20, 26) is obtained. The results are shown in Table 7.7. The results of Table 7.7 show that the TEF method predicts a limit of 3800 MW, which compares very well with the limit of 3935 MW predicted by time simulation.

In another test system comprised of 115 generators, while the MOD test for the stressed conditions analyzed showed a corrected V_{KE} based on 19 generators advanced, the UEP solution gave an interarea mode which included 57 machines with advanced rotor angles [2, 5]. The stability limit was obtained in terms of total generation of six machines located at a critical plant. The limit obtained from time simulation was 4400 MW.

TABLE 7.7
Stability Limited Plant Output
50-Generator System

No.	Output of M/Cs 20, 26 in MW	Normalized Energy Margin	Remarks
1	3750	0.1936	Stable
2	3800	0.001	Critically stable
3	3801	-0.002	Critically unstable
4	3900	-0.3397	Unstable

Figure 7.5 shows the result of the TEF analysis on this system. Corresponding to a zero margin the limit obtained is 4450 MW, which gives excellent comparison with time simulation results.

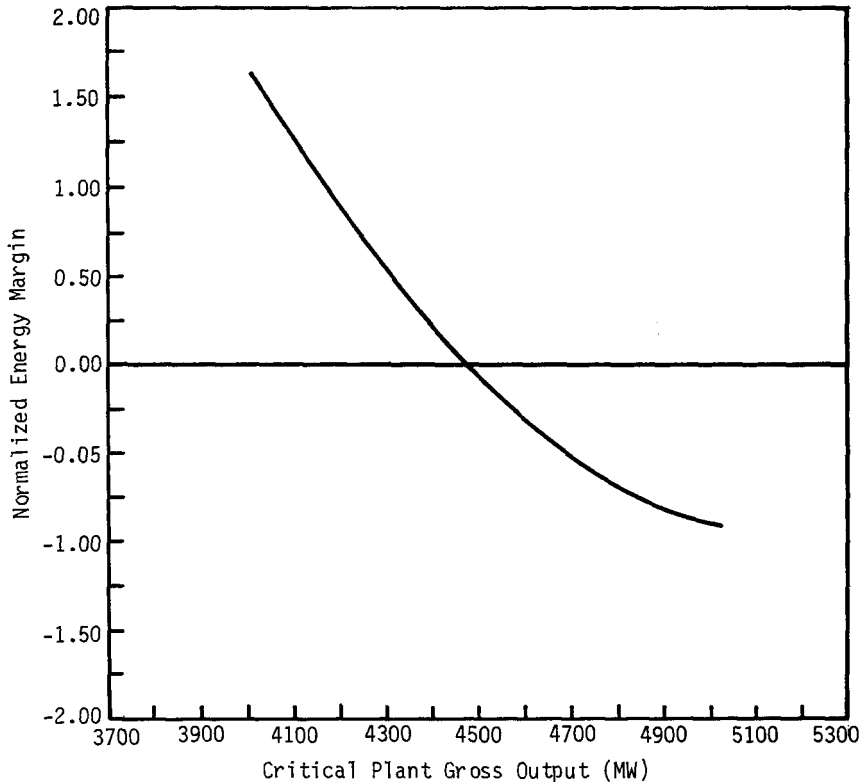


Figure 7.5 Results of TEF analysis on 115-generator system.

7.4.4 Complexity of the System Dynamic Behavior

While in a plant mode stability the critical generators, which have greater influence on the system transient behavior, are clearly identified, in an interarea mode the situation is much more complex. It is true that the generators electrically close to the disturbance

will have a strong influence on the system transient response. They will influence the potential energy, being among the advanced generators in the controlling UEP. They will also greatly influence the corrected kinetic energy, since they invariably form the generator grouping with the greatest corrected kinetic energy.

The influence of the other advanced generators is more complex. The boundary of the area will depend on how extended the heavy loading is in the postdisturbance transmission network, on whether the area where the disturbance is located is importing or exporting power, on whether there are some weak transmission links in the power path, etc. Thus, the number of generators included in the interarea mode, and their location is system as well as disturbance dependent.

Since, depending on the postdisturbance network conditions, any one of a number of interarea modes may occur, the conditions favoring the occurrence of these modes are of great interest to the analysts. Furthermore, there is evidence that these modes may interact in such a manner as to suggest the possibility of energy exchange between the modes. If this is the case, a great deal of complexity is added to power system stability analysis using TEF.

Our experience with the interarea mode is modest since the phenomenon has been encountered only in the last few years. It is now receiving a great deal of attention, and is the subject of numerous investigations.

7.5 APPLICATION IN TRANSMISSION PLANNING

Reference [5] gives an innovative application of the TEF method in analyzing the effect of different transmission expansion plans in a transient stability-limited network. The test system is the Southwestern Ontario part of the Ontario Hydro system. The system comprises part of the Ontario and upper New York areas and their interconnections with the rest of the grid. The system exhibits a complex interarea mode of instability, characterized by the separation of large groups of machines from the main network,

following certain contingencies. The separation pattern depends on a variety of system conditions.

A sample study of two transmission reinforcement plans was made: one double-circuit 500-kilovolt (KV) link to the East and a similar link plus a double-circuit link to the South in the Ontario Hydro system. The variation of the transient energy margin for the same disturbance scenario, as the output of a key power plant (Bruce) is varied, was analyzed. The results obtained showed that the stability limits corresponding to the Bruce plant outputs are original transmission plans 4450 MW, one double circuit to the East 4540 MW, and the circuits to the East plus the circuits to the South 5100 MW.

For planning purposes, the cost of the various transmission reinforcements is weighed against the savings due to increase in economy power.

7.6 DISTURBANCE OTHER THAN FAULTS

Although the most commonly analyzed disturbance scenarios involve faults, they are by no means the only disturbances of interest. There are situations where other types of disturbances may be of concern. Some of these disturbances and the issues involved in analyzing them with the TEF method are dealt with in this section.

When a fault occurs, the sudden change in the power output of generators starts the sequence of events in the transient which generates the excess transient energy (especially the kinetic energy). When the disturbance is removed, the total energy is constant, and the conversion of kinetic energy into potential is the essential ingredient to maintaining stability. Stability assessment is made when the disturbance is removed. For that assessment, both the value of the transient energy (at that instant) and its (projected) critical value are needed.

7.6.1 Opening of a Loaded Line

Let us now consider a situation where the initiation of the disturbance and the

postdisturbance conditions occur at the same instant. The disturbance is initiated with a sudden change in potential energy, which is the case of the sudden opening of a loaded transmission line.

Let superscripts (1) and (2) refer to the conditions prior to and after the opening of the line. The equation of motion for generator i is given by (2.13)

$$\begin{aligned} M_i \ddot{\omega}_i &= P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} \\ \dot{\theta}_i &= \tilde{\omega}_i \end{aligned} \quad (7.6)$$

This is also given by

$$\begin{aligned} M_i \ddot{\omega}_i &= P_{mi} - E_i^2 G_{ii} - \sum_{\substack{j=1 \\ j \neq i}}^n (C_{ij} \sin \theta_{ij} + D_{ij} \cos \theta_{ij}) \\ &\quad - \frac{M_i}{M_T} \left[\sum_{i=1}^n (P_{mi} - E_i^2 G_{ii}) - 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n D_{ij} \cos \theta_{ij} \right] \\ \dot{\theta}_i &= \tilde{\omega}_i \end{aligned} \quad (7.7)$$

Consider the situation at the instant the loaded line is removed, i.e., at $t = 0^+$

$$\dot{\theta}_i = \tilde{\omega}_i = 0$$

$G_{ii}^{(1)}, C_{ij}^{(1)}, D_{ij}^{(1)}$ change suddenly to $G_{ii}^{(2)}, C_{ij}^{(2)}, D_{ij}^{(2)}$

$\theta_i, \theta_j, \theta_{ij}$ cannot change instantly

Therefore, generator i will experience a sudden change in power and will begin to accelerate. It will gain kinetic energy, which in turn must be converted into potential energy for stability to be maintained. However, for stability assessment, the kinetic energy does not enter into the picture, since the assessment will be made at an instant when the kinetic energy is zero.

In essence the disturbance scenario starting with the operating of a loaded line, is initiated with a *sudden change of potential energy*. The transient energy margin computation follows the same procedure as with fault disturbances:

- θ^{s2} is computed
- An MOD procedure is followed to determine the critical generators
- θ^u is determined
- $\Delta V = V_{PE} \Big|_{\theta^{s1}}^{\theta^u}$ is computed

7.6.2 Loss of Generation Disturbance

The loss of a generator affects the mechanical power, the network configuration (due to the loss of a generator terminal node), and the inertia (hence the COI position). All these changes take place instantly, and since the remaining generators' angles and speeds cannot change suddenly, there is no instantaneous change in kinetic energy, and again the disturbance is initiated by a sudden change in potential energy as described in the previous section. However, unlike the case of opening a loaded line (Section 7.6.1), in the case of losing a generator, there is a longer-term effect which must be accounted for in the TEF method: what the postdisturbance equilibrium conditions are and how they can be determined. The transient energy margin computation procedure is outlined below:

- The conditions of the rotor angles at the instant the generator is dropped are calculated. Assume that there are n -generators operating at equilibrium. With

respect to the COI, their rotor angles will be given by θ_i^{s1} , $i = 1, 2, \dots, n$, and their velocities will be zero. Generator x is suddenly dropped at $t = o^+$. The angles and the velocities cannot change immediately, but the COI position changes (since δ_o will change abruptly with the removal of the term $M_x \delta_x$ from

$\delta_o = \sum_{j=1}^n M_j \delta_j$), and the rotor angle positions θ_i of the remaining generators will change to $\theta_i^{s1'}$, $i = 1, 2, \dots, n, i \neq x$.

- The postdisturbance equilibrium angles θ^{s2} are computed. To determine the postdisturbance equilibrium conditions, we must account for the fact that there will be an imbalance between generation and load, due to the loss of power of one generator. We note that in the first swing transient, the generators' dynamic behavior is determined by a combination of effects: initially, by their electrical proximity to the disturbance location and later on by the proportion of their inertias to the total system inertia.⁵ Therefore, for the purpose of TEF computations, the postdisturbance equilibrium conditions are determined by proportioning the power deficiency (due to the loss of generation) among the remaining generators according to their inertias.

The network conditions at $t = o^+$ are obtained by dropping generator x , modifying the network admittance matrix by zeroing the real and reactive power at bus x , and determining ΔG_{ii} , ΔC_{ij} , and ΔD_{ij} which represent the changes in the network admittance matrix due to the removal of generator x . The new mechanical powers accounting for the loss of generator x are given by

$$P_{m_i}^+ = P_{m_i}^- + \frac{M_i}{M_{T-x}} P_{m_x}^- \quad i = 1, 2, \dots, n, i \neq x \quad (7.8)$$

where

$$M_{T-x} = \sum_{i=1}^n M_i - M_x \quad (7.9)$$

Thus, for the postdisturbance equilibrium conditions, the new rotor angles θ^{s2} correspond to the solution of the following equations:

⁵ See Section 3.6 of [6].

$$P_{mi}^+ - P_{e_i}^+ - \frac{M_i}{M_{T-x}} P_{COI}^+ = 0 \quad i = 1, 2, \dots, n, \quad i \neq x \quad (7.10)$$

where

$$P_{e_i}^+ = \sum_{\substack{j=1 \\ j \neq i, x}}^n [(C_{ij}^- + \Delta C_{ij}) \sin \theta_{ij} + (D_{ij}^- + \Delta D_{ij}) \cos \theta_{ij}] + E_i^2 (G_{ii}^- + \Delta G_{ii}) \quad (7.11)$$

$$i = 1, 2, \dots, n, \quad i \neq x$$

$$P_{COI}^+ = \sum_{\substack{i=1 \\ i \neq x}}^n (P_{mi}^+ - P_{e_i}^+) \quad (7.12)$$

- An MOD procedure is followed to determine the critical generators.
- θ^u is determined.

$$\bullet \quad \Delta V = V_{PE} \left| \frac{\theta^u}{\theta^{s1'}} \right| \quad (7.13)$$

is computed.

An important point to remember in the loss of generation transient is that, due to the resulting power deficiency, the remaining generators *are retarded*. The UEP is to be searched for in the direction of decreasing rotor angles.

Another important consideration in the loss of generation transient is that its serious consequences are seldom due to loss of synchronism by some generators, but rather because they cause other disturbances, the effects of which may lead to cascading outages. An example of such a situation is when the loss of generation occurs in a power area which is

importing large amounts of power over some tie lines. The resulting inrush power on these ties, and the swings between the generators on both sides, may cause the out-of-step relays on these ties to trip, with potentially serious consequences. This problem will be discussed later on in this chapter.

7.6.3 Special Case – Fault Resulting in Power Deficiency

This situation may occur when an area, which is part of a large power system, is connected to the system with one transmission circuit. When the area is importing power over this circuit, a fault on the circuit results in an interesting transient. Initially the fault causes the generators in the area to accelerate, as is usually the case with fault disturbances. However, the removal of the circuit connecting the area to the rest of the system (to clear the fault) leaves the faulted area deficient in generation. Thus, while the generators initially accelerate, they will eventually be retarded. Again the UEP will be found in the retarding direction. For stability assessment, the kinetic energy when the fault is removed must be accounted for when the transient energy margin ΔV is computed.

7.7 NETWORK CONDITIONS DURING TRANSIENT

As explained in Chapter 1, power system transient stability analysis involves more than investigation of whether synchronous generators will maintain synchronism after being subjected to a disturbance. Sometimes the limiting criterion, in a particular power network, is a condition *inside* the network. The most common of these criteria are the minimum apparent impedance "seen" by an out-of-step relay or the minimum transient voltage at a given bus.⁶ Two issues must be resolved if the TEF method is to deal with these problems (1) relating the conditions of interest to an instant in the transient for

⁶ In both these criteria, the scheme used must discriminate between voltage dips during the fault, which are very short in duration, and voltage dips during the swing between groups of generators, which take a longer time to occur. The latter is the situation under discussion here.

which the TEF method parameters are known and (2) overcoming the fact that the network nodes and lines had been removed by network reduction.

The first issue is dealt with in two steps. In the first step the assumption is made that the conditions of interest are at the peak of the swing of the critical generators with respect to the rest of the system. The rationales for this assumption are (1) the area (of the power network) of concern to the analyst is usually near the "electrical center" and hence will experience the most severe voltage dip during the transient and (2) the error which may result will be on the conservative side. In the second step some assumptions are made concerning the system trajectory during the transient.

The second issue is dealt with by manipulating the network equations at the instant of interest, i.e., at the "peak of the swing."

These issues are discussed in the following sections.

7.7.1 Identifying "Peak of the Swing"

Assuming that the system is stable for the particular fault disturbance being analyzed (which is usually the case of interest), the maximum swing is characterized by the point in the angle space θ^m at which the potential energy margin becomes equal to the energy margin at the end of the disturbance. Assuming a linear trajectory approximation,⁷ mathematically this is expressed by

$$\begin{aligned} \frac{1}{2} M_{eq} (\tilde{\omega}_{eq}^{cl})^2 - \sum_{i=1}^n P_i (\theta_i^u - \theta_i^{cl}) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[C_{ij} \left[\cos \theta_{ij}^u - \cos \theta_{ij}^{cl} \right] \right. \\ \left. + D_{ij} \frac{\theta_i^u + \theta_j^u - \theta_i^{cl} - \theta_j^{cl}}{\theta_{ij}^u - \theta_{ij}^{cl}} \left[\sin \theta_{ij}^u - \sin \theta_{ij}^{cl} \right] \right] \end{aligned}$$

⁷ This trajectory approximation is used only as an illustration; for specific applications other approximations may be used (see below).

$$\begin{aligned}
&= - \sum_{i=1}^n P_i (\theta_i^u - \theta_i^m) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[C_{ij} \left[\cos \theta_{ij}^u - \cos \theta_{ij}^m \right] \right. \\
&\quad \left. + D_{ij} \frac{\theta_i^u + \theta_j^u - \theta_i^m - \theta_j^m}{\theta_{ij}^u - \theta_{ij}^m} \left[\sin \theta_{ij}^u - \sin \theta_{ij}^m \right] \right] \quad (7.14)
\end{aligned}$$

The same relationship, with the same linear trajectory approximation, can be expressed as

$$\begin{aligned}
&\frac{1}{2} M_{eq} (\tilde{\omega}_{eq}^{cl})^2 - \sum_{i=1}^n P_i (\theta_i^m - \theta_i^{cl}) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[C_{ij} \left[\cos \theta_{ij}^m - \cos \theta_{ij}^{cl} \right] \right. \\
&\quad \left. + D_{ij} \frac{\theta_i^m + \theta_j^m - \theta_i^{cl} - \theta_j^{cl}}{\theta_{ij}^m - \theta_{ij}^{cl}} \left[\sin \theta_{ij}^m - \sin \theta_{ij}^{cl} \right] \right] = 0 \quad (7.15)
\end{aligned}$$

An alternate formulation is to use a two-machine equivalent representing the gross motion of the multimachine system. One equivalent machine will represent the dynamics of the inertial center of the critical generators, and the other equivalent represents the motion of the inertial center of the rest of the generators in the system [8, 9]. Parameters of these equivalents are deduced only from the multimachine system parameters (i.e., no system reduction is needed).

Assuming that the critical generators are generators 1 to k , the rest of the generators in the system are referred to by the subscript $T - k$. The equations representing the dynamics of the inertial centers of the two groups of machines are given by

$$M_k \ddot{\omega}_k = \sum_{i=1}^k M_i \ddot{\omega}_i = \sum_{i=1}^k \left[P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} \right]$$

$$\Delta = P_{mk} - P_{e_k} - \frac{M_k}{M_T} P_{COI} \quad (7.16)$$

$$M_{T-k} \dot{\omega}_{T-k} = \sum_{i=k+1}^n M_i \dot{\omega}_i = \sum_{i=k+1}^n (P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI})$$

$$\Delta = P_{mT-k} - P_{e_{T-k}} - \frac{M_{T-k}}{M_T} P_{COI} \quad (7.17)$$

where

$$P_{mk} = \sum_{i=1}^k P_{mi} \quad P_{mT-k} = \sum_{i=k+1}^n P_{mi}$$

$$P_{e_k} = \sum_{i=1}^k P_{ei} \quad P_{e_{T-k}} = \sum_{i=k+1}^n P_{ei}$$

$$M_k = \sum_{i=1}^k M_i \quad M_{T-k} = \sum_{i=k+1}^n M_i$$

$$\theta_k = \frac{1}{M_k} \sum_{i=1}^k M_i \theta_i \quad \theta_{T-k} = \frac{1}{M_{T-k}} \sum_{i=k+1}^n M_i \theta_i$$

Again using a linear trajectory approximation (as an illustration), the point in the angle space representing the peak of the swing is given by

$$\Delta V_{eq}^{cl} = \Delta V_{eq}^m$$

where

$$\begin{aligned}
\Delta V_{eq}^{c\ell} = & -\frac{1}{2} M_k \left[\tilde{\omega}_k^{c\ell} \right]^2 - \frac{1}{2} M_{T-k} \left[\tilde{\omega}_{T-k}^{c\ell} \right]^2 - P_k \left[\theta_k^u - \theta_k^{c\ell} \right] - P_{T-k} \left[\theta_{T-k}^u - \theta_{T-k}^{c\ell} \right] \\
& - \sum_{i=1}^k \sum_{j=k+1}^n C_{ij} \frac{\left[\theta_k^u - \theta_{T-k}^u \right] - \left[\theta_k^{c\ell} - \theta_{T-k}^{c\ell} \right]}{\theta_{ij}^u - \theta_{ij}^{c\ell}} \left[\cos \theta_{ij}^u - \cos \theta_{ij}^{c\ell} \right] \\
& + \sum_{i=1}^k \sum_{j=k+1}^n D_{ij} \frac{\left[\theta_k^u + \theta_{T-k}^u \right] - \left[\theta_k^{c\ell} + \theta_{T-k}^{c\ell} \right]}{\theta_{ij}^u - \theta_{ij}^{c\ell}} \left[\sin \theta_{ij}^u - \sin \theta_{ij}^{c\ell} \right] \\
& + 2 \sum_{i=1}^{k-1} \sum_{j=i+1}^k D_{ij} \frac{\theta_k^u - \theta_k^{c\ell}}{\theta_{ij}^u - \theta_{ij}^{c\ell}} \left[\sin \theta_{ij}^u - \sin \theta_{ij}^{c\ell} \right] \\
& + 2 \sum_{i=k+1}^{n-1} \sum_{j=i+1}^n D_{ij} \frac{\theta_{T-k}^u - \theta_{T-k}^{c\ell}}{\theta_{ij}^u - \theta_{ij}^{c\ell}} \left[\sin \theta_{ij}^u - \sin \theta_{ij}^{c\ell} \right] \quad (7.18)
\end{aligned}$$

In any of the energy formulations satisfying θ^m , the form of the system trajectory between $\theta^{c\ell}$ and θ^u is assumed. For example in (7.18), a linear system trajectory approximation is assumed. θ^m is found by a search along that assumed trajectory which satisfies these relations.

7.7.2 Voltage at a Given Bus

To obtain information on the voltage at certain buses, we will assume that the network—reduced Y -bus has been modified as follows:

1. The node buses, the voltage of which are of interest, have been retained.
2. The generator terminal nodes have been retained, with the constant EMF source having internal reactance x'_d converted to a constant current source with an

internal admittance ($\frac{1}{jx'_d}$); this current source admittance is "folded" into the Y -bus.

The resulting network is shown schematically in Fig. 7.6. It has dimensions $(n + k)$. Note that nodes $n + 1, \dots, n + k$ are typically small in number.

For generator i , if the internal EMF is $E_i \angle \theta_i$, the equivalent current source is

$$\bar{I}_i = \frac{E_i}{x'_{di}} \angle \theta_i - \pi/2 \quad (7.19)$$

Let the new (modified) Y -bus of the resulting network be $\bar{Y}$, the node voltage vector be $\bar{V}$, and the node current vector be $\bar{I}$.

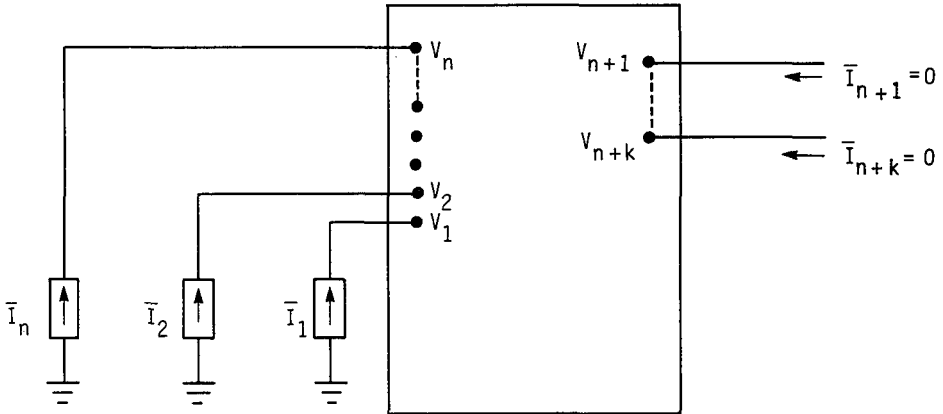


Figure 7.6 Network reduced to generator terminal buses and other key nodes.

$$\bar{I}^t = \begin{bmatrix} \bar{I}_g^t & \bar{I}_k^t \end{bmatrix} = \begin{bmatrix} \bar{I}_g^t & o \end{bmatrix}$$

$$\bar{V}^t = \begin{bmatrix} \bar{V}_g^t & \bar{V}_k^t \end{bmatrix} \quad (7.20)$$

We note that the node current vector is completely known if the angles θ_i , $i = 1, 2, \dots, n$ are known.

The network equations are given by

$$\bar{I} = \bar{Y} \cdot \bar{V}$$

and

$$\bar{V} = \bar{Y}^{-1} \cdot \bar{I} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} \bar{I} \end{bmatrix}$$

or

$$\begin{bmatrix} V_g \\ V_k \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_g \\ o \end{bmatrix}$$

$$\boxed{\bar{V}_k = \bar{Z}_{21} \bar{I}_g} \quad (7.21)$$

The voltages $\bar{V}_k$ are determined with values θ_i , $i = 1, \dots, n$ at the peak of the swing (i.e., $\theta_i = \theta_i^m$).

7.7.3 Apparent Impedance Seen by an Out-of-Step Relay

Figure 7.7 shows a schematic representation of a transmission line monitored by an out-of-step relay. The line is between buses p and q ; it has a series impedance z_{pq} and a shunt susceptance B .

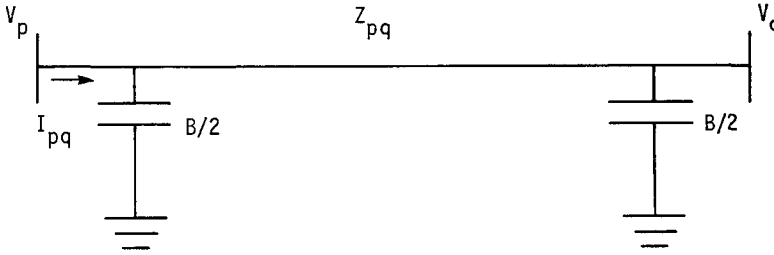


Figure 7.7 Transmission line monitored by an out-of-step relay.

The apparent impedance "seen" by the relay, at node p , is given by

$$z_k = V_p / I_{pq} \quad (7.22)$$

where

$$I_{pq} = \frac{(V_p - V_q)}{z_{pq}} + j \left[\frac{B}{2} \right] V_p \quad (7.23)$$

The voltages V_p and V_q are determined (at the peak of the swing) by the procedure outlined in the previous subsections.

Example

The procedure outlined was tested on a 23-generator, 211-bus equivalent of the Florida Power & Light Co. network derived from a 1983 base case. The network and the case analyzed, described in [9], represented a substantial amount of power imported on a number of 230-KV lines which then existed between the Florida network and the Southern system. Of particular concern for these conditions was the transient initiated by the sudden tripping of a large nuclear unit in the Florida system. In some cases the swing between the generators to the North and South of the 230-KV ties may cause the tripping of the out-of-step relays on the 230-KV ties. The setting of these relays was well below the conditions in which generators rotor angles swings lead to loss of synchronism. The TEF method was used to analyze the disturbances caused by the loss of either (1) 880 MW at St. Lucie or (2) 820 MW at Crystal River. The out-of-step relay zone crossing was to be checked. The results are given in detail in Section 6 of [6], some of which is given below.

Procedure for obtaining swing impedance locus

1. The initial (predisturbance) system conditions, including the internal generator EMFs, are computed. The Y -bus is developed for the network reduced to the generators' internal nodes and terminal buses, and the buses of the line having the out-of-step relay.
2. The disturbance is simulated, and end of disturbance conditions $(\theta, \tilde{\omega})$ are found.
3. The postdisturbance equilibrium conditions θ^{s2} , and the controlling UEP θ^u for the loss of the generator disturbance are computed.
4. The transient energy margin ΔV at the end of the disturbance, and the corresponding value of ΔV_{eq} (for the inertial centers of the equivalent groups of generators) are obtained.
5. The system trajectory in the postdisturbance period is approximated (e.g., by a straight line), and using the energy criteria, the maximum angle swing θ^m is estimated.

6. For θ^m the equivalent current injections at the generators' terminal buses are calculated. The values of V_p and V_k are computed, and the apparent swing impedance at θ^m , or at intermediate points on the system trajectory, are computed.
7. The computed values of the swing impedance are compared with the relay setting.

Results

As a sample of the results reported in [8], the apparent impedance loci obtained by the TEF method using linear trajectory approximation and by the time simulation program are shown in Figs. 7.8, 7.9 for the loss of the St. Lucie and Crystal River units.

7.8 PRACTICAL PROBLEMS

In dealing with transient stability analysis of large power systems, several practical issues are encountered and must be resolved. Among these are the following *types* of problems:

1. A variety of applications are involved. Some of these have been discussed earlier in this chapter; others will be discussed in a later chapter. In some of these applications, loss of synchronism may not even be imminent, yet the threat of cascading outages may result for other reasons. As was seen in the applications discussed in Section 7.6, the use of the TEF method in these applications requires that the TEF parameters (e.g., ΔV) must be related to the power network parameter of interest.
2. The complexity of the manner in which the interconnected system of North America is now operated (more power transfers, more interdependence, etc.) requires bigger systems to study and a greater variety of scenarios to analyze. The prevailing conditions encountered often result in operations which are very close to the steady-state limits. Even steady-state analysis of these conditions is computationally burdensome. With any tool, transient analysis requires robust techniques.

The above must be kept in mind when the use of the TEF method in industry applications is discussed. As a tool of analysis, the TEF method must deal with today's

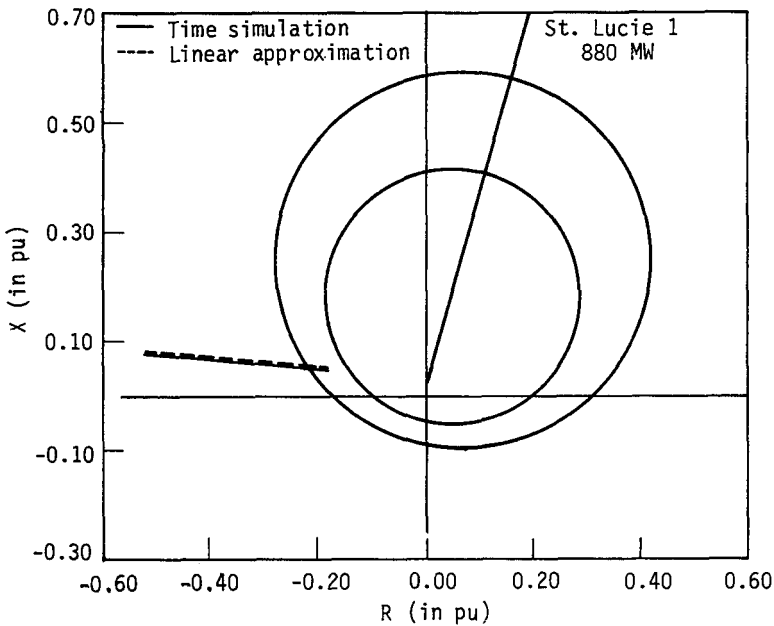


Figure 7.8 θ^m St. Lucie case. Reprinted from © IEEE Transactions on Power Systems 1 (1986).

problems, no matter how complex they may be. Furthermore, it must meet the standards of reliability and accuracy set by the existing tools; i.e., it must give results comparable to those obtained by time simulation.

To meet such exacting standards, the TEF method of transient stability analysis have had, and will continue, to deal with many computational problems. These problems, which are dealt with in other chapters, are outlined below.

7.8.1 Numerical Problems

These problems are mostly associated with obtaining the solutions for the postdisturbance stable equilibrium conditions θ^{s2} and the controlling unstable equilibrium

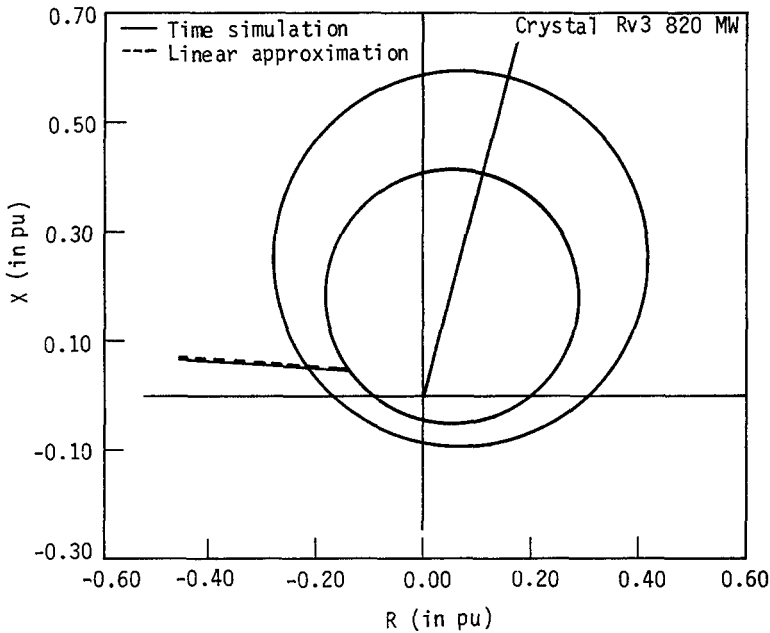


Figure 7.9 θ^m Crystal River case. Reprinted from © IEEE Transactions on Power Systems 1 (1986).

condition θ^u . The system size and the stress caused by heavier transmission loading contribute significantly to these numerical problems. Among the issues dealt with are :

- No SEP solution (because none exists), when the postdisturbance system is steady-state unstable.
- Slow convergence of the solution at θ^{s2} or θ^u ; this is the most common problem encountered.
- Convergence to the "wrong" UEP; when several candidate UEPs are close to each other in the angle space, it is important to start the solution procedure near the desired UEP.

It is important to note that, in the solution of these problems, robust numerical solution techniques are now in use in the TEF program. These techniques have improved both the solution speed and reliability.

7.8.2 Efficiency Problems

These problems are related to the *efficient* use of the computer to obtain results for *one* run. This involves efficient organization of the solution algorithms (e.g., referred to above) as well as other algorithms such as data handling and storage. It also involves the use of efficient programming techniques such as sparsity, coordination of storage with the solution, recognizing when slow convergence is imminent (to switch to more robust algorithms), etc.

7.8.3 Reliability Problems

In many power system applications, the reliability of the answer is as important (perhaps more so) than its accuracy. Some aspects of the reliability are obvious, e.g., to obtain the correct answer concerning stability classification. Other important aspects of reliability are :

- In some applications, e.g., in on-line operation, to always have an answer, if none is available, an explanation and diagnosis must be given.
- If there is an error in the result (i.e., if there is a discrepancy between the TEF and time simulation results), the error is more tolerable when (1) the error is consistent (i.e., consistently low or high) or (2) the error is on the "conservative" side.

7.8.4 Efficiency of a Run Stream

It is to be noted that in power system applications, stability studies are often run many cases at a time. In addition to the efficient running of one case (discussed in Section

7.7.2), there are a great deal of savings in computational time if the following techniques are exploited (see Chapter 9):

- Exploit parallelism in running cases of multidisturbances for the same initial conditions.
- Exploit other parallelism in a multirun stream.
- Exploit sensitivity techniques to obtain power limits.

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Chapter 8

MODELING IMPROVEMENTS IN THE TEF METHOD

8.1 INTRODUCTION

In the previous chapters the basic elements of power system direct transient stability assessment using the TEF method were presented. This included development of the transient energy function and its critical value, identifying the energy responsible for the separation of the critical generators from the rest of the system, the computational issues involved, and applying the TEF method to power system problems. In all this, a relatively simple (i.e., the classical) power system model has been used. In today's power networks this simple model is of limited validity, even in analysis of first swing transients. To overcome this shortcoming, some improvements in the modeling of power system components have been introduced in the TEF method. These improvements are discussed in this chapter.

In developing the new power system models, the following have been kept in mind:

1. The TEF method is a "first swing" tool of analysis. This means that it can be used to determine first swing transient stability or to detect phenomena which may appear during the first swing and which may be indicative of higher probability of occurrence of instability at a later instant in the transient. Therefore, modeling improvements of interest are those which would primarily improve the accuracy of the results during the first swing transient.
2. Conventional transient stability analysis by time simulation is done by "tracking" the various system variables and parameters, especially the generator rotor angles, at different instants during the transient. Desirable or undesirable dynamic system behavior is decided by examining the time response of these variables. In the TEF method, however, the transient system behavior is examined through the behavior of the transient energy responsible for the separation of one or more generators from the rest of the system. (Other system attributes of interest are also examined in relation to that energy.)

Therefore, in the TEF method model improvements are of interest only to the extent that they may give a better estimate of the relevant system transient energy during the first swing transient, i.e., the energy responsible for the separation of some generators from the rest of the system.

The power system components to be discussed in this chapter are excitation control which involves both the synchronous generator and excitation system models, two-terminal HVDC model, and voltage-dependent nonlinear load models. In addition, the following modeling-related issues are presented: sparse formulation and incorporation of (uniform) damping.

8.2 MODELING OF EXCITATION CONTROL EFFECTS

Again we stress the fact that the model improvement is intended to give a better estimate of the transient energy in the first swing transient. Thus, the detailed synchronous generator model which would "capture" the desired effects is the so-called two-axis model or a modified version of the E'' -model (see Section 4.15 of [1]). For the excitation system model, a simple one-gain-one-time-constant – one-limiter model is used. This model is used to standardize the treatment of incorporating excitation control in the TEF method. All other excitation system models are assumed to have been converted into this model format, in the frequency range of interest (see Section 8.2.2). The model developed is applicable to conditions of flux buildup as well as flux decay.¹

In the derivations presented in most of this section, the generator is represented by the two-axis model, and the exciter is represented by the one-gain, one-time-constant, and one-limiter model.

¹ It is often the case in North America that the use of fast, high-ceiling, high-response ratio exciters causes the generator flux to *increase* during the transient. Thus, without modeling of excitation control, *lower* transient stability limited powers would be obtained, especially under conditions of "normal" fault clearing.

8.2.1 Synchronous Generator–Exciter Models

Figure 8.1 shows a schematic representation of a synchronous generator represented by the two-axis model. In that model, x' is the transient reactance used with the appropriate EMF component (x'_d with E'_q and x'_q with E'_d). The voltage $E' \angle \delta$ has the components E'_q and E'_d vary with different time constants (see below). The terminal voltage is V_t .

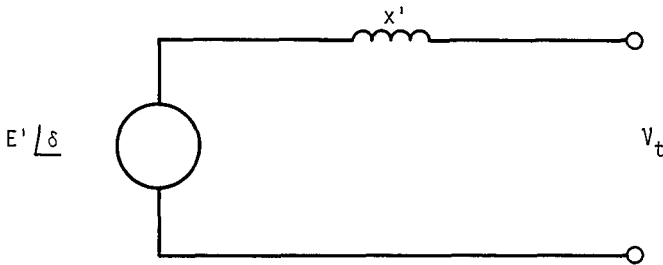


Figure 8.1 Synchronous generator model.

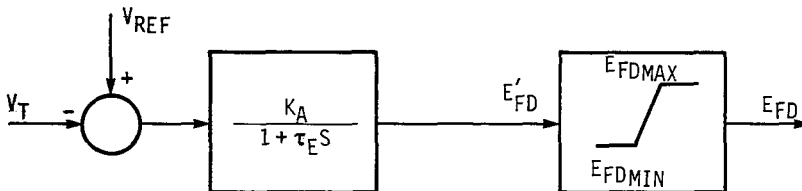


Figure 8.2 Exciter model.

The generator-exciter equations for generator k , represented in detail, in the COI frame of reference are given by²

$$\tau'_{dok} \dot{E}'_{qk} = E_{FDk} - E'_{qk} + (x_{dk} - x'_{dk}) I_{dk} \quad (8.1)$$

$$\tau_{qok} \dot{E}'_{dk} = -E'_{dk} - (x_{qk} - x'_{qk}) I_{qk} \quad (8.2)$$

$$\begin{aligned} \tau_{Ek} \dot{E}'_{FDk} &= g(E'_{FDk}, V_{tk}, V_{REFk}) \\ &= -E'_{FDk} + K_{Ak}(V_{REF} - V_{tk}) \end{aligned} \quad (8.3)$$

$$M_k \dot{\omega}_k = P_{mk} - E'_{dk} I_{dk} - E'_{qk} I_{qk} + (x'_{qk} - x'_{dk}) I_{dk} I_{qk} - \frac{M_k}{M_T} P_{COI} \quad (8.4a)$$

$$k = 1, 2, \dots, m$$

and for a generator ℓ represented classically, it is described by

$$M_\ell \dot{\omega}_\ell = P_{m\ell} - E_\ell I_{q\ell} - \frac{M_\ell}{M_T} P_{COI}$$

$$\ell = m + 1, m + 2, \dots, n \quad (8.4b)$$

where

m = number of generators represented in detail

n = total number of generators

τ'_{do}, τ'_{qo} = open-circuit direct and quadrature axes time constants, respectively

² The use of the two-axis model with salient pole generators presents a problem since for these generators $x'_q = x_q$, and the variation of the d -axis transient EMF will not be correctly represented. An alternate model, more suited for salient pole generators, is presented in Section 8.2.6.

E'_{FD}, E_{FD} = stator EMF corresponding to the field voltage, before and after the limiter, respectively

x_d, x'_d = direct axis synchronous and transient reactances

x_q, x'_q = quadrature axis synchronous and transient reactances

E'_d, E'_q = direct and quadrature axes stator EMFs corresponding to rotor flux components

I_d, I_q = direct and quadrature axes stator currents

V_t = generator terminal voltage

V_{REF} = exciter reference voltage

P_m = mechanical power

E = voltage behind transient reactance (classical model)

$$P_{COI} = \sum_{i=1}^n (P_{mi} - P_{ei})$$

P_e = electrical power

8.2.2 One-Gain, One-Time-Constant Exciter Model

The exciter model is reduced to an equivalent one-gain, one-time constant, and one-limiter model as shown in Fig. 8.2 using the standard "noise-equivalent bandwidth" technique (see Section 3.5 of [2]) in the frequency range of interest (typically 0.1 Hz to 4.0 Hz). An explanation of the procedure, the algorithmic steps involved, and some numerical tests to apply it are given in [3]. An outline of the procedure is given next.

Let $G_{EXACT}(s)$ be the linear transfer function of the exact model to be represented by the first-order model $G_{APPROX}(s) = \frac{K}{1 + Ts}$. The values of K and T are determined by evaluating the mean square response

$$E_{EXACT} = \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} \left| G_{EXACT}(\omega) \right|^2 d\omega \quad (8.5)$$

where ω_1 , ω_2 are the low and high ends of the frequency range.

The value of K is set at

$$K = \left| G_{EXACT}(j\omega_1) \right| \quad (8.6a)$$

$$E_{APPROX} = \frac{K^2}{2\pi T} \left[\tan^{-1}(\omega_2 T) - \tan^{-1}(\omega_1 T) \right] \quad (8.6b)$$

By equating E_{EXACT} and E_{APPROX} , using the value of K in (8.6a), the value of T is obtained by solving the nonlinear equation

$$\frac{(\omega_2 - \omega_1)T}{1 + \omega_1 \omega_2} - \tan \left[\frac{2\pi E_{EXACT} T}{K^2} \right] = 0 \quad (8.7)$$

Example

To illustrate the above procedure, it is applied to a high-speed type DC1 exciter [4].

The model for the exciter is shown in Fig. 8.3.

The data used for the exciter parameters are

$$K_A = 400, \quad K_E = 1.0, \quad K_F = 0.04$$

$$T_A = 0.02, \quad T_B = T_C = 0.0, \quad T_E = 0.015, \quad T_F = 0.05 \text{ s}$$

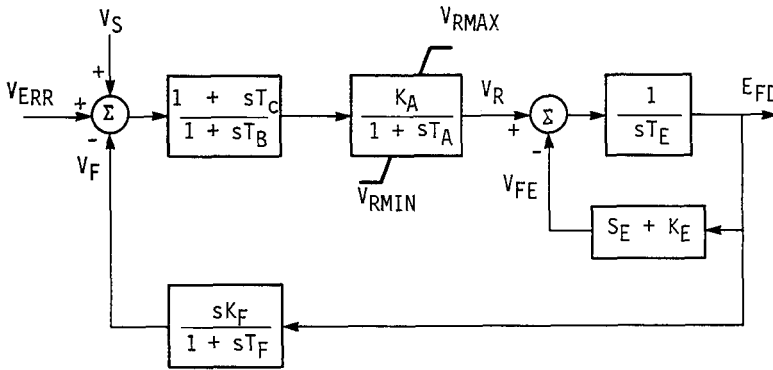


Figure 8.3 Type DC1 exciter.

In obtaining $G_{EXACT}(s)$ for this exciter, the limiter is moved to the output E_{FD} , and the nonlinear function $S_E(E_{FD})$ is linearized around the operating condition. The resulting value is $S_E = 0.1123 E_{FD}$.

Applying the procedure given by equations (8.5 – 8.7), and using the parameters just given, the reduced order model parameters, in the frequency range of 0.1 – 4.0 Hz, are

$$K = 40.018, \quad \text{and} \quad T = 1.4397$$

The parameters were tested against the original exciter model using a standard transient stability computer program. Comparable results were obtained for both models [3].

8.2.3 Network Equations

The power network is assumed to be represented classically: steady-state network parameters, constant impedance load, etc. Assuming that the generator internal reactance is constant (see below), the network can be reduced to the internal generator nodes by

eliminating all the load nodes. A network Y -bus is generated with diagonal elements $Y_{ii}/\theta_{ii} = G_{ii} + j B_{ii}$ and off-diagonal elements $Y_{ij}/\theta_{ij} = G_{ij} + j B_{ij}$.

Following a procedure similar to that used in Chapter 9 of [1], the direct and quadrature axes currents for a generator represented in detail, are given below:

$$I_{di} = \sum_{j=1}^n \left[F_B - G(\theta_{ij}) E'_{qj} + F_G + B(\theta_{ij}) E'_{dj} \right] \quad (8.8)$$

$$I_{qi} = \sum_{j=1}^n \left[F_G + B(\theta_{ij}) E'_{qj} - F_B - G(\theta_{ij}) E'_{dj} \right]$$

where $F_G + B(\theta_{ij})$ and $F_B - G(\theta_{ij})$ are given by

$$F_G + B(\theta_{ij}) = G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}$$

$$F_B - G(\theta_{ij}) = B_{ij} \cos \theta_{ij} - G_{ij} \sin \theta_{ij}$$

It is to be noted that in the general case some generators are represented in detail, while others are represented classically, e.g., as in (8.1) – (8.4). In this case (8.8) should be modified such that for the classical machines $E'_d = 0$ and $E'_q = E = \text{constant}$. Thus the generator currents are given by

$$I_{di} = \sum_{j=1}^m \left[F_B - G(\theta_{ij}) E'_{qj} + F_G + B(\theta_{ij}) E'_{dj} \right] + \sum_{k=m+1}^n F_B - G(\theta_{ik}) E_k$$

$$I_{qi} = \sum_{j=1}^m \left[F_G + B(\theta_{ij}) E'_{qj} - F_B - G(\theta_{ij}) E'_{dj} \right] + \sum_{k=m+1}^n F_G + B(\theta_{ik}) E_k$$

$$I_k = \sum_{j=1}^m \left[F_G + B(\theta_{kj}) E'_{qj} - F_B - G(\theta_{kj}) E'_{dj} \right] + \sum_{\ell=m+1}^n F_G + B(\theta_{k\ell}) E_\ell$$

$$i = 1, 2, \dots, m$$

$$k, \ell = m + 1, \dots, n \quad (8.9)$$

In some situations (see Section 8.2.4), the internal impedance of the generator represented in detail is not considered constant. In this case, the network equations are given to the generators' terminal nodes. A special procedure is used to interface the generators and network equations.

8.2.4 Modeling of the Generator Internal Impedance

It was stated earlier that in order that the network may be reduced to the internal generator nodes, the generator internal reactance $\left[x' = \frac{1}{2} (x'_d + x'_q) \right]$ is usually used. For the prefault and postfault networks, this value of x' is considered accurate enough, since its influence on the total network impedance "seen" from the internal generator node is usually quite small. This may not be true, however, for the faulted network, especially when the fault is close to a generator terminal. In this case a more accurately computed generator impedance z_g is needed. Calculation of the generator effective internal impedance for this case is given below.

Let

$$E' = E'_q + j E'_d$$

where

$$E'_d = V_d + x'_q I_q$$

and

$$E'_q = V_q - x'_d I_d$$

Substituting, we get

$$E' = V_q + j V_d - x'_d I_d + j x'_q I_q$$

$$\Delta = (V_q + j V_d) + z_g (I_q + j I_d)$$

Substituting and rearranging, we have

$$r_g \left[\frac{I_q^2 + I_d^2}{I_q I_d} \right] = x'_q - x'_d$$

or

$$r_g = (x'_q - x'_d) \frac{I_d I_q}{I_q^2 + I_d^2} \quad (8.10a)$$

$$x_g = x'_d + r_g \frac{I_q}{I_d}$$

or

$$\begin{aligned} x_g &= x'_q \frac{I_q^2}{I_q^2 + I_d^2} + x'_d \frac{I_d^2}{I_q^2 + I_d^2} \\ &= \frac{1}{I_q^2 + I_d^2} \left[x'_q I_q^2 + x'_d I_d^2 \right] \end{aligned} \quad (8.10b)$$

As stated, this load-dependent internal generator impedance is needed only during the fault period, and mainly when the fault is near the generator terminal. Since a variable internal generator impedance introduces complexities to the TEF program, an alternate procedure is suggested for use during the fault. The procedure is outlined in the following steps.

1. For generator i represented in detail ($i = 1, 2, \dots, k$), the terminal voltage components are converted to the generator (d, q) reference frame from the network (D, Q) reference frame (see Chapter 9 of [1]), using the relation

$$\begin{bmatrix} \vartheta_{qi} \\ \vartheta_{di} \end{bmatrix} = \begin{bmatrix} \cos \delta_i & \sin \delta_i \\ -\sin \delta_i & \cos \delta_i \end{bmatrix} \begin{bmatrix} \vartheta_{Qi} \\ \vartheta_{Di} \end{bmatrix}$$

2. The d - and q - components of the generator i current are obtained from

$$E'_{di} = \vartheta_{di} + x'_{di} I_{qi} + (x'_{qi} - x'_{di}) I_{qi} \quad (8.11)$$

$$E'_{qi} = \vartheta_{qi} - x'_{di} I_{di}$$

3. The generator i current components are converted to the network reference frame using the relation

$$\begin{bmatrix} I_{Qi} \\ I_{Di} \end{bmatrix} = \begin{bmatrix} \cos \delta_i & -\sin \delta_i \\ \sin \delta_i & \cos \delta_i \end{bmatrix} \begin{bmatrix} I_{qi} \\ I_{di} \end{bmatrix} \quad (8.12a)$$

From these components, an injected current at the terminal node of generator i is obtained

$$I_i = I_{Qi} + j I_{Di} \quad (8.12b)$$

4. The terminal voltages of the generators represented in detail are obtained from

$$I_i = \sum_{j=1}^k Y'_{ij} \vartheta_j + \sum_{j=k+1}^n Y'_{ij} E_j \quad (8.13)$$

$$i = 1, 2, \dots, k$$

where Y'_{ij} are the faulted bus parameters for the network having as nodes: the *terminal* voltages of the generators represented in detail and the *internal* nodes of the generators represented classically.

The set of k equations (8.13) have k terminal voltage unknowns ($\theta_1, \theta_2, \dots, \theta_k$). These voltages are solved for.

5. With the new values of terminal voltages, steps 1–4 are repeated until two successive values of the injection current vectors are within the selected tolerance.
6. Using the updated values of θ_i , $i = 1, 2, \dots, k$, the internal EMFs of the generators represented in detail are also updated, using one integration step.
7. New values of I_{di} and I_{qi} are computed using (8.11). New current injections are computed and iterated upon to obtain updated values of θ_i .
8. The process is repeated until the end of the faulted period. (Usually only a few integration steps are required.)

8.2.5 Effect of Excitation Control on the Transient Energy

Introduction

Following the reasoning given in Section 8.1, incorporation of the excitation control in the TEF method is accomplished by focusing on the energy in the first swing transient, which is primarily an inertial transient. In such a transient the generators' motion is determined by the interaction between the synchronizing torques and the inertias of the generators. The exciters primarily change the generator's internal EMF (mainly E'_q) and hence alter their synchronizing torques.³

In obtaining the system transient energy in the first swing transient, the following assumptions are made:

³ There is a similarity between this view of the first swing transient and the approach used by the authors of reference [5] in viewing the change of E' , due to excitation control, as a parameter variation resulting in an "elastic" potential energy boundary surface (PEBS).

1. The generator flux does not change abruptly; i.e., a reasonably well-behaved function can describe the flux variation in the first swing. It is also assumed that the flux varies smoothly between conditions at fault clearing and at the peak of the first swing.
2. The key parameters needed to compute the transient energy can be obtained from conditions at fault clearing and conditions at the point where the system potential energy is maximum on the critically stable trajectory (also the peak of the critical angle swing for the severely disturbed generators). The last conditions will be referred to as the "peak point."
3. The transient energy in the first swing can be evaluated using a constant value of E' . This value is estimated from the two values at clearing and at the peak point (a simple average value has been used).
4. When the disturbance is removed, the total transient energy is constant.

The transient energy function

The new transient energy function is obtained from the integral

$$V = \int \sum_{i=1}^n \left[M_i \dot{\omega}_i - P_{mi} + P_{ei} - \frac{M_i}{M_T} P_{COI} \right] \dot{\theta}_i dt \quad (8.14)$$

In (8.14), the following expression is used for P_{ei} (for the generator represented in detail):

$$P_{ei} = E'_{di} I_{di} + E'_{qi} I_{qi}$$

where I_{di} , I_{qi} are given by (8.8) and the term $(x'_{qi} - x'_{di}) I_{di} I_{qi}$ is neglected in P_{ei} .

The resulting expression for P_{ei} can be put in the form

$$P_{ei} = \alpha_{ii} G_{ii} + \sum_{\substack{j=1 \\ j \neq i}}^n \left[B_{ij} \left[\alpha_{ij} \sin \theta_{ij} + \beta_{ij} \cos \theta_{ij} \right] + G_{ij} \left[\alpha_{ij} \cos \theta_{ij} - \beta_{ij} \sin \theta_{ij} \right] \right] \quad (8.15)$$

where

$$\alpha_{ij} = E'_{di} E'_{dj} + E'_{qi} E'_{qj} \quad (8.16)$$

$$\beta_{ij} = E'_{di} E'_{qj} - E'_{qi} E'_{dj}$$

we note that $\alpha_{ij} = \alpha_{ji}$, $\beta_{ij} = -\beta_{ji}$.

To obtain the new TEF, equation (8.14) is integrated between the postdisturbance equilibrium condition ($\theta = \theta^{s2}$, $\tilde{\omega} = 0$) and the system state of interest (θ , $\tilde{\omega}$). The transient energy function has two components: kinetic energy and potential energy. They are given below.

Kinetic energy

Strictly speaking the system kinetic energy is given by

$$V_{KE} = \frac{1}{2} \sum_{i=1}^n M_i \tilde{\omega}_i^2 \quad (8.17)$$

However, the kinetic energy responsible for the separation of the group of severely disturbed (critical) generators from the rest of the system is the corrected kinetic energy given by [6]

$$V_{KEcorr} = \frac{1}{2} M_{eq} \omega_{eq}^2 \quad (8.18)$$

where M_{eq} , ω_{eq} are the inertia constant and speed of the equivalent one-generator-infinite-bus system formed by the two groups of generators: the critical group and the rest of the system.

Potential energy

The potential energy, with respect to the stable equilibrium conditions, is given by

$$\begin{aligned}
 V_{PE} = & \sum_{i=1}^n - \int_{\theta_i^{s2}}^{\theta_i} (P_{mi} - \alpha_{ii} G_{ii}) d\theta_i + \sum_{i=1}^{n-1} \sum_{j=i+1}^n \int_{\theta_{ij}^{s2}}^{\theta_{ij}} B_{ij} \left[\alpha_{ij} \sin \theta_{ij} \right. \\
 & \left. + \beta_{ij} \cos \theta_{ij} \right] d\theta_{ij} + \sum_{i=1}^{n-1} \sum_{j=i+1}^n \int_{\theta_i^{s2} + \theta_j^{s2}}^{\theta_i + \theta_j} G_{ij} \left[\alpha_{ij} \cos \theta_{ij} \right. \\
 & \left. - \beta_{ij} \sin \theta_{ij} \right] d(\theta_i + \theta_j) \quad (8.19)
 \end{aligned}$$

The above integrals are evaluated using the usual assumption of a linear trajectory approximation [6, 7] and a constant value for each of E'_{di} and E'_{qi} (average of the values at fault clearing and at the peak point). The α and β parameters in (8.19) are thus treated as constant. For a generator modeled classically, $E'_{qi} = E_i$, and $E'_{di} = \beta_{ij} = 0$. The resulting potential energy expression is given by

$$\begin{aligned}
 V_{PE} = & \sum_{i=1}^n - (P_{mi} - \alpha_{ii} G_{ii})(\theta_i - \theta_i^{s2}) + \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[B_{ij} \alpha_{ij} \left[-\cos \theta_{ij} + \cos \theta_{ij}^{s2} \right. \right. \\
 & \left. \left. - \beta_{ij} B_{ij} \left[\sin \theta_{ij} - \sin \theta_{ij}^{s2} \right] \right] \right] \\
 & + \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[G_{ij} \alpha_{ij} \frac{\theta_i + \theta_j - \theta_i^{s2} - \theta_j^{s2}}{\theta_{ij} - \theta_{ij}^{s2}} \left[\sin \theta_{ij} - \sin \theta_{ij}^{s2} \right] \right]
 \end{aligned}$$

$$+ G_{ij} \beta_{ij} \frac{\theta_i - \theta_j - \theta_i^{s2} - \theta_j^{s2}}{\theta_{ij} - \theta_{ij}^{s2}} \left[\cos \theta_{ij} - \cos \theta_{ij}^{s2} \right] \quad (8.20)$$

Critical value of the transient energy and the energy margin

When the transient stability assessment is made with the TEF method, a threshold value of the transient energy (called $V_{critical}$) is evaluated. Transient stability assessment is made by computing the difference between this threshold value and the value of the transient energy at the end of the disturbance (i.e., at fault clearing). This energy difference is called the energy margin ΔV . The system is stable or unstable depending on whether $\Delta V \geq 0$.

For the classical power system model, the critical value of the transient energy is the value of the potential energy at the controlling unstable equilibrium point (UEP). This UEP represents a system equilibrium condition, attainable only at infinite time, which separates stable and unstable system trajectories. We note that at the UEP, all derivatives of the state variables are zero. For this model, the state variables are the generator angles and speeds, the derivatives of which are zero at the peak of the system trajectory. Since the UEP separates the stable and unstable trajectories, it represents the limiting stable conditions. Thus, its transient energy, which is made up of potential energy only, is the critical value of transient energy against which stability assessment is made. The following questions arise: Will the same situation apply to the generators equipped with exciters? Will the critical energy still be the potential energy at the UEP?⁴

Careful investigation of the system trajectories in the postdisturbance period was made by the authors and their coworkers [8]. These investigations indicated that at the peak of the stable trajectories, while the derivatives of the generator angles and speeds are

⁴ Actually the Iowa State University researchers initially assumed this to be the case. In [7], the critical energy was determined by first solving for the UEP. In later work [8], this error was corrected, and the peak point concept was introduced.

zero, derivatives of other state variables (e.g., E'_q , E'_d , E_{FD}) are usually not zero. This is found to be true even in the critically stable system conditions. Thus, at the critically stable trajectory the peak point is characterized by the following (1) for a critical generator i , $\partial_i^p = 0$ and $\dot{\tilde{\omega}}_i = 0$ and (2) for the system V_{PE}^p is maximum. A procedure for determining the peak point has been developed and is given below.

Combining the characterization of the peak point with the assumptions given earlier in this section, the following view emerges: an elastic potential energy boundary surface (caused by a changing E' due to excitation control) nearly "grazed" at the peak point in the critically stable case.

Critical value of the transient energy

Strictly speaking, the critical energy is the potential energy at the peak point $[\theta^{cl}, \tilde{\omega}^{cl}]$ and $[\theta^p, 0]$. However, the transient energy in the first swing, as given by (8.26), assumes an average value of E'_q and E'_d for the generators represented in detail. These average EMF values are treated as constant in the first swing.

The critical value of the transient energy used in the final transient stability assessment is the value of the potential energy (8.20) evaluated at the rotor angles θ^p , and using E_q^{av} ; E_d^{av} given by

$$\begin{aligned} E_q^{av} &= \frac{1}{2} \left[E_q^{cl} + E_q^p \right] \\ E_d^{av} &= \frac{1}{2} \left[E_d^{cl} + E_d^p \right] \end{aligned} \quad (8.21)$$

where E_q^p , E_d^p are the values at the peak point.

Number of generators modeled in detail

Since the procedure of incorporating excitation control in the TEF method involves a great deal of computation, it is of considerable interest to know the minimum number of generators which need to be modeled in detail, while giving the correct transient stability assessment. It is reasoned that the severely disturbed generators are those whose excitation systems are most affected during the transient. This assumption was investigated in [10] and was found to be true.

Therefore, customarily it is assumed that the generators to be represented in detail are the advanced generators in the MOD. The rest of the generators can be represented classically.

Determination of the peak point with MOD procedure

This procedure involves determination of (1) the mode of disturbance (MOD) when some generators are equipped with excitation control and (2) the conditions at the peak point.

MOD determination

When some of the generators are modeled in detail, the following procedure is used [8]:

1. The candidate modes to be checked are identified by inspection (e.g., if the possible severely disturbed generators can be readily identified) or by using the automatic mode selection procedure.
2. The rotor angles at the (approximate) corner point are computed, for all the generators, using (8.22). For the generators represented in detail, values of E'_q and E'_d at fault clearing are used.

For a generator j in the critical (advanced) generator group, the rotor angles at the corner point are given by

$$\theta_j^{CRN} = \theta_j^{s2} + (\pi - 2 \theta_{eq}^{s2}) \frac{M_{T-k}}{M_T}$$

and for a generator j in the rest of the system (8.22)

$$\theta_j^{CRN} = \theta_j^{s2} - (\pi - 2 \theta_{eq}^{s2}) \frac{M_k}{M_T}$$

where

$$\theta_{eq}^{s2} = \theta_k^s - \theta_{T-k}^{s2}$$

3. For each of the candidate modes, the corrected kinetic energy at fault clearing and the potential energy margin (between the angles at the approximate corner point and the angles at fault clearing) are computed.
4. The normalized potential energy margin, $\Delta V_{PEN} = \Delta V_{PE} / V_{KEcorr}$, is computed for each candidate mode. The mode with the lowest ΔV_{PEN} is selected.

Note: Usually only two or three modes need to be checked for the above steps.

5. For the selected mode, the "actual" corner point voltages and angles, for the generators represented in detail, the corresponding ray point and the exact peak point are computed as outlined below.

Determination of the peak point

The difficulty which must be overcome is that a potential energy maximum is sought while the generator EMF is changing (especially E'_q); the former depends on the latter. This difficulty is overcome by assuming an exponential variation of E'_q toward a limiting value which can be readily estimated, as outlined below.

Approximate function for $E'_q(\theta)$

It is assumed that $E'_q(\theta)$ varies exponentially between conditions at fault clearing and those at the limiting conditions, or

$$E'_q(\theta) = E'_q{}^{\lim} - (E'_q{}^{\lim} - E'_q{}^{cl}) e^{\frac{\theta - \theta^{cl}}{\Delta\theta}}$$

where $\Delta\theta$ is obtained from

$$\Delta\theta = (E'_q{}^{\lim} - E'_q{}^{cl}) / \left[\frac{dE'_q}{d\theta} \right]_{cl} \quad (8.23)$$

Experimentally, it was found that for the best estimate of $\Delta\theta$, $\left[\frac{dE'_q}{d\theta} \right]$ is to be evaluated just prior to the instant of fault clearing.

The proper choice of $E'_q{}^{\lim}$ would be the value of E'_q at the UEP. However, an approximate value of $E'_q{}^{\lim}$ is obtained at the so-called "corner point" [9]. For a known mode of disturbance, i.e., assuming that the most severely disturbed generators (whose rotor angles are to be advanced), the corner point determination involves determining the angles of all the generators and estimating E'_q , E'_d , and E_{FD} for the generators equipped with exciters.

The rotor angles at the corner point are obtained as follows:

$$\theta_k^{s2} = \left[\sum_{i \in k} M_i \theta_i^{s2} \right] / M_k$$

$$\theta_{T-k}^{s2} = \left[\sum_{i \in T-k} M_i \theta_i^{s2} \right] / M_{T-k}$$

$$M_k = \sum_{i \in k} M_i$$

$$M_{T-k} = \sum_{i \in (T-k)} M_i$$

and

$$M_T = M_k + M_{T-k}$$

The detailed generator angles and voltages are obtained by solving (8.24) for $E'_q{}^{CRN}$, $E'_d{}^{CRN}$, $E'_{FD}{}^{CRN}$, and θ^{CRN} , holding the angles of the classical generator as obtained by (8.22). Thus, for a detailed generator i , the solution procedure is started with the angles computed by (8.22) and the predisturbance values of EMFs. The corner point solution is obtained by solving the following set of equations, with the detailed generator angles adjusted to satisfy $\dot{\tilde{\omega}}_i = 0$.

$$\dot{E}'_{qi} = 0$$

$$\dot{E}'_{di} = 0$$

$$\dot{E}'_{FDi} = 0$$

and

$$\dot{\tilde{\omega}}_i = 0$$

$$i = 1, 2, \dots, m \quad (8.24)$$

Conditions at the peak point are determined by the following two-step procedure. First, an approximate peak point, called the ray point, is obtained by maximizing the potential energy on the ray from the postfault stable equilibrium point to the corner point. A one-dimensional search is used

$$\frac{dV_{PE}}{d\alpha} \simeq \sum_{i=1}^n P_{ai} \left[E'_q \left[\theta^{ray} \right], E'_d{}^{ray}, \theta^{ray} \right] \cdot \left[\theta_i^{CRN} - \theta_i^{s2} \right] = 0 \quad (8.25)$$

where

P_{ai} = accelerating power of generator i , given by (8.4a)

$$\theta^{ray} = \theta^{s2} + \alpha (\theta^{CRN} - \theta^{s2})$$

$$E'_q \text{ is given in (8.23) and } E'_d{}^{ray} \simeq \frac{1}{2} \left[E_d'^2 + E_d'^{CRN} \right] = \text{constant}$$

In the second step the exact peak point is determined by solving

$$P_{ai} \left[E'(\theta^p), \theta^p \right] = 0 \quad (8.26)$$

$$i = 1, 2, \dots, m$$

In (8.26) the generators represented in detail are actually modeled classically. However, their internal EMF E' are adjusted according to the updated values of E'_q obtained by (8.23).

Again, it is to be emphasized that at the peak point $\dot{E}'_{qi}$, $\dot{E}'_{di}$, and $\dot{E}_{FDi}$ may not be equal to zero.

Determination of the peak point using the exit point procedure

If the exit point method is to be used to determine the peak point condition, two issues need to be addressed (1) determination of the first potential energy maximum on the faulted system trajectory and (2) the appropriate gradient system for the energy function used.

For the faulted trajectory we assume that while the state variables are integrated during the fault period, the potential energy, to first-order approximation, varies only with the generator angles and speeds. Thus the exit point is determined by locating the first potential energy maximum by examining $\Sigma f_i \cdot \tilde{\omega}_i$ for zero crossing, where for the generators represented in detail the mismatch function, f_i is computed using the detailed generator model for the postdisturbance network.

Regarding the gradient system, we note that the energy function given in (8.20) assumes a generator two-axis model but with *constant* values of E'_q and E'_d . Therefore, the gradient system for this energy function will be angle dependent only, i.e., in it the values of E'_q and E'_d are held constant.

From this, the procedure to be used for determining the exit point is as follows:

1. For the faulted system trajectory, the first maximum of the potential energy is found, assuming that the potential energy varies only with the generator angles and speeds (but allowing E'_q and E'_d to vary).
2. The values of E'_q and E'_d to be used in the gradient system equations are determined by a procedure similar to that suggested in [11], in which the effect of the postdisturbance network is to be accounted for in computing the values of E'_q and E'_d at the exit point. The procedure is outlined below.

Equations (8.1) – (8.3) are integrated in parallel with the faulted trajectory simulation, and at each instant, the currents I_d and I_q (for generators represented in detail) are obtained using the angles from the faulted trajectory solution and the postdisturbance network admittance matrix. The values of E'_q and E'_d , obtained with this procedure, at the exit point are used in the gradient system solution.

3. The gradient system equation for the postfault network is given by

$$\dot{\theta}_i = \left[P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} \right] \quad (8.27)$$

where for the generators represented in detail, P_{ei} is determined from (8.15) and (8.16), with the values of E'_q and E'_d held constant at their exit point values calculated in step 2.

Equations (8.27) are then integrated from the projection of the exit point to the point where $\sum_{i=1}^n |f_i|$ reaches its first minimum. Again the assumption is made that the potential energy varies only with the generator angle.

4. Based on the values of E'_{q_k} and E'_{d_k} obtained in step 2 and using the angles obtained at the point of minimum $\left| \sum_{i=1}^n f_i \right|$, the peak point given by (8.26) is obtained and critical energy is evaluated.

8.2.6 Modeling of Salient Pole Generators

For a salient pole generator, e.g., a generator driven by a hydroturbine, $x'_q = x_q$. In this case equation (8.2) will not represent the correct variation of the d -axis transient EMF. To remedy this situation, a simplified version of the so-called E'' -model (see Section 4.15.2 of [1]) is proposed. The model is similar to that given in Fig. 8.1, except that (1) the generator transient reactance x' is replaced with the subtransient reactance x'' and (2) the voltage E' / δ is replaced with the internal voltage E'' / δ . The generator equations are given by

$$\tau'_{qo} \dot{E}'_d = -E'_d - (x_q - x'_q) I_q \quad (8.28a)$$

$$\tau'_{do} \dot{\Lambda}_D = -\Lambda_D + E'_q + (x'_d - x_\ell) I_d \quad (8.28b)$$

$$\tau'_{do} \dot{E}'_q = E_{FD} - E'_q (1 + K_d) + K_d \Lambda_D + X_{xd} I_d \quad (8.28c)$$

$$E'_q = K_1 E'_q + K_2 \Lambda_D \quad (8.28d)$$

where

$$K_1 = \frac{x'_d - x_\ell}{x'_d - x_\ell}$$

$$K_2 = 1 - K_1$$

$$K_d = (x_d - x'_d)(x'_d - x'_{d'}) / (x'_d - x_\ell)^2 \quad (8.29)$$

$$X_{xd} = (x_d - x'_d)(x'_{d'} - x_\ell) / (x'_d - x_\ell)$$

and

Λ_D = direct axis amortisseur flux

The proposed simplification is to assume that $\tau'_{do} \approx 0$, thus reducing the order of the system by one. We can easily show that, with this assumption, we get

$$\Lambda_D = E'_q + (x'_d - x_\ell) I_d \quad (8.30a)$$

and

$$E'_{q'} = E'_q + (x'_d - x'_{d'}) I_d \quad (8.30b)$$

The resulting generator equations are

$$\tau'_{qo} \dot{E}'_{d'} = -E'_{d'} - (x_q - x'_{q'}) I_q \quad (8.31a)$$

$$\tau'_{do} \dot{E}'_q = -E'_q + (x_d - x'_d) I_d \quad (8.31b)$$

and

$$E'_{q'} = E'_q + (x'_d - x'_{d'}) I_d \quad (8.31c)$$

Thus the order of the system is the same as with the two-axis model. The network Y-bus, reduced to the internal generator nodes, now includes the generator reactance $x'_{d'}$.

The generator currents I_d and I_q are obtained with equations similar to (8.8) except the EMF of a salient pole generator would be $E'' = E'_q + j E'_d$. The power of a salient pole generator i is given by

$$P_{ei} = E'_{qi} I_{di} + E'_{di} I_{qi} \quad (8.32)$$

and the α and β terms in (8.16) become

$$\alpha_{ii} = E'^2_{di} + E'^2_{qi}$$

$$\alpha_{ij} = E'_{di} E'_{dj} + E'_{qi} E'_{qj}$$

and

$$\beta_{ij} = E'_{di} E'_{qj} - E'_{qi} E'_{dj} \quad (8.33)$$

With this definition of terms, the expressions for P_{ei} and for V_{PE} are the same as (8.15) and (8.20), respectively.

Finally, we should point out that the use of this simplified version of the E'' model is not limited only to salient pole generators. It can also be used to model round rotor generators as well.

8.3 MODELING OF NONLINEAR LOADS

8.3.1 Introduction

Again, as we consider load models other than those represented by constant impedance, we must keep in mind that the TEF method is a "first swing" tool of analysis.

Our interest will then be in the changes which can be made in load models to improve the accuracy of the results during the first swing transient.

In general, power system loads are both voltage and frequency dependent. In a multiswing transient, it is expected that accurate load models would reflect this dependence, as well as model the special controls which may cause disconnection of loads (e.g., low-voltage relays). In a first swing transient, however, the magnitude of the change in the power system frequency is not likely to be large enough to have a substantial affect on the power system load. Therefore we will concern ourselves with load models which tend to reflect the voltage dependence of loads.

The experience of the authors, supported by the experience of engineers from several utilities in North America, indicates that for the first swing transient, the power system load can be adequately represented by a mixture of constant current, constant impedance, and constant MVA components. This representation will be used when nonlinear representation is called for. Since the representation of the constant impedance loads had been addressed, discussion of the constant current and constant MVA load models is presented.

8.3.2 General Approach [12]

If the load at bus k is represented as a constant current or a constant MVA load, then

$$P_{Lk} + j Q_{Lk} = V_k / \underline{\Psi}_k \bar{I}_{Lk}^*$$

or

$$\bar{I}_{Lk} = (P_{Lk} - j Q_{Lk}) / V_k / \underline{\Psi}_k \quad (8.34)$$

where

$$V_k = V_k / \underline{\Psi}_k = \text{voltage at bus } k$$

P_{LK}, Q_{LK} = power and reactive power at bus k

The current $\bar{I}_{Lk}$ must be preserved during the transient. This is accomplished by determining equivalent current injections at the internal generator nodes. In [13] Pai et al., use similar current injections at the internal generator nodes, with the assumption that the complex ratio of internal generator voltage to load bus voltage is constant. This assumption could be quite restrictive and is not invoked in the development presented here. The current injections are recalculated at each iteration of the SEP and UEP solution procedure based on the updated values of the load bus voltages. Hence, these current injections reflect the nonlinear load components at the internal generator nodes. This is shown schematically for generator i in Fig. 8.4.

In Fig. 8.4, the EMF $\bar{E}'_i$ is the constant voltage behind transient reactance E_i when the classical generator model is used, or it can be the internal EMF E'_i , the components of which are E'_{qi} and E'_{di} as given by equations (8.1) and (8.2). The currents are as follows: $\bar{I}_i$ is the generator current, $\bar{I}_{ai}$ is the current into the classical model portion of the

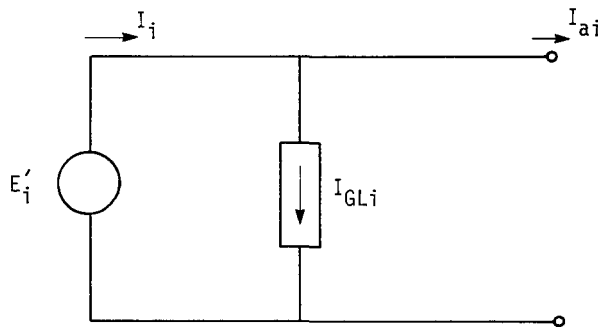


Figure 8.4 Synchronous generator model with current injection representing nonlinear load.

network, and $\bar{I}_{GLi}$ is the constant current or constant MVA load component reflected at the internal node of generator i .

8.3.3 Determining I_{GL} for Classical Generator Model

It is assumed that the constant impedance loads have been folded into the postdisturbance Y_{BUS} . For the constant current and constant MVA portions of the load, the initial current components are evaluated by

$$\begin{aligned} I_{CI}^{\circ} &= \left[(P_L^{\circ} p_2 + j Q_L^{\circ} q_2) / V_L^{\circ} \right]^* \\ I_{CM}^{\circ} &= \left[(P_L^{\circ} p_3 + j Q_L^{\circ} q_3) / V_L^{\circ} \right]^* \end{aligned} \quad (8.35)$$

where

P_L°, Q_L° = predisturbance MW and MVAR load

V_L° = predisturbance load bus voltage

p_2, q_2 = multipliers for constant current portions of the load

p_3, q_3 = multipliers for constant MVA portions of the load

A nonlinear load current vector I_1 , for initial conditions, is formed by

$$I_1 = I_{CI} + I_{CM} \quad (8.36)$$

The current injections at the internal generator nodes are determined, for the classical generator model, by the following procedure:

1. A vector of currents injected in the generator buses is computed from

$$I_2 = E / \theta / j x'_d \quad (8.37)$$

We note that the angle θ is used, indicating that for the generator rotor angles a COI reference frame is used.

2. The set of voltages $\begin{bmatrix} V_L \\ V_G \end{bmatrix}$ is solved for using the relation

$$\begin{bmatrix} -I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{BUS}^A \end{bmatrix} \begin{bmatrix} V_L \\ V_G \end{bmatrix} \quad (8.38)$$

where Y_{BUS}^A includes the load buses and the generator terminal buses, with the terms $\frac{1}{jx'_d}$ incorporated on the diagonal entries of the generator terminal buses.

3. The nonlinear load current components I_{CI} and I_{CM} are updated according to the new V_L by

$$I_{CI} = \left[\frac{P_L^\circ p_2 \frac{|V_{Lnew}|}{|V_L^\circ|} + j Q_L^\circ q_2 \frac{|V_{Lnew}|}{|V_L^\circ|}}{V_{Lnew}} \right]^* \quad (8.39)$$

$$I_{CM} = \left[\frac{P_L^\circ p_3 + j Q_L^\circ q_3}{V_{Lnew}} \right]^*$$

Equations (8.38) and (8.39) are used iteratively until successive iterations give nearly the same magnitude of the current vector (i.e., within a given tolerance).

4. The set of current injections I_{GL} is obtained from the final values of $-I_1$ using

$$I_{GL} = \begin{bmatrix} Y_{21}^B \end{bmatrix} \begin{bmatrix} Y_{11}^B \end{bmatrix}^{-1} \begin{bmatrix} -I_1 \end{bmatrix} \quad (8.40)$$

where Y_{BUS}^B is the same as Y_{BUS}^A augmented with the internal generator buses so that the voltage vector is $\begin{bmatrix} V_L \\ E \end{bmatrix}$.

8.3.4 Incorporation of Nonlinear Loads in the TEF Method

Incorporation of the nonlinear load models in the TEF method affects the following steps in the TEF procedure (1) the SEP and UEP solution, (2) conditions at fault clearing, (3) mode of disturbance evaluation, and (4) energy margin computation. These steps are modified as follows.

1. SEP and UEP solution:

The electrical power for generator i becomes

$$P_{ei} = P'_{ei} + E_i I_{GLi} \cos(\theta_i - \phi_i) \quad (8.41)$$

where

P'_{ei} = electrical power with classical network model

I_{GLi}/ϕ_i = complex current, representing nonlinear loads, reflected at generator i

The SEP and UEP solutions are obtained iteratively from (8.42)

$$P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} = 0 \quad i = 1, 2, \dots, n \quad (8.42)$$

where

$$P_{COI} = \sum_{i=1}^n [P_{mi} - P_{ei}]$$

The iteration on the SEP and UEP solution is performed as follows. The left hand side of (8.42) is evaluated to obtain the mismatch function. Using the procedure in Section 6.7, the value of the mismatch function is used to update the vector θ , and a new value of the mismatch function is evaluated. Assuming that no convergence to a solution is obtained, the injected current vector I_2 in (8.37) is updated, and a new value of I_{GL} is computed by following steps 2–4 of Section 8.3.3. This process is repeated until convergence to a SEP or UEP solution is obtained.

2. Conditions at fault clearing:

The injected currents, at the generator terminals, are obtained using the faulted Y_{BUS} . The constant acceleration assumption is also used during each time step in the fault-on period.

3. Mode of disturbance evaluation:

In the MOD, the effect of the nonlinear load model comes in the SEP solution, which affects the ray point solution.

4. Energy margin computation:

The expression for the energy margin is obtained using a linear trajectory approximation [12]:

$$\begin{aligned} \Delta V = & -\frac{1}{2} M_{eq} \tilde{\omega}_{eq}^{c\ell 2} - \sum_{i=1}^n P_i (\theta_i^u - \theta_i^{c\ell}) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[C_{ij} \left[\cos \theta_{ij}^u - \cos \theta_{ij}^{c\ell} \right] \right. \\ & \left. - \left[\frac{\theta_i^u + \theta_j^u - \theta_i^{c\ell} - \theta_j^{c\ell}}{\theta_{ij}^{c\ell} - \theta_{ij}^{c\ell}} \right] D_{ij} \left[\sin \theta_{ij}^u - \sin \theta_{ij}^{c\ell} \right] \right] \\ & + \sum_{i=1}^n \int_{\theta_i^{c\ell}}^{\theta_i^u} E_i I_{GLi} \cos (\theta_i - \phi_i) d\theta_i \end{aligned} \quad (8.43)$$

All the terms in (8.43) have been previously defined. The last term, which is path dependent, is also evaluated assuming a linear trajectory between $\theta^{c\ell}$ and θ^u . Several points are selected on this trajectory. At each point, the injected currents I_{GL} are determined using the procedure in the previous section and the values of θ at the selected point. The I_{GLi} terms in (8.43) are evaluated using trapezoidal integration.

8.3.5 Incorporation of the Nonlinear Load Model and Excitation Control in the TEF Method

Reference is made to Fig. 8.4 and to the symbols used for the currents and voltages in that figure. In this discussion, the generator internal reactance x'_i is considered constant and is incorporated in the network Y_{BUS} .

As is the custom when detailed generator representation is used (e.g., as in Chapter 9 of [1]), two frames of reference are used: a synchronous reference frame for the network and individual generator reference frame. For the former the capital subscripts D and Q will be used, and for the latter lowercase symbols (e.g., d and q) are used for the generator rotor reference frame. These symbols will be used with the network and generator equations.

The phasor notation $\bar{I}_i = I_{qi} + j I_{di}$ or $\bar{I}_i = I_{Qi} + j I_{Di}$ will be used to express the current of generator i in the appropriate reference frame.

To simplify the discussion in this subsection, the term $\left[(x'_{qi} - x'_{di}) I_{di} I_{qi} \right]$ in the expression for the generator i power is neglected.⁵ We also define P'_e as the electrical power corresponding to the portion of the network in which the loads are represented by constant impedance.

In general, the generator current $\bar{I}_i$ is given by

$$\begin{aligned} I_i &= I_{ai} + I_{GLi} \\ &= (I_{aqi} + j I_{adi}) + I_{GLi} e^{j\phi_i} \end{aligned} \quad (8.44)$$

where I_{aqi} , I_{adi} can be determined from (8.8).

The electrical power for generator i is given by

⁵ This does not represent any loss of generality, since the term can be easily added.

$$\begin{aligned}
 P_{ei} &= P'_{ei} + P_{eGLi} \\
 &= P'_{ei} + (E'_{qi} I_{GLqi} + E'_{di} I_{GLdi})
 \end{aligned} \tag{8.45}$$

and the power to accelerate the COI is given by

$$P_{COI} = P'_{COI} - \sum_{i=1}^n P_{eGLi} \tag{8.46}$$

where

$$P'_{ei} = E'_{qi} I_{aqi} + E'_{di} I_{adi}$$

and

$$P'_{COI} = \sum_{i=1}^n (P_{mi} - P'_{ei}) \tag{8.47}$$

For generator k represented in detail, and with portions of the system loads represented by nonlinear load models, the generator equations become

$$\tau'_{dok} \dot{E}'_{qi} = E_{FDi} - E'_{qi} + (x_{di} - x'_{di}) I_{adi} \tag{8.48}$$

$$\tau'_{qok} \dot{E}'_{di} = -E'_{di} - (x_{qi} - x'_{qi}) I_{aqi} \tag{8.49}$$

$$\begin{aligned}
 M_i \ddot{\omega}_i &= P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} \\
 &= \left[P_{mi} - P'_{ei} - \frac{M_i}{M_T} P'_{COI} \right] - \left[P_{eGLi} + \frac{M_i}{M_T} \sum_{j=1}^n P_{eGLj} \right]
 \end{aligned} \tag{8.50}$$

The resulting expression for the transient potential energy function becomes

$$\begin{aligned}
V_{PE} = & - \sum_{i=1}^n (P_{mi} - \alpha_{ii} G_{ii})(\theta_i - \theta_i^{s2}) + \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[\beta_{ij} \alpha_{ij} \left[-\cos \theta_{ij} + \cos \theta_{ij}^{s2} \right] \right. \\
& + \beta_{ij} B_{ij} \left[\sin \theta_{ij} - \sin \theta_{ij}^{s2} \right] \left. \right] \\
& + \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[G_{ij} \alpha_{ij} \frac{\theta_i + \theta_j - \theta_i^{s2} - \theta_j^{s2}}{\theta_{ij} - \theta_{ij}^{s2}} \left[\sin \theta_{ij} - \sin \theta_{ij}^{s2} \right] \right. \\
& + G_{ij} \beta_{ij} \frac{\theta_i + \theta_j - \theta_i^{s2} - \theta_j^{s2}}{\theta_{ij} - \theta_{ij}^{s2}} \left[\cos \theta_{ij} - \cos \theta_{ij}^{s2} \right] \left. \right] \\
& + \sum_{i=1}^n \int_{\theta_i^{s2}}^{\theta_i} E'_i I_{GLi} \cos(\theta_i - \phi_i) d\theta_i
\end{aligned} \tag{8.51}$$

In (8.51), θ_{ii} , α_{ij} , and β_{ij} are the same as in (8.20), and in the last term are evaluated as in (8.43).

8.4 INCORPORATING TWO-TERMINAL HVDC IN THE TRANSIENT ENERGY FUNCTION METHOD

As has been repeatedly stressed in this chapter, when a new model is incorporated in the TEF method, emphasis is given to how this model influences the system transient energy in the first swing transient. This is also kept in mind when the HVDC model is introduced in the TEF method.

Interfacing an ac/dc system involves a complex arrangement of controls and equipment. In transient stability studies, usually a high-order model is used to model these controls. Furthermore, the equations describing the HVDC system vary according to the dc system operating modes. Therefore, as part of the procedure to incorporate the

HVDC in the TEF method, a simplified—yet accurate—model which is considered valid for the first swing transient had to be developed. In addition, a simplified procedure to determine the dc control mode is also needed.

The process of incorporating the two—terminal HVDC in the TEF method involves the following steps:

1. Development of a simplified two—terminal HVDC model suited for first swing transient and a method to determine the dc control model (when this model is used).
2. Development of a procedure to interface the ac/dc systems. The procedure assumes that the ac network is modeled classically.
3. Incorporation of the HVDC model in the expression for the transient energy.
4. Introducing a procedure for transient stability assessment using the new TEF expression.

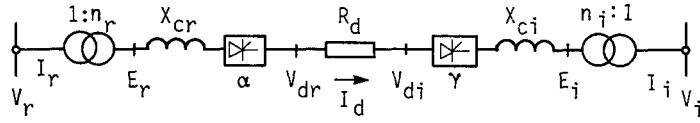
These steps are outlined in the following subsections.

8.4.1 A Simplified Two—Terminal HVDC Model [14, 15]

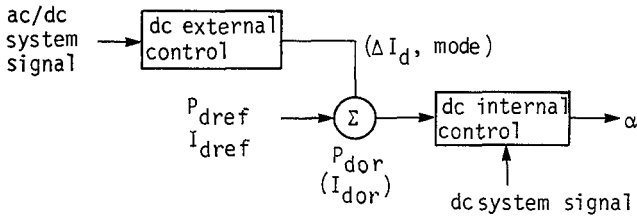
A schematic diagram of the dc line and the associated controls is shown in Fig. 8.5. The model developed below is valid for the first swing transient, for a disturbance in the ac network.

The dc model to be used is the steady—state converter model with static dc line representation, and with the dynamics of the dc internal controls neglected.

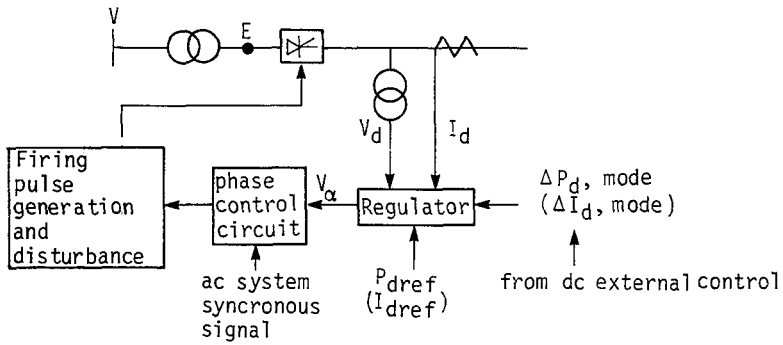
The external dc power or current modulation are assumed to have little effect on the first swing transient and are neglected here. The voltage—dependent current order limit (VDCOL) is included in the model. With the dynamics of its internal controls neglected, the dc model will have discontinuity in its dc variables and control angles. Thus, the dc mode of operation will first need to be determined, especially when VDCOL is considered.



(a)



(b)



(c)

Figure 8.5 Simplified diagram of a two-terminal HVDC:

- (a) The line and associated equipment.
- (b) Control arrangements.
- (c) Internal controls.

A procedure is used by which the dc control mode is determined directly by inference from the possible modes of operation.

Determination of the control mode

The dynamics of the dc internal controls are neglected, since their time constants are much smaller than the period of the first swing transient. In addition, control of the limiting value of the inverter extinction angle (γ_{stop}) is neglected, since its effect on the value of the dc power is small. With these assumptions, the rules developed to determine the dc control mode are based on the following control strategies:

1. Under normal conditions, three modes are possible.

Mode 1: Constant power (P_g) or current (I_g) control at the rectifier, and constant extinction angles control at the inverter.

Mode 2: Minimum firing angles control at the rectifier and constant power or current control at the inverter with current margin.

Mode 3: A transition mode between modes 1 and 2.

2. Voltage-dependent current order limit (VDCOL), derived from the rectifier dc voltage is considered.

The above is illustrated for the constant power control case. The dc steady-state characteristics with constant power control and VDCOL are shown in Fig. 8.6. The operating state along $A_1B_1C_1D_1E_1$ is in mode 1, while the operation along $A_2B_2C_2D_2E_2$ is in mode 2; the region between the two boundaries represents operation in mode 3. Operation outside the region enclosed between the two ABCDE lines is not feasible because of the assumptions on the dc controls. Note also that because of the assumption on γ_{stop} , the line F_2G_2 is replaced with F_2E_2 .

In Fig. 8.6, V_{d1} and V_{d2} are the upper and lower limits for VDCOL; ΔI_d is the current margin. The constant power control characteristics intersects the VDCOL characteristics at V_{d3} . (V_{d3} may be less than V_{d2} .)

Determination of the control mode and corresponding dc current is completed in two steps, assuming that the converter ac bus voltages V_r and V_i are known.

Step 1: To separate control modes of 1, 2, and 3.

First, calculate a predicted operating point for the inverter (see Fig. 8.5a):

$$I_d^* = \frac{1.35n_r V_r \cos \alpha_{min} - 1.35n_i V_i \cos \gamma_N}{R_{cr} - R_{ci} + R_d} \quad (8.52)$$

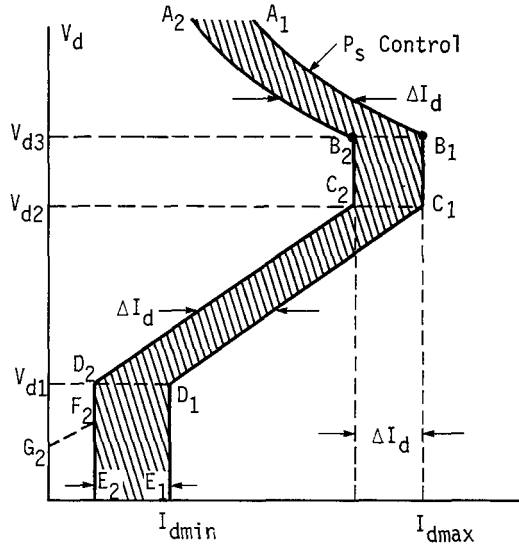


Figure 8.6 DC steady-state characteristics.

where

$$R_{cr} = \frac{3}{\pi} X_{cr}$$

$$R_{ci} = \frac{3}{\pi} X_{ci}$$

The mode of operation is determined as follows:

1. If $I_d^* \leq 0$, then operation is in mode 2.
2. If $I_d^* > I_{dmax}$, then operation is in mode 1.
3. If $0 < I_d^* \leq I_{dmax}$, we compute

$$V_d^* = 1.35 n_r V_r \cos \alpha_{min} - R_{cr} I_d^* \quad (8.53)$$

If $V_d^* > V_{d3}$ and $I_d^* \cdot V_d^* \geq P_s$: mode = 1.

$$I_d^* \leq \frac{P_s}{V_d^*} - \Delta I_d: \text{ mode} = 2. \quad (8.54)$$

Otherwise: mode = 3.

Similar rules can be derived for other voltage conditions (see [13, 14]).

Step 2: To obtain the dc current for the different modes.

1. For mode 3, $\alpha = \alpha_{min}$, $\gamma = \gamma_N$, $I_d = I_d^*$, and $V_{dr} = V_d^*$, where γ_N is the normal extinction angle.
2. For mode 1, let V_{i1} , V_{i2} be the inverter ac bus voltage corresponding to the lower and upper limits of VDCOL respectively, and let VDCOL be such that V_{i3} be at the intersection of the constant power characteristics with the I_{dmax} line (see Fig. 8.7a), then

$$\begin{aligned}
 V_{i1} &= \frac{V_{d1} + (R_{ci} - R_d) I_{dmin}}{1.35 n_i \cos \gamma_N} \\
 V_{i2} &= \frac{V_{d2} + (R_{ci} - R_d) I_{dmax}}{1.35 n_i \cos \gamma_N} \\
 \text{and } V_{i3} &= \frac{(P_s / I_{dmax}) + (R_{ci} - R_d) I_{dmax}}{1.35 n_i \cos \gamma_N}
 \end{aligned} \tag{8.55}$$

we obtain $I_d = I_{dL,1}$, corresponding to the predicted value of V_i .

- For mode 2, we follow a procedure similar to that in (b) to obtain $I_d = I_{dL,2}$ corresponding to the predicted value of V_r (see Fig. 8.7b).

Following the determination of the control mode, and the corresponding dc current, the rest of the dc variables can be readily computed.

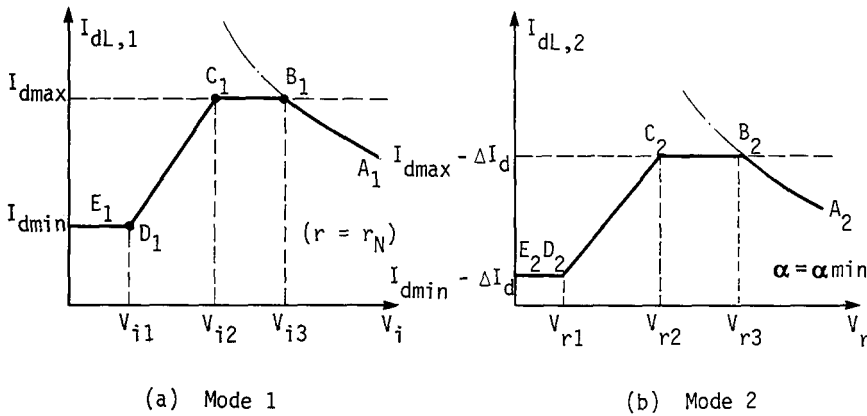


Figure 8.7 DC current for modes 1 and 2.

8.4.2 Interfacing the ac/dc Systems

A classical model is assumed for the ac power network. When the dc line is interfaced with such a system, all the nodes except the internal generator nodes and the (dc) converter ac buses (V_D) can be eliminated. The resulting reduced system equations are

$$\begin{bmatrix} Y_{GG} & Y_{GD} \\ Y_{DG} & Y_{DD} \end{bmatrix} \begin{bmatrix} E_G \\ V_D \end{bmatrix} = \begin{bmatrix} I_G \\ I_D \end{bmatrix} \quad (8.56)$$

where E_G , I_G are vectors of generator voltages and currents and V_D , I_D are the converter ac terminal voltages and currents. The subscripts G and D in the Y matrix denote machine internal buses and converter ac buses, respectively.

The ac/dc interface is calculated as follows:

Step 1: V_D at the time interval t_n (denoted by $V_{D,n}$) is first estimated from the value in the previous time interval

$$V_{D,n}^{(o)} = V_{D,n-1} \quad (8.57)$$

Step 2: The dc control mode is determined using the logical inference method (described in the previous section); the dc variables and the power injected in the ac system at the converter ac buses are thus computed.

Step 3: Treating the dc power injection as a load, the ac load flow is solved for.

Step 4: The estimate of $V_{D,n}$ is updated using steps 2 and 3, until two successive estimates are within the desired precision.

It is to be noted that step 3 involves very simple computations since $E_{G,n}$ (in equation (8.56) is known); only Y_{DD} , which is small in size, is involved in the ac load flow solution.

When the ac system is operating

$$V_D \Big|_{t_n^+} \neq V_D \Big|_{t_n^-}$$

$V_D \Big|_{t_n^+}$ is predicted from the dc open circuit voltage; i.e.,

$$V_D \Big|_{t_n^+}^{(o)} = - Y_{DD}^{-1} Y_{DG} E_G \Big|_{t_n} \quad (8.58)$$

8.4.3 Incorporation of the HVDC Model in the TEF

The effect of the two-terminal HVDC on the expression for the transient energy will be only in the calculation of the potential energy. Since the dc converter buses are retained, new terms which are related to the converter bus voltages should be included.

The new transient energy function is developed as follows. For generator j , the accelerating power in the COI coordinate is given by

$$P_{aj}(COI) = P_{mj} - P_{ej} - \frac{M_j}{M_T} P_{COI} \quad (8.59)$$

where P_m , P_e are the mechanical and electrical powers, respectively,

$$P_{COI} = \sum_{j=1}^{NR} (P_{mj} - P_{ej}), \quad (8.60)$$

M = rotor inertia, $M_T = \sum_{j=1}^{NR} M_j$, and NR = the number of rotors.

For the classical power system model, the rotor accelerating power is given by

$$P_{aj} = P_j - \sum_{\substack{k=1 \\ k \neq j}}^{NR} (C_{jk} \sin \theta_{jk} + D_{jk} \cos \theta_{jk}) - \sum_{\ell \in 2ND} (C_{j\ell} \sin \theta_{j\ell} + D_{j\ell} \cos \theta_{j\ell}) \quad (8.61)$$

where

$$P_j = P_{mj} - E_j^2 G_{jj}$$

$$C_{jk} = E_j E_k B_{jk}, \quad D_{jk} = E_j E_k G_{jk}$$

$$C_{j\ell} = E_j V_\ell B_{j\ell}, \quad D_{j\ell} = E_j V_\ell G_{j\ell}$$

$$\theta_{jk} = \theta_j - \theta_k, \quad \theta_{j\ell} = \theta_j - \theta_\ell$$

ND = number of two-terminal dc lines

$2ND$ = number of converters ($2ND = 2 \times ND$)

j, k = generator internal nodes

ℓ = converter bus

Using the stable equilibrium point (SEP) as reference, the energy function at any point along the trajectory (denoted as $*$) is defined as:

$$V \Big|_{s2}^* = \sum_{j=1}^{NR} \frac{1}{2} M_j \tilde{\omega}_j^{*2} - \sum_{j=1}^{NR} P_j (\theta_j^* - \theta_j^{s2}) + \int_{s2}^* \sum_{j=1}^{NR} \sum_{\substack{k=1 \\ k \neq j}}^{NR} (C_{jk} \sin \theta_{jk} + D_{jk} \cos \theta_{jk}) d\theta_j$$

$$+ \int_{s2}^* \sum_{j=1}^{NR} \sum_{\ell \in 2ND} (C_{j\ell} \sin \theta_{j\ell} + D_{j\ell} \cos \theta_{j\ell}) d\theta_j \quad (8.62)$$

We note that in equation (8.62), the last terms are the dc converter bus voltage-related magnetic and dissipative potential energy contributions.

Since some of the terms in (8.62) are path dependent, a linear trajectory approximation is used to evaluate these terms.

The resulting TEF expression is given below.

$$\begin{aligned} V = & \sum_{i=1}^n \frac{1}{2} M_i \tilde{\omega}_i^2 - \sum_{i=1}^n P_i (\theta_i - \theta_i^{s2}) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n C_{ij} (\cos \theta_{ij} - \cos \theta_{ij}^{s2}) \\ & + \sum_{i=1}^{n-1} \sum_{j=i+1}^n I_{ij} (\sin \theta_{ij} - \sin \theta_{ij}^{s2}) \\ & - \sum_{i=1}^n E_i (\theta_i - \theta_i^{s2}) \sum_{\ell \in 2ND} \frac{B_{i\ell}}{\theta_{i\ell} - \theta_{i\ell}^{s2}} \left[(V_\ell \cos \theta_{i\ell} - V_\ell^{s2} \cos \theta_{i\ell}^{s2}) \right. \\ & \left. - \frac{V_\ell - V_\ell^{s2}}{\theta_{i\ell} - \theta_{i\ell}^{s2}} (\sin \theta_{i\ell} - \sin \theta_{i\ell}^{s2}) \right] \\ & + \sum_{i=1}^n E_i (\theta_i - \theta_i^{s2}) \sum_{\ell \in 2ND} \frac{G_{i\ell}}{\theta_{i\ell} - \theta_{i\ell}^{s2}} \left[(V_\ell \sin \theta_{i\ell} - V_\ell^{s2} \sin \theta_{i\ell}^{s2}) \right. \\ & \left. + \frac{V_\ell - V_\ell^{s2}}{\theta_{i\ell} - \theta_{i\ell}^{s2}} (\cos \theta_{i\ell} - \cos \theta_{i\ell}^{s2}) \right] \end{aligned} \quad (8.63)$$

where

$s2$ = stable equilibrium condition

n = number of generators in ac system

ND = number of HVDC terminals

$2ND$ = number of converters

i, ℓ = generator and converter, respectively

$$I_{ij} = D_{ij} \frac{\theta_i + \theta_j - \theta_i^{s2} - \theta_j^{s2}}{\theta_{ij} - \theta_{ij}^{s2}}$$

Equilibrium conditions

Stable equilibrium conditions (θ^{s2}) and the unstable equilibrium conditions (θ^u) are obtained from

$$P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} = 0$$

$$i = 1, 2, \dots, n - 1$$

$$P_{el}^* - P_{el} = 0 \quad (8.64)$$

$$\ell \in 2ND$$

$$Q_{el}^* - Q_{el} = 0$$

where

P_{el}, Q_{el} = dc power and reactive power injections from the converter buses to the ac system calculated using bus voltages and the Y -matrix

P_{el}^*, Q_{el}^* = dc power and reactive power injections calculated from the dc part load flow

8.4.4 Procedure for Transient Stability Assessment

Transient stability assessment of an ac/dc system using TEF is accomplished by a six-step procedure similar to that of an ac system:

- Step 1: The generator rotor angles and speeds at fault-clearing time are obtained by time simulation.
- Step 2: Starting with the prefault equilibrium point, solution for the postfault stable equilibrium point (SEP) is obtained by a suitable solution technique, e.g., the Newton-Raphson method, with the same dc model as in step 1.
- Step 3: The same procedure as in the ac system TEF method is used to determine the mode of disturbance and to obtain the starting point for calculation of the UEP (see Section 6.5).
- Step 4: Similar to step 2, the controlling UEP is solved by using the starting point obtained from step 3.
- Step 5: The kinetic energy at clearing time and the potential energy margin are computed.
- Step 6: Transient stability assessment is made according to the value of the normalized potential energy margin.

8.5 INCORPORATION OF UNIFORM DAMPING IN TEF

8.5.1 Introduction

Keeping in mind that we are primarily interested in the first swing transient, the question may be raised as to what extent damping may affect the transient energy in that period. Normally, the answer would be that since damping is a function of the speed, which does not (normally) change much in the first swing, damping is not a major factor to be considered. On the other hand, as was discussed in Section 8.1, the interconnected transmission network of North America is often stressed by heavy loading. In addition, the conditions leading to the occurrence of the interarea mode of instability are becoming more common. Under these conditions, the period of the swing may be long and the transient energy level may be low enough to make the effect of the damping on the transient energy nonnegligible.

Incorporation of damping in its most general form (i.e., nonuniform damping) may be too complex to warrant its inclusion in the TEF method for practical use in power systems applications. However, as an indication of the effect of damping, the special and simplified case of uniform damping will be treated in this section. For this case, if D_i is the per unit damping and M_i is the inertia constant for generator i , then the ratio (D_i/M_i) is the same for all the generators in the system.

8.5.2 System Equations

Starting with the system equations, in the synchronous frame of reference, with the generators and the network represented classically, the system equations are (2.3) and (2.4). Using $M_i = 2H_i/\omega_R$, we get

$$M_i \dot{\omega}_i + D_i \omega_i = P_{mi} - P_{ei} \quad (8.65a)$$

$$i = 1, 2, \dots, n$$

Summing all equations (8.64) for all the generators

$$\sum_{i=1}^n M_i \dot{\omega}_i + \sum_{i=1}^n D_i \omega_i = \sum_{i=1}^n (P_{mi} - P_{ei})$$

$$\equiv P_{COI} \quad (8.65b)$$

Using $M_T = \sum_{i=1}^n M_i$, the position of the COI is defined by

$$\delta_o = \frac{1}{M_T} \sum_{i=1}^n M_i \delta_i \quad (8.66)$$

$$\dot{\delta}_o = \frac{1}{M_T} \sum_{i=1}^n M_i \dot{\delta}_i, \text{ etc.}$$

From equations (8.65), (8.66) we obtain

$$M_T \ddot{\delta}_o + \sum_{i=1}^n \frac{D_i}{M_i} \frac{d}{dt} (M_i \dot{\delta}_i) = P_{COI} \quad (8.67)$$

Using $D_i / M_i = \text{constant} = D/M$ we get the equation of motion of the COI

$$\dot{\omega}_o + \frac{D}{M} \omega_o = \frac{1}{M_T} P_{COI} \quad (8.68)$$

from (8.64) we get

$$\dot{\omega}_i + \frac{D}{M} \omega_i = \frac{1}{M_i} (P_{mi} - P_{ei}) \quad (8.69)$$

defining $\theta_i = \delta_i - \delta_o$ and $\tilde{\omega}_i = \omega_i - \omega_o$.

Subtracting (8.68) from (8.69), the equation of motion for generator i in the COI frame is obtained by

$$\dot{\tilde{\omega}}_i + \frac{D}{M} \tilde{\omega}_i = \frac{1}{M_i} (P_{mi} - P_{ei}) - \frac{1}{M_T} P_{COI}$$

$$M_i \dot{\tilde{\omega}}_i + D_i \tilde{\omega}_i = P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} \quad (8.70a)$$

$$\text{or } M_i \ddot{\tilde{\omega}}_i + D_i \tilde{\omega}_i \equiv f_i \quad (8.70b)$$

8.5.3 The Damping Energy

Following the derivation of the transient energy function in Chapter 5, we will consider that the transient energy is made up of three components: kinetic energy, potential energy, and damping energy or

$$V = V_{KE} + V_{PE} + V_{\text{damping}} \quad (8.71)$$

The kinetic and potential energy components will be the same as previously derived. We will now focus on the derivation of V_{damping} .

Multiplying the damping term in (8.70b) by $\frac{d\theta_i}{dt}$, integrating with respect to the time t , and summing over all the generators, we get

$$\begin{aligned} V_{\text{damping}} &= \sum_{i=1}^n \int D_i \tilde{\omega}_i \frac{d\theta_i}{dt} dt \\ &= \sum_{i=1}^n \int_{\theta_i^s}^{\theta_i} D_i \tilde{\omega}_i d\theta_i \end{aligned} \quad (8.72)$$

We note that the integral in (8.72) is path dependent. As is usually the case in the TEF method, we will use a suitable function to approximate the behavior of θ_i as a function of time.

8.5.4 Approximation of θ_i as a Sine Curve

We note at the outset that a straight-line approximation of θ versus t results in a

constant velocity, which would likely give a poor modeling of the contribution of damping to the transient energy. Therefore, a sine curve approximation of θ is used.

Figure 8.8 depicts the approximated behavior of θ_i as a function of time. It is assumed that θ_i follows a sine function between fault clearing and the peak of its swing.

Let

$$\bar{\theta}_i = \theta_i - \theta_i^{cl} \quad (8.73)$$

$$\tau = t - t_{cl}$$

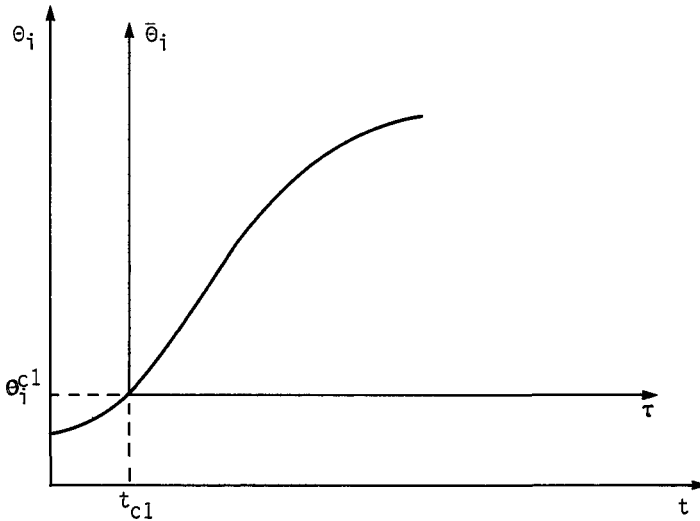


Figure 8.8 Approximation of θ_i versus t .

Then

$$d\bar{\theta}_i = d\theta_i \quad (8.74)$$

$$d\tau = dt$$

and

$$\frac{d\bar{\theta}_i}{d\tau} = \frac{d\theta_i}{dt} = \tilde{\omega}_i \quad (8.75)$$

Boundary conditions

$$1. \quad \tau = 0, \quad \bar{\theta}_i = 0, \quad \left. \frac{d\bar{\theta}_i}{d\tau} = \frac{d\theta_i}{dt} \right|_{t=c\ell} = \tilde{\omega}_i^{c\ell}$$

$$2. \quad \tau = \tau_p, \quad \bar{\theta}_i = \theta_i^p, \quad \frac{d\bar{\theta}_i}{d\tau} = 0$$

Let

$$\bar{\theta}_i = A_i \sin \alpha_i \tau \quad (8.76)$$

then

$$\frac{d\bar{\theta}_i}{d\tau} = A_i \alpha_i \cos \alpha_i \tau \quad (8.77)$$

$$\tau = 0, \quad A_i \alpha_i = \tilde{\omega}_i^{c\ell}$$

$$\tau = \tau_p, \quad A_i \alpha_i \cos \alpha_i \tau_p = 0$$

then

$$\tau_p = \pi / 2\alpha_1$$

and

$$\theta_i^p = A_i \sin \left[\alpha_i \frac{\pi}{2\alpha_i} \right] = A_i$$

Summary

$$A_i = \bar{\theta}_i^p = \theta_i^p - \theta_i^{cl}$$

$$\alpha_i = \tilde{\omega}_i^{cl} / (\theta_i^p - \theta_i^{cl}) \quad (8.78)$$

$$\tau_p = \frac{\pi}{2} \left[\frac{\theta_i^p - \theta_i^{cl}}{\tilde{\omega}_i^{cl}} \right]$$

Using (8.73 – 8.75), and substituting for θ_i in (8.72), we get

$$\begin{aligned} V_{\text{damping}} &= \sum_{i=1}^n \int_0^{\bar{\theta}_i^p} D_i \frac{d\bar{\theta}_i}{d\tau} d\bar{\theta}_i \\ &= \sum_{i=1}^n \int_0^{\bar{\theta}_i^p} D_i A_i \alpha_i \cos \alpha_i \tau d\bar{\theta}_i \\ &= \sum_{i=1}^n D_i \alpha_i \int_0^{\bar{\theta}_i^p} A_i \left[1 - \frac{\bar{\theta}_i^2}{A_i^2} \right]^{1/2} d\bar{\theta}_i \end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^n D_i \alpha_i \int_0^{\bar{\theta}_i^p} \sqrt{A_i^2 - \bar{\theta}_i^2} d\bar{\theta}_i \\
&= \sum_{i=1}^n \frac{D_i \alpha_i}{2} \left[\bar{\theta}_i \sqrt{A_i^2 - \bar{\theta}_i^2} + A_i^2 \arcsin \frac{\bar{\theta}_i}{|A_i|} \right]_0^{\bar{\theta}_i^p} \\
&= \sum_{i=1}^n \frac{1}{2} D_i \alpha_i A_i^2 \arcsin \frac{\bar{\theta}_i^p}{|A_i|} \tag{8.79}
\end{aligned}$$

where A_i and α_i are obtained from (8.78)

Example

The effect of uniform damping was investigated in the four-generator system, described in [4], for the case when the fault is cleared at 0.15 sec. Approximating θ_i by a sine curve, we obtain

$$V_{\text{damping}} = 0.07377 \text{ pu}$$

We note that for this case when the effect of damping is not included $\Delta V = 0.01486$.

8.6 OBTAINING THE CONTROLLING UEP WITH SPARSE FORMULATION

8.6.1 Introduction

The transient energy function developed in the previous chapters and discussed throughout this book is based on the so-called reduced formulation of the power networks. In that formulation the generator is represented by an EMF behind a generator reactance or impedance. The generator EMF may be treated as constant, as in the classical power

system model, or with varying quadrature- and direct-axes components, as in the higher-order generator models. The node to which the generator EMF is connected is referred to as the internal-generator node. In the reduced formulation, all the nodes other than the internal-generator nodes are eliminated.⁶ For constant impedance loads a simple Gauss elimination procedure is used, resulting in a Y -bus the elements of which include the admittance of the transmission network as well as the load admittance. When nonlinear loads are included, there will also be some current injections at the internal generator nodes (see Section 8.4).

The inherent drawbacks of the reduced formulation, often cited in the literature, are (1) the reduction process may be time consuming, (2) the resulting Y -bus is dense (i.e., all the elements are full), (3) the identity of the network nodes is lost, (4) certain load characteristics are not modeled, and (5) the resulting energy function includes path-dependent terms which are to be approximated. As has been documented in this chapter, these drawbacks have been largely overcome. However, there have been several attempts to use energy functions with nonreduced networks, i.e., with the identity of the network nodes retained. These will be referred to here as sparse formulation of the energy functions.

The efforts to generate energy functions with sparse formulation and use them in power system transient stability assessment fall into three main categories. The first category, usually listed in the literature as "structure preserving," focuses attention on the structure of the network and how it behaves during the transient. Of particular interest are the so-called critical network cutsets along which the network tends to separate if the disturbance is large enough to cause loss of synchronism by some generators. References [16, 17] are representative of this category. In [16] the generators are represented classically while the loads can be represented as functions of both voltage and frequency.

⁶ Some network nodes, which are involved in the network changes as part of the switching sequence, are also retained.

In [17] higher-order generator models are used with flux decay, i.e., with no excitation control. These energy functions, while representing significant analytical advancement in the state of the art, have yet to deal with the practical questions associated with transient stability analysis of practical power systems; the major stumbling block appears to be the determination of the critical value of the energy function against which stability assessment is made.

The second category of investigations involved the use of the so-called PEBS method to determine the critical value of the transient energy. The authors of [18, 19] have applied the PEBS method to the sparse formulation in an attempt to incorporate such modeling features as excitation control, nonlinear loads, HVDC, etc.

The third category of investigation uses the controlling UEP concept to determine the critical value of the transient energy. These investigations use the sparse formulation mainly as a means of overcoming the modeling limitation and to expedite the SEP and UEP solution procedure, in large power systems, through the use of sparsity techniques. References [20 – 22] are representative of these investigations. Their findings are discussed below.

8.6.2 Sparse Formulation of the Transient Energy Function

In the COI frame of reference, the system equations can be written in the form

$$V_\ell \sum_{j=1}^N [V_j Y_{\ell j} \cos(\gamma_{\ell j} - \phi_\ell + \phi_j)] = P_\ell \quad (8.80)$$

for all ℓ

$$V_g \sum_{j=1}^N [V_j Y_{gj} \cos(\gamma_{gj} - \phi_g + \phi_j)] - P_{eg} = P_g \quad (8.81)$$

for all g

$$\begin{aligned}
 -V_\ell \sum_{j=1}^N [V_j Y_{\ell j} \sin(\gamma_{\ell j} - \phi_\ell + \phi_j)] &= Q_\ell \\
 \text{for all } \ell &
 \end{aligned} \tag{8.82}$$

$$\begin{aligned}
 -V_g \sum_{j=1}^N [V_j Y_{gj} \sin(\gamma_{gj} - \phi_g + \phi_j)] \\
 -B_g [E_g V_g \cos(\theta_g - \phi_g) - V_g^2] &= Q_g \\
 \text{for all } g &
 \end{aligned} \tag{8.83}$$

$$\begin{aligned}
 B_g E_g V_g \sin(\theta_g - \phi_g) - P_{eg} &= 0 \\
 \text{for all } g &
 \end{aligned} \tag{8.84}$$

$$\begin{aligned}
 \frac{P_{eg}}{M_g} - \frac{P_{eg+1}}{M_{g+1}} &= \frac{P_{mg}}{M_g} - \frac{P_{mg+1}}{M_{g+1}} \\
 \text{for all } g \neq N_g &
 \end{aligned} \tag{8.85}$$

$$\sum_{g=1}^N M_g \theta_g = 0 \tag{8.86}$$

where

N = the total number of buses

N_ℓ = the number of load buses

N_g = the number of generator buses

ℓ = an index for the load buses: 1, 2, ..., N_ℓ

g = an index for the generator buses: 1, 2, ..., N_g

V_i = the magnitude of bus voltage

ϕ_i = the angle of bus voltage in COI

Y_{ij} = the magnitude of element ij of the bus admittance matrix

γ_{ij} = the angle of element ij of the bus admittance matrix

P_i = the real load power injected at bus i

Q_i = the imaginary load power injected at bus i

P_{eg} = the generator electrical output power of generator g

x'_{dg} = the direct-axis transient reactance of generator g

B_g = the reciprocal of x'_{dg}

P_{mg} = the mechanical input power of generator g

Equations (8.80 – 8.86) are nonlinear algebraic equations in the variables ϕ_ℓ , ϕ_g , V_ℓ , V_g , P_{eg} , and θ_g . They are solved for to obtain the SEP and UEP.

The transient energy function

Following a procedure similar to that used in Section 5.2, the transient energy function $V(\theta, \tilde{\omega})$ is obtained from

$$V(\theta, \tilde{\omega}) = \sum_{g=1}^{N_g} \int_{\theta_g^s}^{\theta_g} \left[M_g \dot{\tilde{\omega}}_g - \left[P_{mg} - P_{eg} - \frac{M_g}{M_T} P_{COI} \right] \right] d\theta_g \quad (8.87)$$

where

$$P_{COI} = \sum_{g=1}^{N_g} (P_{mg} - P_{eg}) \quad (8.88)$$

$\tilde{\omega}$ = the vector of the rotor speeds in the COI frame

$$\theta^{s2} = \text{the SEP}$$

Performing the integration, we get the following expression for V :

$$\begin{aligned} V(\theta, \tilde{\omega}) = & 0.5 \sum_{g=1}^N M_g \tilde{\omega}_g^2 - \sum_{g=1}^N P_{mg} (\theta_g - \theta_g^{s2}) \\ & + \sum_{g=1}^N \int_{\theta_g^{s2}}^{\theta_g} P_{eg} d\theta_g \end{aligned} \quad (8.89)$$

Again, as previously discussed in Chapter 5, the proper kinetic energy to be used is the corrected kinetic energy given by

$$V_{KE}|_{corr} = 0.5 M_{eq} \tilde{\omega}_{eq}^2 \quad (8.90)$$

where

$$M_{eq} = \frac{M_{cr} \cdot M_{sys}}{M_T}$$

$$\tilde{\omega}_{eq} = \tilde{\omega}_{cr} - \tilde{\omega}_{sys}$$

The energy margin is the value of V evaluated between $V(\theta^{cl}, \tilde{\omega}^{cl})$ and $V(\theta^u, 0)$.

Calculation of SEP, UEP

The following procedure is suggested in [20 – 22]:

1. Angles and speeds, in the COI frame, at fault clearing are determined by a suitable technique.

2. The set of equations (8.80 – 8.86) are solved to obtain the postdisturbance θ^{s2} , using the postdisturbance network bus admittance matrix.
3. Assuming the mode of disturbance is known (see Section 6.5) an initial estimate of the UEP is made as follows:
 - a. The "corner point" is calculated and corrected for the motion of the COI.
 - b. The ray point is found (i.e., the point where the ray between θ^{s2} and θ^{corner} intersects the PEBS).
 - c. The initial bus voltages are obtained by solving the following sparse nodal equations:

$$Y_{bus} \cdot V_{bus} = I_{bus} \quad (8.91)$$

where Y_{bus} is the bus admittance matrix after adding $j x'_{dg}$ to the diagonal element corresponding to the g th generator and I_{bus} is a vector of zero elements for load buses and elements E_g/jx'_{dg} (complex quantities) for generator buses at the ray point.

- d. The initial guesses of the generator output powers at the ray point are calculated using:

$$P_{eg} = \text{Re} (\bar{E}_g \cdot \bar{I}_g^*) \quad (8.92)$$

where $\bar{I}_g = [\bar{E}_g - \bar{V}_g] / jx'_{dg}$ and $*$ stands for conjugate.

4. Using the above estimates of P_{eg} and bus voltages, equations (8.80 – 8.86) are solved iteratively to obtain the UEP.
5. The energy function, or the energy margin, is computed assuming a linear trajectory approximation between θ^{cl} and θ^u :
 - a. The angles between θ^{cl} and θ^u are divided into a number of equidistant points.
 - b. At each point the network is solved for to obtain the bus voltages and the generator output powers, as done in step 3.
 - c. The integral terms in (8.89) is evaluated by the trapezoidal rule.

8.6.3 Experience with Sparse Formulation

At the time of the writing of this manuscript there is only limited experience reported in the literature on the use of sparse formulation. This experience can be summarized as follows:

1. Good transient stability assessment has been reported in [19 – 21] for systems of sizes up to about 100 generators. Furthermore, [20] reports good results with stressed, ill-conditioned systems.
2. The experience of the authors and their coworkers, some of which reported in [21], indicates that the hoped for savings in computation time for large power systems did not materialize.

8.7 GENERAL COMMENTS ON THE APPROXIMATIONS IN THE TEF METHOD AND ON THE MODELS USED

Having introduced the readers to a wide range of modeling issues, and various approximations invoked in the TEF method, a discussion on the philosophy of these issues and the authors' perspectives on these topics is in order.

Examining the history of power system transient stability analysis by direct methods indicates that work in this area followed two distinct routes: one is analytical which is system-theoretic based; the other is practical and pragmatic which uses reasoning based on simulations to achieve results. Both routes have been well traveled by researchers and engineers. Some, and the authors are among them, argue that both routes have reinforced each other. Indeed, the current state of the art could not have been reached without key contributions achieved in each route. Yet the following questions are continually raised. How valid are the results obtained using simulation based reasoning, which have yet to receive solid analytical proofs? How good are the analytical theorems which give

impractical results? and so on. The authors have been confronted with these questions often enough.⁷ We will attempt to provide an answer below.

The first point which should be made, and perhaps is not generally recognized, is how the simulation-based and analytical contributions have reinforced each other. The concept of a critical energy associated with the potential energy at a saddle point (or a UEP) is very old. The correct determination of that energy was very slow to come by. The key steps, however, came through simulation-based reasoning. Examples are that the system trajectory is nearer to a UEP different from that with the lowest energy (1975)...that the "correct" UEP is in the general direction of the disturbed system trajectory (1977)...that the controlling UEP is among several candidate UEPs in that direction, and it includes the angles of some generators which do not initially lose synchronism (1980 – 1981)...that the controlling UEP is determined by both disturbed and postdisturbance system conditions (1983)...and so on. All these ideas were initially met with considerable skepticism by some theorists and have received analytical justification later. Their validity is now generally accepted.

Examples of analytical development include the application of the invariance principle to overcome the requirement of the time derivative of the Lyapunov function being strictly negative definite and the characterization of the region of attraction in terms of the stable manifolds of the unstable equilibrium points. These two developments have provided strong justification for the transient energy function in its current form.

The second point to be made is that the contributions obtained using simulation-based reasoning are made by persons who have competence in power system analysis as well as familiarity with system theory. Those engineers have used their knowledge, i.e., of understanding power system dynamic response and knowledge of the

⁷ When confronted with questions such as these, the authors often feel that this statement, by Sir Arthur Conan Doyle in *The Adventures of Sherlock Holmes*, is appropriate: "It is a capital mistake to theorize without data. Invariably one begins to twist facts to suit theories rather than theories to suit facts."

theory, to explain the dynamic system behavior and to reconcile the theory with observation of actual system behavior as revealed by simulation. The test of course has always been meeting performance criteria established by other accepted tools of analysis for the same power system model (usually conventional time simulation programs). The authors' experience with many power systems is that this test has been met numerous times.

The third point to be addressed is that the results obtained using simulation-based reasoning involves a number of approximations: some pertain to the power system models used, others are related to the approximations in some of the transient energy terms. The "purists" point to these approximations as an inherent weakness in the simulation-based approach which may lead to serious errors. These points, which are often lumped together, are dealt with separately below.

Regarding the modeling, the authors not only defend the soundness of the approach used, they even recommend using it in many situations which do not involve direct stability analysis. All mathematical models are approximations of the actual system performance. The critical question is whether the model used captures the important phenomena which govern the system dynamic behavior in the period of interest. This principle has been, rather carefully, adhered to in the work reported upon in this book. There is not inherent redeeming value in using a more complex model when a simpler model can do the job. Also an experienced analyst knows that a simpler model for which the parameters are known with confidence is far better than a more complicated model with inaccurate data or with questionable assumptions. (It is rather surprising, but quite true, that both situations occur not infrequently.)

Finally the one remaining question to be addressed is the approximation used in the energy function expression, e.g., the linear trajectory approximation. Only with this approximation has an analytical basis for the TEF been established. How serious, and limiting, is this approximation? The answer lies in the context in which transient stability

studies are made in the industry. Limiting conditions are established for planning new facilities or operating an existing power system. When these limits are found, a margin of safety is used to guard against uncertainties in the method used to determine these limits: in the model, in the scenario, or in the tool of analysis used. Is there a good reason to suspect that the approximation used in the TEF can take us outside this margin of safety? The authors' (extensive) experience to date indicates that the answer to this question is no.

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Chapter 9

ENERGY MARGIN SENSITIVITY TO CHANGES IN SYSTEM CONDITIONS

9.1 INTRODUCTION

In transient stability analysis of power systems the following question is often raised: given the stability results for a set of initial system conditions and disturbance scenario, how would these results be affected by a change in a certain key parameter or an initial system condition? The answer to this question is at the heart of many transient stability studies. To illustrate this point, let us assume that the analyst is seeking the transient stability limit for the loading of a particular power plant (1) by the conventional time simulation technique and (2) by using the TEF method. If the former method is used, the analyst will select a convenient indicator of the severity of the disturbance, e.g., magnitude of the swing of a critical generator. Guided by this indicator, the analyst will conduct a series of transient stability runs until the limit is "bracketed" by two cases: a critically stable and a critically unstable case.

If the TEF method is used, we recall that an inherent advantage of the TEF method is the availability of a qualitative measure of the degree of stability (or instability) in terms of the transient energy margin (ΔV). This was considered an important step toward reducing the number of transient stability runs needed to obtain transient stability information. It is likely, therefore, that a fewer number of TEF runs will be needed to obtain the limit. The information, when displayed, may look like that shown in Fig. 9.1, where it is assumed that four TEF runs were made to determine the limit, which corresponds to the plant loading for $\Delta V = 0$.

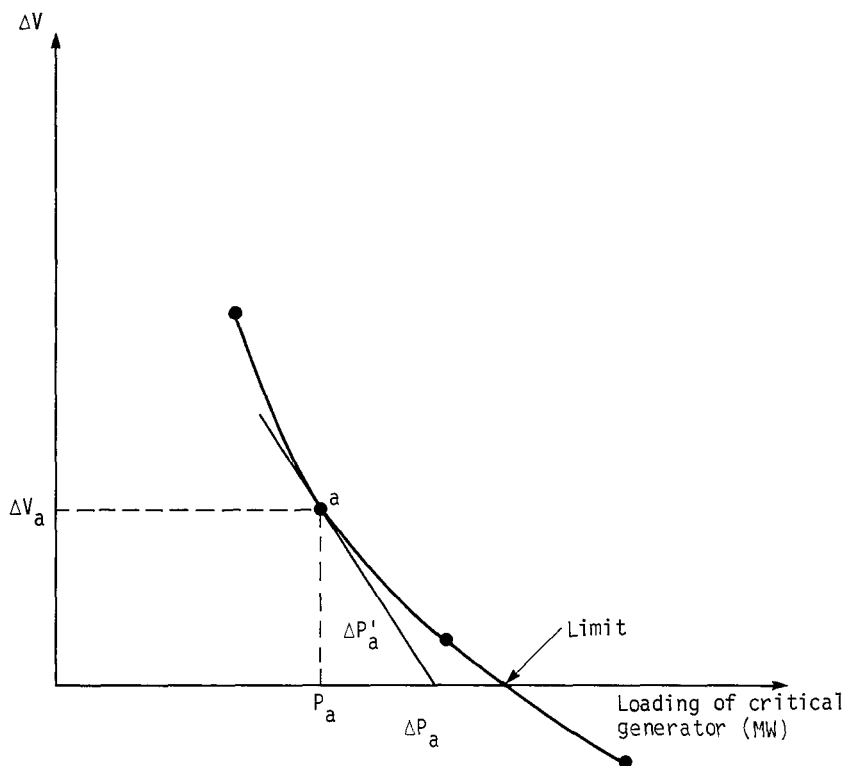


Figure 9.1 *Determination of stability-limited loading of a plant by successive use of the TEF method.*

The use of the sensitivity techniques in the TEF method is illustrated in Fig. 9.1. If the TEF run is made for conditions shown as point a , and if the sensitivity of ΔV_a to changes in P_a can be obtained with the desired accuracy, the limit can be determined quickly from only one stability run. This feature considerably increases the versatility of the TEF method.

Figure 9.1 also shows the limitation of linearized sensitivity techniques [1–3]. Again, if the system conditions initially analyzed are shown as point a in Fig. 9.1, linear techniques give the limit as $(P_a + \Delta P'_a)$, while the true stability-limited plant loading is

$(P_a + \Delta P_a)$ as shown. These techniques require analytically derived sensitivity coefficients and may require the inclusion of higher-order terms.

Analytically derived sensitivity techniques for power system transient stability analysis are based on the techniques used with nonlinear systems, discussed in [4–6]. For example, [4] proposed an analytical approach to energy margin sensitivity determination. In this approach, the transfer conductances were neglected, and only self-clearing faults were considered. The technique discussed here is based on the authors' work discussed in [7].

9.2 CALCULATION OF SENSITIVITY FACTORS

The Energy Margin

For the classical power system model, with n machines, the equations of motion of the synchronous generators written with respect to the center of inertia (COI) reference frame are given by

$$\begin{aligned} M_i \ddot{\omega}_i &= P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI} \\ \dot{\theta}_i &= \tilde{\omega}_i \quad i = 1, 2, \dots, n \end{aligned} \tag{9.1}$$

where

$$\begin{aligned} P_{COI} &= \sum_{i=1}^n \left[P_{mi} - P_{ei} \right] \\ M_T &= \sum_{i=1}^n M_i \\ C_{ij} &= E_i E_j B_{ij}, \quad D_{ij} = E_i E_j G_{ij} \end{aligned}$$

and, for generator i

$$P_{mi} = \text{mechanical power input}$$

G_{ii} = driving point conductance

E_i = constant voltage behind direct axis transient reactance

$\tilde{\omega}_i, \theta_i$ = generator rotor speed and angle, respectively, with respect to the COI

M_i = moment of inertia

$B_{ij}, (G_{ij})$ = transfer susceptance (conductance) in the reduced bus admittance matrix

For the system governed by equation (9.1), the transient energy margin is given by

$$\begin{aligned} \Delta V = & -\frac{1}{2} M_{eq} \tilde{\omega}_{eq}^{cl^2} - \sum_{i=1}^n P_i \left[\theta_i^u - \theta_i^{cl} \right] \\ & - \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[C_{ij} \left[\cos \theta_{ij}^u - \cos \theta_{ij}^{cl} \right] - \beta_{ij} D_{ij} \left[\sin \theta_{ij}^u - \sin \theta_{ij}^{cl} \right] \right] \end{aligned} \quad (9.2)$$

where

$$\beta_{ij} = \frac{\theta_i^u + \theta_j^u - \theta_i^{cl} - \theta_j^{cl}}{\theta_{ij}^u - \theta_{ij}^{cl}}$$

θ^{cl} = rotor angle positions at the end of the disturbance

θ^u = controlling unstable equilibrium point (UEP)

$$M_{eq} = M_{cr} M_{sys} / \left[M_{cr} + M_{sys} \right]$$

$$\tilde{\omega}_{eq}^{cl} = \left[\tilde{\omega}_{cr}^{cl} - \tilde{\omega}_{sys}^{cl} \right]$$

$$P_i = P_{m_i} - E_i^2 G_{ii}$$

M_{cr}, M_{sys} = inertia constants of the critical generators (advanced generators in the controlling UEP) and the rest of the generators, respectively

$\tilde{\omega}_{cr}^{cl}, \tilde{\omega}_{sys}^{cl}$ = speed of inertial centers of the critical generators and the rest of the generators, respectively, at the end of the disturbance

In equation (9.2), the parameters P_i , C_{ij} , and D_{ij} are all evaluated with respect to the postdisturbance network.

From equation (9.2), assuming that the controlling UEP does not change, it is seen that the energy margin ΔV is a function of several variables which can be expressed as

$$\Delta V = \Delta V \left[\theta^{cl}, \tilde{\omega}^{cl}, \theta^u, \left[P_i, i = 1, 2, \dots, n \right], \right. \\ \left. \left[E_i E_j B_{ij}, E_i E_j G_{ij}, i, j = 1, 2, \dots, n \right] \right] \quad (9.3)$$

9.2.1 Functional Dependence on System Parameters

The effect of the change of a given system parameter α_k on the energy margin (ΔV) is obtained from the partial derivatives of (ΔV) with respect to the system parameters, since in general a change in a given parameter introduces other changes in the system conditions. The process is illustrated below (see [5, 6] for a detailed treatment of the subject):

Let $\Delta V = \Delta V(\alpha_1, \alpha_2, \dots, \alpha_N)$, where the α 's are the relevant system parameters of interest.

In general a change in ΔV is given by

$$\Delta(\Delta V) \simeq \sum_{k=1}^N \frac{\partial(\Delta V)}{\partial \alpha_k} \Delta \alpha_k + \frac{1}{2} \sum_{j=1}^N \sum_{k=1}^N \frac{\partial^2(\Delta V)}{\partial \alpha_j \partial \alpha_k} \Delta \alpha_j \Delta \alpha_k \\ + \text{higher-order terms} \quad (9.4)$$

The partial derivatives $\frac{\partial(\Delta V)}{\partial \alpha_k}$ must be strictly defined in terms of limit operations such as

$$\frac{\partial(\Delta V)}{\partial \alpha_k} = \lim_{\Delta \alpha_k \rightarrow 0} \frac{\Delta V(\alpha_k + \Delta \alpha_k) - \Delta V(\alpha_k)}{\Delta \alpha_k}$$

and

$$\frac{\partial(\Delta V)}{\partial \alpha_j \partial \alpha_k} = \lim_{\Delta \alpha_j \rightarrow 0} \lim_{\Delta \alpha_k \rightarrow 0} \frac{\Delta V(\Delta \alpha_j) - \Delta V(\Delta \alpha_j) - \Delta V(\Delta \alpha_k) + \Delta V(\alpha_k)}{\Delta \alpha_j \Delta \alpha_k} \quad (9.5)$$

where

$$\Delta V(\Delta \alpha_j) = \Delta V(\Delta \alpha_j, \Delta \alpha_k, \alpha_k) \quad (9.6)$$

9.3 FIRST-ORDER SENSITIVITY OF ΔV

By using only the first term in (9.4), the first-order sensitivity factor of the energy margin to a change in any parameter α_k is given by the partial derivative of ΔV with respect to α_k . Applying the chain rule of differentiation, this can be obtained as

$$\begin{aligned} \frac{\partial(\Delta V)}{\partial \alpha_k} = & -M_{eq} \tilde{\omega}_{eq}^{cl} \cdot \frac{\partial[\omega_{eq}^{cl}]}{\partial \alpha_k} - \sum_{i=1}^n \left[P_i \left[\frac{\partial \theta_i^u}{\partial \alpha_k} - \frac{\partial \theta_i^{cl}}{\partial \alpha_k} \right] \right. \\ & + \left[\theta_i^u - \theta_i^{cl} \right] \frac{\partial P_{mi}}{\partial \alpha_k} - \left[\theta_i^u - \theta_i^{cl} \right] \left[E_i^2 \frac{\partial G_{ii}}{\partial \alpha_k} + 2 E_i G_{ii} \frac{\partial E_i}{\partial \alpha_k} \right] \\ & - \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[\left[\cos \theta_{ij}^u - \cos \theta_{ij}^{cl} \right] \frac{\partial C_{ij}}{\partial \alpha_k} - \beta_{ij} \left[\sin \theta_{ij}^u - \sin \theta_{ij}^{cl} \right] \frac{\partial D_{ij}}{\partial \alpha_k} \right] \end{aligned}$$

$$\begin{aligned}
& + \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[\left[\beta_{ij} D_{ij} \cos \theta_{ij}^u + C_{ij} \sin \theta_{ij}^u \right] \left[\frac{\partial \theta_i^u}{\partial \alpha_k} - \frac{\partial \theta_j^u}{\partial \alpha_k} \right] \right. \\
& \quad \left. - \left[\beta_{ij} D_{ij} \cos \theta_{ij}^{cl} + C_{ij} \sin \theta_{ij}^{cl} \right] \left[\frac{\partial \theta_i^{cl}}{\partial \alpha_k} - \frac{\partial \theta_j^{cl}}{\partial \alpha_k} \right] \right] \\
& + \sum_{i=1}^{n-1} \sum_{j=i+1}^n D_{ij} \left[\sin \theta_{ij}^u - \sin \theta_{ij}^{cl} \right] \left[\gamma_{ij}^a \left[\frac{\partial \theta_i^u}{\partial \alpha_k} - \frac{\partial \theta_i^{cl}}{\partial \alpha_k} \right] + \gamma_{ij}^b \left[\frac{\partial \theta_j^u}{\partial \alpha_k} - \frac{\partial \theta_j^{cl}}{\partial \alpha_k} \right] \right]
\end{aligned} \tag{9.7}$$

where

$$\begin{aligned}
\gamma_{ij}^a &= \frac{-2 \left[\theta_j^u - \theta_j^{cl} \right]}{\left[\theta_{ij}^u - \theta_{ij}^{cl} \right]^2} \quad \gamma_{ij}^b = \frac{2 \left[\theta_i^u - \theta_i^{cl} \right]}{\left[\theta_{ij}^u - \theta_{ij}^{cl} \right]^2} \\
\frac{\partial C_{ij}}{\partial \alpha_k} &= E_i E_j \frac{\partial B_{ij}}{\partial \alpha_k} + E_i B_{ij} \frac{\partial E_j}{\partial \alpha_k} + E_j B_{ij} \frac{\partial E_i}{\partial \alpha_k} \\
\frac{\partial D_{ij}}{\partial \alpha_k} &= E_i E_j \frac{\partial G_{ij}}{\partial \alpha_k} + E_i G_{ij} \frac{\partial E_j}{\partial \alpha_k} + E_j G_{ij} \frac{\partial E_i}{\partial \alpha_k}
\end{aligned} \tag{9.8}$$

In determining the energy margin sensitivity via (9.3), the sensitivity of a number of dependent variables to the parameter α_k also needs to be determined. These include:

$$\frac{\partial \theta_i^{cl}}{\partial \alpha_k}, \quad \frac{\partial \tilde{\omega}_i^{cl}}{\partial \alpha_k}, \quad \text{and} \quad \frac{\partial \theta_i^u}{\partial \alpha_k} \quad i = 1, 2, \dots, n$$

The sensitivity of the conditions at the end of the disturbance to the change in parameters α_k is obtained using dynamic sensitivity equations [5, 6]. These are computed

by considering the system dynamic equations [5] for the *disturbed* system and considering partial derivatives as follows:

$$\begin{aligned}
 M_i \frac{\partial \tilde{\omega}_i}{\partial \alpha_k} &= \frac{\partial P_{mi}}{\partial \alpha_k} - 2E_i \frac{\partial E_i}{\partial \alpha_k} G_{ii}^f - E_i^2 \frac{\partial G_{ii}^f}{\partial \alpha_k} \\
 &\quad - \sum_{\substack{j=1 \\ j \neq i}}^n \left[G_{ij}^f \cos \theta_{ij} \left[\frac{\partial \theta_i}{\partial \alpha_k} - \frac{\partial \theta_j}{\partial \alpha_k} \right] - D_{ij}^f \sin \theta_{ij} \left[\frac{\partial \theta_i}{\partial \alpha_k} - \frac{\partial \theta_j}{\partial \alpha_k} \right] \right] \\
 &\quad - \sum_{\substack{j=1 \\ j \neq i}}^n \left[\left[\frac{\partial E_i}{\partial \alpha_k} E_j + \frac{\partial E_j}{\partial \alpha_k} E_i \right] \left[B_{ij}^f \sin \theta_{ij} + G_{ij}^f \cos \theta_{ij} \right] \right] \\
 &\quad - \sum_{\substack{j=1 \\ j \neq i}}^n \left[E_i E_j \sin \theta_{ij} \frac{\partial B_{ij}^f}{\partial \alpha_k} + E_i E_j \cos \theta_{ij} \frac{\partial G_{ij}^f}{\partial \alpha_k} \right] \\
 &\quad - \frac{M_i}{M_T} \left[\sum_{j=1}^n \left[\frac{\partial P_{mj}}{\partial \alpha_k} - 2E_j \frac{\partial E_j}{\partial \alpha_k} G_{jj}^f - E_j^2 \frac{\partial G_{jj}^f}{\partial \alpha_k} \right] \right. \\
 &\quad \left. + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \left[D_{ij}^f \sin \theta_{ij} \left[\frac{\partial \theta_i}{\partial \alpha_k} - \frac{\partial \theta_j}{\partial \alpha_k} \right] \right. \right. \\
 &\quad \left. \left. - \left[\frac{\partial E_i}{\partial \alpha_k} E_j + \frac{\partial E_j}{\partial \alpha_k} E_i \right] G_{ij}^f \cos \theta_{ij} - E_i E_j \cos \theta_{ij} \frac{\partial G_{ij}^f}{\partial \alpha_k} \right] \right] \\
 \frac{\partial \theta_i}{\partial \alpha_k} &= \frac{\partial \tilde{\omega}_i}{\partial \alpha_k} \quad i = 1, 2, \dots, n-1
 \end{aligned} \tag{9.9}$$

This is a system of linear differential equations which can be integrated knowing the values of θ_i and $\tilde{\omega}_i$, $i = 1, \dots, n$, at each time step which are available from the procedure used to determine the conditions at the end of the disturbance in the base case. Thus

$$\frac{\partial \theta_i^{cl}}{\partial \alpha_k}, \quad \frac{\partial \tilde{\omega}_i^{cl}}{\partial \alpha_k} \quad i = 1, 2, \dots, n$$

can be determined.

The sensitivity of the controlling UEP to the change in parameter α_k is determined as follows. Consider the dynamic sensitivity equations corresponding to (9.9) for the postdisturbance network. At the UEP, the derivatives of the variables will be zero, resulting in a set of linear algebraic equations given by:

$$\begin{aligned} 0 = & \frac{\partial P_{mi}}{\partial \alpha_k} - 2E_i \frac{\partial E_i}{\partial \alpha_k} G_{ii} - E_i^2 \frac{\partial G_{ii}}{\partial \alpha_k} \\ & - \sum_{\substack{j=1 \\ j \neq i}}^n \left[C_{ij} \cos \theta_{ij}^u \left[\frac{\partial \theta_i^u}{\partial \alpha_k} - \frac{\partial \theta_j^u}{\partial \alpha_k} \right] - D_{ij} \sin \theta_{ij}^u \left[\frac{\partial \theta_i^u}{\partial \alpha_k} - \frac{\partial \theta_j^u}{\partial \alpha_k} \right] \right] \\ & - \sum_{\substack{j=1 \\ j \neq i}}^n \left[\left[\frac{\partial E_i}{\partial \alpha_k} E_j + \frac{\partial E_j}{\partial \alpha_k} E_i \right] \left[B_{ij}^f \sin \theta_{ij} + G_{ij}^f \cos \theta_{ij} \right] \right] \\ & - \sum_{\substack{j=1 \\ j \neq i}}^n \left[\left[E_i E_j \sin \theta_{ij}^u \frac{\partial B_{ij}}{\partial \alpha_k} + E_i E_j \cos \theta_{ij}^u \frac{\partial G_{ij}}{\partial \alpha_k} \right] \right] \\ & - \frac{M_i}{M_T} \left[\sum_{j=1}^n \left[\frac{\partial P_{mj}}{\partial \alpha_k} - 2E_j \frac{\partial E_j}{\partial \alpha_k} G_{jj} - E_j^2 \frac{\partial G_{jj}}{\partial \alpha_k} \right] \right] \end{aligned}$$

$$\begin{aligned}
& + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \left[D_{ij} \sin \theta_{ij}^u \left[\frac{\partial \theta_i^u}{\partial \alpha_k} - \frac{\partial \theta_j^u}{\partial \alpha_k} \right] - \left[\frac{\partial E_i}{\partial \alpha_k} E_j + \frac{\partial E_j}{\partial \alpha_k} E_i \right] G_{ij} \cos \theta_{ij}^u \right. \\
& \left. - E_i E_j \cos \theta_{ij}^u \frac{\partial G_{ij}}{\partial \alpha_k} \right] \Bigg\} \quad i = 1, 2, \dots, n-1 \quad (9.10)
\end{aligned}$$

Equation (9.10) is solved knowing θ^u to obtain

$$\frac{\partial \theta_i^u}{\partial \alpha_k}, \quad i = 1, 2, \dots, n$$

It is to be noted that in the COI frame of reference only $(n-1)$ of the variables

$$\frac{\partial \theta_i^{cl}}{\partial \alpha_k}, \quad \frac{\partial \theta_i^u}{\partial \alpha_k}, \quad \text{and} \quad \frac{\partial \tilde{\omega}_i^{cl}}{\partial \alpha_k} \quad \text{are independent.}$$

Equations similar to (9.10) are also solved with the predisturbance network parameters to obtain changes in the predisturbance operating point due to a change in parameter α_k . This is then used to obtain initial conditions for the system of equations (9.9)

$$\text{The terms } \frac{\partial P_{mi}}{\partial \alpha_k}, \quad i = 1, 2, \dots, n \text{ are calculated if the change in parameter } \alpha_k \text{ affects}$$

the mechanical power input of the machine.

$$\text{The terms } \frac{\partial E_i}{\partial \alpha_k}, \quad i = 1, 2, \dots, n \text{ are evaluated by determining the effect of } \alpha_k \text{ on the}$$

predisturbance terminal voltage, V_i of each generator. Knowing the complex power

generation at each generator bus and the transient reactance, the change in internal voltage E_i is determined.

To determine the terms $\frac{\partial B_{ij}}{\partial \alpha_k}$ and $\frac{\partial G_{ij}}{\partial \alpha_k}$, a procedure based on the Householder method (see Section 4 of [8]) is used to estimate the changes in Y_{BUS} parameters due to changes in α_k . This completes the determination of all parameters required to evaluate the energy margin sensitivity.

There could be a change in more than one parameter, e.g., α_k , $k = 1, 2, \dots, m$. In this case the sensitivities to all the parameters are derived.

In arriving at the change in energy margin due to changes in parameters, the following expression is used

$$\Delta (\Delta V) = \sum_{k=1}^m \frac{\partial (\Delta V)}{\partial \alpha_k} (\Delta \alpha_k) \quad (9.11a)$$

In arriving at (9.11a) it is assumed that only first-order changes of ΔV with respect to the parameters α_k are considered.

We note that the expression for $\frac{\partial (\Delta V)}{\partial \alpha_k}$ may include several terms. If ΔV is a function of the parameters $(x_1, x_2, \dots, x_m)$, which are functions of α_k , then

$$\frac{\partial (\Delta V)}{\partial \alpha_k} = \frac{\partial (\Delta V)}{\partial x_1} \frac{dx_1}{d\alpha_k} + \frac{\partial (\Delta V)}{\partial x_2} \frac{dx_2}{d\alpha_k} + \dots + \frac{\partial (\Delta V)}{\partial x_m} \frac{dx_m}{d\alpha_k} \quad (9.11b)$$

9.4 SECOND-ORDER SENSITIVITY OF ΔV

To evaluate the second-order term in (9.4), two approaches have been used (1) numerical and (2) analytical.

9.4.1 Numerical Method [9]

This technique simply adds a second-order correction term to the energy margin sensitivity. This term is determined numerically in three steps:

1. Assume a small change in the parameter α , e.g., from α_B to $(\alpha_B + \Delta\alpha)$.
2. For each of the variables in equation (9.2), calculate the new value of that variable at $(\alpha_B + \Delta\alpha)$.

For example, $\Delta\theta^u$ due to the assumed change $\Delta\alpha$ is evaluated as follows:

1. Using linear sensitivity expression, $\Delta\theta_o^u$ is computed:

$$\Delta\theta_o^u = \frac{\partial\theta^u}{\partial\alpha} \Delta\alpha$$

2. The mismatch function $\Delta f_i = P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI}$ is computed and its magnitude is checked for convergence (i.e., $|\Delta f_i| < \epsilon$). If convergence is not achieved, a correction is introduced in step 3.
3. Expressing the sensitivity of θ^u as

$$[A] \frac{\partial\theta^u}{\partial\alpha} = \mathbf{b} \quad (9.12)$$

where $[A]$ is a $n \times n$ matrix, $\mathbf{b}$ is a $n \times 1$ vector, and

$$\frac{\partial\theta^u}{\partial\alpha} = \left[\frac{\partial\theta_1^u}{\partial\alpha}, \dots, \frac{\partial\theta_n^u}{\partial\alpha} \right]^T$$

$$\text{then } \Delta\theta_k^u = \Delta\theta_{k-1}^u + [A]^{-1} \alpha f \quad (9.13)$$

where $[A]$ is Jacobian matrix in equation (9.12)

$$\Delta f = [f_1, \dots, f_n]^T. \text{ Return to step 2.}$$

4. After the value of each variable in equation (9.2) at $\alpha = \alpha_B + \Delta\alpha$ is obtained in step 2, the value of $\Delta V \Big|_{\alpha = \alpha_B + \Delta\alpha}$ can be evaluated by equation (9.2). Then the second-order term coefficient $\frac{\partial^2(\Delta V)}{\partial \alpha^2}$ in equation (9.4) can be evaluated by the curve approximation technique.

9.4.2 Analytical Method [10]

Computation of second order sensitivity of ΔV involves the evaluation of the second-order terms in (9.4). Let $\Delta V = \Delta V(\alpha_1, \alpha_2, \dots, \alpha_N)$. The second-order sensitivity terms are of the form:

$$\frac{1}{2} \sum_{j=1}^N \sum_{k=1}^N \frac{\partial^2(\Delta V)}{\partial \alpha_j \partial \alpha_k} \Delta \alpha_j \Delta \alpha_k$$

The procedure for computing the second-order sensitivity terms is illustrated for the case of computing the sensitivity to generation changes [10].

The second-order analytic sensitivity formulation for generation changes is based on the following assumptions:

- The second-order changes in internal voltages with respect to generation shifts are neglected.
- The critical generators at which generation is shifted are known.
- The mode of disturbance does not change when generation is shifted among the generators.
- The total system load is constant, thus shift of generation at the critical machines is countered by an equal and opposite shift at the reference machine.
- The amount of generation that can be shifted at each machine is constrained by the equipment and machine ratings.

In addition, to avoid numerical computational problems associated with the conductance terms in (9.2), the $\sin \theta_{ij}$ terms are replaced with a series expansion. The second-order sensitivity expressions, with respect to a change in the generation P_{mk} , are derived in [10]. A summary of these expressions is given below.

The second-order sensitivity of the *kinetic energy* component of the energy margin (ΔV^{KE}) with respect to generation shift is given by

$$\frac{\partial^2 (\Delta V^{KE})}{\partial P_{mk} \partial P_{m\ell}} = -M_{eq} \left[\frac{\partial \tilde{\omega}_{eq}^{cl}}{\partial P_{mk}} \frac{\partial \tilde{\omega}_{eq}^{cl}}{\partial P_{m\ell}} + \tilde{\omega}_{eq}^{cl} \frac{\partial^2 \tilde{\omega}_{eq}^{cl}}{\partial P_{m\ell} \partial P_{mk}} \right] \quad (9.14)$$

In equation (9.14) the first- and second-order clearing speed sensitivities are given as follows:

$$\frac{\partial \tilde{\omega}_{eq}^{cl}}{\partial P_{mk}} = i_{cr,k}^{cl} - i_{sys,k}^{cl}$$

$$i_{cr,k}^{cl} = \frac{1}{M_{cr}} \sum_{i \notin cr} M_i \frac{\partial \tilde{\omega}_i^{cl}}{\partial P_{mk}}$$

$$i_{sys,k}^{cl} = \frac{1}{(M_T - M_{cr})} \sum_{i \in cr} M_i \frac{\partial \tilde{\omega}_i^{cl}}{\partial P_{mk}}$$

$$\frac{\partial \tilde{\omega}_i^{cl}}{\partial P_{mk}} = \frac{d}{dt} \frac{\partial \theta_i^{cl}}{\partial P_{mk}} = \lim_{\Delta P_{mk} \rightarrow 0} \frac{\tilde{\omega}_i^{cl}(P_{mk} + \Delta P_{mk}) - \tilde{\omega}_i^{cl}(P_{mk})}{\Delta P_{mk}}$$

where

$\tilde{\omega}_i^{cl}(P_{mk} + \Delta P_{mk})$ = i th machine angular speed at fault clearing for the perturbed power flow

$\tilde{\omega}_i^{cl}(P_{mk})$ = i th machine angular speed at fault clearing for the base power flow case

$$\frac{\partial^2 \tilde{\omega}_{eq}^{cl}}{\partial P_{m\ell} \partial P_{mk}} = \dot{u}_{cr,kl}^{cl} - \dot{u}_{sys,kl}^{cl}$$

$$\dot{u}_{cr,kl}^{cl} = \frac{1}{M_{cr}} \sum_{i \in cr} M_i \frac{\partial^2 \tilde{\omega}_i^{cl}}{\partial P_{m\ell} \partial P_{mk}}$$

$$\dot{u}_{sys,kl}^{cl} = \frac{1}{(M_T - M_{cr})} \sum_{i \notin cr} M_i \frac{\partial^2 \tilde{\omega}_i^{cl}}{\partial P_{m\ell} \partial P_{mk}}$$

The second-order partial derivative of the *clearing speed* is given by

$$\frac{\partial^2 \tilde{\omega}_i^{cl}}{\partial P_{m\ell} \partial P_{mk}} = \frac{d}{dt} \frac{\partial^2 \theta_i^{cl}}{\partial P_{m\ell} \partial P_{mk}} = \lim_{\Delta P_{m\ell} \rightarrow 0} \lim_{\Delta P_{mk} \rightarrow 0} \quad (9.16)$$

$$\frac{\tilde{\omega}_i^{cl}(\Delta P_{mk}, \Delta P_{m\ell}) - \tilde{\omega}_i^{cl}(\Delta P_{m\ell}) - \tilde{\omega}_i^{cl}(\Delta P_{mk}) + \tilde{\omega}_i^{cl}(P_{mk})}{\Delta P_{m\ell} \Delta P_{mk}}$$

The second-order sensitivity of the *positional energy* component (ΔV^{PE}) of the energy margin ΔV with respect to generation shift at the ℓ th machine can be found by taking the second partial of those terms which contribute toward the rotor positional energy.

$$\begin{aligned}
\frac{\partial^2 (\Delta V^{PE})}{\partial P_{m\ell} \partial P_{mk}} = & - \sum_{i=1}^n \left\{ P_i \left[\frac{\partial^2 \theta_i^u}{\partial P_{m\ell} \partial P_{mk}} - \frac{\partial^2 \theta_i^{c\ell}}{\partial P_{m\ell} \partial P_{mk}} \right] \right. \\
& + \frac{\partial P_{mi}}{\partial P_{m\ell}} \left[u_{ik}^u - u_{ik}^{c\ell} \right] + \frac{\partial P_{mi}}{\partial P_{mk}} \left[u_{i\ell}^u - u_{i\ell}^{c\ell} \right] \\
& - 2E_i G_{ii} \left[\frac{\partial |E_i|}{\partial P_{mk}} \left[u_{i\ell}^u - u_{i\ell}^{c\ell} \right] + \frac{\partial |E_i|}{\partial P_{m\ell}} \left[u_{ik}^u - u_{ik}^{c\ell} \right] \right] \\
& \left. + \frac{\partial^2 P_{mi}}{\partial P_{m\ell} \partial P_{mk}} \left[\theta_i^u - \theta_i^{c\ell} \right] \right\} \quad (9.17)
\end{aligned}$$

and the UEP sensitivity coefficients are given by

$$u_{ik}^u = \frac{\partial \theta_i^u}{\partial P_{mk}} = \lim_{\Delta P_{mk} \rightarrow 0} \frac{\theta_i^u(P_{mk} + \Delta P_{mk}) - \theta_i^u(P_{mk})}{\Delta P_{mk}} \quad (9.18)$$

where

$\theta_i^u(P_{mk} + \Delta P_{mk})$ = i th machine angular position at the UEP for the perturbed power flow

$\theta_i^u(P_{mk})$ = i th machine angular speed at the UEP for the base power flow case

The second-order sensitivity of the *magnetic energy* component (ΔV^{mag}) of the energy margin with respect to generation shift at the ℓ th machine is given by

$$\begin{aligned}
\frac{\partial^2 (\Delta V^{mag})}{\partial P_{m\ell} \partial P_{mk}} &= \sum_{i=1}^n \sum_{j=i+1}^n - \frac{\partial^2 C_{ij}}{\partial P_{m\ell} \partial P_{mk}} \left[\cos \theta_{ij}^u - \cos \theta_{ij}^{cl} \right] \\
&\quad - \frac{\partial C_{ij}}{\partial P_{mk}} \left[\sin \theta_{ij}^u \left[u_{i\ell}^{cl} - u_{jk}^{cl} \right] - \sin \theta_{ij}^u \left[u_{i\ell}^u - u_{jl}^u \right] \right] \\
&\quad + C_{ij} \cos \theta_{ij}^u \left[u_{ik}^u - u_{jk}^u \right] \left[u_{i\ell}^u - u_{jl}^u \right] \\
&\quad - C_{ij} \cos \theta_{ij}^{cl} \left[u_{ik}^{cl} - u_{jk}^{cl} \right] \left[u_{i\ell}^{cl} - u_{jl}^{cl} \right] \\
&\quad + \frac{\partial C_{ij}}{\partial P_{m\ell}} \left[\sin \theta_{ij}^u \left[u_{ik}^u - u_{jk}^u \right] - \sin \theta_{ij}^{cl} \left[u_{ik}^{cl} - u_{jk}^{cl} \right] \right] \\
&\quad + C_{ij} \left[\sin \theta_{ij}^u \left[\hat{u}_{ik\ell}^u - \hat{u}_{jkl}^u \right] - \sin \theta_{ij}^{cl} \left[\hat{u}_{ik}^{cl} - \hat{u}_{jk}^{cl} \right] \right]
\end{aligned} \tag{9.19}$$

where

$$\begin{aligned}
u_{ik}^{cl} &= \frac{\partial \theta_i^{cl}}{\partial P_{mk}} \\
\hat{u}_{ik\ell}^{cl} &= \frac{\partial^2 \theta_i^{cl}}{\partial P_{m\ell} \partial P_{mk}} \\
\frac{\partial C_{ij}^{pf}}{\partial P_{mk}} &= B_{ij}^{pf} \left[|E_j| \frac{\partial |E_i|}{\partial P_{mk}} + |E_i| \frac{\partial |E_j|}{\partial P_{mk}} \right] \\
\frac{\partial^2 C_{ij}^{pf}}{\partial P_{m\ell} \partial P_{mk}} &= B_{ij}^{pf} \left[\frac{\partial |E_j|}{\partial P_{mk}} \frac{\partial |E_i|}{\partial P_{m\ell}} + \frac{\partial |E_i|}{\partial P_{mk}} \frac{\partial |E_j|}{\partial P_{m\ell}} \right]
\end{aligned} \tag{9.20}$$

The second-order sensitivity of the *dissipation energy* component (ΔV^{dissp}) of the energy margin is given as follows:

$$\begin{aligned}
 \frac{\partial^2 (V^{dissp})}{\partial P_{mk} \partial P_{ml}} = & \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[\left[\frac{\partial^2 D_{ij}}{\partial P_{mk} \partial P_{ml}} \alpha \beta \cos \gamma + \frac{\partial D_{ij}}{\partial P_{mk}} \alpha \frac{\partial \beta}{\partial P_{ml}} \cos \gamma \right. \right. \\
 & \left. \left. - \frac{1}{2} \frac{\partial D_{ij}}{\partial P_{mk}} \alpha \beta \sin \gamma \frac{\partial \beta'}{\partial P_{ml}} + \frac{\partial D_{ij}}{\partial P_{mk}} \beta \cos \gamma \frac{\partial \alpha}{\partial P_{ml}} \right] \right. \\
 & + \left[\frac{\partial D_{ij}}{\partial P_{ml}} \frac{\partial \beta}{\partial P_{mk}} \alpha \cos \gamma + D_{ij} \alpha \frac{\partial^2 \beta}{\partial P_{mk} \partial P_{ml}} \cos \gamma \right. \\
 & \left. + D_{ij} \frac{\partial \alpha}{\partial P_{ml}} \frac{\partial \beta}{\partial P_{mk}} \cos \gamma \right] - \left[\frac{1}{2} \frac{\partial D_{ij}}{\partial P_{ml}} \alpha \beta \sin \gamma \frac{\partial \beta'}{\partial P_{ml}} \right. \\
 & + \frac{1}{2} D_{ij} \alpha \frac{\partial \beta}{\partial P_{ml}} \sin \gamma \frac{\partial \beta'}{\partial P_{mk}} + \frac{1}{4} D_{ij} \alpha \beta \cos \gamma \frac{\partial \beta'}{\partial P_{mk}} \frac{\partial \beta'}{\partial P_{ml}} \\
 & \left. + \frac{1}{2} D_{ij} \frac{\partial \alpha}{\partial P_{ml}} \beta \sin \gamma \frac{\partial \beta'}{\partial P_{mk}} + \frac{1}{2} D_{ij} \alpha \beta \sin \gamma \frac{\partial^2 \beta'}{\partial P_{mk} \partial P_{ml}} \right] \\
 & + \left[\frac{\partial D_{ij}}{\partial P_{ml}} \frac{\partial \alpha}{\partial P_{mk}} \beta \cos \gamma + D_{ij} \frac{\partial \alpha}{\partial P_{mk}} \frac{\partial \beta}{\partial P_{ml}} \cos \gamma \right. \\
 & \left. - \frac{1}{2} D_{ij} \frac{\partial \alpha}{\partial P_{mk}} \beta \sin \gamma \frac{\partial \beta'}{\partial P_{ml}} + D_{ij} \frac{\partial^2 \alpha}{\partial P_{mk} \partial P_{ml}} \beta \cos \gamma \right]
 \end{aligned}$$

$$-\frac{1}{2} D_{ij} \alpha \frac{\partial \beta}{\partial P_{mk}} \sin \gamma \frac{\partial \beta'}{\partial P_{m\ell}} \quad (9.21)$$

where

$$\alpha = \left[1 - \frac{\left[\frac{\theta_{ij}^u - \theta_{ij}^{c\ell}}{2} \right]^2}{3!} + \frac{\left[\frac{\theta_{ij}^u - \theta_{ij}^{c\ell}}{2} \right]^4}{5!} \right]$$

$$\beta = (\theta_i^u + \theta_j^u - \theta_i^{c\ell} - \theta_j^{c\ell})$$

$$\beta' = (\theta_i^u - \theta_j^u + \theta_i^{c\ell} - \theta_j^{c\ell})$$

$$\gamma = \frac{(\theta_{ij}^u + \theta_{ij}^{c\ell})}{2}$$

$$\frac{\partial D_{ij}}{\partial P_{mk}} = G_{ij} \left[|E_j| \frac{\partial |E_i|}{\partial P_{mk}} + |E_i| \frac{\partial |E_j|}{\partial P_{mk}} \right]$$

$$\frac{\partial^2 D_{ij}}{\partial P_{m\ell} \partial P_{mk}} = G_{ij} \left[\frac{\partial |E_j|}{\partial P_{mk}} \frac{\partial |E_i|}{\partial P_{m\ell}} + \frac{\partial |E_i|}{\partial P_{mk}} \frac{\partial |E_j|}{\partial P_{m\ell}} \right] \quad (9.22)$$

The second-order sensitivity equation for some j, k ($1 \leq j, k \leq N$) is given by

$$M_i \frac{d^2}{dt^2} (\hat{u}_{ik\ell}) = -H_{ik\ell}^f + \sum_{j=1}^n A_{ij}^f \hat{u}_{jkl} \quad i = 1, 2, \dots, n-1 \quad (9.23)$$

where

$$\hat{u}_{ik\ell} = \frac{\partial^2 \theta_i}{\partial P_{m\ell} \partial P_{mk}}$$

$$A_{ii}^f = \left[1 - \frac{2M_i}{M_T} \right] \sum_{\substack{j=1 \\ j \neq i}}^n D_{ij}^f \sin \theta_{ij} - \sum_{\substack{j=1 \\ j \neq i}}^n C_{ij}^f \sin \theta_{ij}$$

$$A_{ij}^f = C_{ij}^f \cos \theta_{ij} - D_{ij}^f \sin \theta_{ij} + \frac{2M_i}{M_T} \sum_{\substack{\ell=1 \\ \ell \neq j}}^n D_{\ell j}^f \sin \theta_{\ell j}$$

$$H_{ik\ell}^f = \frac{\partial}{\partial P_{m\ell}} \left[Q_{ik}^f - \sum_{j=1}^n A_{ij}^f u_{jk} \right] - \sum_{j=1}^n A_{ij}^f \hat{u}_{jkl}$$

$$Q_{ik}^f = \frac{M_i}{M_T} - \delta_{ik}$$

$$+ \sum_{j=1}^n \left[\frac{\partial |E_i|}{\partial P_{mk}} |E_j| + |E_i| \frac{\partial |E_j|}{\partial P_{mk}} \right] \left[B_{ij}^f \sin \theta_{ij} + G_{ij}^f \cos \theta_{ij} \right]$$

$$- \frac{M_i}{M_T} \sum_{j=1}^n \sum_{\ell=1}^n \left[\frac{\partial |E_\ell|}{\partial P_{mk}} |E_j| + |E_\ell| \frac{\partial |E_j|}{\partial P_{mk}} \right] G_{\ell j}^f \cos \theta_{\ell j}$$

$$\delta_{ik}^2 = \text{Kronecker } \delta \quad (9.24)$$

9.5 VALIDATION STUDIES

The sensitivity factors, derived in the previous subsection, were used to obtain stability limits for various changes in system parameters and conditions. The procedure used is shown in Fig. 9.2.

9.5.1 First-Order Sensitivity Results

First-order sensitivity factors derived in Section 9.3, and using the procedure in Fig. 9.2, were applied to the 17-generator test system (see the appendix for description of the system). Reference [7] presented data for obtaining the stability limits for three types of system changes (1) plant generation (assuming total system generation and load are constant), (2) load changes, and (3) network changes. In incorporating these changes, it is assumed that the magnitude of the changes are small and that the advanced generators in the MOD are unchanged. The results obtained using the sensitivity techniques are compared with those obtained with repeated stability analysis using the TEF method. A summary of the results is given below.

1. Plant generation change

The procedure was applied to determine for a given fault:

- a. The amount of generation to be increased at the critical generator, before instability is reached, i.e., when the energy margin (ΔV) becomes zero. These data are presented in Table 9.1a.
- b. The amount of generation to be reduced at the critical generators to make an unstable case stable, i.e., to make $\Delta V = 0$. These data are presented in Table 9.1b.

The change in generation is implemented at the generators which are severely disturbed, i.e., the advanced generators in the MOD. These generators provide a greater effect on the energy margin sensitivity.

The results in Table 9.1 show that the first-order analytical sensitivity technique gives an accurate estimate of the stability limits, i.e., comparable to those obtained by repetitive application of the TEF analysis. The largest error is the Ft. Calhoun case for the generation shift at generator 17 where the maximum error is less than 4%.

2. Load changes

In this analysis, the load on bus 471 in the eastern part of the system is to be increased. The increase in load (ΔP_L) at this bus is picked up in equal ratio by generators 12 and 6. These are the critical generators for the fault considered. The limit on the load increase is determined using the proposed technique for two different three phase faults. Table 9.2 shows results for the load change limit and its comparison with repetitive TEF results.

TABLE 9.1 Generation Limits Obtained with First-Order Sensitivity Technique: 17-Generator Test System

a. Generation change limits to make system unstable

Fault Location	t^{cl} (sec)	Generation Change Limit; First-Order Sensitivity	Generation Change Limit; Repetitive TEF Method	% Error
Ft. Cal. (Bus 773) (Clear 773-339)	0.342	$\Delta P_{16} = 37.88 \text{ MW}$	$\Delta P_{16} = 37.64 \text{ MW}$	-0.63
		$\Delta P_{17} = 55.94 \text{ MW}$	$\Delta P_{17} = 54.88$	-1.9
Cooper (Bus 6) (Clear 6-774)	0.191	$\Delta P_2 = 57.51 \text{ MW}$	$\Delta P_2 = 56.69 \text{ MW}$	-1.4
C. Bluffs (Bus 436) (Clear 436-771)	0.187	$\Delta P_{10} = 45.85 \text{ MW}$	$\Delta P_{10} = 45.95$	0.22
		$\Delta P_{12} = 37.54 \text{ MW}$	$\Delta P_{12} = 37.38 \text{ MW}$	-0.42

b. Generation change limits to stabilize system

Fault Location	t^{cl} (sec)	Generation Change Limit; First-Order Sensitivity	Generation Change Limit; Repetitive TEF Method	% Error
Ft. Cal. (Bus 773) (Clear 773-339)	0.357	$\Delta P_{16} = -8.379 \text{ MW}$	$\Delta P_{16} = -8.18 \text{ MW}$	-2.4
		$\Delta P_{17} = -12.68 \text{ MW}$	$\Delta P_{17} = -12.21$	-3.8
Cooper (Bus 6) (Clear 6-774)	1.51	$\Delta P_2 = -55.56 \text{ MW}$	$\Delta P_2 = -54.65 \text{ MW}$	-1.66
C. Bluffs (Bus 436) (Clear 436-771)	0.209	$\Delta P_{10} = -45.44 \text{ MW}$	$\Delta P_{10} = -44.69$	-1.7
		$\Delta P_{12} = -35.94 \text{ MW}$	$\Delta P_{12} = -35.52 \text{ MW}$	-1.2

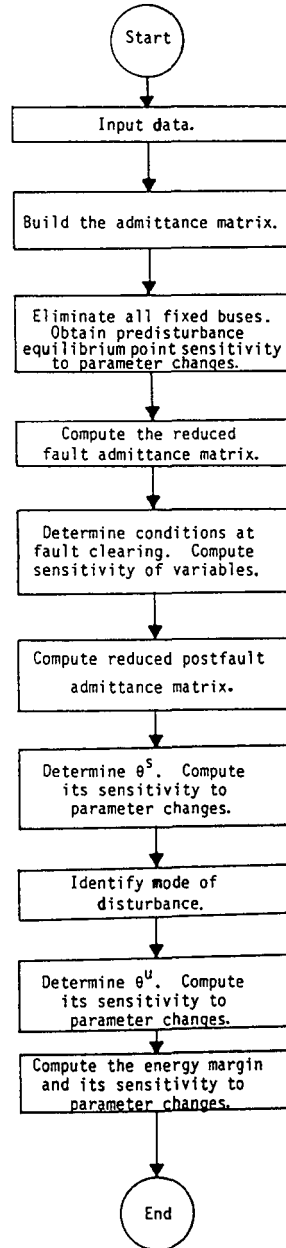


Figure 9.2 Flow chart of sensitivity computation procedure.

TABLE 9.2 Load Change Limits Obtained with First-Order Sensitivity Techniques: 17-Generator Test System

Fault Location	t^{cl} (sec)	Load Change Limit First-Order Sensitivity	Load Change Limit Repetitive TEF Method	% Error
C. Bluffs (Bus 436) (Clear 436-771)	0.190	$\Delta P_L = 52.97$ MW	$\Delta P_L = 53.36$ MW	0.73
Raun (Bus 372) (Clear 372-773)	0.150	$\Delta P_L = 191.96$ MW	$\Delta P_L = 190.70$ MW	-0.66

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TABLE 9.3 Energy Margin Sensitivity to Network Changes

$\Delta(\Delta V)$ pu First-Order Sensitivity	$\Delta(\Delta V)$ pu Second-Order Sensitivity	$\Delta(\Delta V)$ pu Repetitive TEF Method
-0.43120	-0.62903	-0.64320

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It is to be noted that in this study, unlike the change in plant generation, the total load in the system does *not* remain constant. The plant generations are varied in a fixed proportion depending on their rating, to supply the load. Different combinations of plant generation variation could be analyzed to study the effect on stability behavior. The results in Table 9.2 show that the first-order sensitivity technique predicts the load change limit with reasonable accuracy.

3. Network changes

The effect of change of line impedance on the energy margin for a specific contingency is investigated in this study. The particular case analyzed consists of a three-phase fault at Raun (bus 372) cleared in 0.155 by opening line 372–773. The energy margin sensitivity to doubling the impedance of line 482–435 is investigated. In this case the first-order sensitivity method did not provide very satisfactory results compared to the repetitive TEF method. These results are shown in Table 9.3. Careful analysis shows that the change in network configuration results in a nonlinear change in the energy margin. Thus first-order sensitivity analysis may not be sufficient to estimate the stability limit accurately. Second-order terms, such as those discussed in Section 9.4, need to be introduced.

9.5.2 Second-Order Sensitivity Results

Numerical method

As outlined in Section 9.4.1 a second-order correction can be added to the first-order analytical sensitivities by numerical methods. In [9] this technique is applied to the 17-generator test system. The stability-limited power flow which can be transmitted from the western part of the network to the east is evaluated using the sensitivity of the energy margin with respect to the line power.

Analytical method

In [10], second-order analytical expressions for the energy margin sensitivity to changes in system conditions have been applied to the 11-generator and 50-generator test systems. The procedure shown in Fig. 9.1 has been used to determine for a given fault:

1. The amount of generation to be increased at the critical generator(s), before instability is reached, i.e., to make $\Delta V = 0$, and
2. The amount of generation to be reduced at the critical generator(s) to make an unstable case stable, i.e., to make $\Delta V = 0$.

It is to be noted that the change in generation is introduced where it would result in the greatest change in the energy margin, i.e., at the severely disturbed generators. These

generators are easily identified by their advanced angles at the UEP. These same generators would also have greater influence on the energy margin sensitivities.

Results for 11-generator test system

The disturbance is a three-phase fault at bus 226, cleared at 0.068 second by opening lines 226–145, 144–145, 146–226, and 144–146. The MOD test shows that the critical machines are generators 6, 7, and 8. Generation changes to obtain the stability limits are introduced at these generators. The results for the stable case is shown in Table 9.4, and for the unstable case is shown in Table 9.5. For both cases the reference generator is generator 11, and the base case generation at generator 8 is 1300 MW.

The results in Table 9.4 show the amount of generation change limits for shifts at different generators to "absorb" the margin in a stable base case. They also provide a comparison of the results obtained using first-order and second-order sensitivities. The results of Table 9.4 show that the proposed procedure provides results which compare accurately with those obtained by repetitive application of the TEF method with the power flows obtained at the new generation levels. The results also show the improvement obtained in the limits using the second order approximation over the first-order terms alone.

In Table 9.5 the results obtained for the initially unstable base case are presented. (The cases had been made unstable by increasing the fault-clearing time.) In these cases the generation is reduced at the critical generations to make the system stable. These results clearly demonstrate the improvement in accuracy obtained using second-order sensitivities when the generation shifts, to obtain the stability limits, are large.

Results for 50-generator test system

The disturbance is a three-phase fault at the HV bus near the Nanticoke plant, cleared by opening the Nanticoke–Milton line. The critical generators are 20 and 26, and

TABLE 9.4 Generation loading limits – (stable case):
11-generator test system

Actual TEF Run (MW)	Generator Loading Limit ΔP_{mk} for $\Delta V=0$			
	First-Order Sensitivity Analysis (MW)	Relative Error (%)	Second-Order Sensitivity Analysis (MW)	Relative Error (%)
$\Delta P_{m8} = 322.65$	$\Delta P_{m8} = 280.02$	13.21	$\Delta P_{m8} = 332.11$	-2.94
$\Delta P_{m7} = 327.30$	$\Delta P_{m7} = 287.35$	12.21	$\Delta P_{m7} = 340.81$	-4.13
$\Delta P_{m6} = 329.00$	$\Delta P_{m6} = 268.54$	18.37	$\Delta P_{m6} = 322.30$	2.04
$\Delta P_{m7} = 167.75$ $\Delta P_{m8} = 162.75$	$\Delta P_{m7} = 141.83$ $\Delta P_{m8} = 141.83$	12.85	$\Delta P_{m7} = 167.32$ $\Delta P_{m8} = 167.32$	2.81
$\Delta P_{m6} = 164.75$ $\Delta P_{m7} = 164.75$	$\Delta P_{m6} = 138.80$ $\Delta P_{m7} = 138.80$	15.75	$\Delta P_{m6} = 167.26$ $\Delta P_{m7} = 167.26$	0.3

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TABLE 9.5 Generation loading limits – (unstable case):
11-generator test system

Fault Clearing Time (sec)	Generator Loading Limit ΔP_{mk} for $\Delta V=0$				
	Actual TEF Run (MW)	First-Order Sensitivity Analysis (MW)	Relative Error (%)	Second-Order Sensitivity Analysis (MW)	Relative Error (%)
0.167	$\Delta P_{m7} = -100.35$	$\Delta P_{m7} = -112.75$	-12.36	$\Delta P_{m7} = -103.94$	-3.57
	$\Delta P_{m7} = -47.38$	$\Delta P_{m7} = -51.42$	-8.53	$\Delta P_{m7} = -47.58$	-0.42
	$\Delta P_{m8} = -47.38$	$\Delta P_{m8} = -51.42$	-8.70	$\Delta P_{m8} = -47.58$	-0.42
0.218	$\Delta P_{m8} = -174.79$	$\Delta P_{m8} = -190.01$	-8.70	$\Delta P_{m8} = -167.91$	3.93
	$\Delta P_{m7} = -95.67$	$\Delta P_{m7} = -108.73$	-13.65	$\Delta P_{m7} = -94.92$	0.78
	$\Delta P_{m8} = -95.67$	$\Delta P_{m8} = -108.73$	-13.65	$\Delta P_{m8} = -94.992$	0.78

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the reference generator is 50. The base case is for Bruce generation of 1400 MW. For the stable case the fault is cleared in 0.0667 second, and for the unstable case the fault duration is 0.15 second.

The results for the 50-generator test system stable and unstable cases for large generation shifts are given in Table 9.6. These results again illustrate that improved accuracy is obtained with the second-order approximation over that obtained with the first-order technique.

It should be pointed out that these results have been obtained for the situations where the generation shifts to obtain the power limits do not change the MOD; i.e., the critical generators are unchanged.

TABLE 9.6 Generation loading limits:
50-generator test system

Classification	Generator Loading Limit ΔP_{mk} for $\Delta V=0$				
	Actual TEF Run (MW)	First-Order Sensitivity Analysis (MW)	Relative Error (%)	Second-Order Sensitivity Analysis (MW)	Relative Error (%)
Stable	$\Delta P_{m20} = 197.70$	$\Delta P_{m20} = 185.74$	6.05	$\Delta P_{m20} = 202.91$	-2.63
	$\Delta P_{m26} = 197.70$	$\Delta P_{m26} = 185.74$		$\Delta P_{m26} = 202.91$	
Unstable	$\Delta P_{m20} = -180.51$	$\Delta P_{m20} = -188.07$	-4.18	$\Delta P_{m20} = -178.15$	1.30
	$\Delta P_{m26} = -180.51$	$\Delta P_{m26} = -188.07$		$\Delta P_{m26} = -178.15$	

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9.6 DETERMINATION OF INTERFACE FLOW STABILITY LIMITS

For a stability-limited power system the limiting condition is often the power flow on a critical transmission interface. Transient stability limits are determined by

conducting many off-line transient stability studies for different operating conditions and for different contingencies. The limits determined from these studies establish the safe regimes of operation. During system operation, the flow on the critical transmission interfaces are monitored to check that their flows are within the established limits.

As it has been often stated in this book, today's interconnected power networks are operated closer to their stability limits. The conditions which need to be analyzed to establish the safe operating regimes are much more numerous than for yesterday's more robust power networks. Whether the analysis is conducted off line by the operations planning engineers, or at or near real time for on-line applications, the need for faster derivation of the limits is greater than ever. With the help of sensitivity techniques it will be shown that the TEF method makes it possible to obtain stability limits with fewer stability runs. In this section the analytical sensitivity procedures previously discussed are extended to determine critical interface power flow limits. Only first-order analytical sensitivities are used.

In determining the interface flow limit, an important factor must be kept in mind. In contrast with other system parameter changes, a given transmission interface power flow change can be obtained by using different combinations of generation. These combinations, however, would result in different amounts of changes in the transient energy margin.

In the derivation below it is assumed that the total generation is constant, i.e., a change ΔP_{mk} in the initial power of generator k is accompanied by an equal and opposite change in generation of the reference generator. For a particular contingency, the change of energy margin with respect to a generation shift can be obtained using the assumption that the total generation remains constant and that the energy margin varies linearly with respect to the generation shifts

$$\Delta(\Delta V) = \sum_{k=1}^{N-1} \alpha_k \Delta P_{mk} \quad (9.25)$$

where

$$\alpha_k = \frac{\partial(\Delta V)}{\partial P_{mk}} \quad k = 1, 2, \dots, N - 1$$

$N - 1$ = number of generator at which generation can be shifted with respect to the reference generator (N th machine is the reference generator)

A linear relationship also exists between a line power flow change and generation shift [12] given by

$$\Delta \ell_1 = \sum_{k=1}^{N-1} S_{\ell k} \Delta P_{mk} \quad (9.26)$$

where $S_{\ell k}$ is the distribution factor of line ℓ due to generation shift at generator k , $k = 1, 2, \dots, N - 1$.

Using the relationship shown in (9.25) and (9.26), a procedure to relate the change in interface power flow for a specific change in the base case margin will be developed. In doing so the ranges of transmission interface power flow limits are obtained as a result of generation shifts.

Assumptions

In this development the following assumptions are made.

1. The critical generators at which generation can be shifted with respect to the reference machine are known.
2. When generation is shifted at the critical generators, the sensitivity of the energy margin to the change at these generators α_k , $k = 1, 2, \dots, N - 1$ have the same sign. In order to obtain a unique power flow over a given line ℓ , the distribution factors of this line $S_{\ell k}$, $k = 1, 2, \dots, N - 1$ with respect to the critical generators should all have the same sign. On a critical line this is normally the case.

3. The range of allowed generation shift at each generator is of such a value as to result in zero energy margin. If the system is stable (or unstable) for the base case load flow, generation shifts considered will result in zero energy margin.
4. The range of generation shifts assumed in 3 do not exceed the machine or equipment ratings.

Approach

The interface flow limits are determined as the minimum and maximum values of interface power flow limits satisfying the given constraints. This problem is formulated mathematically as follows:

1. Minimize or maximize

$$\Delta I_\ell = \sum_{k=1}^{N-1} S_{\ell k} \Delta P_{mk} \quad (9.27)$$

subject to

$$\sum_{k=1}^{N-1} \alpha_k \Delta P_{mk} = -\Delta V$$

and (9.28)

$$-\Delta V \leq \alpha_k \Delta P_{mk} \leq 0, \quad \text{if } \Delta V > 0$$

$$0 \leq \alpha_k \Delta P_{mk} \leq -\Delta V, \quad \text{if } \Delta V < 0 \quad \text{for } k = 1, 2, \dots, N-1$$

where ΔV is the system energy margin, which has to be adjusted to result in zero energy margin for the stable or unstable case.

This is a linear programming problem with a simple objective function and constraint equations.

2. The linear programming problem is solved by defining the generation shift vector as ΔP_m

$$\Delta P_m = (\Delta P_{m1}, \Delta P_{m2}, \dots, \Delta P_{m,N-1})^T \quad (9.29)$$

transforming equation (9.29) by $\Delta P'_{mk} = \alpha_k \Delta P_{mk}$, $k = 1, 2, \dots, (N - 1)$, we then have

$$\Delta P'_m = (\Delta P'_{m1}, \Delta P'_{m2}, \dots, \Delta P'_{m,N-1})^T$$

Using the theorem of the simplex method [13] the minimum and maximum amount of line power flow change ΔI_1 and ΔI_1^* are, respectively, given as follows,

$$\Delta I_{\ell*} = -\Delta V a_{i*} \quad (9.30)$$

when $\Delta P'_m = -\Delta V e_i$.

$$\Delta I_{\ell}^* = -\Delta V a_j^* \quad (9.31)$$

when $\Delta P'_m = -\Delta V e_j$,

where

$e_i = (N-1)$ dimensional unit vector whose i th component is 1 and other components 0

$$|a_{i*}| = \min \{ |a_i| \}$$

$$|a_j^*| = \max \{ |a_j| \}$$

$$a_k = \frac{S_{\ell k}}{\alpha_k}, \quad k = 1, 2, \dots, N - 1$$

3. For a particular contingency, we can pick an appropriate line which shows a narrow range of the energy margin change for observed line power flow change using the following rules.
 - a. For the assumed generation shifts, all the a_k should be of the same sign.
 - b. The difference between $\Delta I_{\ell*}$ and ΔI_{ℓ}^* should be small.
 - c. For the assumed generation shifts, the distribution factors of the line should be large.

The three conditions stated are well satisfied for a radial system. In an interconnected system, however, the three conditions are satisfied only for a critical line which is a major passage of power flow change caused from generation shifts between critical generators and the reference generator.

Example

The proposed procedure has been applied to the 17-generator test system, described in the appendix, in [11]. The contingencies investigated are three-phase faults at Ft. Calhoun, Council Bluffs, and Cooper buses, respectively. For each fault location, two fault durations were selected to give a stable and an unstable case. First, the plant generation is shifted between the critical generators¹ and the reference generator (which is among the noncritical* groups) to obtain zero energy margin. These results are given in Table 9.7, together with the results obtained by repetitive applications of the TEF method for initial conditions corresponding to the different distribution of generation.

To obtain the amount of change in line flows corresponding to changes in generation, distribution factors are obtained using a standard power flow computer program developed

¹ These generators are identified in the TEF procedure, as given in Chapter 5.

TABLE 9.7 Generator Loading Limits (G_0 : Reference Machine)

Fault Location	Clearing Time (sec)	Contingency	Shifting Generator	Loading Limit Sensitivity Analysis (MW)	Loading Limit Full TEF (MW)	Relative Error (%)
Ft. C. Fault	0.342	Stable	G_{16}	$\Delta P_{m16} = 37.87$	$\Delta P_{m16} = 37.26$	1.6
			G_{12}	$\Delta P_{m12} = 41.66$	$\Delta P_{m12} = 40.85$	2.0
			G_{17}	$\Delta P_{m17} = 55.94$	$\Delta P_{m17} = 54.31$	3.0
			G_{17}	$\Delta P_{m17} = 56.03$	$\Delta P_{m17} = 54.58$	
			G_{13}	$\Delta P_{m13} = -56.03$	$\Delta P_{m13} = -54.58$	-2.7
	0.367	Unstable	G_{16}	$\Delta P_{m16} = -36.89$	$\Delta P_{m16} = -37.53$	-1.7
			G_{12}	$\Delta P_{m12} = -41.02$	$\Delta P_{m12} = -42.79$	-4.1
			G_{17}	$\Delta P_{m17} = -55.79$	$\Delta P_{m17} = -56.80$	-1.8
C.B. Fault	0.187	Stable	G_{12}	$\Delta P_{m12} = 37.54$	$\Delta P_{m12} = 37.33$	0.6
			G_{10}	$\Delta P_{m10} = 45.85$	$\Delta P_{m10} = 45.88$	-0.1
	0.209	Unstable	G_{12}	$\Delta P_{m12} = -45.44$	$\Delta P_{m12} = -44.69$	1.7
			G_{10}	$\Delta P_{m10} = -35.94$	$\Delta P_{m10} = -35.52$	1.2
Cooper Fault	0.191	Stable	G_2	$\Delta P_{m2} = 57.51$	$\Delta P_{m2} = 56.63$	1.6
	0.219	Unstable	G_2	$\Delta P_{m2} = -55.06$	$\Delta P_{m2} = -54.65$	0.8

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by the Philadelphia Electric Co. The distribution factors for the selected lines are given in Table 9.8.

The data in Table 9.8 show that the line between buses 471 and 435 (Hill-Sycamore) has the highest distribution factors when the generation change is introduced at the critical generators. It is selected for determination of stability limited flow, for the three selected contingencies. The results are shown in Table 9.9.

To show the validity of rule 3 for selecting a critical line, three noncritical lines are chosen. The sensitivity analysis for these lines are shown in Table 9.10. These results clearly show the effect of the distribution factors on the sensitivity analysis for these lines.

TABLE 9.8 Distribution Factors of Monitored Lines (G_9 : Reference)

Monitored Line	Distribution Factor				
	G_2	G_{10}	G_{12}	G_{17}	G_{16}
471 Hill 435 Sycamore	-0.296	-0.313	-0.316	-0.302	-0.301
193 Lakefield - 372 Raun	-0.101	-0.115	-0.116	-0.109	-0.138
7 Lincoln - 779 Sub. 3454	0.069	-0.052	-0.056	0.007	-0.182
6 Cooper - 393 St. Joe	0.266	0.159	0.176	0.226	0.174

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TABLE 9.9 Line Power Flow Limits at Line 471 Hill - 435 Sycamore

Contingency			Range of Line Power Flow Limits			
Fault Location	Clearing Time (sec)	Critical Generators	Sensitivity Analysis	Full TEF Results	Error (%) Results	
					Lower Limit	Upper Limit
Ft. Cal.	0.342	$G_{10} G_{12}$	$\Delta I_t > -16.89$	$\Delta I_t > -16.22$	4.1	2.0
		$G_{16} G_{17}$	$\Delta I_t < -11.40$	$\Delta I_t > -11.09$		
C. Bluffs	0.187	$G_{10} G_{12}$	$\Delta I_t > -14.35$ $\Delta I_t < -11.86$	$\Delta I_t > -14.27$ $\Delta I_t < -11.75$	0.6	0.0
Cooper	0.191	G_2	$\Delta I_t = -17.02$	$\Delta I_t = -16.02$	2.4	

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TABLE 9.10 Line Power Flow Limits at Noncritical Line

Monitored Line	Contingency	Range of Line Power Flow Limits			
		Sensitivity Analysis	Full TEF Results	Error (%)	
				Lower Limit	Upper Limit
193 Lakefd.	Ft. Cal.	$-6.1 \leq \Delta I_L \leq -4.83$	$-5.67 \leq \Delta I_L \leq -4.59$	7.6	5.2
—	C. Bluffs	$-5.27 \leq \Delta I_L \leq -4.35$	$-5.06 \leq \Delta I_L \leq -4.19$	4.2	3.8
372 Raun	Cooper	$\Delta I_L = -5.89$	$\Delta I_L = -5.46$	7.9	
7 Lincoln	Ft. Cal.	$-6.89 \leq \Delta I_L \leq 0.39$	$-6.64 \leq \Delta I_L \leq 0.41$	3.8	-4.9
—	C. Bluffs	$-2.39 \leq \Delta I_L \leq -2.10$	$2.37 \leq \Delta I_L \leq 2.07$	0.4	1.4
779 S. 3454	Cooper	$\Delta I_L = 3.97$	$\Delta I_L = 3.85$	3.1	
6 Cooper	Ft. Cal.	$6.59 \leq \Delta I_L \leq 12.64$	$6.6 \leq \Delta I_L \leq 12.33$	-0.1	2.4
—	C. Bluffs	$6.61 \leq \Delta I_L \leq 7.29$	$6.74 \leq \Delta I_L \leq 7.55$	-1.9	3.4
393 S. Joe	Cooper	$\Delta I_L = 15.30$	$\Delta I_L = 15.07$	1.5	

Critical generators for each contingency:

Ft. Cal. fault: G_{10} , G_{12} , G_{16} , G_{17}

C. Bluffs fault: G_{10} , G_{12}

Cooper fault: G_2

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9.7 ON-LINE DERIVATION OF STABILITY LIMITS

9.7.1 Introduction

The North American Electric Reliability Council (NERC) and its regional councils establish reliability criteria which must be met by all member utilities. These criteria require that the system be operated so that it can withstand certain contingencies (or disturbances) under normal conditions, as well as withstanding more severe contingencies

under emergency conditions. Depending on the condition of the transmission network in the system, its load level and operating condition, the limiting parameter or operating condition will differ for different parts of the North American interconnected system. Nowadays with most systems operating closer to their capacity than before, and with heavy transfers of power between networks, many systems are stability limited. This means that in order to meet NERC criteria, contingencies involving large disturbances and resulting in the separation of one or more generators from the rest of the system, causing possible cascading outages in the system, must be withstood.

To operate within the reliability criteria established by NERC, the North American power systems determine the limiting conditions for various contingencies or disturbances. For these contingencies, the most restrictive limits are identified together with the associated critical power system element or parameter. Examples of such limiting conditions are the loading of a key power plant, the flow on a particular transmission interface, the voltage at a critical bus, the operation of a certain relay, etc. The critical power element is then monitored at all times to ascertain that the established limits are not exceeded.

The key to secure power system operation, therefore, is to *know* the limits for all possible situations which may be encountered. To obtain these limits for a stability-limited power network, a lot of information is required. To illustrate the complexity of this undertaking, the authors of [14] discuss such requirements for an essentially radial portion of the Ontario Hydro network, connecting the Bruce generating power plant to the main high-voltage grid, as follows:

The DACS² approach to monitoring power system security in areas other than the Bruce area is based on the results of off-line computer studies which identify predefined precontingency stability limits for normal-condition criteria faults and

² Data acquisition system.

emergency-condition criteria faults. These limits are input to the computer using a decision table language and are monitored by the System Security Monitor program (SYSM).

The stability limits are determined for a wide range of loading patterns. In addition they cover network patterns for all elements in service and any one element out of service. This coverage requires an extremely large number of studies particularly in areas of large impact to Ontario Hydro's fuel bill.

In the same paper, the authors go on to point out that "It is impractical to cover all combinations of outages using the current technique. For this reason, on-line capability to determine transient stability-based operating limits has been developed to drive the closed loop control process of arming the Bruce Load and Generation Rejection Scheme."

The conditions described are becoming more typical of situations encountered in large interconnected networks. First, emergency control measures to maintain stability, such as generation tripping, insertion of braking resistors, and load shedding, are now commonly used. Second, it is becoming increasingly difficult to predetermine, for stability-limited power networks, the stability limits for all possible operating conditions, all possible network configurations, and all credible contingencies [15]. Thus the interest in on-line determination of stability limits.

9.7.2 Stability Limits for an Area in a Large Network

Reference [14] discusses the on-line derivation of stability limits needed for an essentially radial portion of the Ontario Hydro (O.H.) network connecting the Bruce generating station to the rest of the O.H. network, and the load and generation tripping scheme to maintain stability. The Bruce area network is shown in Fig. 9.3. The stability assessment/load and generation rejection scheme is outlined below, as described in [14].

A manageably large number of postcontingency transmission configurations are selected, and for each of these, accurate stability limits are determined by detailed off-line

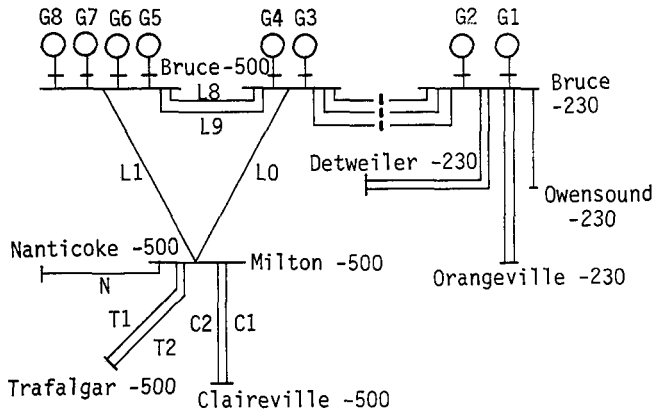


Figure 9.3 Bruce area transmission system [14]. Reprinted from the *Proceedings of the 8th Power System Computer Conference* (1984); used with permission.

studies for different precontingency generation levels. These are tabulated for on-line usage and are designed to cover the majority of actual situations.

The network postcontingency configuration is determined by a special software for each possible contingency required by the reliability criteria. The steps followed to obtain the stability limits are as follows:

For each contingency,

1. Establish the postcontingency configuration (the resultant parallel and series elements connecting Bruce to the main system).
2. Identify which generation rejection modules will operate for that configuration.
3. Try to find a match for the resulting postcontingency system in the data base (cases studied off line).
4. If a match is found, get the maximum generation limit and the required generation rejection. If there is no match, use the on-line stability assessment technique to compute them.

The transient stability assessment scheme described in [14] is based on the equal area criteria. This is computed for the postcontingency configuration for which the data acquisition system cannot find a match in the data base obtained from the off-line studies.

It is reported in [16] that this scheme is being adapted to the use of the TEF method for on-line determination of stability limits for the network configurations for which precomputed stability limits are not available. An outline of the scheme used, as given in [16], is given below. The overall scheme for on-line transient stability monitoring is shown in Fig. 9.4.

According to the scheme in Fig. 9.4, the steps for computing the transient stability limit are:

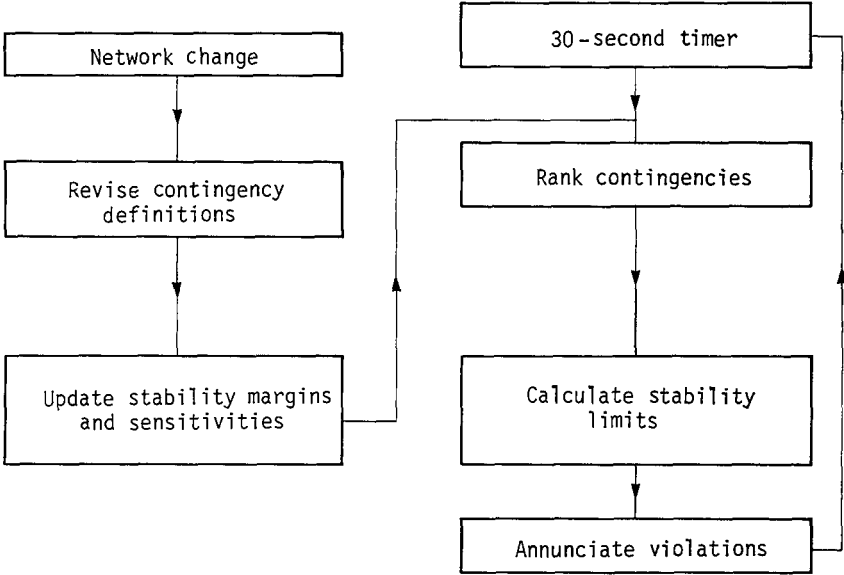


Figure 9.4 System for on-line transient stability monitoring.

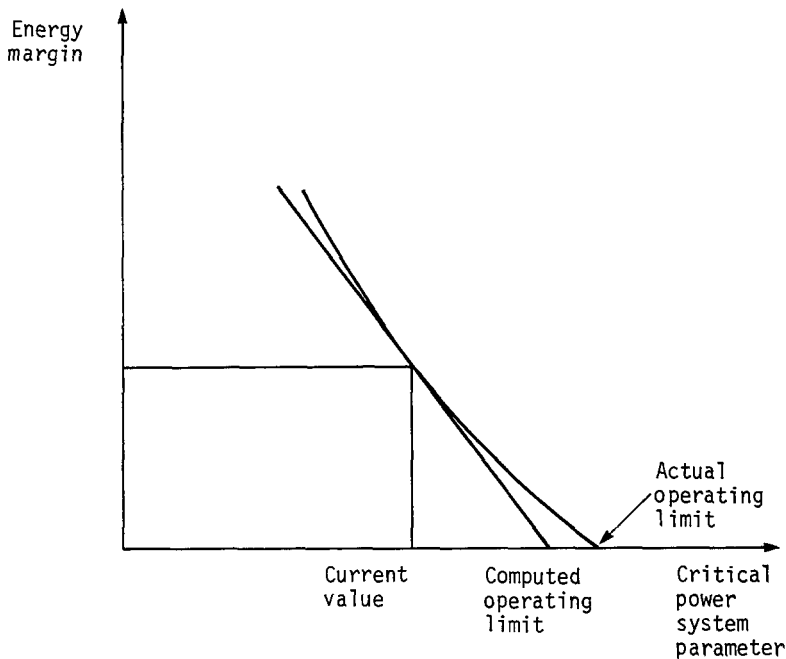


Figure 9.5 Use of sensitivity technique to obtain the stability limit for a given contingency.

1. For the present network (i.e., after a change or every 30 seconds) and for the first contingency, run the TEF method to obtain the transient energy margin.
2. Using sensitivity technique, obtain the stability limit for the critical power system parameter (e.g., in the O.H. network, the flow away from the Bruce complex). This is depicted in Fig. 9.5 for the case when the initial assessment gives positive energy margin.
3. Repeat step 2 for all the contingencies on the contingency list.
4. The stability limit is the most restrictive limit (typically the lowest) obtained in steps 2 and 3.

Note: If the worst contingency is known, the limit is computed directly from the stability assessment of that contingency and the energy margin sensitivity.

It should be pointed out that in the example there is only *one* critical power system parameter to be concerned with. Even then, a large amount of data is required to obtain the stability limits for all possible operating conditions and contingencies. The use of energy margin analytical sensitivities will substantially reduce the computational burden involved to obtain these limits.

9.7.3 Stability Limits for More Complex System Conditions

In this section we will deal with a situation sometimes encountered in the North American interconnected system. The power system (or part of it) is stability limited, and there are several critical parameters involved. To obtain stability information covering all possible operating conditions for various combinations of the different critical parameters is a major undertaking which may involve computation of numerous (usually thousands) of transient stability limits, with each stability limit requiring several stability runs. If the stability assessment is made with the TEF method, the task can be greatly simplified by using energy margin sensitivity to the critical parameter. This is illustrated by an example based on conditions encountered in the 1980s in a North American power network.

The information used in this example has been obtained from a number of reports received by one of the authors in his capacity as chairman of an IEEE Power Engineering Society Working Group. These reports were written by one of the working group members to explain the dynamic security assessment practices at his company. The stability criterion for that system is the transient voltage at a critical bus. Operating security limits are derived such that the voltage at that bus does not fall below a certain level for a specified length of time under the worst contingency. The transient voltage criterion is used because it represents the voltage at the bus of a nuclear power plant. The specified transient voltage dip results in shutting down the plant, causing a voltage collapse in the system. This voltage transient occurs before the occurrence of angular instability.

The power system of this example serves a metropolitan area in the United States. The complexity of the problem comes about from the fact that there are *six* main variables which affect the stability results. They are:

1. Current major lines out
2. Amount of generation in the region of the system where voltage collapse might occur
3. The level of generation at a certain region of the system
4. Main source of reactive power in the area
5. The flow on a certain dc line
6. The import from a neighboring system on a certain 500-KV line

All these variables are combined into one critical parameter to be carefully monitored (the flow on a certain group of transmission lines). This flow is used as an indicator of security. Its value is carefully monitored and is compared with the limits arrived at by off-line studies, using conventional time simulation methods. Because of the large number of variables involved, numerous approximations and simplifying assumptions are used to obtain the desired limits from the off-line studies.

A problem such as the one described in this section would be greatly simplified if the TEF method is used to make the transient stability assessment. The procedure would be to obtain the energy margin ΔV for the initial operating conditions and the specified contingency.³ Analytical sensitivity of the energy margin to different parameters can be used to obtain the desired limits. For example if the variables described are $\alpha_1, \alpha_2, \dots, \alpha_6$, the limit can be obtained (using equation (9.11)) by

³ It is to be noted that ΔV must be related to the transient voltage criterion of interest in this example.

$$\Delta(\Delta V) = \sum_{k=1}^6 \frac{\partial(\Delta V)}{\partial \alpha_k} (\Delta \alpha_k) + \text{higher-order terms if needed}$$

We also note that in general $\frac{\partial(\Delta V)}{\partial \alpha_k}$ would be made up of several terms (see Section 9.3),

each arising from the dependence of a variable x_j in ΔV on α_k .

Thus the effect of changes in the critical variables on the energy margin can be determined with fewer stability assessment runs. However, determination of the transient stability limits and the changes in system conditions to reach those limits (i.e., to make $\Delta V = 0$) are a much more complex undertaking. Numerous combinations of changes in generation's schedule, in flows on key lines, etc., to make $\Delta V = 0$ are possible. This issue is discussed in Section 10.2.

9.7.4 Use of Sensitivity Methods to Track Security Trend

There is another concept in power system security which has increasingly become of concern. To illustrate the idea behind it, let us arbitrarily use a stability measure such as the transient energy margin. Let us consider two operating points 1 and 2 with different levels of energy margin ΔV such that $\Delta V_1 > \Delta V_2$, for the same contingency. It would appear that the operating condition 1 represents a more secure operation than that for condition 2. Let us now consider the situation depicted in Fig. 9.6 where the sensitivity to a critical parameter p_1 is substantially greater than that with respect to another critical parameter p_2 , and the critical parameter of concern during the operating condition 1 is p_1 , while the parameter p_2 is of concern in condition 2.

Examining the situation depicted in Fig. 9.6, we note that the *level* of system security (in this case as indicated by the value of ΔV) is not meaningful in itself. Rather, it should be viewed in the context of the *changing* system conditions. In other words, the trend in system security is of concern in the system operations environment. This is the concept of

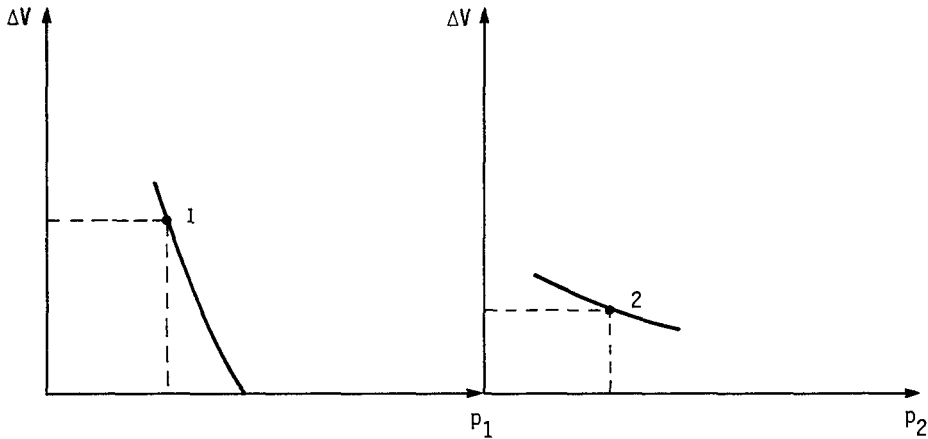


Figure 9.6 Different operating conditions with different security levels, but also different sensitivities.

system vulnerability. When the TEF method is used in transient stability assessment, the sensitivity methods can be used to assess the system vulnerability, especially in a changing environment, e.g., changing weather or major changes in system dispatch.

By the same reasoning, if corrective action is needed to avert undesirable situations, sensitivity techniques can help direct where such action can be most effective.

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Chapter 10

ADVANCED APPLICATIONS OF THE TEF METHOD

10.1 INTRODUCTION

The transient energy function method of power system transient stability analysis is a relatively "young" method. Most of the research and development effort on the method has been devoted to the improvement of the method itself as a tool of analysis and to the investigation of its usefulness for certain power system applications. These applications took precedence over others because of industry needs (and hence industry support of the work). The previous chapters were devoted to these improvements and applications. The potential applications of the TEF method, however, are much greater than what has been presented in these chapters. Since these potential applications are at various levels of development and investigation, they are presented in this chapter to indicate to the reader that, at this stage, they should be considered plausible applications worthy of consideration.

The authors recognize that the designation that some applications are "advanced" is rather arbitrary. The implication, however, is that a certain degree of sophistication is involved beyond being a new application of the TEF as a tool. In some cases, the application itself is novel; in others, the marriage of two emerging technologies is involved.

10.2 CORRECTIVE ACTIONS

It has been repeatedly mentioned in this book that power systems in North America are designed and operated to meet reliability criteria established by the North American Electric Reliability Council and the appropriate regional reliability councils. When these

criteria are adhered to, stability limits may be encountered in some systems. These limits may be in the form of a limit on the loading of a certain power plant, on the power flow on a key group of transmission lines, on the voltage at a given bus, etc. In almost all these situations the stability limits are caused by inadequate transmission capacity for the operating conditions encountered and for the contingencies which the system must withstand.

When the stability-limited conditions are recognized, the engineers supporting system operation and/or operations planning functions are confronted with dealing with these conditions. One obvious answer is that whenever a system operating condition is reached for which the contingencies specified by the reliability criteria will not be adequately handled, this operation would not be permitted. In other words, the distribution of plant generations and transmission line flows would be rescheduled such that the established reliability criteria would be met if these contingencies were to occur. If this answer is deemed desirable, and it may very well be a rather costly one ([1] quotes a \$1 billion cost for such a solution on the Ontario Hydro system), the engineer is still faced with the question of how to do this rescheduling most effectively.

If for some compelling reasons (e.g., prohibitively large cost) the above solution is not desirable, some other measures must be used to prevent unacceptable damage to the system or to its major components should these contingencies occur. These measures are in the form of discrete controllers which tend to reduce the "shock" to the system when disturbances occur. If this solution is resorted to, the question still remains as to how these controllers are to be selected, etc.

These questions are addressed in this section.

10.2.1 Predisturbance Action

The problem is formulated as follows. Given an initial operating condition and a designated contingency (or disturbance sequence) which would make the system unstable,

the changes in the initial conditions which can be implemented to make the system stable (for the same contingency) are to be identified. In the TEF terminology, the contingency for the original system conditions results in a negative energy margin. The changes in the initial conditions sought are those which would make the energy margin zero or positive. The answer to this problem can be simple or exceedingly complex, depending on the type of changes to be implemented.

Let us first consider a very simple problem in which the possible instability will be averted by a change in the generation schedule. To simplify the problem further, we can assume that the TEF procedure indicates that there is only one critical generator in the MOD test, say, generator k . It is obvious that the most effective (i.e., minimum) change in the generation schedule would be in changing P_{mk} . Furthermore, we know that by using the sensitivity coefficient $\frac{\partial(\Delta V)}{\partial P_{mk}}$, we can estimate the amount of generation change ΔP_{mk} needed to make the system stable when subjected to the given disturbance sequence. This is the essence of the treatment of the subject given in Chapter 9. The assumption is made that while the change ΔP_{mk} may result in changes in other system variables, and hence the change in ΔV may include several terms (such as given in equation (9.11b)), no other constraints are affected, e.g., the rest of the generation schedule is unaffected except for balancing ΔP_{mk} with an equal and opposite change at a remote generator.

Let us now consider a slightly more complicated problem. Let us assume that the TEF procedure indicates that there are three critical generators: i , j , and k . To make the system stable, for the disturbance scenario under consideration, it is obvious that there is a large number of possible combinations of ΔP_{mi} , ΔP_{mj} , and ΔP_{mk} which can make the system stable; which one should be chosen? Another practical consideration can be introduced as well: some of the changes introduced at some or all of these generators may be constrained. Obviously this problem is much more complicated than the one previously

considered. Not only are there many possible solutions which are more difficult to find, selecting the "right one" among them may also be difficult.

Let us now introduce one more complexity to this same problem. Let us assume that the generation shift at generators i , j , and k will cause heavier loading at some critical transmission circuits ℓ and m . Again the problem of rescheduling the generation becomes more complicated because there are additional conditions to satisfy. We can go on and introduce additional complications or conditions to satisfy.

The category of problems discussed here is known in the literature as "vector optimization," or "multicriterion optimization," or "generalized goal programming." The problem is formulated mathematically as follows:

Given a vector of decision variables $\mathbf{x}$, $\mathbf{x}^T = [x_1, x_2, \dots, x_n]$, and a vector of objective functions $\mathbf{f}$, $\mathbf{f}^T = [f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_k(\mathbf{x})]$, a solution $\mathbf{x}^*$ is sought such that

$$\mathbf{f}(\mathbf{x}^*) = \text{opt. } \mathbf{f}(\mathbf{x}) \quad (10.1)$$

subject to constraints

$$\begin{aligned} \text{inequality:} \quad & g_j(\mathbf{x}) \geq 0 \quad j = 1, 2, \dots, m \\ \text{equality:} \quad & h_j(\mathbf{x}) = 0 \quad j = 1, 2, \dots, p < n \end{aligned} \quad (10.2)$$

The vectors $\mathbf{x}$ and $\mathbf{f}$ are defined in the Euclidean spaces E^n and E^k , respectively. The constraints (10.2) define the feasible region of solution X in which the so-called set of optimal solutions $\mathbf{x}^*$ lie.

There is a large body of literature on this optimization problem, the treatment of which is beyond the scope of this book. Only a brief discussion of the subject is given here (see, for example, [2–4] for more detailed treatment).

The traditional approach to this problem is to seek what is known as a number of Pareto optimums from which the final solution is selected. A point $\mathbf{x}^* \in X$ is Pareto optimal if for every $\mathbf{x} \in X$ either

$$\bigwedge_{i \in I} (f_i(\mathbf{x}) = f_i(\mathbf{x}^*)) \quad (10.3a)$$

or there is at least one $i \in I$ such that

$$f_i(\mathbf{x}) > f_i(\mathbf{x}^*) \quad (10.3b)$$

where I is the set of indices $(1, 2, \dots, k)$ of the objective function.

Among the commonly used techniques to obtain Pareto optimal solution(s) of a vector function are function scalarization techniques. The vector function $f(\mathbf{x})$ is transformed into a scalar function $f(\mathbf{x})$. The techniques most commonly used are:

1. To use weighting coefficients, e.g.,

$$f(\mathbf{x}) = \sum_{i=1}^k \omega_i f_i(\mathbf{x}) \quad (10.4)$$

The weighting coefficients assign different importance to the different objective functions.

2. Another method of scalarization is to use hierarchical optimization. In this method the criteria to be satisfied are ordered according to their importance, with the most important on top of the list, then the next important, and so on. Each objective function is minimized separately and in the order of importance.

Example

The above approach is illustrated with a simple example. The generation is to be rescheduled to reach the stability limit, with the total load held constant. In addition, possible restrictions on the transmission line flows may exist because of stability limitations.

Let the changes in initial generation be ΔP_{mi} , where $i = 1, 2, \dots, k$ are the critical generators.

Let us assume that the change in the power of generator i results in a corresponding change in the flow in transmission line j given by $(S_{ji} \Delta P_{mi})$. Thus, the flow in line ℓ is given by

$$\Delta I_{\ell} = \sum_{i=1}^k S_{\ell i} \Delta P_{mi} \quad (10.5)$$

The energy margin sensitivity to a change in generation ΔP_{mi} is $\frac{\partial(\Delta V)}{\partial P_{mi}}$, and the change in the energy margin is given by

$$\Delta(\Delta V) = \sum_{i=1}^k \frac{\partial(\Delta V)}{\partial P_{mi}} \Delta P_{mi} \quad (10.6)$$

Equations (10.5) and (10.6) can be combined in the form

$$\begin{bmatrix} \frac{\partial(\Delta V)}{\partial P_{m1}} & \dots & \frac{\partial(\Delta V)}{\partial P_{mk}} & 0 & \dots & 0 \\ S_{11} & \dots & S_{1k} & -1 & 0 & 0 & 0 & 0 \\ \vdots & & \vdots & 0 & -1 & 0 & 0 & 0 \\ S_{m1} & & S_{mk} & 0 & 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} \Delta P_{m1} \\ \vdots \\ \Delta P_{mk} \\ \Delta I_1 \\ \vdots \\ \Delta I_m \end{bmatrix} = \begin{bmatrix} \Delta(\Delta V) \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (10.7)$$

Obviously there are many combinations of ΔP_{mi} , $i = 1, 2, \dots, k$ which satisfy (10.7); which solution should be adopted? Mathematically the problem can be cast as follows.

Find $\min (C_{m1} \Delta P_{m1} \dots C_{mk} \Delta P_{mk} \ C_{\ell 1} \Delta I_1 \dots C_{\ell m} \Delta I_m)$ such that (10.5) and (10.7) are satisfied, and the following constraints are also satisfied:

$$\Delta P_{mi_{\min}} \leq \Delta P_{mi} \leq \Delta P_{mi_{\max}} \quad (10.8a)$$

$$\Delta I_{lj} \leq \Delta I_{lj_{\max}} \quad (10.8b)$$

Other constraints, e.g., economic constraints, can also be included in (10.8).

The problem can be put in the form (10.4) where the weighting factors ω_i , $i = 1, 2, \dots, k$ help to determine the optimum solution by using goal programming. This goal programming can set suitable (preselected) desired goals. The program would then satisfy these goals in the order of their importance. Deviations from the different goals are multiplied by weighting factors. Thus the program attempts to obtain a solution for the problem

$$\text{minimize } \left[d = \sum_{j=1}^K P_j \left(\sum_{i \in \Omega_j} (w_i^+ d_i^+ + w_i^- d_i^-) \right) \right]$$

$$\text{subject to } C_i X_i - d_i^+ + d_i^- = F_i^*$$

$$AX = B$$

$$X_{\min} \leq X \leq X_{\max}$$

$$d_i^-, d_i^+, w_i^+, w_i^-, \geq 0, \quad i = 1, 2, \dots, N + L$$

where P_j is the j th priority factor, Ω_j is the number of objectives on the j th priority level, F_i^* is a desired goal of the i th objective, d_i^+ and d_i^- represent deviational variables from the desired goal, and w_i^+ and w_i^- are weighting factors.

10.2.2 Emergency Actions

In some power networks the system configuration is such that a large block of power is generated at locations which are remote from the load centers and/or the major portion of the high-voltage network. A limited-capacity transmission corridor is used to transmit this power to the rest of the network. A fault on this transmission corridor, however, may make the power system vulnerable to instability, which may result in cascading outages. Preventive action, as discussed in Section 10.2.1, requiring generation rescheduling to reduce the power transmitted over the limited-capacity corridor may be prohibitively costly. For this reason, emergency control actions are sometimes used to prevent cascading outages. The most common of these controls may be one or combinations of the following: generation tripping (or rejection), load rejection, and braking resistor insertion. They will be treated separately.

Generation tripping

The idea of tripping generation to maintain stability aims at reducing the excess transient energy, tending to separate a group of generators from the rest of the system. It is a rather drastic measure which is resorted to when the electrical power cannot be produced at a level high enough to maintain stability because of the fault on the limited-capacity transmission corridor. When it is used, there are several questions which need to be resolved. Among them are the following (1) when to initiate the tripping, (2) how much generation to trip, and (3) whether it is to be used with other controls.

If the generation tripping is implemented *after* fault clearing, another switching event is created so that stability assessment must be deferred to that instant. Thus, system conditions at that instant (e.g., generators' angles and speeds, network admittances) must be computed before the TEF method is used. The amount of generation to be tripped has to be in blocks of power equal to generation on specific units. For example, if six generators are at a plant, with each producing 375 MW, and the stability computations revealed that

425 MW needs to be tripped to maintain stability, two generators will have to be tripped. (Other controls, such as controlled load rejection, may thus be required as well.)

To illustrate some of the issues to be dealt with, a simple case will be investigated below.

Let us assume that at the initial operating condition, TEF stability assessment revealed that $\Delta V < 0$. Because of system configuration, the potential instability will be averted by generation tripping. The situation to be considered is that of reduction of mechanical power at node k which is preserved. (This is equivalent to having a number of generators at node k some of which are to be tripped.)

With the representation of node k , all the transfer admittances B_{ij} , G_{ij} , $i, j = 1, 2, \dots, n$ are unchanged. Only the diagonal element Y_{kk} may change (e.g., because of the change in x'_{dk}).

For the various system parameters the following subscripts are used:

k refers to generator k where generation is to be tripped

cr refers to the critical generators in the MOD

sys refers to the rest of the generators in the system

T refers to all the generators in the system

eq refers to the two generator equivalent

The following assumptions are made:

1. Generation tripping (at node k) occurs at the end of the disturbance, i.e., at fault clearing.
2. The inertia of the tripped generators is small compared to the inertia of the rest of the generators, i.e., M_{sys} and M_{eq} are approximately constant. This also implies that $\Delta M_k \ll M_{sys}$, and $\Delta M_k \ll M_T$, and that δ_o does not change appreciably due to the tripping of generation at node k . Thus, θ^{cl} is also unchanged due to generation tripping.

3. The change ΔP_{mk} is small compared to the total system generation. Thus

$$\left[\Delta M_k / M_T \right] \propto \left[\Delta P_{mk} / P_{mT} \right]$$

4. The change ΔP_{mk} due to generation tripping will not change the MOD. However, θ^u will change.

In addition, it is also assumed that the following information, required for the initial TEF stability assessment, is available: θ^u , θ^{cl} , $\tilde{\omega}^{cl}$; generator data; and network admittances. The generation tripping is used to change ΔV from a negative quantity to a quantity ≥ 0 .

To estimate the change in (ΔV) we note that the parameters which will change due to generation tripping, under the assumptions stated, are θ_i^u , $i = 1, 2, \dots, n$, and M_{eq} .

The change in M_{eq} due to generation tripping is

$$\frac{\Delta M_{eq}}{\Delta P_{mk}} = M_k / P_{mk} \quad (10.9)$$

Next, we estimate the change $\Delta \theta^u$ due to generation tripping. Using J as the Jacobian at the equilibrium point without generation tripping, and noting that the change in the mismatch functions $\Delta f_i = 0$, $i \neq k$, and $\Delta f_k = \Delta P_{mk}$, a first estimate of the change in the equilibrium point is given by

$$\begin{bmatrix} \Delta \theta_1 \\ \vdots \\ \Delta \theta_k \\ \vdots \\ \Delta \theta_n \end{bmatrix} = J^{-1} \begin{bmatrix} 0 \\ \vdots \\ \Delta P_{mk} \\ \vdots \\ 0 \end{bmatrix} \quad (10.10)$$

The estimate of $\Delta\theta$ can be improved by evaluating the Jacobian at the new value of θ ; then recompute $\Delta\theta$ with the revised value of J^{-1} .

In situations where accuracy is very important, the above procedure to evaluate $\Delta\theta^u$ due to generation tripping may not provide sufficient accuracy. In such cases, it is suggested that the UEP should be recomputed using the value of the UEP obtained without generation tripping as the starting point and incorporating the appropriate changes relating to P_{mk} in the mismatch functions.

Change in the energy margin (ΔV)

The change in the energy margin is given by

$$\frac{\Delta(\Delta V)}{\Delta P_{mk}} = \frac{\partial(\Delta V)}{\partial M_{eq}} \frac{\Delta M_{eq}}{\Delta P_{mk}} + \sum_{i=1}^n \frac{\partial(\Delta V)}{\partial \theta_i^u} \frac{\Delta \theta_i^u}{\Delta P_{mk}} + \frac{\partial(\Delta V)}{\partial P_{ik}} \frac{\Delta P_{ik}}{\Delta P_{mk}} \quad (10.11)$$

where

$$\frac{\partial(\Delta V)}{\partial M_{eq}} = \frac{1}{2} (\omega_{eq}^{cl})^2 \quad (10.12a)$$

$$\frac{\partial(\Delta V)}{\partial P_{ik}} = -(\theta_k^u - \theta_k^{cl}) \quad (10.12b)$$

$$\begin{aligned} \frac{\partial(\Delta V)}{\partial \theta_i^u} = & -P_i + \sum_{j=i+1}^n \left\{ C_{ij} \sin \theta_{ij}^u - D_{ij} [(\sin \theta_{ij}^u \right. \\ & \left. - \sin \theta_{ij}^{cl}) \frac{(\theta_{ij}^u - \theta_{ij}^{cl} - \theta_{i+j}^u + \theta_{i+j}^{cl})}{(\theta_{ij}^u - \theta_{ij}^{cl})^2} \right\} \end{aligned}$$

$$+ \frac{(\theta_{i+j}^u - \theta_{i+j}^{cl})}{(\theta_i^u - \theta_{ij}^{cl})} \cos \theta_{ij}^u \} \quad (10.12c)$$

The change in the transient energy margin can now be estimated by computing the change in each of the transient energy components.

- Kinetic energy

Let the values of M_{eq} and ω_{eq} , after generation tripping, be $\hat{M}_{eq}$ and $\hat{\omega}_{eq}$, and let $K = \Delta M_k / M_{cr}$.

We can easily show that

$$\hat{M}_{eq} = (1 - K) M_{eq}$$

and

$$\hat{\omega}_{eq} \simeq \omega_{eq} + K(\omega_{cr} - \omega_k) \quad (10.13)$$

The change in the kinetic energy is given by

$$\Delta V_{KE} = \left(\frac{1}{2} \hat{M}_{eq} \hat{\omega}_{eq}^2 - \frac{1}{2} M_{eq} \omega_{eq}^2 \right) \quad (10.14)$$

- Position energy

The change in the position energy component is given by

$$-[P_{ko} \Delta \theta_k^u + \Delta P_k \theta_{ko}^u + \Delta P_k \Delta \theta_k^u + \sum_{i \neq k} P_{io} \Delta \theta_i^u] \quad (10.15)$$

- Magnetic energy

Assuming that the transfer admittances (C_{ij}) are unchanged the change in the magnetic energy component is given by

$$-\sum_{i=1}^{n-1} \sum_{j=i+1}^n C_{ij} \sin \theta_{ijo}^u \Delta \theta_{ij}^u \quad (10.16)$$

- Dissipation energy

Again with the same assumptions, the change in the dissipation energy is given by

$$\begin{aligned} & -\sum_{i=1}^{n-1} \sum_{j=i+1}^n D_{ij} \left\{ \left[\frac{\theta_{ijo}^u - \theta_{ij}^{cl} - \theta_{i+j}^u + \theta_{i+j}^{cl}}{(\theta_{ijo}^u - \theta_{ij}^{cl})^2} \right] \left[\sin \theta_{ijo}^u - \sin \theta_{ij}^{cl} \right] \right. \\ & \left. + \left[\frac{\theta_{i+j}^u - \theta_{i+j}^{cl}}{\theta_{ijo}^u - \theta_{ij}^{cl}} \right] \left[\cos \theta_{ijo}^u - \cos \theta_{ij}^{cl} \right] \right\} \Delta \theta_{ij}^u \end{aligned} \quad (10.17)$$

- $\Delta(\Delta V)$ = sum of the changes in all the energy components

Amount of generation to be tripped

Let us assume, as above, that the generation tripping takes place at fault clearing at plant k . The number of units to be tripped to change ΔV from a negative quantity to a quantity ≥ 0 , say, A , is to be estimated. Let P_u = initial power of *one* generator at plant k . A simple procedure to estimate the number of units to be tripped would be as follows:

1. For $\Delta P_{mk} = P_u$, compute $\Delta \theta^u$, $\hat{M}_{eq}$, $\hat{\omega}_{eq}$

2. Compute $\Delta(\Delta V)'$, the change in the energy margin due to ΔP_{mk} .

Let $\lambda = | \Delta(\Delta V)' | / P_u$, and compute

$$\frac{| \Delta(\Delta V) |}{\lambda}$$

where $\Delta(\Delta V) = (A - \Delta V)$.

3. If $\frac{| \Delta(\Delta V) |}{\lambda} < P_u \rightarrow$ trip one unit

$$P_u < \frac{| \Delta(\Delta V) |}{\lambda} < \gamma P_u \rightarrow \text{trip two units}$$

where γ is a number slightly less than 2.0.

4. If $\frac{| \Delta(\Delta V) |}{\lambda} > \gamma P_u$

repeat the above steps for $\Delta P_{mk} = 3P_u$ to determine whether three or four units should be tripped and so on.

Load rejection

Assuming that the load is represented by constant impedance, load rejection is modeled by a sudden change in the impedance connected to some load bus, which in turn represents a sudden change in some elements of the Y -bus, i.e., in some of the G_{ii} , C_{ij} , and D_{ij} terms. This will result in a change in θ^u .

With no change in P_{mi} , M_i , and again assuming that the load rejection occurs at the instant of fault clearing (i.e., θ_i^{cl} and ω_i^{cl} will not change), the change in the energy margin will be due to the change in potential energy.

$\Delta \theta^u$ due to load rejection

$$\begin{aligned}
 f_i &= P_{mi} - P_{ei} \\
 &= (P_{mi} - E_i^2 G_{ii}) - \sum_{j \neq i} (C_{ij} \sin \theta_{ij} + D_{ij} \cos \theta_{ij}) \\
 \Delta f_i &= -E_i^2 \Delta G_{ii} - \sum_{j \neq i} (\Delta C_{ij} \sin \theta_{ij} + \Delta D_{ij} \cos \theta_{ij})
 \end{aligned} \tag{10.18}$$

Let J = Jacobian for conditions without load rejection

The change $\Delta \theta^u$ due to load rejection is given by

$$\begin{bmatrix} \Delta \theta_1^u \\ \Delta \theta_2^u \\ \vdots \\ \Delta \theta_n^u \end{bmatrix} = J^{-1} \begin{bmatrix} \Delta f_1 \\ \Delta f_2 \\ \vdots \\ \Delta f_n \end{bmatrix} \tag{10.19}$$

Again the estimate of $\Delta \theta$ can be improved by another iteration involving reevaluating the Jacobian at $(\theta^u + \Delta \theta^u)$ and repeating the computation of $\Delta \theta^u$.

It should also be pointed out that when greater accuracy is warranted, a procedure similar to that suggested in connection with generation tripping can be used. Evaluation of the change in θ^u can be made by including ΔP_{mk} in the mismatch function.

In the above derivation, the total change $\Delta \theta_i^u$ due to all the changes G_{ii} , C_{ij} , and D_{ij} are computed. To compute the change in (ΔV) , it might be more convenient to compute the change $\Delta \theta_i^u$ due to a change in only some of these parameters. Such a change in θ_i^u will be designated by the appropriate subscripts. For example, $\Delta \theta_{i, G_{ii}}^u$ is the change in θ_i^u due to ΔG_{ii} only; $\Delta \theta_{i, C_{ij}}^u$ is the change in $\theta_{i, C_{ij}}^u$ due to all the changes ΔC_{ij} , $j = 1, 2, \dots, n$ ($j \neq i$).

$$\begin{aligned}
\frac{\Delta(\Delta V)}{\Delta G_{ii}} &= \frac{\partial(\Delta V)}{\partial P_i} \frac{\Delta P_i}{\Delta G_{ii}} + \sum_{i=1}^n \frac{\partial(\Delta V)}{\partial \theta_i^u} \frac{\Delta \theta_{i,G_{ii}}^u}{\Delta G_{ii}} \\
&= (\theta_{io}^u - \theta_{io}^{cl})(-E_i^2) + \theta_{io}^u \frac{\Delta \theta_{i,G_{ii}}^u}{\Delta G_{ii}}
\end{aligned} \tag{10.20a}$$

$$\begin{aligned}
\frac{\partial(\Delta V)}{\partial \theta_i^u} &= -P_i + \sum_{j=i+1}^n \left\{ C_{ijo} \sin \theta_{ij}^u - D_{ij} [(\sin \theta_{ij}^u - \sin \theta_{ij}^{cl}) \right. \\
&\quad \left. \frac{(\theta_{ij}^u - \theta_{ij}^{cl} - \theta_{i+j}^u + \theta_{i+j}^{cl})}{(\theta_{ij}^u - \theta_{ij}^{cl})^2} + \frac{(\theta_{i+j}^u - \theta_{i+j}^{cl})}{(\theta_{ij}^u - \theta_{ij}^{cl})} \cos \theta_{ij}^u \right\}
\end{aligned} \tag{10.20b}$$

$$\frac{\partial(\Delta V)}{\partial C_{ij}} = \frac{\partial(\Delta V)}{\partial \theta_i^u} \frac{\Delta \theta_{i,C_{ij}}^u}{\Delta C_{ij}} \tag{10.20c}$$

$$\frac{\partial(\Delta V)}{\partial D_{ij}} = \frac{\partial(\Delta V)}{\partial \theta_i^u} \frac{\Delta \theta_{i,D_{ij}}^u}{\Delta D_{ij}} \tag{10.20d}$$

The change in (ΔV) due to load rejection will be given by

$$\Delta(\Delta V) = \frac{\partial(\Delta V)}{\partial G_{ii}} \Delta G_{ii} + \frac{\partial(\Delta V)}{\partial C_{ij}} \Delta C_{ij} + \frac{\partial(\Delta V)}{\partial D_{ij}} \Delta D_{ij} \tag{10.21}$$

Braking resistor insertion

The effect of the insertion of a braking resistor is analogous to that of load rejection

(but acts in the opposite direction). The change in (ΔV) is estimated by the same procedure as above: first as a change in the Y -matrix elements, then as a corresponding change in $\Delta \theta_i^u$ and (ΔV) .

We will next illustrate how the size of the braking resistor, needed to maintain stability, is estimated. Again in this example we will assume that the braking resistor is inserted at fault clearing, and its location is known (e.g., determined by sensitivity techniques). The following procedure is suggested to bring (ΔV) from $-b$ to ≥ 0 .

1. Let $R_{brake} = 1.0$ pu
 Compute ΔG_{ii} , ΔC_{ij} , ΔD_{ij} , $i, j = 1, 2, \dots, n$ due to the insertion of R_{brake} .
2. Compute $\frac{\partial(\Delta V)}{\partial G_{ii}}$, $\frac{\partial(\Delta V)}{\partial C_{ij}}$, $\frac{\partial(\Delta V)}{\partial D_{ij}}$, and the corresponding $\Delta(\Delta V)$ for the 1.0 pu resistor insertion (see equations (10.20)).
3. The desired braking resistor size = $b/|\Delta(\Delta V)|$.

We note that in the above treatment the braking resistor is assumed to be inserted permanently. The common practice, however, is that the braking resistor is inserted for a finite period of time and then removed. In this case determination of the size of the braking resistor and the duration of its insertion become interrelated and more complex.

Combination of controls

The above discussion of the various controls (generation tripping, load rejection, and braking resistor insertion) has been simplified by two factors:

1. The assumption that the control action takes place at the instant of fault clearing. This assumption was made to focus our attention on the computational issues related to estimation of the change in (ΔV) due to the control action itself.

If the control is initiated at some other instant, conditions (i.e., θ , $\tilde{\omega}$, and (ΔV)) at that instant must be computed or estimated.

2. Each control was assumed to take place alone. It should be recognized, however, that combinations of these controls may be used. Whether their effects can be estimated by a simple superposition of the effect of individual controllers may depend on the size of the controls and their action. It is reasonable to assume that an iterative procedure may likely be adequate to account for the combined affects of more than one discrete controller.

In [5] an application of the TEF method is used in a transient which involves generation tripping and braking resistor insertion. The application is further complicated by (1) having the generation tripping occur after fault clearing, (2) reclosing on an end-of-line fault, and (3) braking resistor removal after final fault removal. The study reported in [5] is a good illustration of how the TEF method can be adapted to very complex situations. It should be recognized, however, that it is difficult to generalize the procedure used in such complex situations and project them to other applications.

10.3 EXPERT SYSTEMS AND DYNAMIC SECURITY ASSESSMENT

In Section 9.7 the need for on-line derivation of stability limits in some power networks is discussed. It is stressed that in today's system conditions, it is getting increasingly difficult to study the combinations of outages and disturbances to ensure that NERC reliability criteria will be met at all times. The number of scenarios to be analyzed for prevention of cascading outages would be very large. Reference [1] of this chapter illustrates this complexity well when it points out that in what is essentially a radial transmission network connecting the Bruce generating plant to the rest of the Ontario Hydro network, *45% of the time there are two or more elements out of service*. This is an important characteristic of the problem of the prevention of cascading outages: identifying the safe operating regimes requires analysis of a large number of combination of outages and processing and manipulating of (what amounts to) a massive amount of data.

Even with the large amount of information generated to derive operating stability limits, a great deal of special operator knowledge is used as a guide to operate a stability-limited power system safely. This special knowledge is needed to handle such things as when changing system conditions, due to external (e.g., the weather) or internal (e.g., load pickup), may appreciably affect system security; when certain events are expected to precipitate others;¹ when recognizable trends which affect system security occur; which events or contingencies to guard against in certain situations; and so on. In essence the special operator knowledge helps overcome the large data requirement for decision making by recognizing which contingencies to analyze and which trends and patterns to focus upon. The contingencies, and their consequences, are often grouped into classes for classification purposes. Based on these classifications system security is assessed, and security-related decisions are made.

The above seems to indicate that, for stability-limited power networks, conventional stability analysis tools are presenting some problems to engineers supporting operating functions. Each stability study analyzes only one scenario (or disturbance sequence) at a time when many such studies are needed. The amount of the data required is too voluminous to be generated quickly and needs good organization and classification for swift decision making regarding security. Thus, it is sometimes felt that operations planning functions are supported by tools that are inadequate for the job at hand. What is needed are tools which would (1) allow quick determination of stability limits, (2) have the ability to compute the effect of changing system parameters and/or conditions on these limits, (3) and are suited for organization for group classification. Such tools would bring security-related analysis and decision-making closer to real time and exploit the special knowledge about the power network which the system operators have. The new tools call

¹ The expression "sympathetic outages" is sometimes used by system operators.

for an imaginative use of a combination of the TEF method and expert systems to obtain a new framework for dynamic security assessment.

The TEF method attributes which are particularly valuable to the issue at hand are the method's ability to express the degree of stability in terms of a quantifiable index (the transient energy margin ΔV) and its flexibility to obtain analytical sensitivity information, for the simulated contingencies, on how the energy margin is affected by changing system parameters or conditions.

The use of the expert systems technique offers the opportunity to capture the thinking of the power system operators and how they view power system security (and how they deal with it). In addition, the technique gives the analyst a good framework for organizing a large amount of data to provide information on trends in system security and in grouping of system dynamic behavior patterns for classification purposes for decision making.

Use of expert systems in power system security assessment has been proposed by several researchers, mostly in connection with static security assessment (i.e., to determine whether in the postdisturbance operating conditions no constraints are violated). However, only a few applications of expert systems to dynamic security assessment have been proposed in the literature [6]; even a recent survey by a Task Force of the CIGRE SC38 suggested that an on-line expert system security scheme is very much needed [7]. Among the few applications suggested in the literature are the use of inductive learning to construct decision trees for classification of the transient stability status based on critical variables such as plant generations and bus voltages [8], determination of the locations and amounts of load shedding in order to stabilize the power system following a fault [9], and the use of rules to help decide the complexity of the system model for transient energy function calculations [10]. In the next subsection a proposed approach, in which the authors have been involved, is outlined.

An expert systems approach

In [11] C. C. Liu has raised two issues which must be dealt with when expert systems are to be used for on-line dynamic security assessment. The first is whether expert systems technology is sufficiently mature for an on-line implementation of security assessment functions. The second is whether there is sufficient logic, or expertise component, in security assessment to justify an expert systems approach. The approach proposed below addresses these two issues and outlines the structure of the required knowledge base and how its contents are processed for security-related decisions. In this approach, the operator requirements in dynamic security assessment which can be addressed by expert systems were first identified and studied, followed by an assessment of the TEF method capabilities which can be utilized to meet such requirements. The development of the TEF expert system was then carried out which includes description of the structure and contents of the TEF knowledge base, the application rules, parameters, etc.

Use of TEF to build the knowledge base

The following two figures illustrate the use of the transient energy margin for security analysis.

The first, Fig. 10.1, is a one-dimensional example showing the variation of the energy margin with respect to a key operating parameter, in this case the flow on a key transmission interface. The second, Fig. 10.2, shows the security domain as a function of two operating parameters. There are four sets of knowledge bases, to be generated with the help of the TEF method, which can identify the security domain. These knowledge bases are:

- *The dynamic security advisor performs a complete assessment of the system security status and provides suitable advice and means to deal with potential "insecurities."*

- *The what-if study analyzer* helps the operator/operations planner to simulate and analyze specific system situations and operating regimes from the security point of view.
- *The security domain calculator* assembles/defines the security boundary (domain) associated with a particular operating regime.
- *The policy development manager* provides a set of high-level tools which help in defining operating policies and determining the relative impact (degree of damage) of various "classes" of contingencies.

In the development of these knowledge bases, two concepts have been identified as important for dynamic security assessment (1) system vulnerability and (2) macro expert analysis.

System vulnerability

An operating regime may be secure for some modes of operation (for example, the current mode) but vulnerable with respect to some potential contingencies and/or changes in some operating parameters.

Figure 10.3 shows two operating regimes which are equally secure (in the energy margin sense), but regime 2 is more vulnerable to changes in the operating parameter (may represent flows, generation availability, weather concerns, etc.).

System security is often related to (or measured by) such criteria as:

- Degree of system strength, as indicated by the value of the critical energy V_{cr}
- Severity of contingency as indicated by the value of the energy at fault clearing (V_{cl}) for a fault disturbance
- Probability of contingency
- $\vdots$

Note that energy margin = $V_{cr} - V_{cl}$.

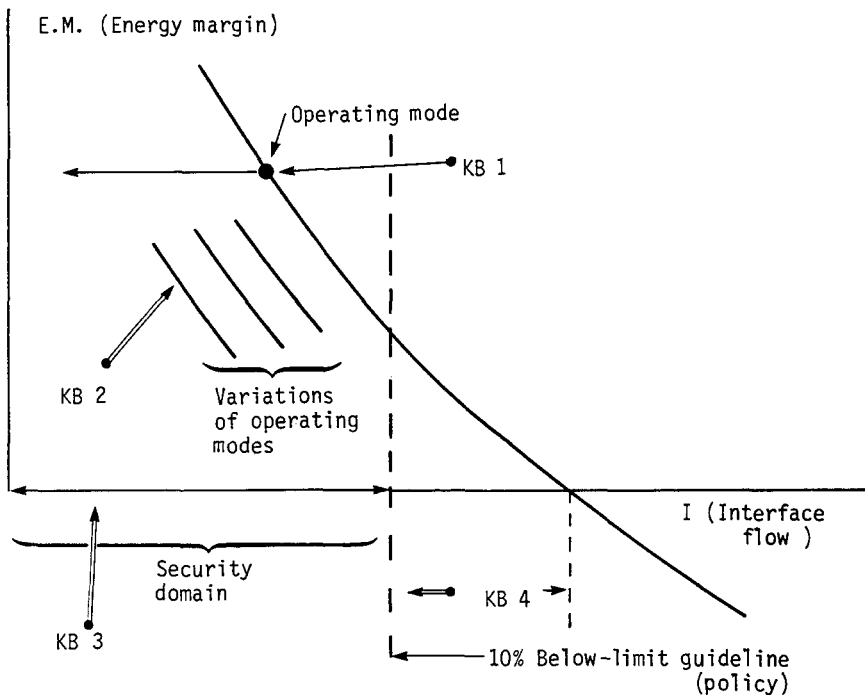


Figure 10.1 One-dimensional example (energy margin as a function of interface flow).

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System vulnerability is often related to (or measured by) such criteria as:

Sensitivity of a "security measure to changes in one or more operating parameters"; for example, for an operating parameter p , this sensitivity is given by (see Chapter 9 for detailed treatment of the subject).

$$\frac{\Delta(\text{energy margin})}{\Delta p}$$

$$\Delta p$$

Next contingency environment (for example, the system may survive a particular contingency but will become insecure for the next possible single or double contingency).

Difficulty of consequent restoration (a contingency may have a relatively smaller impact on security than other contingencies but is more difficult to deal with from the system restoration point of view).

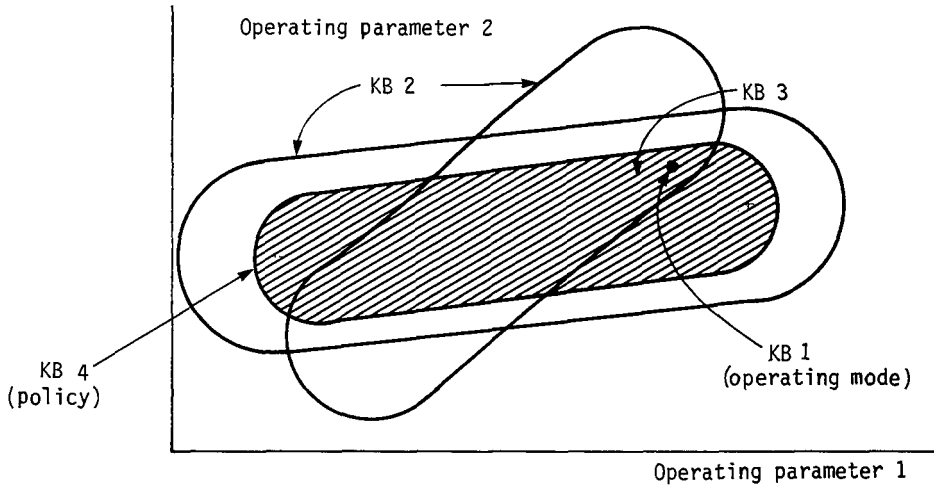


Figure 10.2 Two-dimensional example.

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Macro expert analysis

It is among the concepts which help capture the thinking of the system operator. It is recognized that it is neither feasible nor desirable to build a general expert system which analyzes every possible condition and parameter variation individually in order to make security related decisions. Instead, classes of regimes and ranges of parameter variations are established from knowledge of the system obtained from off-line analysis. This knowledge is checked by observing the system on-line behavior to arrive at the required decisions.

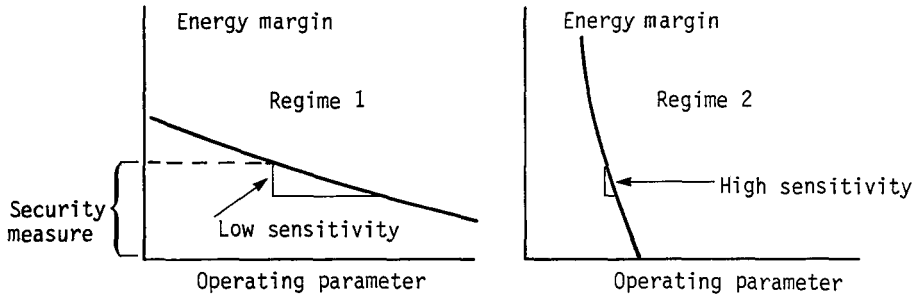


Figure 10.3 Energy margin sensitivities – system vulnerability. Reprinted from the *Proceedings of the 10th Power System Computer Conference* (1990); used with permission.

The authors of [12, 13] illustrate the development of the first knowledge base, i.e., *the dynamic security advisor*. This knowledge base constitutes a chain of five frames which, when developed, offers a framework for dynamic security assessment in the rule base.

In [13, 14] an expert system is proposed for dynamic security assessment of a stability-limited power network in North America. Artificial intelligence techniques are used to analyze the effect of contingencies on the security level and its rate of change with changing system conditions and parameters. These changes are to be computed using the TEF sensitivity analysis programs. Figure 10.4 illustrates a sample application. The expert system determines the loading trend and analyzes the data obtained from TEF to compute the trend of the security index. The trend of the control parameters can then be adjusted to maintain the security index within a safe range of values.

In the proposed application, the expert system knowledge base contains four types of information (1) the loading trend, (2) weather information, (3) off-line stability data, and

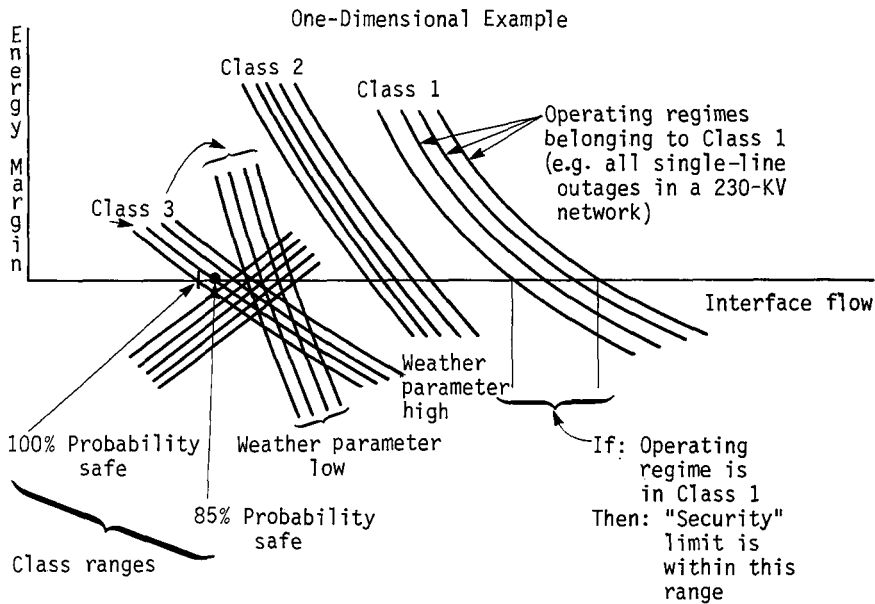


Figure 10.4 *Illustration of contingency classes.* Reprinted from the *Proceedings of the 10th Power System Computer Conference* (1990); used with permission.

(4) TEF sensitivity tables. Heuristic rules provide the control for the execution of the expert system to assess the system security, to detect and alert potential security violations, and to propose corrective steps to maintain a safe level of security.

10.4 PARALLEL COMPUTATION

Transient stability analysis of modern power systems usually requires numerous runs involving many scenarios: different initial network configurations and operating conditions, different disturbances, different postdisturbance system configurations, and so on. Especially when the stability analysis is performed to support operating functions,

there is a strong motivation to perform the stability analysis quickly. One obvious approach is to speed up the time to obtain a transient stability run, e.g., by using a more efficient computer code or a faster computer. Another approach is to have fewer cases to analyze to obtain the information needed to operate the system safely. These are valid concerns which have been dealt with in other sections of this book. In this section, we address the issue of how to expedite the transient stability analysis of a stream of runs by parallel computation to enhance the rate of processing the information and computation.

The use of parallel computing machines in power system analysis applications have received attention in the literature since the late 1970s [15–18]. Recent commercial availability of computing machines with multiprocessors (e.g., the Intel Hypercube) and the establishment of large computing centers with such facilities as the supercomputers have increased the interest in such applications. The problem which received the most attention has been the transient stability problem using time simulation (or related) techniques [19–24].

In performing power system transient stability analysis with the TEF method using parallel computer architecture, we must first deal with the question of whether an essentially serial program, written for a sequential machine, can be used with parallel computer architecture. We note that there has been an effort reported in the literature in which a computer code written for a sequential machine was converted to be used on an Intel Hypercube machine [24]. If a similar effort is to be used in connection with the TEF method, not only the parallelism in the TEF procedure is to be identified and exploited, but also the built-in parallelism in the disturbance scenarios included in a group of stability studies must be recognized and taken advantage of as well. These issues have been addressed in a recently completed study in which the authors have been involved [25]. The findings of this study, conducted on the IBM 3090–600E supercomputer at Cornell University [26], are included in the discussion below as a sample of issues to be dealt with when parallel computations are used with TEF.

10.4.1 Parallel FORTRAN Environment on the IBM 3090

The parallel FORTRAN environment on the IBM 3090 supercomputer consists of three features:

1. A parallel FORTRAN programming language [27] based on VS FORTRAN Version 2.1. This language has the capability to provide facilities for writing and executing programs in parallel on the IBM 3090. Parallelism can be achieved through (1) automatically generated parallel loops and (2) explicitly programmed parallel FORTRAN code by the user.
2. Parallel tasks, where two or more tasks of a single program are executed simultaneously and independently of each other. These tasks can be created by (a) parallel loops, where iterations in the loop can be executed independently (b) parallel cases, where groups of statements can be executed independently, and (c) parallel subroutines, where a group of subroutines, copying or sharing data in common, can be executed independently.
3. FORTRAN processors are created in the user's own virtual machine and specified by the user at execution time. During execution, tasks waiting to do work are assigned to the next available FORTRAN processor. Parallel execution is achieved when two or more tasks execute concurrently on different processors. Each FORTRAN processor gets mapped on its corresponding virtual processor in the virtual machine. An additional virtual processor processes all the operating system commands. The virtual processors are assigned to real processor for execution by an operating system scheduler. Figure 10.5 shows a schematic description of the virtual machine configuration.

10.4.2 Modifying the TEF Program Code

As a first step toward efficiently modifying the TEF program to suit the parallel environment, an IBM version of the code was obtained from the EPRI software center, and executed serially on the IBM 3090-600E machine; i.e., only one processor was requested. Various options available on the PF compiler were used to compile the existing TEF code. No code modification was done to the TEF program at this stage. The main reason for conducting this step was to determine the execution time performance of the existing program and to identify the TEF program subroutines which are candidates for parallelism.

Based on the control flow of the TEF program, a dependence graph was drawn. From the dependence graph, it was observed that maximum parallelism can be achieved by parallelism at the subroutine level because it is quite coarse grained.

Since having more than six parallel tasks (on a six-processor system) at any time will not improve speedup, the objective was to have a maximum of six parallel tasks at any particular time depending on the system saturation level. In addition, parallel constructs were identified which would minimize overhead as much as possible.

With these criteria in mind, two task management methods were identified. Type (1) parallelism represents the parallelism that exists in a single TEF program run for a given fault scenario and a specified postfault configuration. Type (2) parallelism represents the parallelism that exists when the TEF program is used for assessing transient stability of several fault scenarios and postfault configurations starting from the same base case pre-fault power flow.

Type (2) parallelism was chosen because in type (2) parallelism, the parallel tasks are originated and terminated only once, and no data are copied between tasks. In addition in type (2) parallelism, the common data in the existing sequential program are examined and found to be quite easily modified and rearranged such that some of it can be shared among tasks, and none of it has to be copied between tasks. Considering that the total size of the common data are more than 1 megabyte, this data management method can reduce a tremendous amount of CPU overhead.

Based on this discussion, a new flowchart of the TEF procedure is drawn and shown in Fig. 10.6. From the new flowchart, it is apparent that after a given base case power flow and machine dynamic data are read, and the pre-fault network admittance matrix (Y -matrix) is built, the program can be split into different independent run streams executing on different fault scenarios and postfault configuration data. This arrangement is also ideally suited for conducting multiple stability run streams starting from the same base case as a part of dynamic security assessment.

For the parallel scheme implemented in Fig. 10.6, some of the salient features regarding the parallel implementation are as follows:

1. Data Management

We sort data into two groups according to their usage in the TEF program. These groups are then put into two separate common blocks as discussed:

a. Shared data

This is the type of data computed before parallel subroutines are created and is not to be changed anywhere in the parallel subroutines.

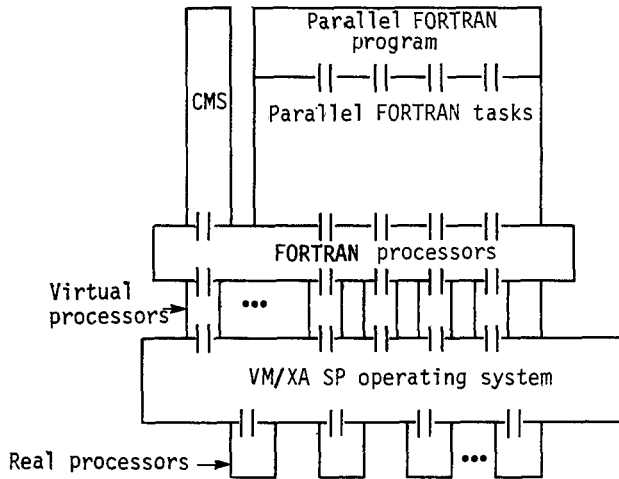


Figure 10.5 IBM 3090 virtual machine configuration. Reprinted from © *IEEE Transactions on Power Systems* 6 (1991); used with permission.

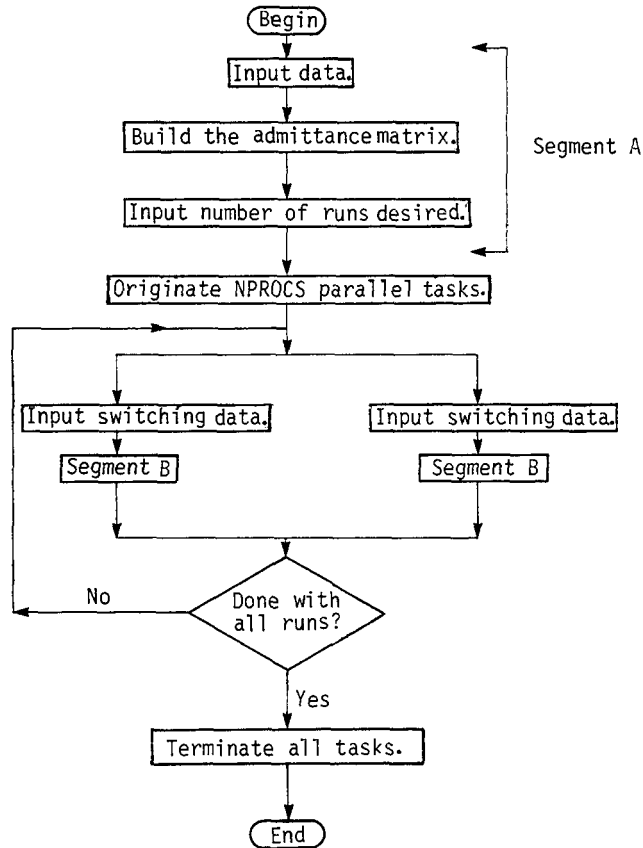


Figure 10.6 Flow chart of the new parallel TEF program. Reprinted from ©IEEE Transactions on Power Systems 6 (1991); used with permission.

b. Local data

This type of data reside in the local or private memory area of each independent or parallel task. When one task completes its operations, the local data area will be reinitialized before a new task is performed.

2. Task Management

In segment A of the parallelized TEF program, no parallel tasks are originated. To implement the proposed parallelism, a new program is written which creates a number of PF tasks according to the number of FORTRAN processors required at execution time. Each parallel task is then assigned to work on one of the replications whose total number is always bounded by six to minimize overhead. The synchronism between (among) tasks depends on the availability of the six physical processors. Because the operating system is time shared, all tasks may not begin executing at the same time, there may be some delay between tasks. As a result, in computing the elapsed time speedups the overhead incurred in waiting for free processors is added to the elapsed time needed to execute the parallel tasks.

3. Overhead Involved

One of the goals in parallelizing the TEF code is to minimize both CPU time and elapsed time overheads. This is done by choosing the optimum task and data management procedure as follows:

a. Parallel construct overhead

CPU time overhead in parallel constructs is proportional to the number of times tasks are originated and terminated, and also to the number of time data communication takes place between the master task and its originated task.

In the TEF program used, the control flow depicted in Figure 10.6 allows us to use the same task to analyze all the data without having to terminate and reoriginate any task at any point in the program.

b. Data management overhead

This overhead is taken care of using the procedure described in item a.

Test results

The new TEF parallel program was tested on a 50-generator, 145-bus, 649-line test system (see the appendix). Starting from a given base case power flow, several fault locations to analyze three-phase faults were selected. Faults of different duration and postfault configurations were analyzed. Timers were inserted in the sequential and parallel TEF program to record execution times of segments A and B. Two different speedups as described were calculated.

1. Speedup of segment B, S_1 .

$$S_1 = \frac{\text{segment B (sequential)}}{\text{segment B (parallel)}}$$

2. Overall speedup, S_2 . This speedup is expected to depend on the number of runs performed and is bounded by S_1 . It shows the overall real time savings achieved by the parallel TEF program

$$S_2 = \frac{(\text{segment A} + \text{segment B})(\text{sequential})}{(\text{segment A} + \text{segment B})(\text{parallel})}$$

Table 10.1 shows the execution times for both the serial and parallel versions of the code and the two speedups achieved.

TABLE 10.1 Parallel and Sequential Execution Performance and Speedups

Number of Cases Run	Sequential (sec)		Parallel (sec)		Speedup	
	Segment A	Segment B	Segment A	Segment B	S_1	S_2
3	0.54	5.79	0.50	2.14	2.71	2.40
6	0.54	11.80	0.50	2.15	5.49	4.66
10	0.54	19.31	0.50	3.52	5.49	4.94
20	0.54	38.82	0.50	7.09	5.49	5.19
30	0.54	58.06	0.50	10.52	5.50	5.31
50	0.54	96.00	0.50	17.49	5.50	5.36

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used with permission.

The results in Table 10.1 show that in the case when not all processors are utilized (only three processors were requested in this case), the speedup achieved is small. With all

processors utilized and the parallel task and common data management used, the parallelizable part of the TEF program is speeded up by a factor of 5.5 on the IBM 3090-600E machine which is very close to the theoretical maximum of 6. This corresponds to a speedup of 92%.

10.4.3 General Comments on Parallel Computation

The results presented in Section 10.4.2, and the study from which these results were obtained, must be viewed as preliminary. The issues discussed must be considered as representative of the issues to be dealt with when parallel computation of TEF is used. The study showed, however, that even though a significant part of the TEF program was not "parallelized", an overall improvement of 5.36 times in the speed of processing 50 transient runs was achieved over the time it would have taken to process 50 individual cases separately. Missing in the above study are the following important issues which had not been addressed (1) exploiting the inherent parallelism in the TEF procedure, (2) improving the efficiency of the TEF code by parallelizing the remaining segments of the code, and (3) searching for the optimum computer architecture for the transient stability assessment problem using TEF (the six-processor architecture in the above study was fixed).

10.5 ALTERNATE TRANSIENT ENERGY FUNCTIONS

In the literature on direct transient stability analysis many functions have been proposed to assess the transient stability of power networks. Some of these functions have been reviewed in the literature (see, for example, [28] for a review of energy functions). In this section, a selected number of the energy functions, which use concepts similar to those used by the TEF method, are briefly discussed.

10.5.1 Individual Machine Energy Function

The individual machine energy function concept was developed by the Iowa State University researchers [29]. It was reasoned that the separation of one or more generators from the rest of the system may depend primarily on the transient energy of individual generators or groups of generators. Development of a transient energy function for an individual generator was pursued. Analytical justification for using these functions for transient stability assessment has been provided using an invariance principle for ordinary differential equations.

As in the systemwide energy function, the transient energy of machine i includes kinetic and potential energy components after the initiation of the transient. This transient energy depends on the postdisturbance network and the position and speed of machine i in relation to other machines in the system. Thus, the components of transient energy of machine i vary along the postdisturbance trajectory. However, as that machine moves away from the rest of the system, its kinetic energy is being converted into potential energy. When all the kinetic energy is converted into potential energy, stability is maintained. Instability will result if the kinetic energy is not totally absorbed by the network.

The individual machine energy function is presented here for the classical power system model in the COI reference frame (see Section 2.2). Starting with equation (2.13), which is repeated here for convenience, we get

$$M_i \ddot{\omega}_i = P_{mi} - P_{ei} - \frac{M_i}{M_T} P_{COI}$$

$$\dot{\theta}_i = \tilde{\omega}_i \quad i = 1, 2, \dots, n$$
(10.22)

Multiplying the terms of (10.22) by $\dot{\theta}_i$ and rearranging we get

$$\left[M_i \dot{\tilde{\omega}}_i - P_{mi} + P_{ei} + \frac{M_i}{M_T} P_{COI} \right] \dot{\theta}_i = 0 \quad i = 1, 2, \dots, n \quad (10.23)$$

Defining t_s as the time for the system trajectory to reach the postdisturbance stable equilibrium point θ^{s2} , i.e., $\theta(t_s) = \theta^{s2}$, we obtain the transient energy function V_i for machine i by integrating (10.23) with respect to time. Since in the integration process t does not appear explicitly, the limits of integrating are θ_i^{s2} , ω_i^{s2} , and θ_i , $\tilde{\omega}_i$. Noting that $\tilde{\omega}_i^{s2} = 0$, V_i is given by

$$\begin{aligned} V_i = & \frac{1}{2} M_i \tilde{\omega}_i^2 - P_i (\theta_i - \theta_i^{s2}) + \sum_{\substack{j=1 \\ j \neq i}}^n C_{ij} \int_{\theta_i^{s2}}^{\theta_i} \sin \theta_{ij} d\theta_i \\ & + \sum_{\substack{j=1 \\ j \neq i}}^n D_{ij} \int_{\theta_i^{s2}}^{\theta_i} \cos \theta_{ij} d\theta_i + \frac{M_i}{M_T} \int_{\theta_i^{s2}}^{\theta_i} P_{COI} d\theta_i \quad i = 1, 2, \dots, n \end{aligned} \quad (10.24)$$

The transient energy function thus obtained gives the energy interchange between machine i and each other machine in the system.

Critical transient energy for machine i

Since the potential energy varies along the postdisturbance trajectory, the question arises as to whether it has a maximum value along the trajectory, and whether this maximum value (if it exists) varies with different disturbances. In the work performed by the Iowa State University group, numerous simulations were conducted on several systems. From these simulations, the following conclusions were reached.

1. If the fault is not long enough for one (or more) machine(s) to become critically unstable, the potential energy of the critical machine goes through a maximum before instability occurs.
2. This maximum value of the potential energy along the postdisturbance trajectory of a given machine has been found to be essentially independent of the duration of the disturbance and the mode of disturbance.
3. This maximum value of potential energy is fairly "flat," i.e., constant.

Thus, whether the fault is barely sufficient to drive the critical machines to instability, or whether it is sustained so that additional machines may lose synchronism, it is the critical machines that go unstable *first*. In either case, the critical machines "see" essentially the same system, i.e., the same group of machines that remain stable. For this group, the network's ability to absorb their transient energy and convert it to potential energy is essentially constant. Instability will occur only if their initial energy exceeds that limit. This limit is called $V_{i\text{critical}}$.

It should be pointed out that the simulations conducted on the individual machine energy functions involved what is known as plant mode stability phenomenon. No interarea mode data have been given in the literature.

Stability assessment

By simulating a sustained fault (or a fault of long duration), the potential energy terms of equation (10.24) are computed for various instants, V_{PEi} , $i = 1, 2, \dots, n$. The values of V_{PEmax} are noted for the different machines (or groups of machines). These are the same as $V_{critical}$; i.e.,

$$V_{i\text{critical}} = V_{PEi\text{max}}$$

Instability occurs if, at the instant of fault clearing, the total transient energy of the critical machines exceeds their critical energy, or

$$\text{if } V_i^{cl} < V_{i,critical} \quad \text{stable} \\ \text{if } V_i^{cl} > V_{i,critical} \quad \text{unstable} \quad (10.25)$$

V_i for nonclassical model

Consider generator i , the electrical and mechanical torques of which are modeled to any degree of detail desired. Defining the accelerating power for this generator as

$$P_{ai} = P_{mi} - P_{ei} \quad (10.26)$$

the equations of motion for generator i , in the inertial center frame of reference, are given by

$$M_i \ddot{\omega}_i = P_{ai} - \frac{M_i}{M_T} P_{COI} \quad (10.27)$$

Again following the procedure used in Section 2.2, we get for the transient energy function

V_i

$$V_i = \frac{1}{2} M_i \dot{\omega}_i^2 \bigg|_{\omega_{i1}}^{\omega_{i2}} - \int_{\theta_{i1}}^{\theta_{i2}} \left[P_{ai} - \frac{M_i}{M_T} P_{COI} \right] d\theta_i \quad (10.28)$$

where the subscripts 1 and 2 refer to the two instants of time t_1 and t_2 .

10.5.2 Transient Stability Program Output Analysis

Since time simulation of a sustained fault system trajectory is needed to determine $V_{i,critical}$, practical applications of the individual machine energy function concept have been slow to come. One of these applications has been the output analysis of a conventional time simulation transient stability computer program [30]. The basic idea is

in the makeup of the transient energy of individual machines. For a fault disturbance additional kinetic energy is created when the fault is on, but none is created when the fault is removed. Thus, after the disturbance is removed, the rate at which the transient energy of a given machine is converted from kinetic to potential energy is indicative of how much that machine has been affected by the disturbance. Similarly, if not all the transient kinetic energy of a given machine is converted to potential energy, the machine will lose synchronism with the system. The amount of unconverted transient kinetic energy is indicative of how unstable that machine is. This is how the output analysis program determines how the various machines are affected by the disturbance.

Assessment of the degree of stability

The approach of the potential energy to its maximum contains the information about the degree of stress on a disturbed but stable machine. This is best reflected in the slope of the potential energy. The time derivative of the potential energy determines the rate of energy conversion from kinetic to potential form or vice versa. The degree of stress ratio for generator i is defined as

$$R_{si} = \frac{\Delta \dot{V}_{PE \text{ ave } i}}{\dot{V}_{PE \text{ max } i}} \quad i = 1, 2, \dots, n \quad (10.29)$$

Assessment of the degree of instability

An index for the degree of instability is the ratio of the absorbed potential energy, ΔV_{PE} , to the injected kinetic energy. The ratio for an unstable machine is

$$R_{ui} = \frac{\Delta V_{PEi}}{V_{KEi} (t = t_{clearing})} \quad i = 1, 2, \dots, n \quad (10.30)$$

If R_{ui} is equal to 1, it means that the machine is stable and if it is less than 1, it means that part of the injected kinetic energy of that machine has remained unabsorbed.

A computer program, which is designed to examine the pattern of behavior of individual machines in a transient stability run, is available for use with the Philadelphia Electric Co. transient stability computer program. It is distributed by the Electric Power Research Institute to its members.

10.5.3 Individual Machine Energy Function for the Critical Generator and Cutset

The authors of [31] derived an individual machine energy function based on the synchronous reference frame. It has the form given in (10.31) for the classical power system model. All the terms are as defined in Chapter 2.

$$\begin{aligned}
 V_i = & \frac{M_i}{2 M_T} \sum_{j=1}^N M_j (\omega_i - \omega_j)^2 - \frac{1}{M_T} \sum_{j=1}^N (P_i M_j - P_j M_i) (\delta_{ij} - \delta_{ij}^{s1}) \\
 & - \sum_{j=1}^N C_{ij} (\cos \delta_{ij} - \cos \delta_{ij}^{s1}) \\
 & + \sum_{j=1}^N \int_{\delta_i^{s1} + \delta_j^{s1} - 2\delta_o}^{\delta_i + \delta_j - 2\delta_o} D_{ij} \cos \delta_{ij} d(\delta_i + \delta_j - 2\delta_o) \quad i = 1, 2, \dots, n
 \end{aligned} \tag{10.31}$$

This energy function was used in [31] to develop a stability measure based on a multimachine equal-area criterion. In this technique, the critical generator and the

corresponding critical cutset of the network² are identified as the last generator within the critical cutset that crosses its potential energy absorbing capability.

The multimachine equal-area-based stability measure is constructed by finding the accelerating and decelerating energy of a particular machine in the power system.

Authors of [31] applied this stability criterion to the 17-generator test system (see the appendix) and reported good transient stability assessment results.

10.5.4 The Partial Energy Function

The concept of the partial energy function was recently introduced by Stanton [33]. In it the individual machine energy function, similar to that presented in (10.24), is used to estimate the "distance from instability." For a generator i , represented by the classical model, the partial energy function between two system conditions a and b is given by

$$\begin{aligned} \frac{1}{2} M_i (\omega_i^b)^2 - \frac{1}{2} M_i (\omega_i^a)^2 = & (P_{mi} - V_i^2 G_{ii})(\theta_i^b - \theta_i^a) + \sum_{\substack{j=1 \\ j \neq i}}^n (V_i V_j) \left[\frac{\theta_i^b - \theta_i^a}{\theta_{ij}^b - \theta_{ij}^a} \right] \\ & \cdot [B_{ij}(\cos \theta_{ij}^b - \cos \theta_{ij}^a) - G_{ij}(\sin \theta_{ij}^b - \sin \theta_{ij}^a)] \\ & - \left[\frac{M_i}{M_T} \right] \sum_{k=1}^n \left[(P_{mk} - V_k^2 G_{kk})(\theta_i^b - \theta_i^a) \right. \\ & \left. - \sum_{\substack{j=1 \\ j \neq k}}^n (V_k V_j) \left[\frac{\theta_i^b - \theta_i^a}{\theta_{kj}^b - \theta_{kj}^a} \right] G_{kj}(\sin \theta_{kj}^b - \sin \theta_{kj}^a) \right] \end{aligned} \quad (10.32)$$

² The idea of a critical cutset had been previously proposed by Chandrashekhkar and Hill in connection with the so-called structure-preserving model energy function [32]. They proposed a technique based on identifying "candidate" cutsets, by computing $V_{critical}$ for each to obtain the smallest value; this value is used for transient stability assessment.

Thus, if point a is at fault clearing and point b is the point (assumed to be known) at which generator i loses synchronism, then equation (10.31) gives an energy margin for generator i which is indicative of the distance from instability for that generator.

The main point made in [33] is that the loss of synchronism point (i.e., the UEP) represents the parameters of an ill-posed phenomenon. As such it must be treated (and computed) carefully. Thus, in contrast to the procedure used in the systemwide energy function, the energy margin should be evaluated for each generator. This energy margin for generator i is given by (10.31) if point a is at fault clearing and point b is at the UEP. We note that this procedure requires the knowledge (or determination) of the UEP.

Reference [33] suggests a procedure to apply the use of the individual machine energy margin (or partial energy function) for transient stability-related security monitoring. The procedure is applied to the 17-generator reduced Iowa test system, and the results are used to compute the amount of generation change needed to maintain stability.

10.5.5 Hybrid Method

Recently the authors of [34] proposed a new technique which "combines both the time domain method and the TEF evaluation in solving stability problems and producing a stability index similar to the transient energy margin." The method is capable of handling detailed generator representation with exciters and nonlinear load models.

In the hybrid method the actual system trajectory is first computed using time domain simulation, then a transient energy margin is estimated and used as a stability index. The transient energy margin is computed as follows: the point having the maximum potential energy on the system trajectory is identified, a ray is established between the postdisturbance SEP and the maximum potential energy point on the trajectory, the point having maximum potential energy on that ray is found (the authors treat it as the intersection of the ray with the PEBS), the potential energy difference between the maximum potential energy points on the system trajectory and on the ray is

the transient potential energy margin, and finally the transient energy margin is obtained by subtracting the corrected kinetic energy from the potential energy difference.

The authors of [34] report that for the unstable case the system trajectory crosses the PEBS, and when that is detected the time domain simulation is terminated. For the stable case, the end of the first swing is usually detected by a large potential energy difference between the maximum potential energy points on the ray and on the system trajectory.

10.5.6 Modal-Based Transient Energy Function

In [35], a physically based decomposition technique is exploited to perform direct stability analysis of power systems using the energy function method (see also Chapter 3 of [28]). The coherency analysis decomposes the system into r areas. The centers of inertia of these areas form the slow subsystems; this gives $(r - 1)$ slow modes. The fast modes corresponding to the intermachine oscillations are associated with different areas.

The system transient energy function V is decomposed into components V_{slow} associated with the variables of the slow subsystem, components V_{fast} associated with the fast variables of the r -areas, and components $V_{\text{fast-slow}}$, which are a function of both fast and slow variables. The sum of V_{slow} , $V_{\text{fast-slow}}$, and V_{fast} of the disturbed areas provides a good approximation to the system energy function.

10.6 EMERGING APPLICATIONS

The applications which have been presented in the previous sections were selected not only because of their potential merit but also because they have been "screened" by some review process either in the publication stage or by funding agency reviewers. The applications to be presented in this section, however, will meet only the first of these two criteria. They are presented here based on their potential, but only the judgment of the authors is used for their selection. Thus, at this stage, and until they successfully meet the

scrutiny of other engineers, they should be considered plausible ideas that are good candidates for further exploration.

10.6.1 A Framework for Reliability Computation

First let us review some of the reliability concepts to see how TEF may "fit in." The reliability of the electric power supply refers to the degree of assurance in providing the customers with continuous service of satisfactory quality (e.g., voltage and frequency within established limits).

When dealing with reliability, the North American Electric Reliability Council (NERC) and its regional councils distinguish between adequacy and security. The latter has been dealt with in various sections of this book. The former is defined by NERC as "ADEQUACY is the ability of the bulk power electric system to supply the aggregate electrical power and energy requirements of the consumers at all times, taking into account scheduled and unscheduled outages of system components."

This is the subject of interest here, and it invariably deals with unscheduled outages and their probability of occurrence. Thus, probabilistic indices are used as reliability measures. Reliability criteria are established based on these indices, and the bulk power supply components are designed to meet these criteria (e.g., within an expected failure rate).

A computational method frequently used to obtain reliability indices is the so-called "state enumeration" method (see Chapter 4 of [36]). In this method the states of the system under investigation are assumed to be known, along with the rates of all the transitions between them. From these data the long-term probability, the frequency, and the mean duration of system failure are computed. The key steps include (1) defining the criteria for system failure, (2) performing a detailed analysis of the failure effects of *all* system states and classifying the states into work or failed categories, and (3) using the long-run probabilities of these states and their frequencies obtain the system failure

probability and the system failure frequency. It is to be noted that any system failure mode can, at least conceptually, be incorporated in this framework. For a stability-limited power network, however, the computational effort to perform the failure effects of all the possible states has been considered impractical.

Let us now consider that reliability computations are to be performed for a stability-limited power network. If the state enumeration method is to be used, we can envision that the TEF method can be used in the following phases of the computational procedure:

1. To identify the states with higher probability of instability or insecurity such as (a) greater sensitivity to changing system parameters, (b) low critical energy, etc.
2. Efficient computation of stability results for numerous disturbance scenarios (see for example, Section 10.4)
3. Classification and grouping of states according to their energy margin characteristics (see, for example, Section 10.3)

10.6.2 Combining the Use of TEF and Time Solution Methods

Researchers tend to think (and speak) of direct methods of transient stability analysis as a substitute for stability analysis using conventional time simulation programs. Practitioners, who must provide answers to practical problems with accuracy and reliability they can depend upon and defend, are not willing to give up the methods with which they are familiar for a *promise* of something which may be better in the future. The authors, who have considerable experience with both methods, tend to think of the two methods simply as tools which tend to complement each other. Since each tool has its strengths and weaknesses, each is more suited for certain applications which can take advantage of the strengths and at the same time are more tolerant of the weaknesses of that tool. This point of view is likely to have been reflected in the discussion of the applications which have been presented in this book.

To summarize briefly, the strengths of time simulation are no or little modeling or size limitations, user familiarity, good user support system, and "a culture" based on about three decades of industry experience. Its weaknesses are mainly the lack of qualitative information and analysis capability, the dependence on many "data points" to identify regimes of operation, trends, limits, etc. In contrast, the TEF method's strengths tend to overcome the time simulation's weaknesses, and its weakness lies where time simulation is strong.

While one tends to think that the two methods are suited for different applications and/or environments, a valid question can be raised as to whether both methods can be combined such that the user can take advantage of the strengths of each method. The authors feel that the answer should be yes, and we suggest a plausible approach for accomplishing this here. To illustrate, starting with a base case of 3000-bus, 300-generator system with all the controls, loads, etc., represented in any degree of detail desired by the engineer. After making a time simulation stability run for this case, the same initial condition is run with TEF (the system may be represented by a smaller size equivalent, fewer controls, simpler load model, etc.). The TEF method can now be used to provide certain answers which are not conveniently obtained with time simulation, e.g., the effect of certain parameter changes, stability limits, etc. This information can then be translated into equivalent information for the large system used in the time simulation study, and so on. Thus, a few base cases run with time simulation, augmented with TEF runs to obtain supplementary information (e.g., on sensitivities) can provide the analyst with a great deal of practical information on dynamic system behavior. Furthermore, this information can be generated efficiently and can be easily organized for security-related decisions.

The authors are aware that some analytical issues need to be satisfactorily addressed before this approach's potential is proven. Among these are relating a function of several variables (e.g., ΔV) which is used in TEF to a variable or parameter used in time

simulation (e.g., the rotor angle δ_i). However, it is felt that these issues are not insurmountable.

10.6.3 Interaction Between Various Modes of Oscillation

This is a problem which, at first glance, may appear to be an analytical issue of interest only to the researcher. This is only partly true, however, since it has serious practical implications. What is at stake is the understanding of (1) how the interarea mode may occur in a power system, (2) the system conditions favorable to its occurrence, and (3) the energy relations which can account for the network's influence on the system transient energy. As pointed out earlier in this book, occurrence of the interarea mode is not uncommon in power systems with heavily loaded transmission networks.

The first question to be answered is whether the interarea mode is one of system's basic modes of oscillation, or it is a combination of several modes. Recent investigations at Iowa State University [37, 39] seem to indicate that the latter is the case. They also show that nonlinearity, as indicated by second order terms, play an important role in determining the coupling between these modes and how they interact.

If these preliminary investigations are confirmed, it becomes obvious that system loading and fault severity (as indicated by fault-clearing time) will influence which modes become dominant, and hence which groups of generators are more severely affected by the disturbance. Time simulation results, together with modal analysis results, on very stressed networks confirm this observation [37–39].

If different modes are involved in the system's interarea mode then the machine groupings may be time dependent for different system conditions and disturbance scenarios. Again simulation results [37–39] substantiate these observations: fault duration, whether the area where the fault is located is importing or exporting power, location of transmission "bottlenecks," substantially influence which group of generators become the critical group and when this grouping is formed in the transient, and so on.

Pursuing this thought one step further, if the dominant interacting modes, which form the interarea mode are determined, and the machines participating in these modes are found, then the concept of the modal-based energy presented in Section 10.5.6 could be extended to account for the energy interaction among the modes in the interarea mode analysis. The energy of the relevant areas corresponding to the interacting modes could thus be used to determine the critical energy for stability assessment. These issues are of considerable practical significance and are under active investigation.

These concepts have been recently used by the authors to develop a new modal-based transient energy function. The generators belonging to the relevant disturbed areas are determined by the nonlinear interaction among the various natural modes of oscillation and the participation factors of the machines participating in these modes. Preliminary results indicate that this modal-based energy function gives excellent transient stability assessment results for stressed systems exhibiting interarea mode conditions [40].

10.7 CONCLUDING REMARKS

While this chapter is devoted entirely to the so-called advanced applications of the TEF method, the fact remains that TEF is basically a tool of analysis which deals with problems which exist today in a mature industry. There are existing tools with which the industry deals with these problems. It is very difficult to think in "futuristic terms" in such an environment. In this regard, the authors are not so much concerned with the built-in biases in the system which resist change (for commercial reasons, lack of time, personal difficulty in coping with change, etc.). It is implied in our thinking that these biases can be overcome. Therefore, it is very much discounted (or greatly reduced) in our thinking.

The greatest difficulty (we have discovered) which a new technique encounters is that it is hardly ever given a chance to prove its merit in an environment which can do justice

to it. The new technique is constantly forced to do things "the old fashioned way" and show that, in such an environment, it can flourish. This is admittedly an important step in the historical development of a tool such as TEF. It is equally important to recognize, however, that until this historical step is passed, and the TEF is accepted on its own merit, the TEF method's real potential cannot be proved. Only when the TEF is accepted will the truly advanced applications of the method emerge.

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APPENDIX

17—Generator Test System

Figure A-1 shows the major high voltage lines in the 17-generator test system. This test system represents a reduced version of a larger system which was used in a stability study run in 1978 by one of the electric utilities in the state of Iowa for a new generating unit.

The base load-flow system is a model of 862 buses and 1323 lines and transformers. Most of the transmission lines are 345 KV and 161 KV; some of the lines are 230 KV, 115 KV, or 69 KV. The base load-flow model was reduced by a network reduction program (steady state) to a model with 163 buses (of which 30 are terminal buses of the equivalent network) and with 304 lines and transformers (of which 69 are equivalent lines).

The Reduced Iowa System was selected to reproduce the "first swing" characteristic of the original full model. This was done with 17 generators: seven generators correspond directly to generators in the full model, four generators are single-generator equivalents of two-machine pairs from the same generating plants, and six generators are equivalent machines to represent inertia of machines cut off by the load-flow network reduction. The value of inertia constants of the six equivalent machines was reduced from very large values to values of 100 or 200 sec (on a 100-MVA base).

The inertia constant H in seconds and the direct axis transient reactance in per unit for the generators are given in Table A-1 to a 100-MVA base. As indicated Fig. A-1 shows the major high-voltage lines in the Reduced Iowa System; the classical model machine data are given in Table A-1.

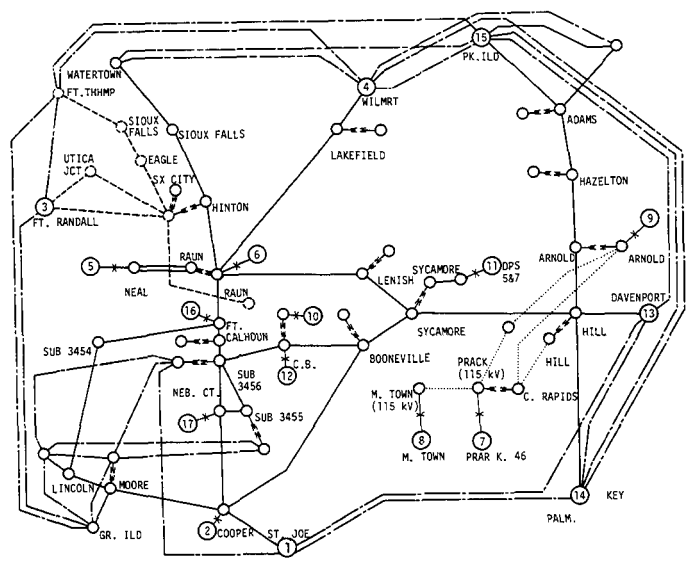


Figure A-1. 17-generator system (Reduced Iowa System).

TABLE A-1. 17-Generator System Machine Data

Power Flow Bus Number	Generator Parameters ^a	
	H (sec)	x'_d (pu)
3	100.00	0.004
6	34.56	0.0437
15	80.00	0.0100
27	80.00	0.0500
73	16.79	0.0507
76	32.49	0.0206
99	6.65	0.1131
101	2.66	0.3115
108	29.60	0.0535
114	5.00	0.1770
118	11.31	0.1049
121	19.79	0.0297
124	200.00	0.0020
125	200.00	0.0020
126	100.00	0.0040
130	28.60	0.0559
131	20.66	0.0544

^a On 100-MVA base.

50-Generator Test System

This test system is derived from an equivalent of realistic power systems in North America. The network captures the main features of the system near two major generator plants consisting of generators at bus 93, bus 99, bus 104, bus 110, and bus 111. These plants are connected to the rest of the network by a number of 500 KV and 230 KV circuits. The capacity of the transmission circuits in the area close to these plants is less than was originally planned for. Thus according to the loading pattern of these plants, stressed system conditions may result, causing a variety of complex system dynamic responses to disturbances. Some of the disturbances result in simple plant mode stability phenomena, while others result in interarea separation nodes where a large group of generators tend to separate from the rest of the system.

The system is comprised of 50 generators, 145 buses, and 453 lines and transformers. Table A-2 provides the machine data for the 50-generator test system; two-axis model machine data are also provided for six generators which could be represented in detail. These six generators are also equipped with the one gain one time constant exciters. The data for the exciters are given in Table A-3.

Network Data

The network power flow data for both systems can be obtained by sending a prepaid request to the following address:

Iowa State University Computation Center
Reference and Supplies Office
195 Durham Center
Iowa State University
Ames, Iowa 50011

The data can be requested on either 3-1/2-inch or 5-1/4-inch double density MS-DOS diskettes.

TABLE A-2. 50-Generator System Machine Data

Power Flow Bus Number	Generator Parameters ^a									
	H (sec)	x'_d (pu)	x'_q (pu)	x_d (pu)	x_q (pu)	x_l (pu)	S (1.0)	S (1.2)	τ'_{do} (sec)	τ'_{qo} (sec)
93	115.0366	0.024	0.03655	0.09842	0.09673	0.01237	0.0654	0.5743	8.50	1.24
104	73.8528	0.0122	0.0144	0.1016	0.0982	0.0081	0.2100	0.5500	10.00	1.50
105	84.3915	0.0208	0.03149	0.1144	0.1092	0.01102	0.1300	0.4096	6.61	1.50
106	56.261	0.03118	0.0472	0.17165	0.16377	0.01653	0.1300	0.4096	6.61	1.50
110	115.05	0.024	0.03655	0.09842	0.09673	0.01237	0.0654	0.5743	8.50	1.24
111	73.8528	0.0122	0.0144	0.1016	0.0982	0.0081	0.2100	0.5500	10.00	1.50
60	1.41	0.4769								
67	52.1796	0.0213								
79	6.65	0.1292								
80	1.2857	0.6648								
82	2.115	0.5291								
89	20.5602	0.0585								
90	0.7628	1.600								
91	1.6848	0.3718								
94	17.3424	0.0839								
95	5.4662	0.1619								
96	2.1216	0.4824								
97	5.4912	0.2125								
98	13.96	0.0795								
99	17.108	0.1146								
100	7.56	0.1386								
101	12.2844	0.0924								
102	78.4366	0.0135								
103	8.16	0.1063								
108	30.432	0.0248								
109	2.6622	0.2029								
112	12.2844	0.0924								
115	97.33	0.0024								
116	105.50	0.0022								
117	102.16	0.0017								
118	162.74	0.0014								
119	348.22	0.0002								
121	116.54	0.0017								
122	39.24	0.0089								
124	116.86	0.0017								
128	503.87	0.0001								
130	230.90	0.0010								
131	1101.72	0.0001								
132	120.35	0.0016								
134	802.12	0.0003								
135	232.63	0.0008								
136	2018.17	0.0001								
137	469.32	0.0004								
139	2210.20	0.0001								
140	899.19	0.0003								
141	1474.22	0.0001								
142	950.80	0.0003								
143	204.30	0.0023								
144	443.22	0.0004								
145	518.08	0.0018								

^a On 100-MVA Base

TABLE A-3. 50-Generator System Exciter Data

Power Flow Bus Number	K_A	τ_A	τ_C	τ_B	$E_{FD\text{MAX}}$	$E_{FD\text{MIN}}$
93	185.0	0.020	1	1	8.89	-2.00
104	253.0	0.015	1	1	8.86	-7.00
105	54.63	0.468	1	1	7.38	0.0
106	54.63	0.468	1	1	7.38	0.0
110	185.0	0.020	1	1	8.89	-2.0
111	253.0	0.015	1	1	8.86	-7.0

In 1991 the data are priced as follows, any change in pricing will be noted when the order is placed.

Diskette	— U.S. \$12.00
Tax	— U.S. \$ 0.60
U.S. shipping	— U.S. \$ 2.00
Overseas shipping	— U.S. \$ 7.00

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